EUROPEAN UNION AGENCY FOR RAILWAYS

Moving Europe towards a sustainable and safe railway system without frontiers.

European Fare Information Profile for Rail

Telematics TSI - Technical document - B16

Version 4.0

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A. Document management

A.1 Document properties

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Disclaimer:

This document is based on the initial working draft prCEN TS 16614-FM Profile FV (E)-v08]. This TD is an intermediate stage before publication of the profile by CEN and will be withdrawn once the relevant standard will be published.

A.2 Change management

Updates to this technical document shall be subject to Change Control Management procedure managed by the Agency pursuant:

- the applicable requirements in the reference TSI
- Art. 23(2) of the Agency Regulation

If necessary, working groups are created in line with Art. 5 of the Agency Regulation.

A.3 Configuration management

A new version of the document will be created if new changes are considered following the Change Control Management Process led by ERA.

More specifically:

- if there is a change in the requirements which influences the implementation
- if information is added to or deleted from the technical document
- adding test cases to the field checking in messages or databases.

Modifications will have to be highlighted, so they can be easily identified.

Disclaimer:

Specific legal references to technical documents and legal acts shall be revised after the enter into force of the Telematics TSI. In some sections this text can be highlighted.

A.4 Availability

The version in force of this document is available on Agency's Gitlab repository. Any printed copy is uncontrolled.

A.5 Application and actors in the scope

Date of entry into force of reference TSI.

This document applies to all the actors in the scope of the reference TSI.

A.6 Document history

Table 1 - Document history

Version	Date	Comments
4.0	06.02.2026	Version **initially released 10.6.2025 and officially published with** the TSI Telematics ***(Commission Implementing Regulation** (EU) 2026/253 of 6 February 2026)*

B. Acronyms, definitions and external references

B.1 Acronyms

Abbreviation	*Description*
IM	Infrastructure Manager
PRM	Person with reduced mobility
PRR	Passenger Rights Regulation
RU	Railway Undertaking
SM	Station Manager
TV	Ticket vendor
UML	Unified Modeling Language
XML	Extensible Markup Language
XSD	XML Schema Definition

B.2 Definitions

Terms contained in this document are defined in the ERA Ontology. Some terms and definitions are given in EN 12896 (Transmodel V6) and CEN/TS 16614 (NeTEx) apply, as below.

Abbreviation	*Description*
attribute	property of an entity

conceptual data model	description of a real world domain in terms of entities, relationships and attributes, in an implementation independent manner. It should provide a structure on which the rest of the development of an application system can be based
data domain	data structure (in this European Standard, a part of the Reference Data Model for Public Transport) made up of data related to each other, through the fact that there is a functional area or group of functions using this data set as a whole
data model	description of a real world domain in terms of data and relationships
entity	object (data) that has its own existence (as opposed to an attribute)
fare management	all activities related to the collection of money from passengers
functional area	arbitrarily defined set of activities, used, in this European Standard, to define the objectives and limits of the data model and exchange profile
interoperability	ability of (sub)systems to interact with other (sub)systems according to a set of predefined rules (interface)
logical normalised model	relational data model that is not fully normalised, i.e. does not completely follow the normalisation rules and thus may be redundant
object-oriented data model	data structure expressed according to principles that allow for a direct implementation as an object-oriented database, where information is represented in form of objects, i.e. respecting the principle of encapsulation meaning in particular that each data is accessed or modified through operations (methods) belonging to it

operations monitoring and control	all activities related to the transportation process, i.e. real-time functions related to the driving and transportation of passengers according to given instructions, including the monitoring of the driving process and its control in case of deviations, as well as all activities that support the driving process (traffic light priority, track switching, bay selection, advance/delay advice, etc.). Such functions are often assisted by computer-aided tools, known as Automated Vehicle Monitoring (AVM)
passenger information	all activities related to informing the users either about the planned or about the actual transportation services.
Profile	subset of a standard selecting the needed concepts (entities and associated attributes) for a specific use cases or set of use cases, and complemented by rules defined to restrict possibilities of divergent interpretations of open parts of the standard.
Access right	right to consume a Public Transport service or other service provided along with the transport
Validation	verification that an access right is valid, has been or is being consumed, and that this consumption is allowed, achieved by checking the validity of the passenger's travel rights and comparing them to the parameters collected by the previous inspections and/or inspection context
Tariff	(Transmodel) A particular tariff , described by a combination of parameters.
Fare product	(Transmodel) An immaterial marketable element (access rights, discount rights, etc.), specific to a CHARGING MOMENT .

B.3 External references

The referenced documents listed below are indispensable for the application of this document:

- For dated references, only the edition cited applies;
- For undated references, if any, the latest edition of the referenced document (including any amendments) applies.

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 12896, *Public transport - Reference data model* (Transmodel) Parts 1 to 10.

CEN/TS 16614-1 Network and Timetable Exchange (NeTEx) - Network description

CEN/TS 16614-2 Network and Timetable Exchange (NeTEx) - Timing information

CEN/TS 16614-3 Network and Timetable Exchange (NeTEx) - Fare description

CEN/TS 16614-PI Profile — *Passenger Information European Profile*

EN 15531-1, *Public transport - Service interface for real-time information relating to public transport operations - Part 1: Context and framework*

EN 15531-2, *Public transport - Service interface for real-time information relating to public transport operations - Part 2: Communications infrastructure*

EN 15531-3, *Public transport - Service interface for real-time information relating to public transport operations - Part 3: Functional service interfaces*

CEN/TS 15531-4, *Public transport - Service interface for real-time information relating to public transport operations - Part 4: Functional service interfaces: Facility Monitoring*

CEN/TS 15531-5, *Public transport - Service interface for real-time information relating to public transport operations - Part 5: Functional service interfaces - Situation Exchange*

European foreword

This document presents a Fare Information European Profile for Rail of the European Technical Specification CEN/TS 16614 known as “NeTEx”.

NeTEx provides a framework for specifying communications and data exchange protocols for organizations wishing to exchange scheduled Information relating to public transport operations.

NeTEx is made up of three parts defining a single European Standard, which provides a complete exchange format for public transport networks, timetable description and fare information.

Part 1 is the description of the public transport network topology exchange format. It also contains use cases shared with part 2, and modelling rules and the description of a framework shared by all parts.

Part 2 is the description of the scheduled timetables exchange format.

Part 3 is the description of the fare information exchange format.

Part 1 is fully standalone, parts 2 and 3 rely on part 1.

The XML schema can be downloaded from https://github.com/NeTEx-CEN/NeTEx and https://transmodel-cen.eu/index.php/netex/ provides available guidance on its use, example XML files, and case studies of national and local deployments.

This document is a so called fare profile, defining those elements of NeTEx to be used for the exchange of fare information.

This document is highly technical, and special care has been taken on keeping the text readable. This has been done through a set of editorial rules enhancing usual CEN writing rules as presented in 4.1

Introduction

Public transport services rely increasingly on information systems to ensure reliable, efficient operation and widely accessible, accurate passenger information. These systems are used for a range of specific purposes: setting schedules and timetables; managing vehicle fleets; issuing tickets and receipts; providing real-time information on service running, and so on.

This European Technical Specification specifies a Profile of NeTEx for fare information, applicable in particular to rail transport. NeTEx is intended to be used to exchange information between Public Transport organisations systems containing scheduled public transport data. It can also be seen as a complement to the SIRI (Service Interface for Real-time Information) standard (EN 15531 series), as SIRI needs reference data exchange in the scope of NeTEx before any possible real-time exchange.

Well-defined, open interfaces have a crucial role in improving the economic and technical viability of public transport Information Systems of all kinds. Using standardized interfaces, systems can be implemented as discrete pluggable modules that can be chosen from a wide variety of suppliers in a competitive market, rather than as monolithic proprietary systems from a single supplier. Interfaces also allow the systematic automated testing of each functional module, vital for managing the complexity of increasing large and dynamic systems. Furthermore, individual functional modules can be replaced or evolved, without unexpected breakages of obscurely dependent function.

NeTEx improves a number of features of public transport information and service management:

- Interoperability – the standard will facilitate interoperability between information processing systems of the transport operators by:

  (i) introducing common architectures for message exchange;
  (ii) introducing a modular set of compatible information services for real-time vehicle information;
  (iii) using common data models and schemas for the messages exchanged for each service;
  (iv) introducing a consistent approach to data management.

- Technical advantages include the following: reusing a common communication layer shared with SIRI for all the various technical services enables cost-effective implementations, and makes the standard readily extensible in future.

Profiles are an additional document to the standard which specifies additional rules for implementation in a given context. The profile contains information such as:

- Details of the objects used in an exchange
- Details on the options proposed by the standard
- Details on optional elements
- Precision on the codifications to be used
- …

The reason behind the detail of the profile specifications is that it facilitates subsequent implementations. A developer intending to implement a certain service or type of service in a given environment would just need to access the corresponding standard profiles, and implement the options and parameters already specified in the document.

The current document is a European NeTEx Profile for rail fare information (European Fare Profile for Rail (EFIP)) , designed to carry the minimum information set to feed passenger information services in a European wide context for rail. The scope of the fare p

1 Scope

This technical specification is a profile of the NeTEx CEN/TS 16614 series. It focuses on information relevant for exchanging fare information to support trip planners and pricing engines. This version of the European Fare Information Profile (EFIP) has a specific focus on rail transport modes, but remains a part of the future European Fare Information Profile.

The advent of mobile technology is transforming both the way fares are delivered, and the products that are offered for example with mobile apps, account based ticketing with consumption based capping. The profile includes capabilities to describe advanced electronic fares.

The overall NeTEx standard is dedicated to the exchange of planned data (network topology, timetable and fare information) and is based on Transmodel V6 (EN 12986) which sets out an extensive conceptual model describing many different functional areas of passenger information and transport operation in addition to fares. This profile focuses specifically on the exchange of fare data, in particular for rail operation.

The scope of the profile is defined according to the Directive (EU) 2016/797 of the European Parliament and of the Council of 11 May 2016 on the interoperability of the rail system within the European Union23 (the 'Rail Interoperability Directive'). The geographical coverage is defined as the Union rail system.

1.1 Profiles

Standards are by their nature and definition, consensus documents. In the case of the CEN and ISO standards, and more particularly in the field of traffic applications, these are established at an international level, with a significant number of different stakeholders involved in their development and discussion. This means that CEN and ISO standards take into account requirements that are much wider than a typical local implementations of such standards. The standard writing procedure itself makes updates of the document a long process, so standards are written to have the longest possible life and stability.

In practice a given implementation will target only a subset of a full standard and further more will typically make certain assumptions, impose further constraints on identifiers and other aspects of the implementation that reduce the work of implementation. The rules as to the contents and constraints can be formally described as a 'profile'.

Additionally, profiles can be accompanied by the definition of specific testing procedures to assess the conformance of the implemented solution with the profile.

A profile remains 100% compliant with the full standard.

From a practical point of view, profiles can be seen as an implementation guidelines. Instead of having to face the challenge of analysing the whole standard, discover the relevant parts for a given application, and then adjusting optional values and parameters for the intended application, profiles can be specified to address the needs of a particular application and then used for any subsequent similar initiative.

Fare structures and fare products vary greatly in their complexity; furthermore, some are very common while others are rare and there are many 'edge cases'. While even very complex fare structures can be expressed in NeTEx, the majority of fares are more straightforward. EFIP (European Fare Profile) focuses on commonly found tariff structures and products such as point to point long-distance, fares, urban zonal fares etc.

1.2 European Fare Information Profile

The European Fare Information Profile (EFIP) complements the European Passenger Information Profile (EPIP) . Common framework and Network elements that are already described in the EPIP are not described.

The European Fare Information Profile is concerned only with data relevant to feed passenger information service with fare informations. It excludes operational and back office information..

Typical use cases for the European Fare Information Profile are:

Provide fare products & fare prices as open data for third party use (e.g.trip planners, fare engines)

- Describe **available fare products** and their eligibility conditions.
- Relate **fare products to network and timetabled journeys** so trip planners can compute fare products and fare prices for trips, show available products for an area, etc.
- Allow the s**eparate exchange of prices** from fare structures & products.
- Expose a **justification of the fare** (Distance, discounts etc)
- Support both **machine readable** & **human readable** representation of validity parameters needed to understand the access rights to travel granted and to control fares.
- Include information about how/where products **can be bought**

- Include fares valid for specific and **multiple operators**.
- Provide information about fare routes, zones and networks for maps and online visualisations.

EFIP requires;

(a) Use of some common NeTEx Framework features (NeTEx Part1) and EPIP
(b) Use of use some common NeTEx network and journey components (NeTEx Part1 & NeTEx Part2), and EPIP .
(c) Use of the NeTEx Fare model (NeTEx Part3).

This profile has been designed to be as concise as possible and focus on commonly agreed needs to fulfil the requirements expressed in the Commission Implementing Regulation (EU) 2026/253 of 6 February 2026 on a technical specification relating to the telematics subsystem of the rail system in the European Union for interoperability of data sharing in rail transport (TEL TSI) and repealing Regulations (EU) No 454/2011 (TAP TSI) and (EU) No 1305/2014 (TAF TSI).

1.3 Type of fare covered by the profile

This document sets a profile for use of the NeTEX schema to describe common European fares. The profile aims to cover basic fare data for commonly found fares, covering all modes of transport with particular attention to the rail transport, including international and cross-border tariffs.

- Fare structures: point-to-point, zonal, distance, unit stage and flat fares.
- Product types:
  - Trips single and return trips, carnets of trips,
  - Period Passes; time passes, day passes , carnets of day passes, season tickets,
  - Supplement products: seat reservations, luggage transport: pets transport, bicycle transport, accompagnied car transport etc
- User types: adult child, senior , concession, group, family, etc
- Anonymous and Account based ticketing,
- Fare capping by period,
- Sales offers: s: single products and product bundles,
- Materalisation: ticketpapaer, plain-paper, smartcard, SMS and electronic,
- Pricing: absolute, derived and dynamic prices (yield managed prices).

Information on sales and conditions

- Distribution channles, fulfilment means, payment means ,
- Exchange and refund conditions.

- Purchase window.

The following fare 'types' are covered by this profile:

- Trip-specific fare
    - Single trip, P2P,
    - Single Trip, Zonal
    - Single Trip , Fare Stage
    - Short Single trip, Fare Stage
    - Day Return trip, P2p,
    - Period Return Trip, P2p
    - NRT tariffs (Series, route restrictions incl. routing areas, allowed trains, allowed Carrier codes, Fare tables, fare station list)
    - IRT tariff (tariffs, tariff ranges, Cards used with market prices, Exclusion types, Sales conditions, After sales conditions, Fare tables, Station/zone list)

- Pass
    - Day Pass, Zonal
    - Hour/Minute Timed Pass, Zonal
    - Season pass, Zonal
    - Season pass ,P2P
    - School term/year Pass, Zonal
    - Interrail pass (allowed trains, allowed Carrier codes, Fare tables)
    - Eurail pass (allowed trains, allowed Carrier codes, Fare tables)

- Carnet
    - Carnet of Single Trips
    - Carnet of Day Passes, Zonal

- Other
    - Trip Supplement (bike, luggage, reservation etc)
    - Discount card (Sales Discount)
    - Usage Based Discount
    - Pay as you go, capping right
    - Routing restrictions?
    - Special Tariffs (offer and its conditions, fares ranges, Supplements for specif trains, Number of passengers/accompanying passengers and their categories, Reduction card types, Exclusion types, Sales conditions, After-sales conditions Reservation fees, allowed routings)

- Modes
    - EPIP is for all modes of Public Transport, but rail is the main focus of this profile
    - additionally services covered by rail tariffs, such as coaches or ferries might are covered partially
    - the proposed selection of objects can also fit some needs for urban/periurban modes: Metro, Bus, Tram, Coach, Ferry or even Bike/E-scooter, Taxi.

- Variants by
    - Media: paper, app, etc,
    - Sales Channel,
    - User Type: adult, child etc,
    - Single Trip, Zonal,
    - restricted times of use.

1.4 (Placeholder)

(Intentionally empty)

1.5 General Approach

NeTEx uses Transmodel ‘access rightbased’ description of fares, rather than a ‘price-based’ or ‘product-based’ approach. This means that 'fares’ are described

- in a first place through access rights to transport services (by sets of rules and related parameters),

- fare products and a fare offer are defined in a second step (sometimes as combination of fare products provided for example by different operators),
- then pricing mechanisms are applied to the fare offer elements to provide the price.

To fully understand this specification it is necessary to take into account this 'philosophy'.

1.6 Data consistency and Quality

One of the goals of a profile is to simplify and enhance interoperability. But interoperability will not be achieved if the data just pretend to be compliant to the profile without fully following it and being of good quality (the data can be read, is consistent, and describes the reality).

Therefore the profile comes together with a set of consistency and quality checking rules specifically designed for NeTEx under this profile. Following these rules will not allow to guarantee a 100% quality of any dataset, but will prevent most of the quality issues usually encountered (when there is a way to automatically control it: for example a rule can control that a SERVICE JOURNEY has PASSING TIMEs, but will not be able to detect that a STOP PLACE name is not the right one).

2 Normative references

See section B.3.

3 Terms and definitions

See section B.2

4 Presentation conventions

The conventions used in this document are the ones used in the main NeTEx document.

4.1 Recap on Transmodel and NeTEx naming and presentation conventions

NeTEx follows certain conventions for presentation of technical terms:

- Transmodel conceptual model elements are shown in UPPER CASE, for example "LINE", "SCHEDULED STOP POINT".
- Concrete XML elements are shown in bold italic, for example, Line. Compound names are camel cased without a space and are never pluralised, e.g. "ScheduledStopPoint".

- Entity and class names are capitalised, as are properties of an element that are implemented as a XML child element, for example, “ScheduledStopPoint / Name”.
- Properties that are implemented as XML attributes are lower cased, for example “id”, “order “.
- One-to-one relationships that are implemented as versioned references are shown as simple attribute names, for example ScheduledStopPointRef, ZoneRef. Sometimes these are qualified by a name to distinguish them for example ParentZoneRef, FromPointRef, ToPointRef, etc.
- One-to-many relationships that are implemented as collections are lower cased and are always in the plural, for example, stopPoints, vehicleJourneys.
- Data types are shown in italic, built-in xml types are shown in lowerCamelcase, for example dateTime, normalizedString, integer. NeTEx Data types are shown in UpperCamelCase, e.g. LengthType, DirectionTypeEnumeration. Compound names are camel cased without a space, for example ScheduledStopPointIdType.

4.2 Use of colour

Elements in Transmodel and NeTEx UML diagrams are coloured according to their functional area. Additionally, in the EFIP abstract super classes are greyed out to rely on the concrete classes corresponding to the XML tag instances.

4.3 Additional presentation conventions

A few additional conventions were added for the specific context of the profile:

Table 2 – Additional presentation convention

«FK»	dataSourceRef	DataSourceIdType	0:1	Data system which originated data instance.
	created	*xsd:dateTime*	0:1	Date and time of creation of ENTITY. This is a specific addition (rule, constraint, etc.) for the profile.

No highlight means that it is identical or similar (same meaning

Light Grey highlight emphasis an additional rule or constraint adde

	changed	*xsd:dateTime*	0:1	Date and time of last change to ENTITY.

Blue highlight means that this attribute is not selected for the profile

Note: Unselected attributes (highlighted in blue) can still be used since they will still be valid for the NeTEx XSD. However they will not be read and understood by the receiver. Moreover they may be the source of issues for those deciding to use a reduced or optimised XSD. Therefore, using them should result of an agreement between involved partner.

4.3.1 *Caution on terminology*

Transmodel & NeTEx use the same systematic vocabulary of terms for specific domain concepts (LINE, NETWORK, VEHICLE, TRANSPORT MODE, etc). Often the usage will be straightforward and the meaning intuitive (POINT OF INTEREST, TIMING POINT, FARE ZONE, etc.).

- In some cases, these terms may appear non-intuitive or cumbersome (e.g. ACCESS RIGHT PARAMETER ASSIGNMENT, TEMPLATE SERVICE JOURNEY, SALES OFFER PACKAGE ELEMENT, etc) and it requires some effort to acquire a ready understanding of their precise meaning.
- In other cases, where the term corresponds to just one of several senses that a term has in common vernacular use (for example LINE, ROUTE, DIRECTION, TARIFF, FARE PRODUCT, etc), care should be taken to avoid confusion with the other vernacular senses.

4.4 XML schema technology

NeTEx uses W3C XML schema, a very well- established mainstream technology for describing data sets. W3C schema are used to describe documents for many of the most commonly exchanged data sets (HTML, Microsoft Office, etc) globally. Although relatively verbose compared to unvalidated formats (such as csv or JSON), it has a number of important advantages for the long-term management of complex data sets.

- Data quality: The schema provides a detailed description of the expected content which can be used to perform many basic validity checks for data quality and referential integrity automatically.
- Flexibility: The self-describing nature of schema documents makes it possible build applications that use data selectively, for many different purposes, and to evolve the schema without breaking backwards compatibility.
- Machine-readable- the self-describing format can be used to automate most of the legwork of implementation, building parser bindings that present the data to a program interface in a ready to use form.

- Widespread adoption: there are a wide range of tools and many people with the necessary technology skill sets to draw on.

5 Overview of access rights and component-based fares

5.1 Use of NeTEx for Fare data

NeTEx relies on the European Reference Data Model also called Transmodel, developed within CEN and which is a European norm (EN12896).

Transmodel provides a rich representational data model for describing both simple and complex fare products and their prices. Fare products are assembled from a set of low-level reusable components, building on other common data sets that describe the stops, lines and timetabled services of a transport network. Here we explain a few key principles of the Transmodel/NeTEx representation before discussing the EFIP scope further.

5.2 Access rights and the Transmodel/NeTEx representation

The essential characteristic of the Transmodel fares data model (EN12896-5) is that it is based on ***access rights*** i.e. the right to consume travel and other services, rather than directly on fare products. As mentioned above, NeTEx adopts this approach and for this reason, the explanations provided use UPPER CASE for the involved data entities and the text below is largely based on EN12896-5.

An **access right** granted to a customer is a part of a service that a user is entitled to consume, and of which the service provider (or another organisation) is able to control the consumption. Familiar examples include: to ride between two stops on a service provided by a specific operator, to travel within a specific zone for a specified period of time, to make a short bus journey of up to three stops, to have meal onboard, to sleep in a couchette, to take a pet or a bicyle as bagage, etc, etc . Various access rights may be combined in order to form immaterial **"fare products**" (for example, a "*single ride*" granted by a fare product called "simple ticket" or "*multiple trips during one month*" granted by a fare product called "monthly pass"), which are marketable sets of access rights.

One or more fare products may be associated with a "**travel document**" and materialised as **sales offer packages** that may be purchased.(e.g. a paper single ticket allowing only a "single ride" or an electronic card containing various fare products, or a mobile app that holds a ticket purchased in electronic form including a token such as bar code that can be used to validate its consumption).

Combinations of fare products and travel documents , with options selected for the customers' specific trip are sold to customers as "**customer purchase packages**". Each sold package is part of an individual "contract" with a particular customer.

A variety of fare **prices** may be attached to a particular access right, each distinguishing a price that applies under particular conditions. The same access right may be sold in a wide range of marketable combinations, and access right can cover not just use of the transport network, but other consumable services such as meals, use of wifi, parking etc.

The advantage of using the concept of an access right is that it allows a flexible composition of many different types of fare products, a clear definition of what is being priced, the rules for consuming access rights and the commercial conditions related to the sale.

The Transmodel / NeTEx component approach can be contrasted with classical models for representing fares, which typically conflate the different concerns of access right, fare product, fare price, type of travel document etc, in a smaller number of elements that cannot be recombined or reused.

In summary: the concept of an access right allows tariff elements to be treated as reusable components which can be assembled in different ways to create different products.

5.3 Defining Tariffs with Transmodel/NeTEx

Tariffs and Fare products are described in several steps (Figure 2), summarised as follows;

a) The relevant elements of the transport network (stops, tariff zones, etc,) and timetabled services (e.g. specific journeys with fare restrictions, etc). which may be used in a product are identified.
b) A tariff structure is defined in terms of the spatial and temporal elements that are available to access, along with any other features or aspects of use that are subject to differentiated pricing (e.g. class of use, eligible types of user, peak or off-peak consumption, etc.).
c) Sets of access rights (e.g. rights to use a zone, or travel between stops, etc.) are defined as validable elements.
d) Fare products are defined by combining the sets of access rights with additional conditions such as commercial conditions for purchase and refund, etc.
e) Fare products are combined into sales offer packages that describe the ticket media on which the products are available, along with any additional commercial conditions and the permitted sales channels.
f) Prices for the allowed combinations of packages and fare product options are defined.
g) Purchases of the fare offer elements can be recorded as sales transactions, each recording a set of fare product choices and a purchase price. (This last step is outside of the scope of the EFIP).

There are uniform common mechanisms for organizing and versioning sets of data, and for describing common such as validity conditions, etc.

EFIPEFIP adopts additional specifc conventions for organising tariffs and fare elements in a document.A fundamental principle of the Transmodel/ NeTEx approach is that prices are held separate to the elements that they price.

Another important principle is to reuse existing data elements from other Transmodel 'layers' wherever possible – thus for example, the same stop data is used for timetables and tariffs.

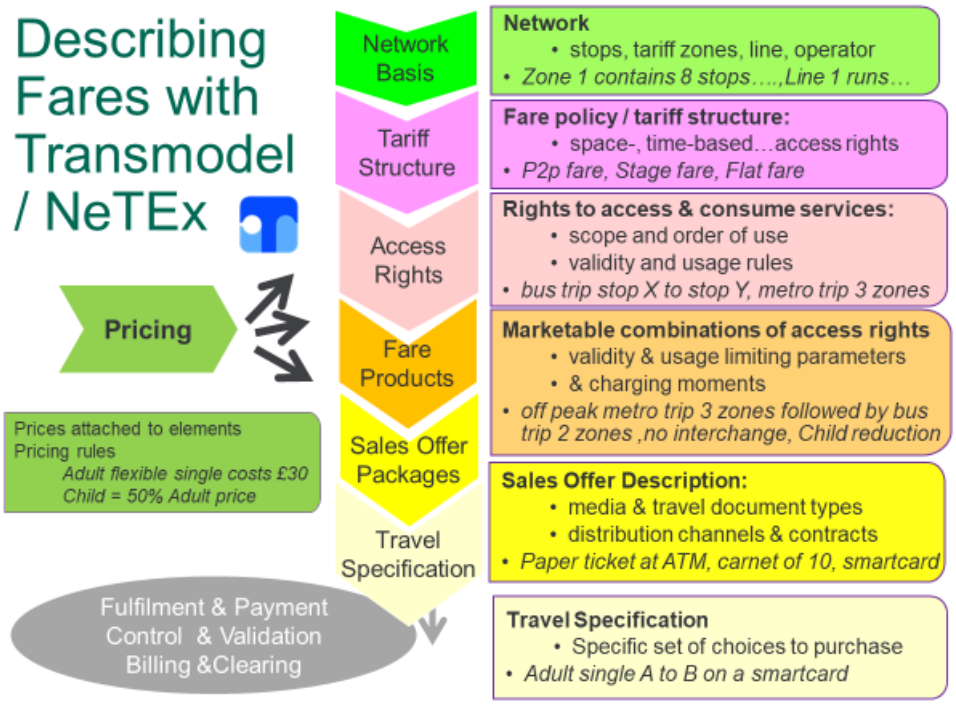


Figure 1 – Transmodel/NeTEx Fare Specification modules with examples

For each of the different stages of defining a fare (access rights, fare product conditions of use, offer commercial conditions etc), NeTEx provides a set of standardized reusable components.

The definition of a given fare product may contain multiple options. The resulting **fare product definition will include all the possible alternatives that someone buying the fare product(s) based on the tariff might select.** For example, a *standard metro ticket* might be available with different options *adult* or *child*; *peak* or *off-peak*; *zone 1* or *zone 2* etc(as opposed say to having different fare products for different combinations of options).

When a customer wants to know the price of a given fare product, a travel specification provides the set of specific choices (out of the available alternatives) for their intended consumption that determine the access rights, parameters and price of the fare. For example, 'An adult trip in zone 1', 'A child trip in zone 2' might represent different choices from the same product that has separate adult/child and zone1/zone2 options. Any set of specific choices can be given a price. This may either be done statically (i.e. by holding a separate price for every permitted combination), or by holding a base price for certain of the tariff components and apply a set of derivation rules to calculate the prices (for example child price is half the adult price).

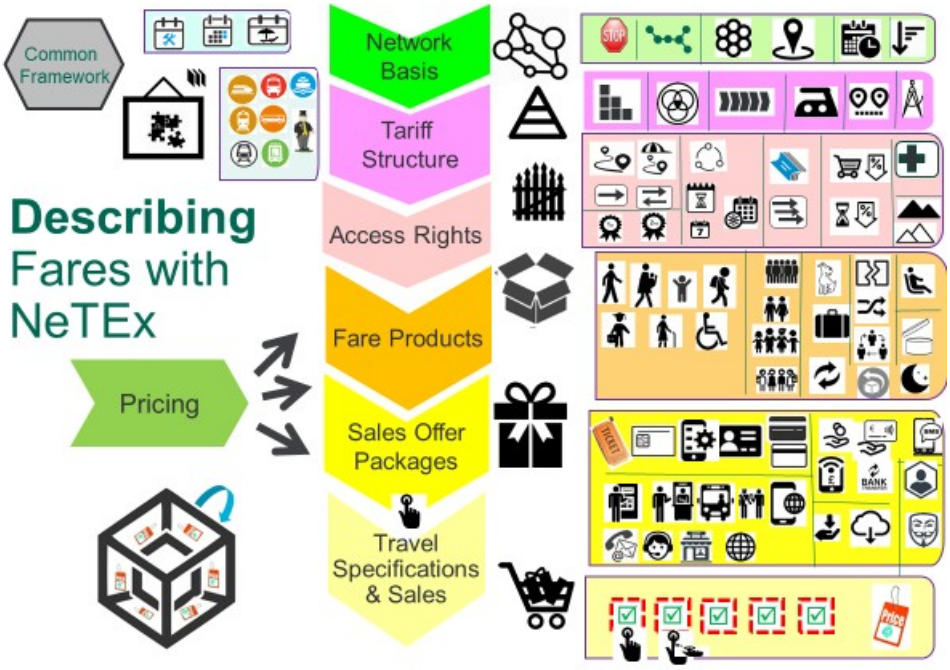


Figure 2 – Transmodel/NeTEx Fare Specification modules

5.4 Notation – Use of Infographics

Figure 2 above summarises the components of a NeTEx fare specification. The figure, as elsewhere in the presentation material, makes use of “infographics”; that is, a consistent icon is assigned to each concept and used to aid the understanding of complex material by readers.

5.5 Combining access rights

A further visualisation of how fares are specified with Transmodel/NeTEx components is given in the following diagram (UPPER CASE is used for Transmodel data elements)

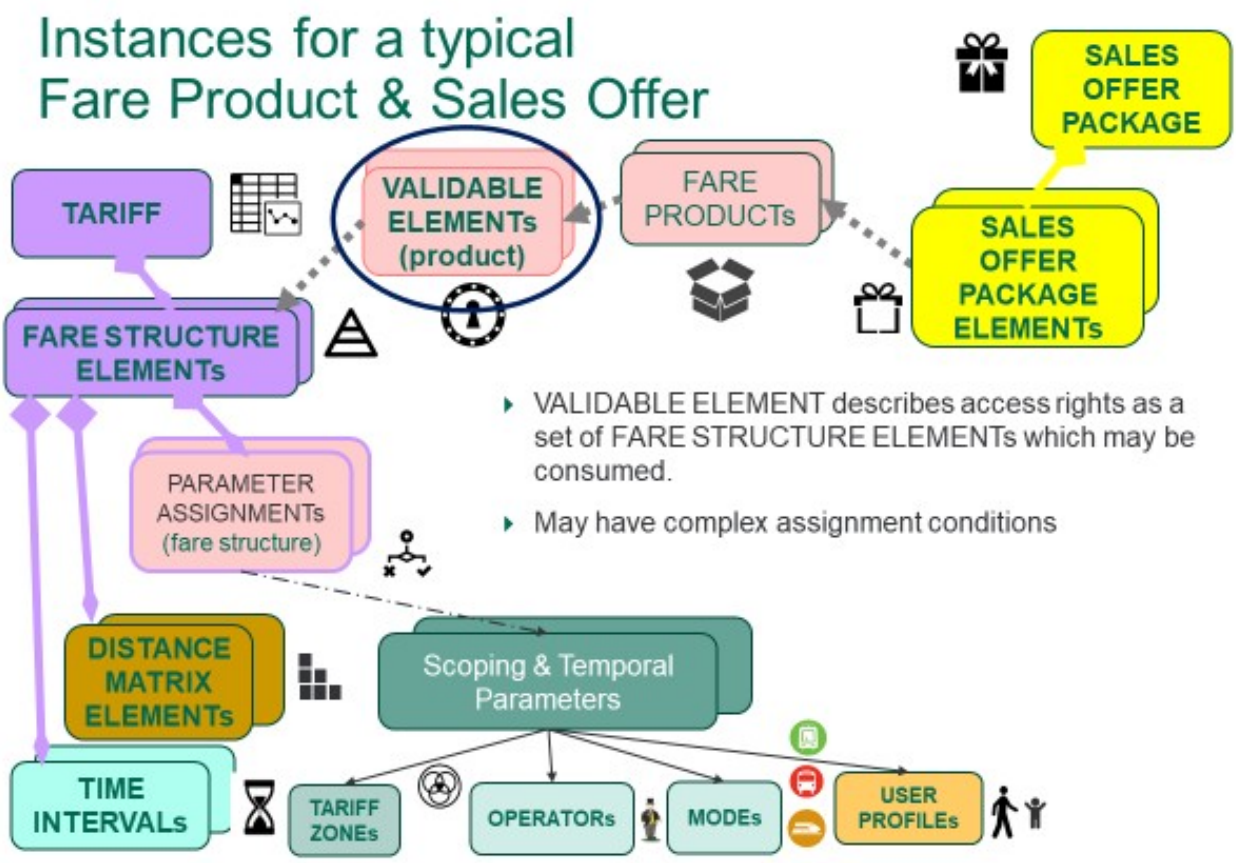


Figure 3 – Composing a product definition from fare structure elements

The figure above shows that to express **several options for one access right** (e.g a VALIDABLE ELEMENT) several types of parameters may be assigned (PARAMETER ASSIGNMENT).

This figure shows several data elements (in UPPER CASE) which are further described in chapter 8 of this document

- A TARIFF is a set of FARE STRUCTURE ELEMENTs that define access rights in terms of network elements, temporal components and other parameters. The access right and FARE STRUCTIURE ELEMENTs can be given individual prices if appropriate;
- A VALIDABLE ELEMENT groups a set of FARE STRUCTURE ELEMENTS that can be checked (e.g. a bus ride in Zone 1) when the passenger is travelling.;
- A FARE PRODUCT is the set of choices of elements and parameters that may be consumed as VALIDABLE ELEMENTS and that are marketed to the user as named product.;
- A SALES OFFER PACKAGE bundles up one or more FARE PRODUCTs as priced offering that may be purchased, and also specifies the media upon which it is available (e.g. paper, magstripe mobile app,

When a user purchases a product (a sales offer element), he chooses a permitted combination of specific options.

4

There may be further conditions that apply to a specific VALIDABLE ELEMENT, FARE PRODUCT or SALES OFFER PACKAGE. These also can be described using the same mechanism of parameter assignment. Transmodel/NeTEx has a general-purpose VALIDITY PARAMETER ASSIGNMENT element that allows further complex validity conditions to be constructed using simple elements related to for example to space or time (for instance ‘access right valid for metro line A form 10 p.m. to 2 a.m.).

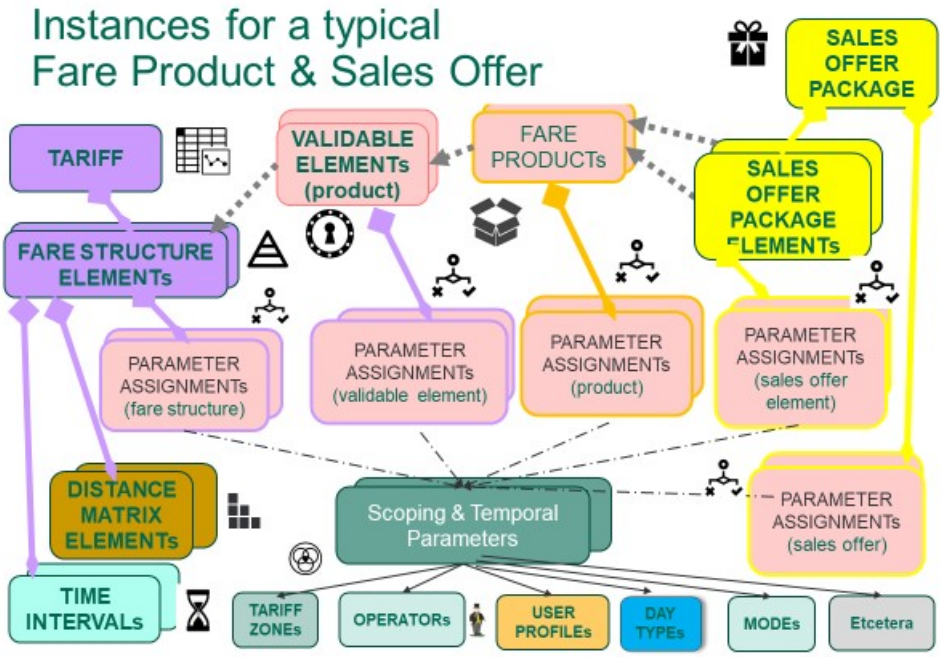


Figure 5 – Specifying complex conditions for a product

5.6 Combining Multiple Tariffs

Often multiple tariffs will apply to an itinerary. For example

i. the simple single trip fares,
ii. zonal fares for an operator's day and season pass products and
iii. tariffs marketed with other fare products, for example related to different modes (train only, train + bus, etc).

here may also be products from multiple different operators for different modes .

In particular, ***where multiple tariffs and fare products are applicable to an itinerary, Transmodel/NeTEx do not say which is "best"; it merely allows them all to be specified.*** Typically, it is the task of journey planning or fare finding software (whether in an online service, ticket machine or station office on-line terminal) to decide which fares it wishes to show as available and in which order they should be presented. Various criteria are possible – cheapest for the user; most flexible for the user; most profitable for the operator; satisfying regulatory strictures, etc.

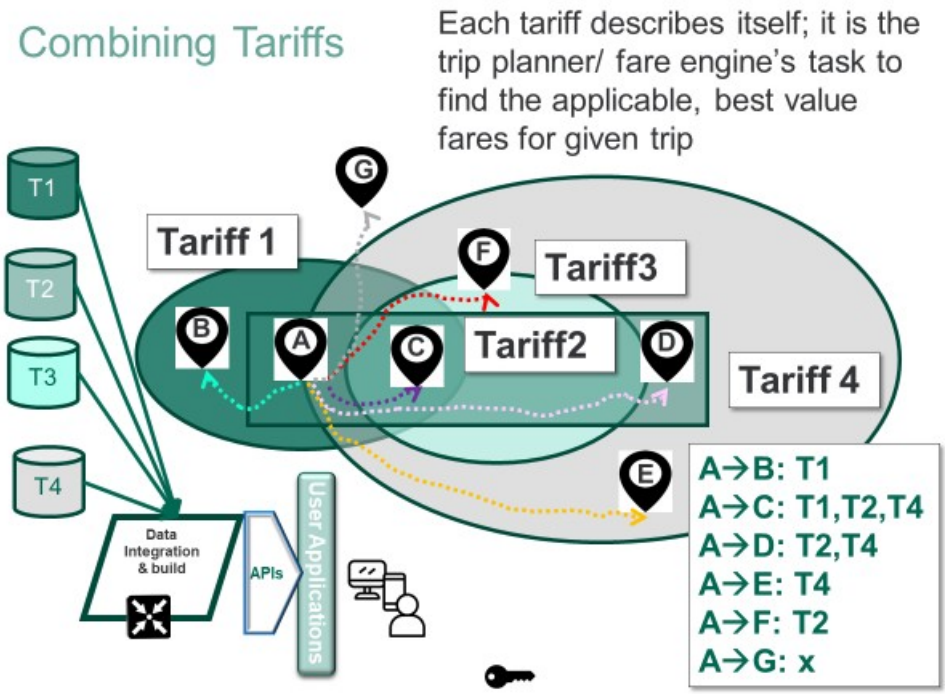


Figure 6 – Combining multiple tariffs

In order for trip planners or fare engines to make use of multiple tariffs, the fare structure, fare products and prices for each tariff should be separately defined, with all relevant conditions and restrictions. For their access rights they should reference the same network elements, for example stops and tariff zones as the trip. They should be packaged as separate XML documents that can be updated and distributed independently of each other. A trip planner/fare engine can then use these structure definitions to find the fare offers that are relevant for a given travel specification.

5.7 Other Introductory comments

5.7.1 *Organisation of elements in profile documents*

Different transport operators will choose to market their fares in different ways. For example, one operator might have separately named fare products for *Adult* and *Child* , with different price options for *peak* and *off peak* travel. Another operator might have four differently named farevproducts: *Adult Peak*, *Adult off peak*, *Child Peak*, and *Child off peak*. Essentially these are offering the same access rights, but are just

marketed in different groupings. The strength of a component based approach is that it allows great flexibility in describing the marketed packages. The same basic components may be used in a myriad of ways. However, in order to reduce the complexity for an implementor it is important to limit the ways in which components are combined to a practical subset and to organise the elements in a uniform manner. This is discussed further below.

5.7.2 *Handling large data sets*

Public transport data sets can be technically challenging to standardise, not only because they cover a complex domain, but also because they can be large, requiring consideration of the scaling implications for processing. Furthermore, data and subsets of data are often exchanged repeatedly between many stakeholders, necessitating a careful coordination of versions in order to achieve coherence (with necessary metadata to achieve this). The NeTEx framework includes mechanisms to assist with the management of large datasets including:

- **Frames**: A container mechanism that allows data to be modularised into coherent subsets. For example, “all the Season ticket products for a specific operator”, or “all the Network elements describing a line and its stops”. There are different types of version frames for exchanging different types of data (Figure 7).
- **Groups**: Groups of elements of a specific type may be grouped in named lists, allowing their repeated reuse as sets, and the assignment of common properties. For example, groups of sales offerings may be given shared properties, so that many similar products can be efficiently defined just by their additional properties.
- **Versioning and Validity conditions**. Fare management especially for multi-operator products will typically involve the exchange and aggregation of data from multiple stakeholders. There will be successive versions of data over time with updates having a specific validity period.
- **Systematic global identifiers**. In order to integrate diverse data sets a persistent, globally unique way of identifying elements of any type is needed. NeTEx employs a powerful uniform system of code spaces based on W3C namespaces.
- **Referential integrity checks:** Typically, a given data set will require a coherent set of elements all to be present in order to be usable. NeTEx makes extensive use of XML’s built in validity-checking mechanisms to validate internal cross-references within a document to ensure completeness. For external references to entities in other documents a uniform approach (including versioned references) is used that facilitates programmatic checking.

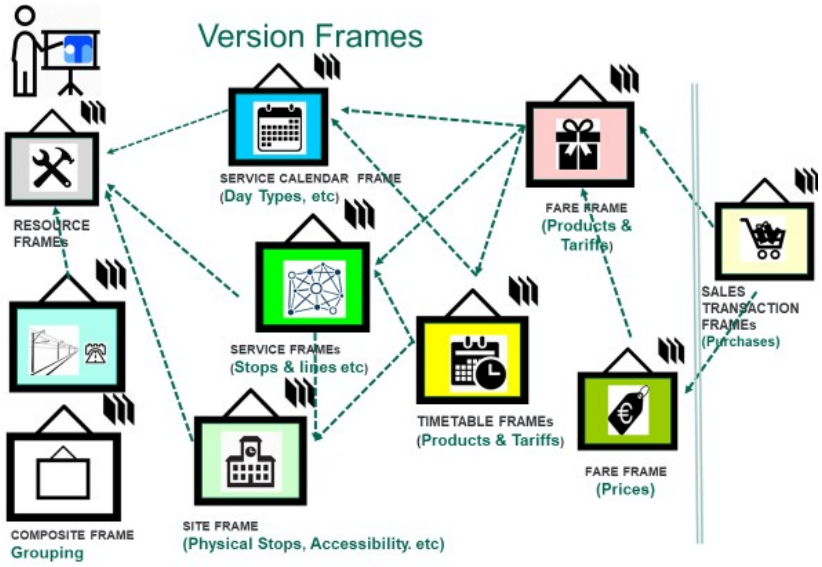


Figure 7 – Types of Version frame

6 EFIP Functional scope

This part describes the functional scope for a EFIP Fares profile, in particular for rail.

6.1 Primary use case for EFIP

Fare and tariff data may be used in different stages of fare management and in many different ways. For example, to plan and set tariff structures and prices; to create published printed fare tables; to provision ticket machines; to inform journey planning engines of trip prices; to support validation and control of passengers, etc. The Transmodel and NeTEx specifications set out a number of potential use cases that characterise the requirements.

Note however that the EFIP is concerned primarily with just one specific use case;

- ***To provide fare products & fare prices as distributed open data for third party use (by journey planners, fare calculation engines, etc.).***

The content of the distributed data shall:

- Describe available fare products and their eligibility conditions for different types of users (adult, child, student, etc),
- Relate fare products to the bus network and the timetabled journeys so trip planners can compute fare products and fare prices for trips, show the tariff zones and the available products for an area, etc,
- Describe available travel documents (printed tickets, smart cards mobile apps) and any tariff differentiation between them),
- Allow the separate exchange of prices independently of the tariff structures & products.
- Expose a justification of the fare (by distance, stage, etc),
- Support both machine readable & human readable representation of validity conditions and terms of carriage. (NB mostly URL to document for MMTIS),
- Include information about how/where products can be bought,
- Include fares valid for specific and multiple operators,
- Be usable in cross border routes, such as in Ireland,
- Include metadata about the document contents, eg operator, type of fare.

6.2 Standard fares versus best value fares

Passenger information applications may be concerned to answer two quite distinct questions from passengers that differ substantially in their data requirements:

- **What is the typical cost of a standard fare?** A useful indication of this can usually be obtained just by considering single use, one-way fares for adults and children that are available for a trip. Although cheaper fares may well exist, the standard fare sets a limit as to the likely cost.
- **What is the best value fare?** To answer this correctly, a knowledge of all the available products is needed (ncluding also returns, day passes, carnets, season passes, etc.) as well their particular restriction on use (peak, off-peak, eligibility, etc), all described in a machine-readable representation. Computation of the 'best' fare may thus also require additional consideration of the frequency and time of travel, the type of user and their eligibility for discounts, etc.

Answering the "best value" question is therefore a much more ambitious target for data suppliers. The 'basic' fare profile is focused primarily on the first case.

6.3 Work flows envisaged

XML Documents satisfying the EFIP may be exchanged in many different workflows. In particular, they can be used for arbitrary peer-to-peer exchange as open data sent between any party. The same data may reach a consumer system by multiple routes and still need to be aggregated and integrated unambiguously;

thus, it is important that element identifiers are stable and globally unique and that they can be managed through distributed processes rather than a central registrar.

Data is likely to be processed according to the following overall scenario (but others are not precluded).

1. Fare data will be originated by operators to complement their timetable data. It will reference stop , location and line data elements already defined and reference by timetables encoded in Netex EPIP .
    - Operators will group route-specific tariffs in a similar manner to the way they group their Timetable data, that is, by line/service. Thus, there will be one or more XML fare documents per timetable, providing the tariffs for the line. This may include both single trip and season pass products.
    - Zonal products that apply to many different lines will be described in one or more separate documents for each operator for a specific network or part of a network. For such products, a definition of the zones will normally exist separately from the products themselves – and may be shared with other operators.
    - For certain products supported by multiple operators there will be a core definition providing a description of the common product and then additional definitions describing the exceptions for specific areas, supplied by each operator. A modular structure is needed to support such a workflow.
2. Operators will distribute their data by making it available on their sites, or by sending to one or more preferred aggregators.
3. Price data can be stated separately from the definition of the products (even if they are included in the same document for the purposes of distribution); typically, the networks and products themselves are fairly static over time; prices may change more often.

A common workflow is for one party to act as an aggregator, collecting data from multiple stakeholders and validating and integrating it into a useful data set for other consumers. In practice, a workflow that allows peer-to-peer exchange can equally well be used for aggregation.

Data will include information as to data source, validity etc that can be used to manage documents.

Certain common definitions will apply to all EFIP fare data. These definitions can be provided as part of the common fare profile (versioned so as to allow for future evolution).

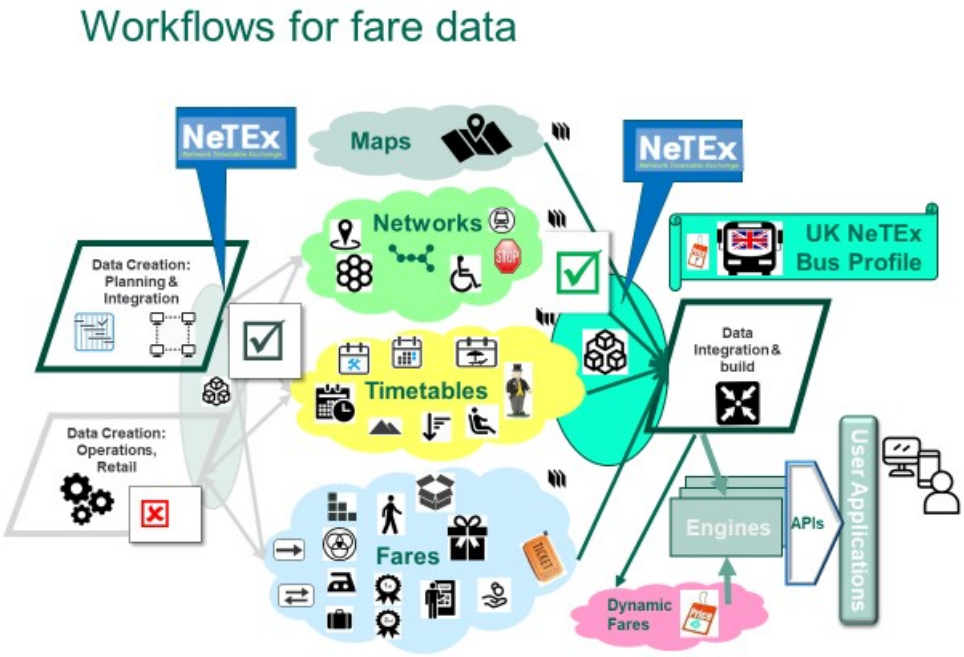


Figure 8 – Example of workflows for fare data

6.4 Overview of EFIP Fare features

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6.5 Summary of scope – Basic Profile

Tariffs may be based on many different factors. The following sections outline each of the most commonly found features supported in the profile

- **Tariff Structures – Spatial & Network**
  - Arbitrary Point to fares (the most common),
  - Zonal fares. (zones may be variously disjointed, overlapping or nested),
  - Section or Stage based fares,
  - Flat Fares.
- **Tariff Structures – Temporal**

- Validity period for making use of a ticket;
- Maximum trip duration,
- Different time intervals for passes (day, month year),
- Start dates for season passes either variable or fixed to a day type or calendar day
- Fare Demand Periods: (Peak, off Peak, etc.).

- **Tariff Structure – Other factors**
  - Inclusion / Exclusion of specific lines,
  - Inclusion / Exclusion of specific journey run within a timetable,
  - Inclusion / Exclusion of specific operators,
  - Routing limitations,
  - Class of use.
- **Eligibility**
  - Standard products for any purchaser,
  - Concessionary products or discounts for different classes of user (Child, Senior, Student, Armed forces, etc.),
  - Accessible products allowing registered disabled travellers and their companions discounts,
  - Group tickets for more than a certain number of travellers travelling together.
- **Products & Sales Packages**
  - Short hop single trips,
  - Single trip (single or return) products for use at any time,
  - Single trip products with travel restrictions as to time, route etc,
  - Carnets or Multi-trip products,
  - Day Passes,
  - Carnets of day passes to be used in a certain period,
  - Season passes,
  - Supplements – pets etc
- **Conditions**
  - Conditions of use (variable time of travel, interchanging, journey breaks, etc.).
  - Conditions of purchase (exchangeability, refunds, Time of purchase etc.).
  - Distribution & Fulfilment conditions (ticket media, sales channels).
- **Units**
  - Prices will generally be in euros but may be in multiple currencies,
  - available facilities on the service (wifi, meal, etc.).

- **Products & Sales Packages**
  - Discount Right products giving the right to purchase other products at a discount,
  - Usage Reward products giving the passenger a rebate or discount based on their previous consumption,
  - Supplement products such as seat reservations, couchettes, bicycle tickets etc.
  - Service products such as accompanying animals, transporting bicycles.

6.6 Round up of EFIP Tariff Features

In practice, although there is considerable variation in how fares are described and presented to the public, only a subset of all the possible combinations of PT tariff structures, products and conditions are typically found in EFIP Tariffs.

The following section summarises the main product types and tariff features suggested for inclusion in EFIP.

Not every tariff feature is found in combination with every possible common product.

Figure 9 introduces the main combinations of features commonly found in EFIP fares. It shows separate groupings of different individual **Trip** and repeated use i.e. **Pass** products, collated against a number of different tariff structures (flat, point-to-point, named zones, and unit section/ stage count), as well as whether they are likely to be offered with time of travel restrictions or with group tickets.

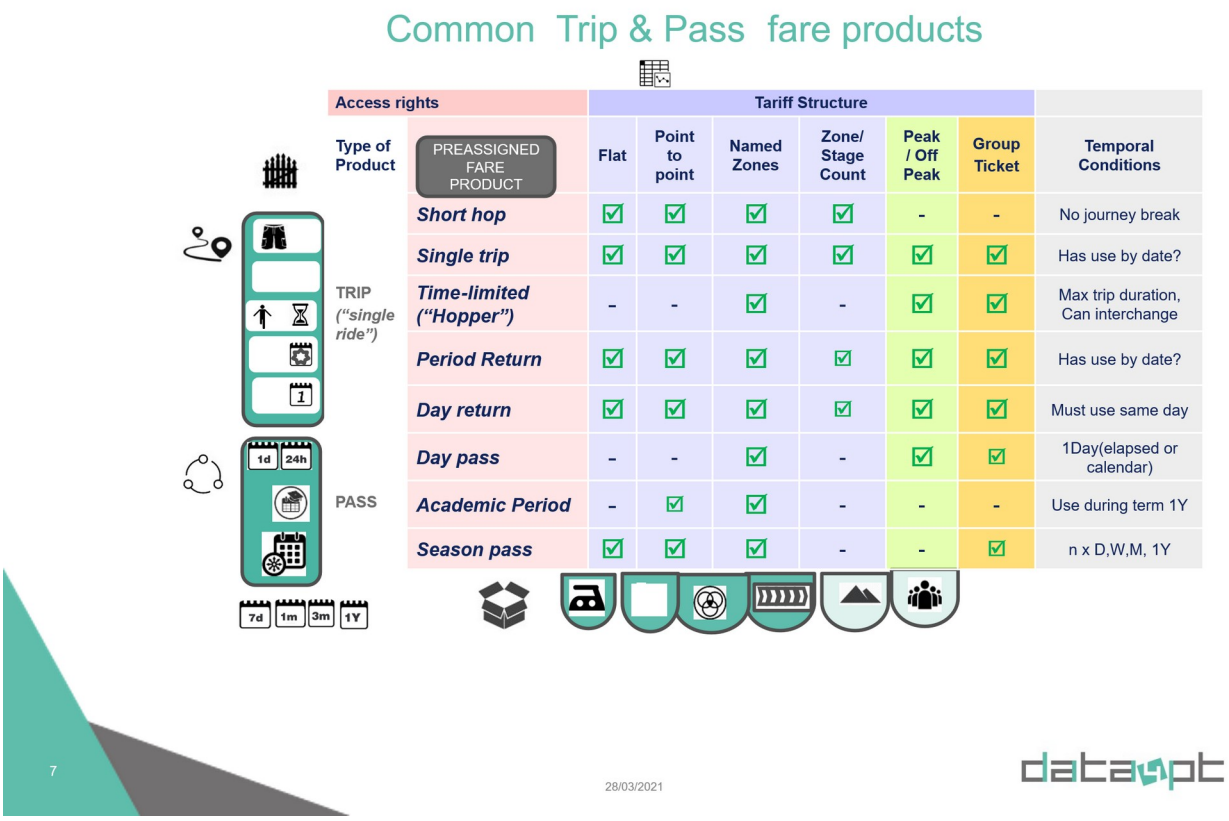


Figure 9 – Common Product / Tariff combinations

For convenience and to save transaction costs, multi-trip products are also found. Such carnet products bundled up multiples of trips or day passes.

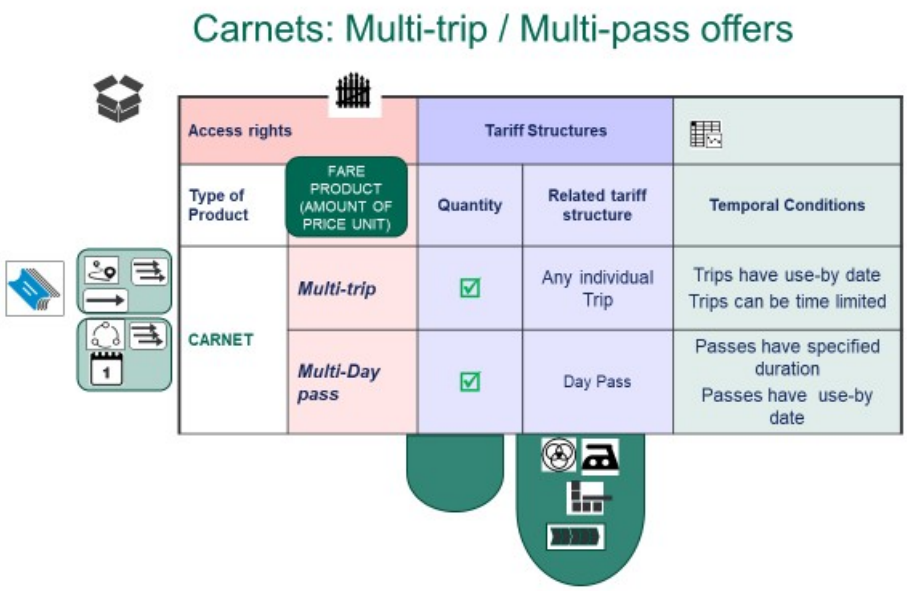


Figure 10 – Multi trip /Carnet Tariff combinations

6.6.1 *Relating the features to NeTEx elements*

Each of the tariff aspects corresponds to specific components in the NeTEx Model; some relevant NeTEx components are noted in each section so as to give an indication as to how the fare is encoded as a NeTEx document.

- A clover leaf bullet is used to indicate the relevant NeTEx elements. These can be ignored by non-technical readers

FA manicule (pointed index finger) symbol is used for specific EFIPEFIP guidelines.

6.6.2 *Commonly found Tariff structures in EFIP*

The fare products given above can be based in a number of different possible tariff structures; point-to-point, zonal, flat, etc.

In general, we may distinguish between those aspects of tariffs representing *access rights*, that is, the rights to consume different types of public transport service, from those of *usage parameters* which further moderate or qualify the consumption of services governed by the access rights.

Table 3 (below) and Table 4 (later) show common spatial and temporal access right concepts respectively. Usage parameters are discussed later below.

Network access rights for tariff structures for EFIP fares

Name	Description	Comment	Note/example
Point-to-point (P2) 	Access right is to travel between two named points.	***Explicit***: Separate prices (or price groups) are given for each named Origin and Destination Pair. ❖ Represented in NeTEx by a DISTANCE MATRIX ELEMENT. Where the network has multiple routes, or operators there may be constraints by route and or by operator.	Widespread – e.g. UK standard fares. Access in different directions may have different price (e.g. on circular route).
Zonal 	Access right is to travel in named zones.	There is an absolute price for each zone or zone combination. The specific origin and destination zones determine the price. ❖ The tariff zones are represented in NeTEx by a TARIFF ZONE or FARE ZONE, which can be used to assign stations to one or more tariff zones. ❖ Each Zone O/D pair is represented in NeTEx by a DISTANCE MATRIX ELEMENT.	
Stage Counting 	Access right is to traverse a certain *number* of stages or zones.	Each stage or zone has a unit cost associated with traversing it. the number of zones traversed is used to determine the price. A given stop may be in several zones (e.g. also a 'near to' zone) so as to smooth the prices of short trips. (A stop may be assigned to a FARE	

		ZONE, or to a specific FARE SECTION within a SERVICE PATTERN). ❖ Represented in NeTEx by one or more GEOGRAPHICAL INTERVALs, where the unit is a zone or section. ❖ A GEOGRAPHICAL INTERVAL PRICE may be given to each interval.	
Flat Rate 	Access right is regardless of distance or time.	Price is fixed. Commonly used for supplements. ❖ Represented in NeTEx by some form of FARE PRODUCT PRICE associated directly with the supplement product.	Bicycle fees, Airport Express fare, some night trains.
Distance 	Access right is to travel between two named points, the pricing being based on a shortest path distance through the network.	***Distance Based:*** A table of distances and a price per kilometre is given. This can be used to compute a P2P price based on the timetable distance. Represented in NeTEx by a set of GEOGRAPHICAL INTERVALs with GEOGRAPHICAL INTERVAL PRICEs that can be used to price the distance from a DISTANCE MATRIX ELEMENT.	

Table 3 – Common Network access aspects of Tariff Structures

6.6.2.1.1 *Why different tariff structures are needed*

Why not just make all fare point to point? The choice of the optimum model for the exchange of fare prices involves making a trade-off between complexity and size; one may choose to exchange either a small data set of parameters requiring a complex set of calculations to interpret, or a large data set requiring a simpler processing by the consumer.

Although the simplest tariff structure for a consuming system to interpret is a point-to-point fare, and although other tariff structures can be processed to transform them into a table of point-to-point prices, such an approach does not scale well for very large numbers of stops. Other price structures such as

distance based prices with choices for bands of dstances are much more compact. Derived prices can also reduce data set size , for example that a child fare that is defined as half an adult fare.

6.6.3 *Temporal fare aspects of Tariff structures*

Temporal constraints may apply the tariff structure (Table 4). Some of these can be regarded as fundamental to the access rights of the tariff structure, for example the period of a season pass, or the duration of a trip, or when travel may take place; others may apply to ancillary factors such as the commercial conditions attached to the fare, for example, the purchase window within which the ticket may be bought. In both cases different prices may attach to different conditions.

Nature	Description	Comment	NeTEx
Pass duration (Access Time Limited) 	The traveller may have the rights access the network for a specified period – ranging from a day to one year	As in day and season passes. Prices typically depend on the duration, with discounts for longer periods. For a day pass, the start and end times may be fixed (the fare day) of variable (from the time of purchase. For a season pass the start day and duration may be fixed (e.g. Monday for seven days, or the 1st of the month for one month) or variable.	❖ Available intervals are represented in NeTEx by a TIME INTERVAL
Trip duration (Access Time Limited) 	*Once started the passenger may be required to complete an individual trip within a certain time*	In a zonal system, different time intervals may be offered at different prices (e.g. 1 hour, 2 hours, 3 hours, etc.), in effect constituting a form of distance-based fare. In a point-to-point system where the destination is already fixed, the time represents the maximum time allowed to make the trip – relevant in particular if journey breaks are allowed.	❖ **Represented in NeTEx by a TIME INTERVAL**
Use by date	Once purchased. the product may be required	Applies mainly to trip products – most standard tickets must be used within a	❖ Represented in NeTEx by a USAGE

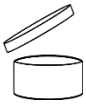
 **(Usage Validity)**	to be used within a certain period.	certain time. A return trip may have to take place within a certain time after the outbound trip.	VALIDITY PERIOD.
Time of Travel (Fare Demand) limited 	Travel may be limited to travel only at certain days or times of day.	Standard tickets can be used anytime, other tickets may be restricted to various Off peak timebands and or day types. There may be different time band bands for different directions of travel.	❖ **Represented in NeTEx by a FARE DEMAND FACTOR.**
Purchase Window 	The product may only be available a certain in advance or up to a certain time before travel	Discount and group products in particular tend to have limits as to when they can be purchased. These may vary by distribution channel and period. There may for example be different prices associated with purchasing or changing a fare within a given time. Different times may apply to initial purchase and renewal.	❖ **Represented in NeTEx by a PURCHASE WINDOW.**
Post-sales conditions 	The time conditions as to when a product may modified, exchanged, refunded, renewed, etc.	Limits may apply to the period in which exchanges and refunds are allowed. For passes	❖ **Represented in NeTEx by an EXCHANGING, or RETURNING.**

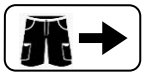
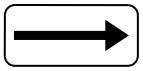
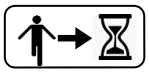
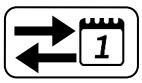
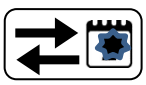
Table 4 – Some temporal concepts applying to fares

6.6.4 *Commonly found Fare Products on EFIP*

Most EFIP products fall into one of a small number of categories of product type as to the overall access rights for travel they grant; for example, single, return, day pass, season pass, etc.

Table 5 shows the main product types found. The products charge for specific trips, based on different possible underlying tariff structures, described later below. They are further modulated by a large number of parameters and sales conditions, also summarised in the following sections.

6.6.4.1 *Common Trip Products*

	Nature	Description	Note	EFIP
Trip 	**Short hop** 	A single trip between an origin and destination limited by the number of stops (Cf Hopper ticket). May or may not allow a journey break (in which case a maximum travel time will apply).	❖ Represented in NeTEx by a PRE-DEFINED ACCESS RIGHT + DESTINATION MATRIX ELEMENT	FBasic
	Single 	A single trip between an origin and destination. May or may not allow a journey break (in which case a maximum travel time will apply).	❖ Represented in NeTEx by a PRE-DEFINED ACCESS RIGHT.	FBasic
	Time-limited Single ("Hopper") 	A single trip whose duration is for a given period (e.g. 30 mins, 1 hour, etc) after activation.	❖ Represented in NeTEx by a PRE-DEFINED ACCESS RIGHT + TIME INTERVAL	FBasic
	Day Return 	A return trip between an origin and destination and back. Must be completed in the same fare day (which might extend into the next calendar day if the fare day.	❖ Represented in NeTEx by a PRE-DEFINED ACCESS RIGHT + DESTINATION MATRIX ELEMENT	FBasic
	Period Return 	A return trip between an origin and a destination and back. Must be completed within a certain period.	❖ Represented in NeTEx by a PRE DEFINED ACCESS RIGHT + + DESTINATION MATRIX ELEMENT	FBasic
Multi	**Carnet of Trips**	A bundle of individual trips sold	Book of 10 tickets, etc.	FBasic

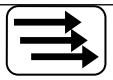
Trip 		together at a discount, typically to be consumed within a certain time limit.	❖ Represented in NeTEx by a NeTEx AMOUNT OF PRICE UNIT	

Table 5 – Commonly found EFIP Products –Trip

6.6.4.2 *Common Pass Products*

	Nature	Description	Note	
Pass 	**Day Pass, zonal** 	The right to make an unlimited number of journeys within a one or zones within a day (Which may run into the next day in certain cases). ❖ Fixed start: For the current fare day ❖ Variable: A 24-hour pass from time of purchase or activation	❖ Represented in NeTEx by a PRE-DEFINED ACCESS RIGHT.	FBasic
	Season, P2P Routed 	The right to make a number of journeys between two points within a certain period. The number may unlimited or of a set frequency e.g. once a day in each direction. May be constrained to a route or routes and operator or operators.	❖ Represented in NeTEx by a PRE-DEFINED ACCESS RIGHT with aone or more TIME INTERVALs	FComplex
	Season, Zonal 	The right to make an unlimited number of journeys within a given area for certain period (e.g. 1 day, 2 days, 1 week, 1 month, 1	❖ Represented in NeTEx by a PRE-DEFINED ACCESS RIGHT	FBasic

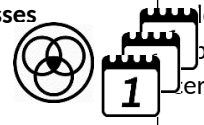
		year, etc.).	with a FARE STRUCYURE ELEMENT referencing the zones.	
	Season Pass Routed+ Extension 	A hybrid pass allowing commuters to travel a route and also in a central zone at their destination.	❖ Urban Metro/ Bus add on to a Rail ticket	FFuture
Multi Pass 	**Carnet of Day Passes** 	A bundle of individual day passes sold together at a discount, typically to be consumed within a certain time limit.	❖ Represented in NeTEx by an AMOUNT OF PRICE UNIT.	Basic

Table 6 – Commonly found EFIP Products – Pass

6.6.4.3 *Common Discount Products*

	Nature	Description	Note	
Discount	**Discount Right (TravelCard)** 	A product that gives the right to buy other products at a discount.	❖ Represented in NeTEx by a SALES DISCOUNT RIGHT	FExtended
	Capped Discount 	A product that gives the right to buy other products at a discount, with a further limit on charges within a given period.	❖ Represented in NeTEx by a CAPPED SALES DISCOUNT RIGHT	FExtended

	Usage Discount 	A product that gives the right to a rebate according actual usage. Requires either a smartcard or an account or both.	❖ Represented in NeTEx by a USAGE DISCOUNT RIGHT	FExtended

Table 7 – Commonly found EFIP Products – Discount

6.6.4.4 *Other Products*

	Nature	Description	Note	
Stored Value	**Predefined Units** 	A bundle of units that may be marked as used to make an arbitrary trip. Relevant for stage fares with a fixed price per stage. If the units are money, corresponds to the balance on a smartcard.	Oyster card stored value	FBasic
Other	**Third Party Entitlement**	A third-party product or membership schema that gives the holder rights to obtain or purchase a travel product. Can be sued for example to represent an identity card issued to authenticate a user as belonging to a specific eligibility group.	❖ Represented in NeTEx by a THIRD PARTY PRODUCT	FExtended

Table 8 – Other EFIP Fare Products

❖ The products described all correspond to various types of NeTEx FARE PRODUCT; a number of different specialisations (PRE-DEFINED ACCESS RIGHT, DISCOUNT RIGHT, USAGE RIGHT, FARE SUPPLEMENT, etc). describe specific types of fare product. In some cases, several different types of NeTEx r

Fare product may be bundled together in a single sales offer. For example, rail tickets and passes may include a supplement to cover urban travel at the destination.

6.6.5 *Permitted combinations of tariff elements*

6.6.5.1 *Likely combinations of tariff types and products*

As discussed in the introduction, not all types of fare structures are relevant for all the possible product types. Furthermore certain possible combinations are not used in actual fares. The following table shows the likely combinations of tariff types: trip, multi-trip and pass products, using the NeTEx TariffType and FareProduct enumerated values as classifiers.

- LGP – Found in long distance products (coach, rail etc), but not local bus.
- n/a – not applicable.

	Fare Product			
	Trip	Multi-trip	Pass	Supplement
				Ì
				
FareProduct TariffType	PreassignedFareProduct	AmountOfPriceUnit	PreassignedFareProduct	SupplementProduct
flat 	singleTrip	tripCarnet passCarnet		supplement
pointToPoint 	shortTrip, singleTrip, timeLimitedSingleTrip, dayReturnTrip, periodReturnTrip	tripCarnet	periodPass (LGP)	Bicycle, etc journeyExtension, journeyAddon,

zoneToZone 	singleTrip. dayReturnTrip	tripCarnet	n/a	n/a
zonal 	singleTrip, timeLimitedSingleTrip	passCarnet	dayPass, periodPass,	animal, topup, parking etc
unitSection 	singleTrip	unitCoupon	n/a	n/a

Table 9 - Tariff types forTrip, Multitrip and Pass products

6.6.5.2 *Likely combinations of products and fare structure elements*

Not all fare structure elements are relevant for all product types and only certain USAGE PARAMETERs are relevant as pricing factors their own right.

The following table shows which parameters are relevant for EFIP Trip, M9ulti-trip and Pass products.

Key

- LGP – Found in long distance products (coach, rail etc), but not local bus.
- () – Applied through by another parameter
- n/a – not applicable
- NFB – Possible to define in NeTEx

	Trip	Multi-trip	Pass	Use as pricing factor?
FareProduct Fare structure element	Preassigned-FareProduct	AmountOf-PriceUnit	Preassigned-FareProduct	
TariffZone	Zonal, zoneToZone	Yes	dayPass, periodPass	(DistanceMatrixElement)

FareSection / Stage	Stage fare	n/a	n/a	(GeographicInterval)
DistanceMatrixElement	pointToPoint zoneToZone	tripCarnet passCarnet	pointToPoint periodPass	Yes
GeographicInterval	unitSection	Stored Unit	pointToPoint perido pass	Yes
TimeInterval	timeLimitedFare	passCarnet	dayPass, periodPass	Yes
FareQualityFactor	NFB	tripCarnet passCarnet	NFB	Yes
FareDemandFactor	pointToPoint zoneToZone	tripCarnet passCarnet	Yes	(SalesOfferPackage)

Table 10 - Network & Fare Structure elements

6.6.5.3 *Likely combinations of products and usage parameters*

Not all USAGE PARAMETERS are relevant for all product types and only certain USAGE PARAMETERs are relevant as pricing factors in their own right. The following table shows which parameters are relevant for EFIP Trip, multi-trip and Pass products.

- NFB – Possible to define in NeTEx .
- LGP – Found in long distance products (coach, rail etc), but not local bus.
- () – Applied through another parameter.
- n/a – not applicable.

Usage Parameter	Trip	Multi-trip	Pass	Use as pricing factor?	EFIP

					scope
UserProfile	Yes	Yes	Yes	Yes	Basic
CompanionProfile	Yes	Yes	Yes	(UserProfile)	Basic
GroupTicket	Single, return	NFB	Yes	Yes	Basic
ResidentialQualification	(UserProfile)	NFB	(UserProfile)	(UserProfile)	Advanced
CommercialProfile	All	NFB	All	Yes	Advanced
RoundTrip	Single, Return, out, back	Single/ Return	Day	(ValidableElement)	Basic
FrequencyOfUse	n/a	n/a	NFB	No	Basic
Interchanging	Yes	NFB	n/a	NFB	Basic
UsageValidityPeriod	Use by	Use by	Start by	No	Basic
StepLimit	Single	NFB	n/a	No	Advanced
Routing	LGP	LGP	LGP	LGP	Advanced
EntitlementRequired	Yes	Yes	Yes	No	Advanced
EntitlementGiven	No	No	Yes	No	Advanced
EntitlementConstraint	Yes	n/a	Yes	No	Extra

LuggageAllowance	LGP	LGP	n/a	(SupplementProduct)	Advanced
Transferability	Yes	Yes	Yes	No	Basic
PurchaseWindow	LGP	LGP	Yes	(SalesOfferPackage)	Basic
Reserving	LGP	LGP	n/a	LGP	Advanced
Cancelling	LGP	LGP	n/a	No	Advanced
BookingArrangements	(Flexible transport)	NFB	n/a	No	Advanced
Exchanging	LGP	LGP	Yes	No	Advanced
Refunding	LGP	LGP	Yes	No	Advanced
Replacing	LGP	LGP	Yes	No	Advanced
Suspending	n/a	n/a	NFB	No	Advanced
ChargingPolicy	Yes	n/a	n/a	No	Advanced
PenaltyPolicy	Yes	n/a	n/a	No	Advanced

Subscribing	n/a	n/a	Yes	(SalesOfferPackage)	Advanced

Table 11 - USAGE PARAMETERs – For Trip, Multi-trip and Pass products

Usage Parameter	Sales Discount Right	Pricing Factor	EFIP scope
UserProfile	All	Yes	Basic
ResidentialQualification	(UserProfile)	(UserProfile)	Advanced
EntitlementRequired	Yes	n/a	Advanced
EntitlementGiven	Yes	n/a	Advanced
Replacing	Yes	n/a	Advanced

Table 12 – USAGE PARAMETERs – SALES DISCOUNT RIGHT

6.6.6 *Classification of fare elements*

Data elements should be classified with the NeTEx built in enumerated values and in some cases also the appropriate EFIP TypeOfValue.

6.6.7 *Classifications of Tariffs*

Each EFIP Tariff should be categorised with a TariffBasis and a TypeOfTariff efp :value

Each EFIP FareStructureElement should be classified with a TypeOfFareStructureElement efp:value

Each EFIP AccessRightAssignments should be classified with a TypeOfAccessRightAssignment

6.6.8 *Relating tariffs to the network*

A fundamental goal of the fare profile is to ensure that the data allows tariffs to be related to the underlying network by journey planners and fare calculation engines, so that appropriate products and prices can be found for trips.

Although it is straightforward in NeTEx to specify that specific tariffs or fare products apply to specific timetabled journeys, this is not the general case; rather the tariff is normally stated in terms of what parts of the network (lines, operators, zones, stops etc) and when (time bands etc) it applies to and a trip planner or fare engine must use the trip parameters to select the available fares.

Different methods are used to relate different spatial tariff structures to the network:

- For **point-to-point** tariff between stops, the origin and destination stops can be related directly to the DISTANCE MATRIX ELEMENT linking the stops using their stop identifiers.
- For a **flat fare** for a single line, the journey pattern can be used to determine all of the stops of the line.
- For a **zonal** tariff, stops can be assigned to a tariff zone; a trip planner first determines fares that apply between the origin and destination zones. In principle, different methods might be used to make the assignment of stops to zones:
    - **Explicitly:** i.e. membership of a stop of a tariff zone is stated by a link.
    - **Implicitly** by projection: thus, if the coordinates of the stop fall within the bounding polygon of the containing tariff zone, it is deemed to be within the zone. This however cannot be done
    - **By boundary stop:** the stops on the boundary are enumerated and all stops enclosed in the polygon are included.

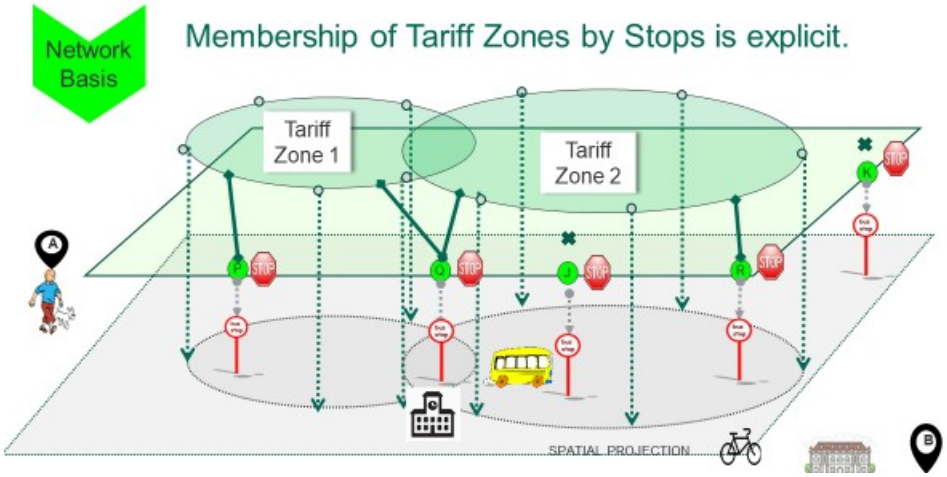


Figure 11 – Relationship of stops to zones

- The data model specifies which stops are in which tariff zone: a stop may be in more than one zone, zones maybe specific to an operator or shared with other operators.
- Both stops and zones have a spatial projection (i.e. spatial coordinates). However spatial containment of a stop within a tariff zone's extent does not invariably imply semantic membership of the zone. Therefore, although in many cases the stop coordinates can be used to compute which stops are in a given tariff zone so as to populate the membership links of the tariff zone, it is nonetheless important to have an explicit representation.

EFIPEFIPEFIP assumes membership is explicitly stated. it is of course possible to use a process to populate a set of explicit memberships from spatial data.

6.6.8.1 *Who is responsible for defining tariff zones?*

For zone-based tariffs, a key question is who is responsible for defining the tariff zones, that is, for allocating to them identifiers and names and specifying which stops and lines are included? Tariff zones, like stops, need to have unique persistent identifiers.

Responsibility may be specific to an operator or authority, or be managed by a wider consortium.

Responsibility type	Description	Example

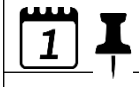
National	Zones are defined by a national organisation	For example, PlusBus, National Rail areas
Authority	Zones are defined by an Authority and used by many operators.	E.g. Tfl London Zones
Group of Operators	Zones are defined by a group of operators. There must be a common process of allocating identifiers E.g. National Rail Network areas	Common organisation e.g. Discovery
Scheme Defined	Zones are defined by an organisation other than an operator	E.g. National Concession Scheme
Other shared zones	Zones are defined by a specific operator and are shared with others.	
Operator specific	Zones are defined by a specific operator and are specific to that operator.	E.g. Metrobus

Table13 – Specification of Tariff Zones

6.6.9 *Commonly found Product Durations*

Although products in principle could be created with a validity period of any duration, in practice certain durations are much more commonly found s (see Table 14Table 14). Note however that some season passes allow the number of days or months to be chosen arbitrarily.

	Nature	Description	Comment
Trip 	**1-Hour**	Ticket valid for up to 1-hour travel.	E.g. Short Trip.
	90-Minutes	Ticket valid for up to 1.53 hours travel.	E.g. Zonal.
	120-Minutes	Ticket valid for up to 2 hours travel.	

	180-Minutes	Ticket valid for up to 3 hours travel.	
Pass 	**24-Hour** 	Ticket valid for up to 24 hours travel starting at the time of activation.	e.g. Variable start Day Pass.
	1-Day 	Ticket valid from start of day (or moment of purchase if later) until the end of fare day.	e.g. Fixed start Day Pass.
	3-Day	Ticket valid for up to 72 hours travel.	
	5-Day	Ticket valid for up to 5 days travel.	May be fixed (Monday) or variable start.
	1 Week 	1 Week pass, fixed or variable start.	May be fixed (e.g. Monday) or variable start day.
	Weekend Only	Pass only valid Saturday and Sunday	
	1 Month 	1-month pass	May be fixed (e.g. 1st of month) or variable (i.e. any day) start day.
	3-Month	2-month pass.	
	6-Month	3-month pass.	
	12-Month 	1-year pass.	May be fixed (e.g. 1st of year) or variable (i.e. any day) start day.
	2-Year	2-year pass.	

	3-Year	3-year pass.	Some Rail cards that have Bus discounts.
Student / Youth Pass 	**Term (semester)**	Valid any day during term-time.	Specific to local academic calendar.
	Term - Schooldays	Valid only on schooldays during term time.	Specific to local academic calendar.
	Academic Year	Valid any day during academic year.	Specific to local academic calendar.
	Holiday	Valid for a holiday period. Summer holiday tickets are most common	Specific to local calendar.

Table 14 – Common Product Duration

- In NeTEx, a TIME INTERVAL, and / or a TIME STRUCTURE FACTOR is used to represent each different duration (such intervals have several uses). A different TIME INTERVAL PRICE can be associated with each interval.
- Calendars for School and University terms and holidays may be defined using a SERVICE CALENDAR FRAME.

6.6.10 *Commonly found Product Charging moments*

An important characteristic of Fare Products is the moment of actual payment for access rights –Table 15Table 15.

Nature	**Description**		
Prepaid	Product is bought outright before travel.	E.g. Classical tickets.	Basic fare profile
Prepaid Subscription	Product is bought by a regular monthly or other period automated subscription at a discounted rate.	E.g. ABO products.	Extended profile

In transit	Product is bought onboard during travel or at exit point; effectively the same as prepaid except that it is permitted to start travel without a ticket.		Extended profile
Pay as you go at start of travel	The user is charged on starting the journey at check in.		Extended profile
Pay as you go Prepayment with adjustment	The user is charged on starting the journey, either for the correct amount, or for an average fare. On completion of the journey, any difference from actual rights consumed is used to make an adjustment. The user may be required to keep a credit balance on the card, either manual or using an auto top up.		Extended profile
Pay as you go post payment	The user is charged on completion of the journey according to the access rights consumed (by explicit or implied check out). Fare calculation may include capping or usage-based rebates.	e.g. Tfl Oyster, Tfl Oyster Contactless.	Extended profile
Prepaid with rebate.	Account based and smart card systems may offer the user rebates if they exceed certain usage thresholds.		Future

Table 15 – Common Product Charging moments

- In NeTEx, a NeTEx CHARGING MOMENT is used to represent the time of payment (pre-/post-payment) and the account location (card, backoffice).

6.6.11 *Commonly found Fare Demand types*

Some tariffs are often restricted to certain periods so as to spread demand on congested networks. A different tariff may apply to night operation. Table 16Table 16 shows some common distinctions found.

	Nature	Description	
	**Peak / Unrestricted**	Tariff may be used at any time including at weekends and on public holidays	E.g. Standard walk u tickets.
	**Off-peak**	Tariff may be used only at off peak times, for example, outside of the morning and / or evening rush hours on working days, and anytime at weekends and on public holidays. Definitions may depend on direction of travel.	E.g. ABO products.
	Weekends and holidays	Tariff may be used only at weekends and on public holidays	
	**Night**	Tariff applies only at evenings or at night;	

Table 16 – Common Time demand types

- In NeTEx, a FareTimeDemandType is used to represent a demand period. The actual days and hours of operation can be specified with DAY TYPEs

6.6.12 *Temporal factors for peak and off-peak demand*

This section presents visualisations of temporal concepts concerning time demand periods – in order to spread demand on the network to less congested periods, differential pricing is common for peak and off-peak periods.

Blackout periods

Figure 12 shows that a travel period may be banded so as to create peak and off-peak periods that meet various conditions

- **Day Type.** Specifying the type of day (week day, holiday, etc.) under which the condition applies.
- **Timeband:** specifying a period within the day.
- **Time at stop.** A further complication is that the time-band may be directional (e.g. into or out of a specific station) and that the specific time may vary station by station, so for example the time band starts at 08:50 at station A, 08:53 at Station B, 08.58 at station C, etc.

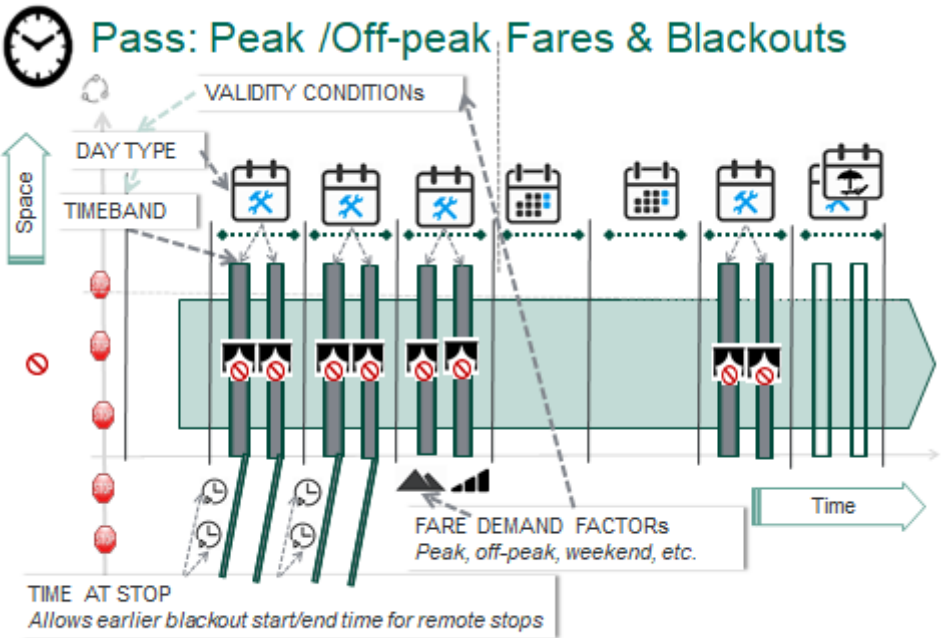


Figure 12 – Blackout day types and timebands

- In NeTEx, different demand periods can be represented as FARE DEMAND FACTORs. These be defined in terms of DAY TYPEs, TIME BANDs and other validity conditions. A TIME AT STOP element can be used to define the time relative to a specific stop.
- A SERVICE CALENDAR can be used to specify which DAY TYPEs apply on a given calendar day.

Time demand restrictions on an individual Trip

Figure 13 shows time demand restrictions as applied to an individual trip. The restrictions may apply to any or all of the start of the trip, the end of the trip or the course of the trip

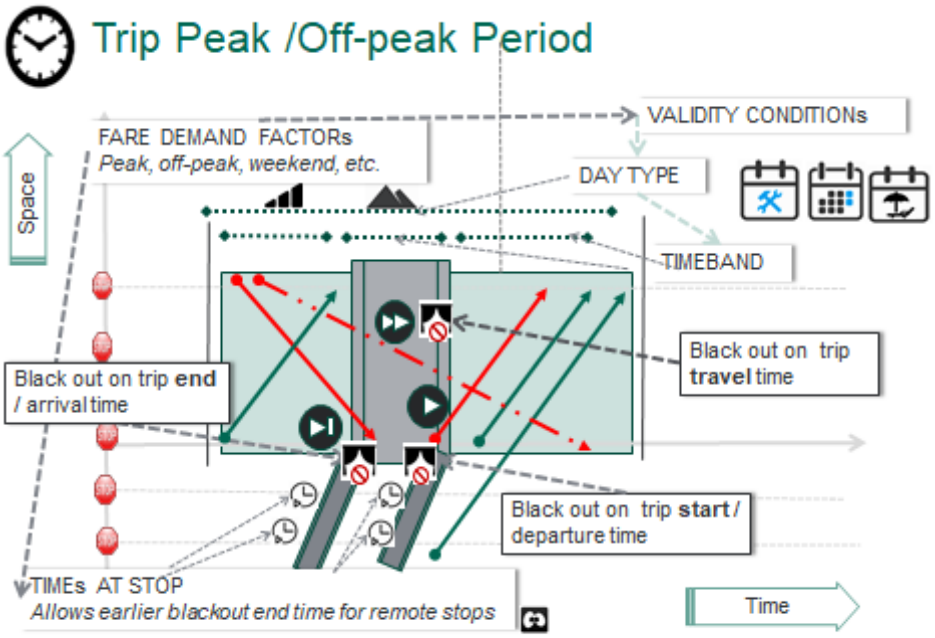


Figure 13 – Time demand restrictions on an individual Trip

6.6.13 *Single and multiple operator access*

A tariff may be specific to an operator, such that it can only be used on that operator's services, or be honoured by multiple operators (for example the UK common zonal addon for rail, *PlusBus*). Or it may be that different operators offer different tariffs covering overlapping areas, or specifically cover journeys that go from one operator's services to another's.

- In NeTEx access rights and tariffs may be specific to an OPERATOR or to a GROUP OF OPERATORs,
- The OPERATOR and GROUP usually, but not always, sell their own tickets. Other distributors may also be used, such as TRAVEL AGENTS, or o a RETAIL CONSORTIUM (a sales organisation representing a group of operators).

6.6.14 *Commonly found Passenger Types in EFIP*

Tariffs usually segment the market, with different prices for different types of users. Certain products or certain discounts to products may be restricted to specific types of user who must meet specified eligibility criteria.

- Some of these categories may be immediately available to anyone (e.g. *Adult*, *child*), others may require documentary proof (e.g. a disabled card) and/or prior registration (e.g. to get a senior discount in the TfL you must have a Senior Oyster card).
- Some profiles entitle the user to take companions (children, assistants, dogs, etc.) either free or at a discount (so can be considered as "group tickets").
- Certain user profiles may be ordained by statutory measures, with eligible users being allowed a concessionary discount.

User profiles may also apply to the purchase of group tickets – see next section.

Commonly found user profiles in many countries are shown in Table 17Table 17. Although broadly the same categories are found in most countries, for the age dependent user types, there are distinct variations as to the ages for eligibility in specific groups.

	Nature	Description
	**Adult**	Any person too old for a child or youth discount. Some products may have a minimum age. E.g. in order to gain the right to take an infant free or a child at a discount.
	**Infant**	Babes in arms and children. Maximum age varies between countries – from 4 to 6 years old. Usually travel free without a seat. Cannot travel unaccompanied.
	**Child (Accompanied)**	Children who are no longer infants, accompanied by a parent or sibling. Usually discounted by between 30-50%, or free. Both the minimum and maximum ages for being eligible for child fares vary considerably between countries an even within countries. Minimum is typically between 5 and 7 years old, maximum varies from 12 to 16 years old. In a few cases there is even variation within a country for specific products (for example, some airport shuttle services). Some countries / products require the child to be a descendant (i.e. son/daughter or grandchild of the adult).

	**Child (Unaccompanied)**	A child (see above) travelling without an adult. Usually eligible for a discounted fare, but not free as for some fares for accompanied children. There is usually a minimum age for being allowed to travel unaccompanied.
	**Youth**	Young person over the minimum age for child discounts (see above - typically 15 years) and below the age for adult (e.g. between 18 and 26 years).
	**Pupil**	School children in full time education. Usually under 19.
	**Student / Trainee**	Person attending an accredited college. Usually under a certain age (e.g. 26), with special provision for PhDs and mature students.
	**Senior**	Persons over a certain age (or formally registered as a pensioner), often statutory. Usually eligible for a discount. Age varies between countries (typically between 60 and 65 years old, but in some cases as high as 70). Some rights may be restricted to local residents or nationals. A few countries have different age bands for seniors (e.g. 0-65, 65-70, 70+) with different rights.
	**Disabled**	Usually Statutory. May be subdivided into different categories, e.g., Blind, severely handicapped, handicapped child, invalid, etc.
	**Disabled Companion**	Companions to disabled, veterans or other specific types of user. Usually discounted or free.
	**Social Benefits**	Unemployed, qualifying for special social benefits, etc. Usually registered membership scheme.

	**Armed forces**	HM armed forces
	**Special Occupation**	Member of an organisation participating in an employee ticket scheme. Discounted, sometimes with tax benefits.
	**Member**	Member of a membership scheme offering discounts to its members.

Table 17 – Commonly found categories of passenger

- In NeTEx, USER PROFILE elements are used to represent different types of individual user and their eligibility conditions such as age, registration, etc. COMMERCIAL PROFILE elements are used to represent different types of corporate and organisational user.
- In specify age conditions, precision is needed to say when the age is actually measured (e.g. actual birthday, age on 1st January, etc), especially in the cases of season passes valid for an extended period during which a user's eligibility may change. NeTEx allows various specific conditions to be expressed.
- Often a profile is distinguished so that a different price can be offered to eligible users (as indicated by a DISCOUNTING RULE).

6.6.15 *Commonly found Group Tickets in EFIP*

Tariffs often offer a discount to groups of users travelling on a single ticket. There may be complex rules as to how these groups are made up of specific types of user, with minimum and maximum numbers of users of a specific type (e.g. Two to five adults with up to three children). Groups commonly found in EFIP Tariffs are shown in Table 18Table 18.

	Nature	Description	Note

	**Family**	Family with at least one child under a qualifying age (e.g. 15). In some cases, children must be related to the adult (child or grandchild).	In NeTEx, specified by the GROUP TICKET usage parameter
	**Group**	Groups travelling together on a single product, usually at least three.	In NeTEx, specified by the GROUP TICKET usage parameter
	**Large Group**	Large groups travelling together on a single product. Usually require longer lead times to book, and conditions for modifying the number of users closer to time of travel may be material.	In NeTEx, specified by the GROUP TICKET usage parameter
	**Student Group**	Groups of school pupils or students, accompanied by supervisors. A certain ratio of adults to students is required. Places for supervisors may be free or discounted.	In NeTEx, specified by the GROUP TICKET usage parameter
	**Duo / Two Together**	Two adults at a discount, usually off peak.	In NeTEx, specified by the GROUP TICKET usage parameter

Table 18 – Commonly found Group tickets passenger

6.6.16 *Commonly found Conditions of Use in EFIP*

Products may be subject to specific conditions about the consumption of access rights. Commonly found distinctions include.

	Condition	Name	Note
	**Can break Journey**	Whether the user can break the journey along the route, as long as the overall travel time limit is met.	E.g. A UK rail single allows a break of journey.

	**Can interchange**	Whether the user can make an interchange to another service or can only use one vehicle. This constraint is meaningful for zonal or stage counting journeys.	❖ In NeTEx, specified by the INTERCHANGING usage parameter.
	**Any route permitted**	Whether the user is limited to a specific route.	❖ In NeTEx, specified by the Exact Routes can be specified using the SERIES CONSTRAINT mechanism.
	**Frequency**	How many times the fare product may be used in a given period. For example, a season pass may give unlimited use, or be limited to two journeys a day.	❖ In NeTEx, specified by the FREQUENCY usage parameter.
	**Is transferable**	Whether another person can use the product or whether it is personal to a user.	❖ In NeTEx, specified by the TRANSFERABILITY usage parameter.
	Maximum distance	Maximum distance allowed for a product.	
	Maximum journey time	See temporal conditions.	

Table 19 – Commonly found conditions of use

- ❖ In NeTEx, a number of different types of USAGE PARAMETER are used to set additional conditions as to how a product may be used. These can include ROUTING, INTERCHANGING, TRANSFERABILITY, etc., etc.
- ❖ Maximum distance can be specified by a DISCOUNTING RULE.

6.6.17 *Possible Supplement Products on EFIP*

Long distance modes such as coach / rail may include additional supplement products which a ticket holder can purchase to complement their basic access rights. For example, an upgrade to first class, or a ticket for an accompanying bicycle.

Table 20 shows some possible product supplement types. The products build on underlying tariff structures described later below.

	Nature	Description	Note/ Example
Return Trip 	**Excursion**	Includes entry price to an attraction or event	
Trip  	**Seat Reservation** 	Seat reservation may be included in price or may be charged extra.	Only relevant for coach
	Animal Ticket 	Extra ticket for a dog or other animal on the trip.	Guide dogs are usually free. Usually free a flat fee.
	Excess Luggage 	Extra for luggage that exceeds free allowance, e.g. outsize sports equipment, extra suitcases, buggy, etc.	Usually a flat fee.
	Parking	Parking at origin stop for duration of return trip at discounted or free price.	

Table 20 – Commonly found Supplement Products

- A supplement product is normally represented in NeTEx by a FARE SUPPLEMENT (If it involves complex access rights a PREASSIGNED FARE PRODUCT may also be used). ENTITLEMENT GIVEN and

ENTITLEMENT REQUIRED usage parameters can be used to indicate the dependencies between fare products. Most supplements apply to long distance travel.

6.7 Overview of other EFIP product features

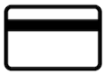
In addition to the tariff aspects discussed above (temporal and spatial, user type etc)EFIP Tariffs may be based on a number of other factors such as the ticket media, distribution channel, etc., etc.), as described below.

Each of these aspects correspond to specific components in the NeTEx Model, some relevant components are noted in each section so as to give an indication as to how the fare might be encoded as a NeTEx document.

6.7.1 *Commonly found Types of Travel Document in EFIP*

In order to travel, passengers require some form of travel document to prove they have purchased the required access rights. Travel documents may take the form of either paper, plastic or electronic media. For either commercial or technical reasons, certain products may be available only on certain media. Electronic and Account Based Ticketing fares are also becoming increasingly common, in some systems a payment card or device may be used as a travel document.

Since there are different levels of operational cost and convenience associated with different media types, sales offer packages may be priced differently according to the media they use.

	Type of Travel Document	Name	Note
	**Paper Ticket**	Printed or handwritten paper ticket.	Not machine readable.
	**Machine readable printed ticket**	Machine printed ticket.	Readable by OCR, barcode, shot code or magnetic stripe
	**Self-print ticket**	Self-print ticket (on paper or held electronically).	Machine readable by OCR/barcode.

	**Smartcard**	Card containing chip able to store apps and /or documents. May be used in conjunction with an online account.	Usually machine readable either by OCR, Barcode or NFC.
	**Device App**	On device application holding ticket.	Machine readable by OCR, barcode, or NFC.
	**EMV**	Credit or Debit card used as an identity token.	Machine readable by magstripe, or NFC.
	**SMS**	Token held as SMS on device.	Not machine readable.
	**Identity Card**	Membership card accepted as document.	
	Other	Pass or other proof of rights.	Not machine readable.

Table 21 – Commonly found Types of Travel Document

- The available ticket formats can be indicated in NeTEx by a TYPE OF TRAVEL DOCUMENT.
  - The EFIPEFIP defines a standard set of instances that should be used
- The available TYPEs OF TRAVEL DOCUMENT that apply to a product are specified on a SALES OFFER PACKAGE

6.7.2 *Commonly found Fulfilment Methods for EFIP*

Fulfilment describes how a travel document certifying the purchase of access rights is delivered to the customer. The methods used will typically vary according to the nature of the channel and the types of travel documents available (electronic, printed etc). or any given channel there will be one or more possible methods.

- There may be surcharges for certain methods, such as post.

	Fulfilment	Description	Note
	**Collect at Counter**	Travel document is collected at ticket counter.	Widest selection.
	**Collect from Ticket Machine**	Travel document is collected from an automatic self-service machine.	Standard products.
	**Collect at barrier / validator**	Product is collected by using a validator or barrier	
	**Issued on Board**	Product is issued on board from conductor or machine on board.	Often at a premium
	**Download and print**	Downloaded and printed onto paper.	
	**Download as e-document**	Downloaded and held on a mobile device for inspection.	
	**Postal mail**	Travel document is sent by post.	
	**Courier**	Travel document is sent by courier.	Usually additional fee.
	**Mobile App**	Travel document is downloaded to a mobile device and held in electronic form.	The app may apply additional intelligence as to how the document is used, e.g. activation or validating it.

	**SMS**	Travel document is sent as an SMS message.	

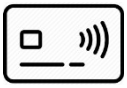
Table 22 – Commonly found Fulfilment Methods

- The available methods be indicated in NeTEx by a FULFILMENT METHOD. Particular payment methods and fulfilment methods can be associated with a specific DISTRIBUTION CHANNEL by means of a DISTRIBUTION ASSIGNMENT.
  - The EFIPEFP defines a standard set of instances that should be used

6.7.3 *Commonly found Payment Methods for EFIP*

The permitted methods of payment may vary from channel to channel and product to product. Most channels will limit payment to one or more methods.

There may be a discount for payment with a specific method.

	Method	Description	Note
	**Cash**	Currency – notes and or coins	On Board May be limited to exact fare
	**Debit/Credit card**	Payment from account by standard EMV card.	Some operators participate in credit card schemes.
	**Contactless card**	Payment from account standard EMV card with NFC to a terminal.	
	**Mobile Payment**	Mobile device Payment system such as Apple pay, Google pay, etc.	May be linked to a Mobile app.

	**SMS**	Payment by SMS from mobile operator account of user.	Low value fares only
	**Cheque**	Cheques are still supported in some countries	May be subject to limit.
	**Direct transfer**	By direct transfer from a bank account.	Used for Season tickets.
	**Standing order**	Products on subscription may be paid for by direct debit at a specified interval (monthly, quarterly, etc)	Used for subscriptions for season tickets.
	**Coupon**	Payment or part payment may be by redeeming a coupon.	Usually additional fee.
	**Mileage Points**	Payment or part payment may be by redeeming accumulated mileage points form a frequent traveller scheme.	Requires an account.
	**Warrant**	Payment may be by submitting a travel warrant (e.g. for armed forces).	
	**Gift voucher**	Payment or part payment may be by redeeming a gift voucher for a specified amount.	

Table 23 – Commonly found Payment Methods

- The available payment methods can be specified in NeTEx by a PAYMENT METHOD. Particular payment methods and fulfilment methods can be associated with a specific DISTRIBUTION CHANNEL by means of a DISTRIBUTION ASSIGNMENT.
- The UK Profile will define a number of different instances of PAYMENT method that should be used by preference.

6.7.4 *Commonly found Distribution Channels for EFIP*

Different products are typically made available for purchase by the public through different sales channels. There may be specific conditions as to which distribution channel can be used to obtain a product. The separate concepts of fulfilment method and payment method are discussed in subsequent sections.

- There will sometimes be a discount for products bought through self-service and advance purchase channels.
- There may be different after-sales conditions attached to different channels. For example, the right to refund or exchange a ticket may only apply if the ticket was bought over the counter.

Channel	Description	Note
At Counter 	In station counter.	Widest selection of products. May exclude online-only offers.
At stop Machine 	Automatic self-service machine at stop.	Standard products.
On Board 	On board from conductor or machine.	Often more expensive and restricted as to products and discounts offered.
Online Web 	One line from a web interface. Subsequent	Fulfilment may be self -print or electronic, or by mail.
Mobile App 	Self-service mobile app using mobile data or wifi.	Fulfilment will normally be handled via the app

SMS 	Self-service using GSM phone with SMS.	
Account Based Ticketing with PAYG 	Pay as you go. May be smartcard or online account based (e.g. for contactless EMV card), or both.	
Central Office 	Certain products may only be available at central ticket offices in person.	Common for Annual passes and special discounts
Call centre 	Certain products may only be available by phone or by emailing a service centre.	May be different prices for different speeds of fulfilment: first class post, courier, etc. Email may also be used for fulfilment
Third Party Retailers 	Including kiosks, newsagents etc with retail footfall.	May charge an additional fee.

Table 24 – Commonly found Distribution Channels

- The available channels can be represented in NeTEx by DISTRIBUTION CHANNELs. Particular payment methods and fulfilment methods can be associated with a specific DISTRIBUTION CHANNEL by means of a DISTRIBUTION ASSIGNMENT that applies to a specific SALES OFFER PACKAGE.

6.7.5 *Commonly found After Sales conditions in EFIP*

Fare Products and/or sales packages may be subject to specific conditions about the after sales commercial rights of the user.

Trip	Single	
Can modify time of travel/ Exchange 	Whether the time of travel or route can be modified.	❖ EXCHANGING usage parameter.
Can Refund 	Whether the user can obtain a refund for and unused or partly used product, either ahead of travel or after having failed to use it. Mainly relevant for season passes.	❖ REFUNDING usage parameter.
Replacement possible 	Whether a replacement can be obtained if lost, possibly retaining the balance on an account-based product. Mainly relevant for season passes and smartcard products	❖ REPLACING usage parameter.

Table 25 – Commonly found After Sales Conditions

❖ In NeTEx, a number of different types of USAGE PARAMETER are used to set additional commercial conditions on post-sales handling of the ticket. These can include EXCHANGING, REFUNDING, REPLACING, etc., etc.

6.8 Summary of EFIP Price features

An important aspect of fares is of course the **fare price**, that is, the monetary amount associated with each selectable combination of tariff features.

- ❖ In NeTEx, prices are represented separately from the elements that are priced, making it possible to have successive prices sets at different times with different validity periods, without altering the tariff structure. It also allows for prices for the same products to be stated in different currencies, or even, for example, be stated in other valuation units such as frequent flier points from a membership scheme.
- ❖ Prices are represented as specialisations of FARE PRICE.
- ❖ Sets of prices can be organised using a FARE TABLE – see later below. Prices may be included in the same document as the fare structure, or be exchanged separately.

- It is also possible to indicate that some or all prices are supplied dynamically through a pricing service, as for many yield managed products.

6.8.1 *Pricing dimensions*

Anything which is a pricing factor for a EFIP Fare needs to be represented by a priceable model element in the profile. Thus, for example for a point to point fare where ther is a separate price for each origin/destination pair, there should be a price for each DISTANCE MATRIX ELEMENT. Other triff elements commonly requiring a price are TARIFF ZONEs, TIME INTERVALs,HEOGRAPHIC INTERVALs USER PROFILEs, TYPES OF TRAVEL DOCUMENT all variously contribute to prices and so are included in the profile.

Other factors may be relevant for passenger information, but do not necessarily need to be modelled for pricing. For example, if different prices are offered for children and seniors for the same product , then a user type (USER PROFILEs - e.g. *adult*, *child*, *senior*) needs to be in the profile and to have an associated price. But if the product is marketed as two separate SALES OFFER PACKAGEs, the different prices can be associated with the SALES OFFER PACKAGE and not with the USER PROFILE i.

The price table for a given SALE OFFER PACKAGE is typically an NxN matrix of all the relevant factors, (possibly with certain combination instances excluded.). The actual choice of factors depends on the tariff structure and product type..

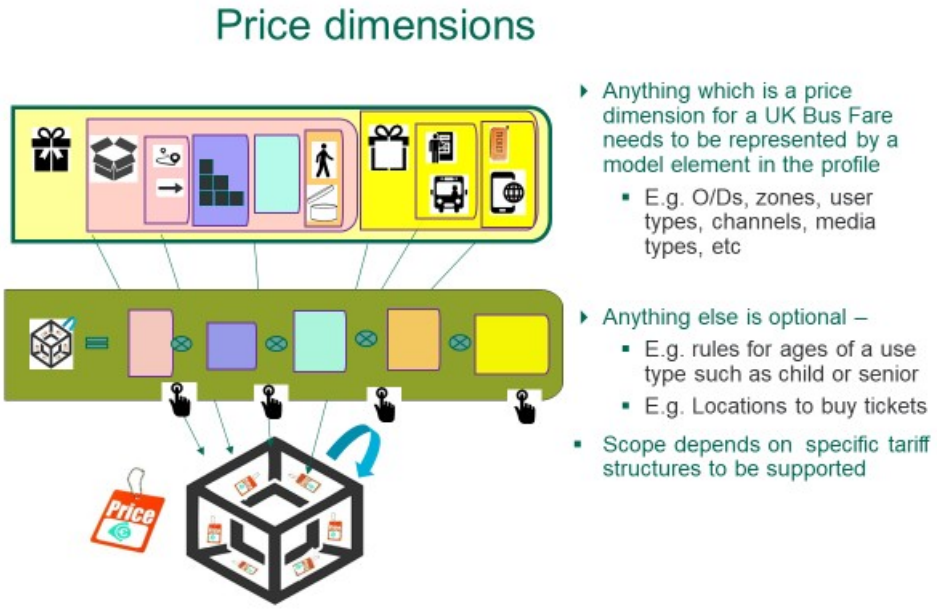


Figure 14 – Pricing dimensions.

The most common combinations of tariff factors affecting prices found in price tables are the following:

6.8.1.1 *Point-to-point trip*

- DISTANCE MATRIX ELEMENT x USER PROFILE x (SALES OFFER PACKAGE)

6.8.1.2 *Stage fare trip*

- GEOGRAPHICAL INTERVAL x USER PROFILE x (SALES OFFER PACKAGE)

6.8.1.3 *Zonal Fare*

- FARE ZONE x TIME INTERVAL x USER PROFILE x (SALES OFFER PACKAGE)

Where the SALES OFFER PACKAGE also embodies a specific media (TYPE OF TRAVEL DOCUMENT) – such as *paper*, *smartcard*, or *mobile app* – and DISTRIBUTION CHANNEL (e.g. *at stop*, *on-board*, *online*).

6.8.2 *Other pricing dimensions*

Other factors may also apply, for example the FULFILMENT METHOD, if an additional charge is charged for delivery by post or courier.

The following figure summarises the NeTEx elements most relevant as price dimensions.

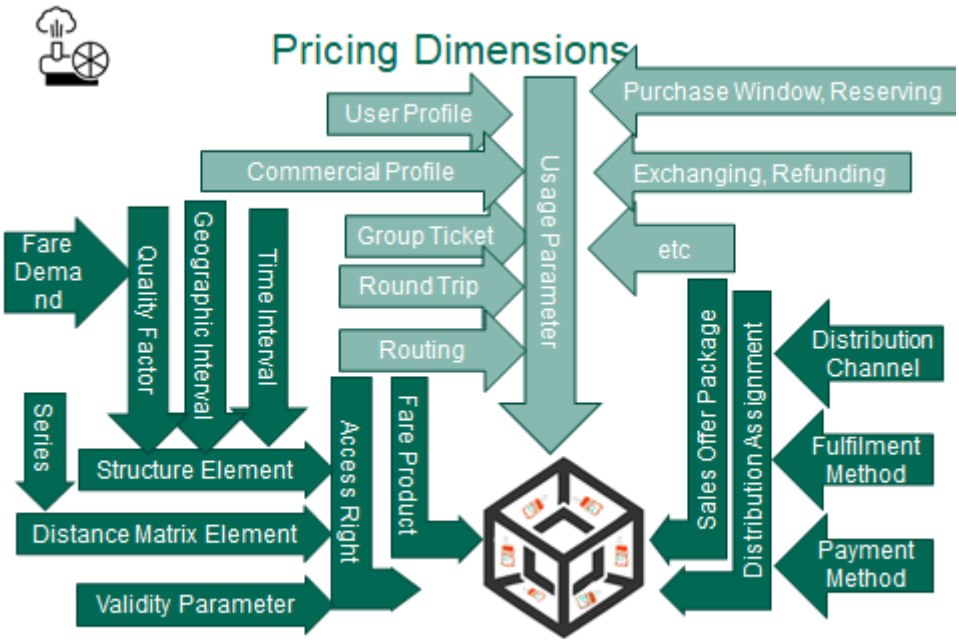


Figure 15 – *NeTEx* Pricing dimension elements

6.8.2.1.1 *Associating fares with prices*

NeTEx allows sevel different ways of attaching prices to PRICEABLE OBJECTs. For simplicity the EFIP uses FARE TABLES and CELLS. Each CELL references all the factors contributing to the CELL PRICE.

Note: NeTEx also supports the use of specific prices, DISTANCE MATRIX ELEMENT PRICEs, USAGE PARAMETER PRICEs etc.

6.8.3 *Absolute and derived prices*

Passenger information for EFIP fares includes both prices stated as absolute amounts (e.g. €4.50) and prices stated as a derivation of another price (E.g. *The child price is 50% of the adult price*), or *each additional person on a group ticket is €1.00*).

- NeTEx supports complex price derivation rules, with discounting factors and minimum and maximum limits, so that one price may be derived from another (using a PRICE RULE, DISCOUNTING RULE or LIMITING RULE).
- ROUNDING rules may be specified for the calculation so that the result is quantized to specified currency units.
- The derivation of a price may be recorded as a RULE STEP.
- Prices may be attached to many different types of tariff structure and fare product element. The allowed combinations of prices of the different types of factor can be used to create an n-dimensional pricing matrix.
- Rules may be chained, so for example one might apply successive discounts and then calculate added tax.

6.8.4 *Dynamic prices*

In NeTEx, prices can also be ***dynamic*** **(***rather than* **static)**, that is, fetched from a named pricing service at the time a fare offer is made. Dynamic pricing is mainly used for yield managed fares on long distance travel, i.e. not on local buses.

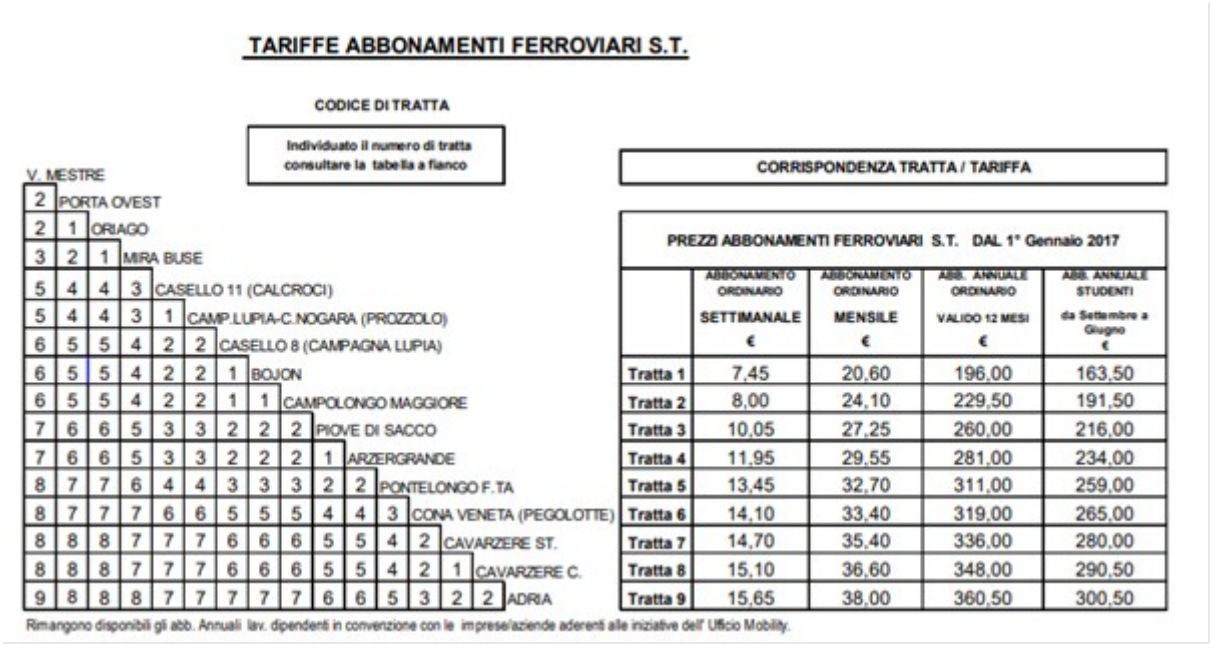
6.8.5 *Prices in the EFIP*

BasicF In the EFIP Basic Fare profile all prices should be absolute – derived prices should not be used.

ExtendedF In the EFIP Extended Fare profile derived prices are allowed. Derived prices are specified using PRICING RULES, and a base price. ROUNDING parameters may also be specified.

6.8.6 *Indirection -Use of Price groups*

It is common practice to simplify tariffs by grouping prices into price bands (For example *Band A = €1.00*, *Band B=€1.50*, *Band C = €1.80.* etc). Sometimes these will be used as an internal mechanism that is not exposed to passengers, in other cases they will be shown on published tariffs (e.g. see Figure 16).



TARIFFE ABBONAMENTI FERROVIARI S.T.

CODICE DI TRATTA

Individuato il numero di tratta consultare la tabella a fianco

V. MESTRE													
2	PORTA OVEST												
2	1	ORIAGO											
3	2	1	MIRA BUSE										
5	4	4	3	CASELLO 11 (CALCROCI)									
5	4	4	3	1	CAMP.LUPIA-C.NOGARA (PROZZOLO)								
6	5	5	4	2	2	CASELLO 8 (CAMPAGNA LUPIA)							
6	5	5	4	2	2	1	BOJON						
6	5	5	4	2	2	1	1	CAMPOLONGO MAGGIORE					
7	6	6	5	3	3	2	2	2	PIOVE DI SACCO				
7	6	6	5	3	3	2	2	2	1	ARZERGRANDE			
8	7	7	6	4	4	3	3	3	2	2	PONTELONGO F.TA		
8	7	7	7	6	6	5	5	5	4	4	3	CONA VENETA (PEGOLOTTE)	
8	8	8	7	7	7	6	6	6	5	5	4	2	CAVARZERE ST.
8	8	8	7	7	7	6	6	6	5	5	4	2	1	CAVARZERE C.
9	8	8	8	7	7	7	7	7	6	6	5	3	2	2	ADRIA

CORRISPONDENZA TRATTA / TARIFFA

PREZZI ABBONAMENTI FERROVIARI S.T. DAL 1° Gennaio 2017

	ABBONAMENTO ORDINARIO SETTIMANALE €	ABBONAMENTO ORDINARIO MENSILE €	ABB. ANNUALE ORDINARIO VALIDO 12 MESI €	ABB. ANNUALE STUDENTI da Settembre a Giugno €
Tratta 1	7,45	20,60	196,00	163,50
Tratta 2	8,00	24,10	229,50	191,50
Tratta 3	10,05	27,25	260,00	216,00
Tratta 4	11,95	29,55	281,00	234,00
Tratta 5	13,45	32,70	311,00	259,00
Tratta 6	14,10	33,40	319,00	265,00
Tratta 7	14,70	35,40	336,00	280,00
Tratta 8	15,10	36,60	348,00	290,50
Tratta 9	15,65	38,00	360,50	300,50

Rimangono disponibili gli abb. Annuali lav. dipendenti in convenzione con le imprese/aziende aderenti alle iniziative dell' Ufficio Mobility.

Figure 16 – Example of use of price groups in a tariff (Adria Mestre)

- The PRICE GROUP element can be used in NeTEx to represent a shared price.
- If the price bands are based on a number or range of units, a GEOGRAPHICAL INTERVAL PRICE can be used directly as a “price band”

6.8.7 *Organisation of EFIP prices*

The exchange of fare data typically involves the exchange of large numbers of prices. Typically, there will be a separate price for each permitted combination of the discrete tariffs and fare product elements.

- NeTEx provides a structure, the FARE TABLE to allow the efficient organisation of prices as nested tables. A fare table is made up of CELLs.

  Prices can be arranged as hierarchically nested rows, with tariff elements attached to each nesting level, as shown in Figure 17.

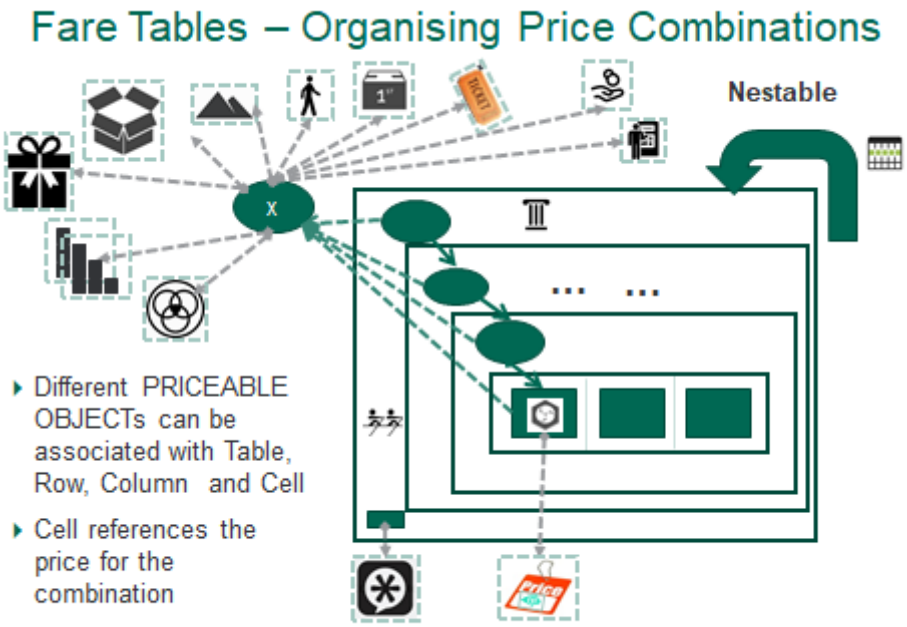


Figure 17 – Nesting prices for fare elements

The optimum hierarchy can be chosen for any given application

The following table (Table 26) gives some examples of price data for some products and their permitted combinations of parameters that might be nested in various alternate hierarchies, for example, as ordered in the table, it could be nested:

- *A to B / Product / Class of Use / User Type / Channel / Media*

However, it would so be possible to nest the data according to any one of several arbitrary alternative hierarchies, for example:

- *A to B / Product / User Type / Class of Use / Channel / Media*
- *A to B / Product / Channel / Media / User Type / Class of Use*
- *Class of Use / User Type / Product / Channel / Media / A to B*
- *etc*

O/D	Product	Type	Class of Use	User Type	Channel	Media	Amount
A+B	Single	Trip	1st Class	Adult	online	selfprint	10.00
A+B	Single	Trip	1st Class	Adult	online	mobile app	9.00
A+B	Single	Trip	1st Class	Adult	at stop	smart card	9.00
A+B	Single	Trip	1st Class	Adult	at stop	printed	11.00
A+B	Single	Trip	1st Class	Adult	onboard	printed	12.00
A+B	Single	Trip	2nd Class	Adult	online	selfprint	7.00
A+B	Single	Trip	2nd Class	Adult	online	mobile app	6.00
A+B	Single	Trip	2nd Class	Adult	at stop	smart card	6.00
A+B	Single	Trip	2nd Class	Adult	at stop	printed	8.00
A+B	Single	Trip	2nd Class	Adult	onboard	printed	8.50
A+B/B+A	Return	Trip	1st Class	Adult	online	selfprint	18.00
A+B/B+A	Return	Trip	1st Class	Adult	online	mobile app	16.00
A+B/B+A	Return	Trip	1st Class	Adult	at stop	smart card	17.00
A+B/B+A	Return	Trip	1st Class	Adult	at stop	printed	19.00
A+B/B+A	Return	Trip	1st Class	Adult	onboard	printed	n/a
A+B/B+A	Return	Trip	2nd Class	Adult	online	selfprint	12.00
A+B/B+A	Return	Trip	2nd Class	Adult	online	mobile app	10.00
A+B/B+A	Return	Trip	2nd Class	Adult	at stop	smart card	10.00

A+B/B+A	Return	Trip	2nd Class	Adult	at stop	printed	11.00
A+B/B+A	Return	Trip	2nd Class	Adult	onboard	printed	n/a
A+B	Pass	1 week	1st Class	Adult	At counter	smart card	100.00
A+B	Pass	1 month	1st Class	Adult	At counter	smart card	350.00
A+B	Pass	1 week	1st Class	Child	At counter	smart card	50.00
A+B	Pass	1 month	1st Class	Child	At counter	smart card	175.00
A+B	Pass	1 week	2nd Class	Adult	At counter	smart card	60.00
A+B	Pass	1 month	2nd Class	Adult	At counter	smart card	200.00
A+B	Pass	1 week	2nd Class	Child	At counter	smart card	40.00
A+B	Pass	1 month	2nd Class	Child	At counter	smart card	120.00

Table 26 – Sample Product Price data for trip from A to B

6.8.8 *Presentation of fares and fares prices*

NeTEx is concerned primarily with the semantic aspects of representing fare, rather than with presentation aspects such as layout, formatting and fonts of tariff tables. It does however have some basic layout capabilities that are useful to guide rendering of data in a tabular form, as indicated in Figure 18.

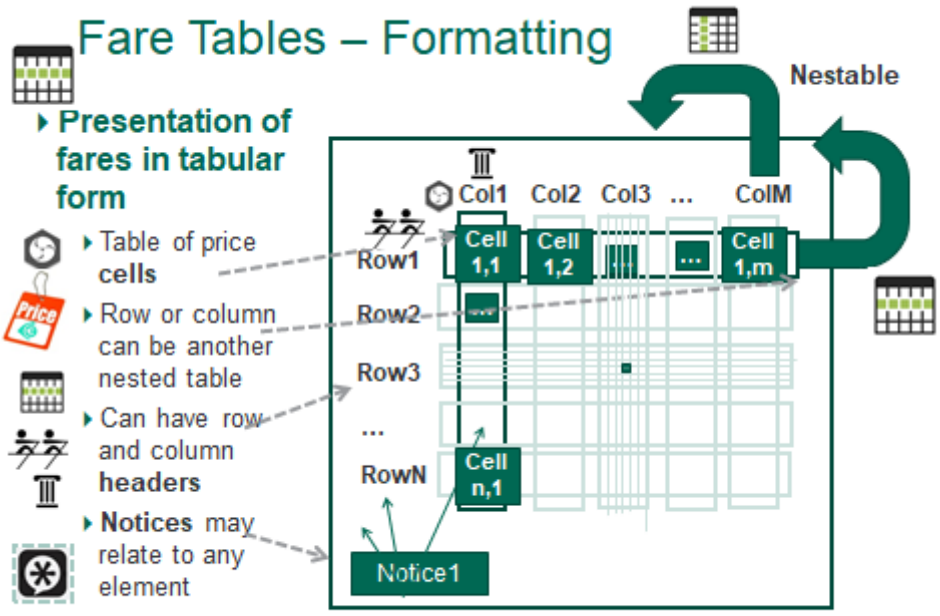


Figure 18 – Formatting a fare table

- In NeTEx, the PRESENTATION element can be used to set standard colours, fonts, etc for consistent presentation for zones, lines etc.
- The rows and columns of a FARE TABLE can have headings, nested recursively.
- The BRANDING element can be used to associate logos and branding names with products.
- The NOTICE element can be used to associate footnote with the cells, to the rows and columns of a fare table.

6.8.9 *Visibility of underlying Tariff structures*

We note that the pricing structure presented to the user may not necessarily expose the underlying fare structure use by the operator – for example:

- What are actually kilometre distance-based fares may be translated into simple point-to-point fares.
- Price groups may be used to give the same fare to many different journeys, but users will not necessarily be aware of them as they will only see the resolved prices assigned to each individual trip.
- Separately priced product elements (such as parking, a trip and travel in a destination zone) may be compounded into a single sales offer so that the underlying discrete pricing of elements is not visible.

6.8.10 *Other factors used in Tariff Structures*

There may be other factors relevant for tariff structures, either limiting tariffs more specifically, or offering a set of prices based on a further factor (see Table 27).

Nature	Description	Comment	Examples
Specific Journey limited **(non-flexible vs flexible)**	The passenger may be required to make the journey on a specific service at a certain time.	Normal for coach and fares requiring a reservation (though if a flexible fare this may be changeable). Also normal for advance fares sold at a discount to the standard fare, which require travel by a specific train. Certain tariffs may include or exclude one or more specific journeys.	❖ In NETEX the tariff may be restricted to specific journeys, or a purchase may be restricted to the service indicated by a reservation.
Quota steps	Operators may place quotas on the numbers of seats available for a specific product at a specific price.	Operators may simply wish to indicate that supply is limited i.e. quota applies (but not reveal what the specific thresholds are). In other cases, the operator may want to provide the size of the quotas and /or the different pricing steps so as to indicate the range of possible prices that may be found for a	❖ In NeTEx FARE QUALITY FACTOR can be used to indicate quota steps ❖ Also, in NETEX the fare product

		given product. Where price comparison engines are used, the quota price range may be used to populate a "Prices from x" value. May be relevant for coach or rail, not relevant for bus Fû Only in EFIP Extended fare profile	summary conditions can indicate if a quota applies
Entitlements required (Prerequisites)	Some products may require the owner to already have another product (	The prerequisite may be another fare product, or a third-party non-travel scheme (e.g. a registered disabled pass) A discount may be associated with a particular entitlement.	❖ In NeTEx stated can be stated with an ENTITLEMENT REQUIRED usage parameter.
Entitlements required (Prerequisites)	Some products may give rights to purchase specific other products, sometimes at a discount.	The entitlement may be another fare product, or a third-party non-travel scheme (e.g. a ticket to an event)	❖ In NeTEx entitlements can be stated with an ENTITLEMENT GIVEN usage parameter.
Corporate Schemes and "Frequent flier " dscounts	Operators may offer a discount on price to Employers according to the number of Employees.	In some cases, the operator will publish a scale of prices – in others it is by negotiation.	❖ In NeTEx Can use Fare Quality Factors to define Pricing steps for different numbers of employees. ❖ COMMERCIAL PROFILE parameter can represent employee and membership products

Table 27 – Other tariff aspects of tariff structures

- NeTEx allows a number of different validity parameters (to be used in fare structures These can be used to reference network and service elements (for example, SERVICE JOURNEY, OPERATOR, TRAIN CATEGORY, etc etc.) to specify conditions and exclusions. A GENERIC PARAMETER ASSIGNMENT can be used to associate various logical combinations of conditions with specific FARE STRUCTURE ELEMENTs and VALIDABLE ELEMENTs, as well as FARE PRODUCTs.
- NeTEx also provides a general-purpose FARE QUALITY FACTOR that can be used to include any arbitrary additional quantitative or range-based criteria (such as the quota given above) in a tariff structure.

7 EFIP Framework

7.1 Basic elements

The Transmodel NeTEx fare model used by the EFIP makes use of a number of common NeTEx Framework elements that are described in the NeTEx specifications and in the EPIP, so are not described again here. The framework provides common mechanisms for all NeTEx objects to support effective data exchange, such as versioning, validity, grouping, and responsibility tracking. The EFIP also makes use of a number of resuable components suchas calendar, acceesibility etc, etc. The elements are summarised here – refer to the other specifcatiosn for further details.

7.1.1 *Core Framework elements (NeTEx Part1)*

- DATA MANAGED OBJECT,
- VALIDITY CONDITIO,N
- RESPONSIBILITY SET, ORGANISATION, CONTACT DETAILS,
- VERSION FRAME,
- GROUP OF ENTITIES.

7.1.2 *Generic components (NeTEx Part1)*

- LOCATION, POINT, LINK, ZONE LINK SEQUENCE, PLACE, SECTION, PROJECTION,
- SECURITY LIST,
- LOGGABLE OBJECT, EVENT.

7.1.3 *Reuseable Components for Public Transport (NeTEx Part1)*

- ACCESSIBILITY ASSESSMENT,
- FACILITY SET, FACILITY,
- TRANSPORT MODE,
- NOTICE, ALTERNATIVE TEXT,
- SERVICE CALENDAR, DAY TYPE, PROPERTY OF DAY, OPERATING DAY, DAY TYPE ASSIGNMENT.

7.1.4 *Network elements also used in EPIP (NeTEx Part1/Transmodel Part 1&2)*

The Transmodel/NeTEx fare model used by the EFIP also makes use of the same descriptive elements that are used to describe the network topology (as stops, lines, zones etc).

The coloured data elements in the figure below show the concrete entities (abstract classes are greyed).

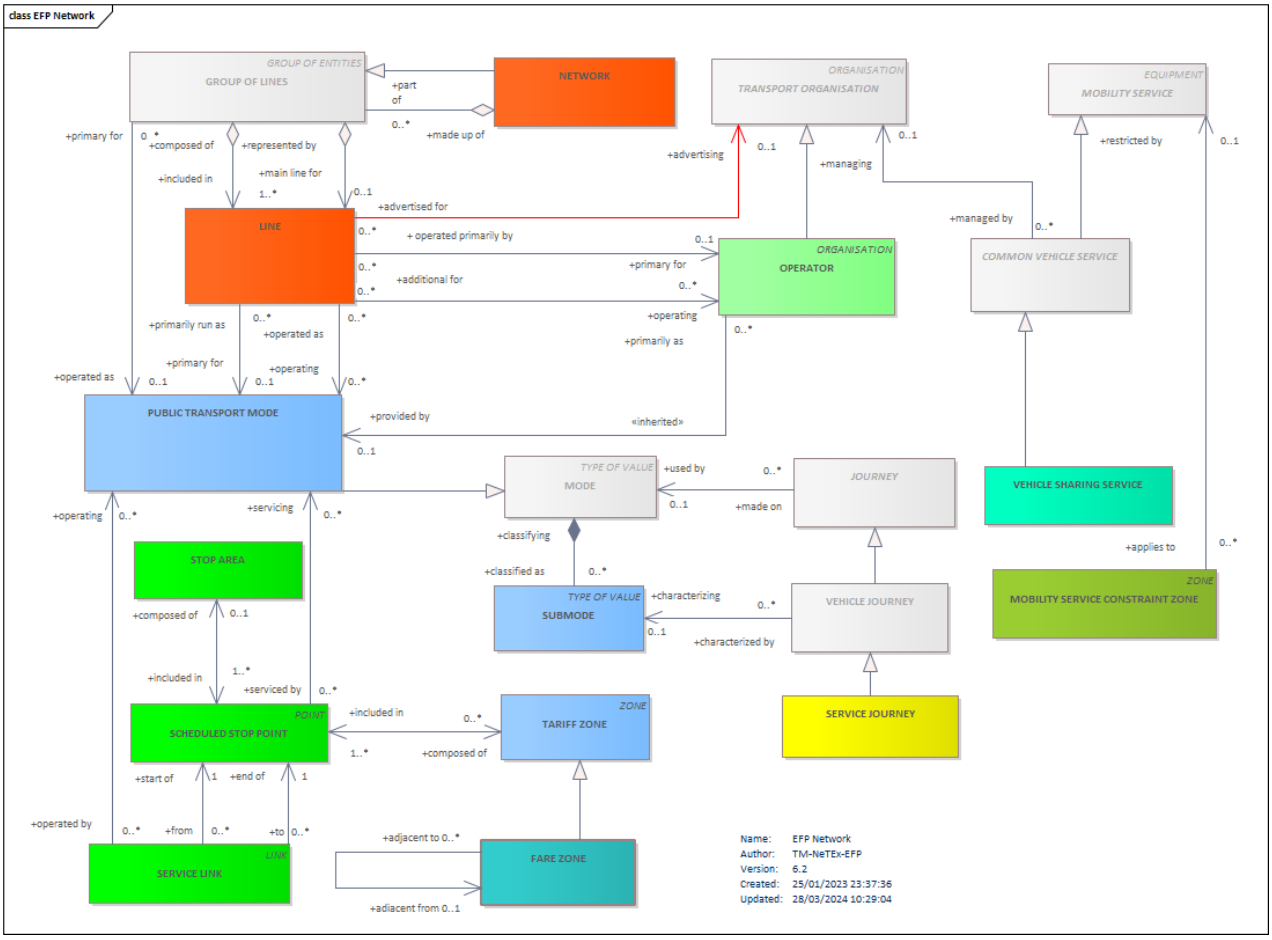


Figure 19 - High level Overview of Network Model (UML)

Network elements include

- LINE, GROUP OF LINES, NETWORK
- OPERATOR, AUTHORITY, GROUP OF OPERATORS
- VEHICLE MODE
- STOP PLACE, SCHEDULED STOP POINT and STOP ASSIGNMENT

See EPIP for further details.

7.1.5 *Timetable elements also used in EPIP (NeTEx Part2/Transmodel Part 3)*

The Transmodel / NeTEx fare model used by the EFIP also makes use of some of the same descriptive elements that are used to describe the timetables (service journeys, etc). These include

SERVICE JORUNEY, TRAIN NUMBER

ee EPIP for further details.

8 EFIP: Overview

The remaining parts of this document give a summary of the elements in the EFIP fare profile. Extracts of Transmodel / NeTEx data models show in colours concrete data elements (abstract classes are greyed).

8.1 Fare Specification aspects

NeTEx defines fare structures and prices using a layered set of reusable components. The following diagram summarises the main layers as four separate columns : Sales Offer, Fare Product, Fare Structure and Fare Price (visualised from left to right).

- A TARIFF is a coherent set of FARE STRUCTURE ELEMENTs which serve to describe the different aspects of the fare structure and relate them to the underlying transport network - for example the available tariff zones, stop-to-stop transitions, pass durations, etc.
- A FARE PRODUCT is a marketable set of access rights, specified as one or more VALIDABLE ELEMENTs, each referencing one or more FARE STRUCTURE ELEMENT.
- A SALES OFFER PACKAGE packages one or more FARE PRODUCTs as an offering which may be purchased by a customer; it adds information of the types of ticket and distribution channels and other commercial conditions. It also specifies what TYPE OF TRAVEL DOCUMENT (i.e. how the product is materialised as a ticket).
- FARE PRICEs specify the prices for the different elements to which a price may attach. Prices may be held at many different levels and there will be separate prices for each allowed combination of product options.

Each of these aspects is described further in the chapters later below.

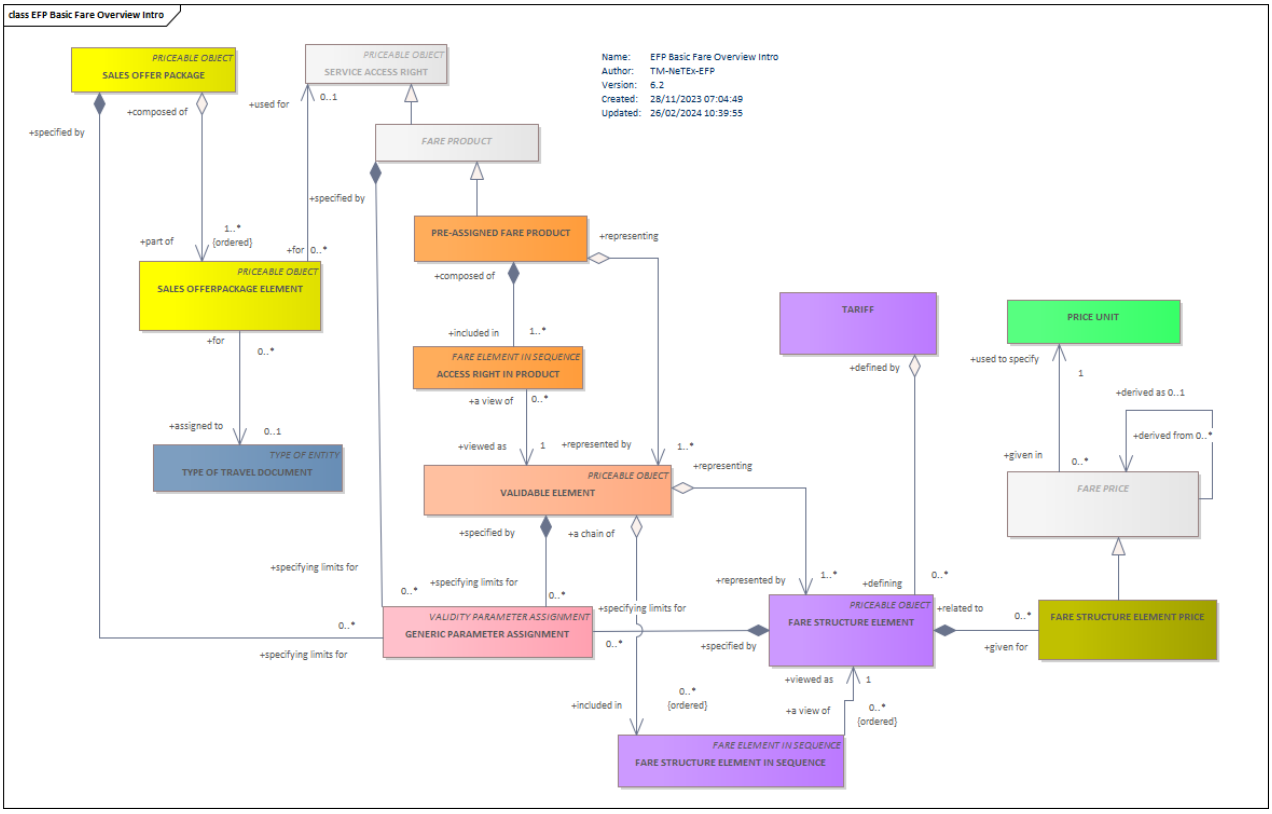


Figure 20 - High level Overview of Network Model (UML)

8.2 Fare Model Submodels

NeTEx Fare model is modularised into a number of submodels defined as UML packages, these in turn depend on packages in NeTEx Part1 and Part2 that define framework and transport network components.

- The FARE ZONE Models describe the network related fare constructs.
- The FARE PRICE Model present FARE PRICEs and PRICE GROUPs; the FARE CALCULATION Model holds common pricing parameters; The FARE TABLE Model presents FARE TABLEs for organising large groups of prices.
- The FARE STRUCTURE Models provide the various types of element used to represent different tariff structures.

- The FARE PRODUCT Models describes the available FARE PRODUCTs.
- The USAGE PARAMETER Models describe the limiting conditions for the fare products.
- The FARE ACCESS RIGHT PARAMETER Models assign the access rights to specific products and limiting parameters.
- The SALE OFFER PACKAGE Model describes how the fare elements are combined as marketable components using various TYPES OF TRAVEL DOCUMENT
- The FARE FRAME Model describes the elements used to group fare data for exchange.

9 EFIP:Fare Network

9.1 General

NeTEx Part1 includes the concept of a TARIFF ZONE, which can be used to define the permanent fare zones of a network. A given SCHEDULED STOP POINT can belong to one or more TARIFF ZONEs.

- Many networks will use TARIFF ZONEs. A TARIFF ZONE is a view of a ZONE, specifically defined for fare calculation. It is composed of SCHEDULED STOP POINTs. A TARIFF ZONE may have specific points on its borders, the TARIFF POINTs. Some such points activate an automatic detection of the NeTEx Part3 FARE ZONE MODEL extends these elements and also adds further concepts relating to the network that can be used additionally to underpin fare structures.
- FARE SCHEDULED STOP POINT specialises SCHEDULED STOP POINT, extending its definition with additional fare related attributes
- A FARE ZONE is a specialization of TARIFF ZONE that may have additional properties, in particular FARE SECTIONs associated with it.
- A FARE SECTION is a subdivision of a JOURNEY PATTERN, consisting of consecutive FARE POINTs in PATTERN, each referencing a SCHEDULED STOP POINT in that JOURNEY PATTERN. FARE SECTIONs allow arbitrary sections of the network to be associated with a specific FARE ZONE or staging points.
- A SERIES CONSTRAINT allows a constraint on a routing between an origin and destination to be specified, for example that a passenger may travel on any routes that pass through particular intermediate stops, or that TARIFF ZONES must only be traversed in a specifc order. Alternative SERIES CONSTRAINTS can be used to specify different distances for pricing different routings.

- To be noted: The following sections use either the Transmodel notation (UPPER CASE) or the NeTEx notation (CamelCase). For example: FARE SCHEDULED STOP POINT (Transmodel entitiy) becomes *FareScheduledStopPoint (XML Element).*

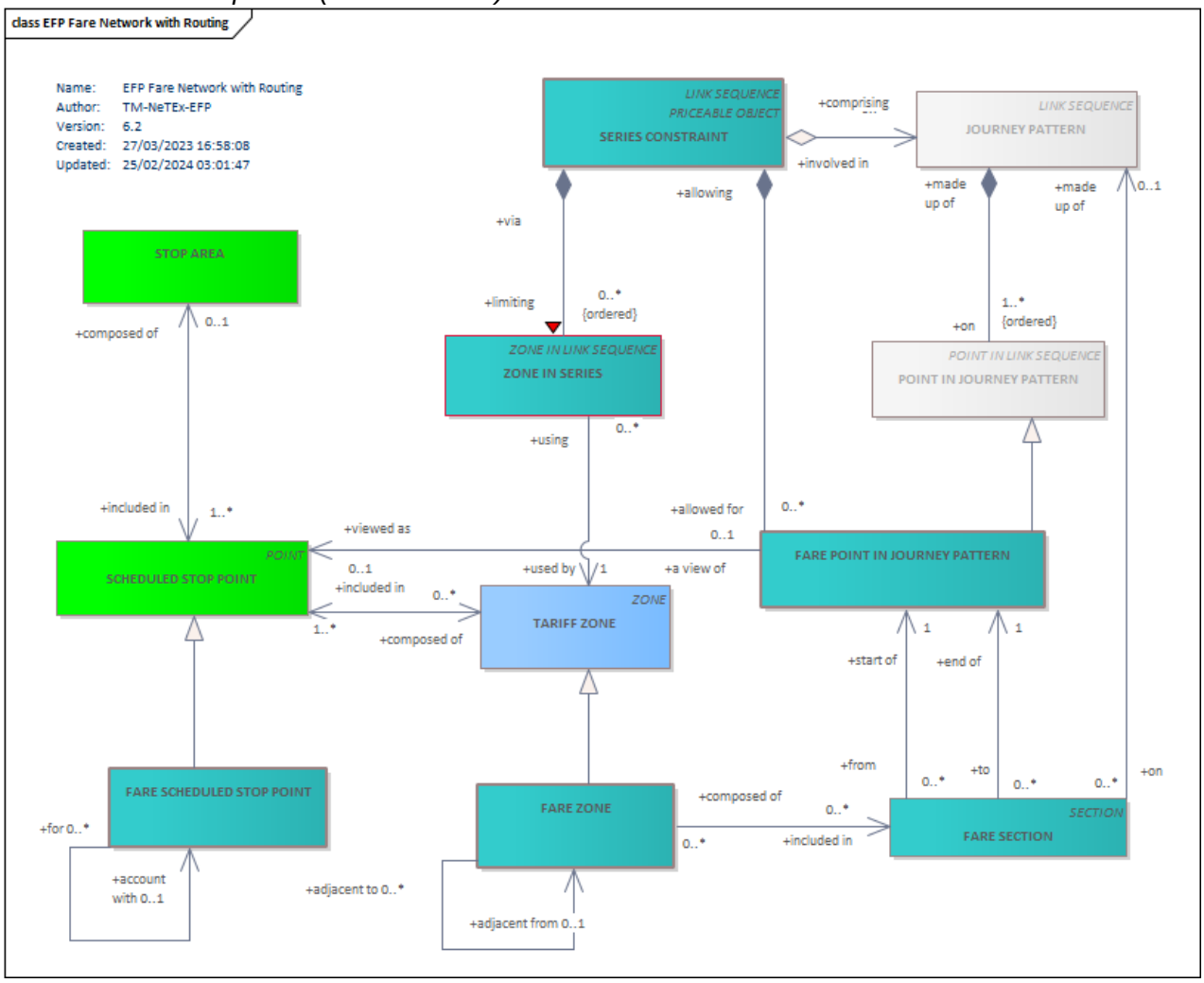


Figure 21 — Basic use of Network elements for fares (UML)

9.2 FareScheduledStopPoint

A specialisation of SCHEDULED STOP POINT describing a stop with fare accounting and routing characteristics.

A FARE SCHEDULED STOP POINT is used just as if it were a SCHEDULED STOP POINT, but has some extra attributes.

Classification	Name	Type	Cardinality	Description
::>	::>	*ScheduledStopPoint*	::>	FARE SCHEDULED STOP POINT inherits from SCHEDULED STOP POINT. See NeTEx Part1.
«PK»	***id***	*FareStopPointIdType*	1:1	Identifier of FARE SCHEDULED STOP POINT.
	SiteFacilitySet	*SiteFacilitySetRef*	0:1	Set of Facilities available at the stop.
	NameOnRouting	*MultilingualString*	0:1	Name to use to indicate station on routings and itineraries.
«FK»	***AccountingStop-PointRef***	*FareScheduled-StopPointRef*	0:1	Identifier of another station to use for accounting purposes for this station.
«FK»	***BorderPointRef***	*BorderPointRef*	0:1	BORDER POINT associated with FARE SCHEDULED STOP POINT.

Table 28 – *FareScheduledStopPoint* – XML Element

9.3 Border point

A Point on the Network marking a boundary for fare calculation. May or may not be a SCHEDULED STOP POINT.

Classification	Name	Type	Cardinality	Description
::>	::>	*TimingPoint*	::>	BORDER POINT inherits from TIMING POINT. See NeTEx Part1.
«PK»	***id***	*BorderPointIdType*	1:1	Identifier of BORDER POINT.

	ShortName	*MultilingualString*	0:1	Short Name of BORDER POINT.
	Description	*MultilingualString*	0:1	Description of BORDER POINT.
«FK»	***GroupOf-OperatorsRef***	*GroupOfOperatorsRef*	0:1	OPERATORs related to BORDER POINT.

Table 29 – BorderPoint – Element

A border point is a conventional location used to indicate where the responsibility of the passenger is passed from one RU to the next one in case of successive carriers.

9.4 FareZone

A specialization of TARIFF ZONE to include designated FARE SECTIONs.

A FARE ZONE is used just as if it were a SCHEDULED STOP POINT, but has some extra attributes.

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*TariffZone*	::>	FARE ZONE inherits from TARIFF ZONE. See NeTEx Part1.
«PK»	***id***	*FareZoneIdType*	1:1	Identifier of FARE ZONE.
«FK»	***ParentFareZone Ref***	*FareZoneRef*	0:1	Parent FARE ZONE of which this is part.
«enum»	***ZoneTopology***	*ZoneTopologyEnum*	0:1	Topology of FARE ZONE with regard to other zones. See allowed values below. • *ring;* • *annular;* • *nested;* • *tiled;* • *sequence;*

				• *overlappingSequence;* • *other*
«enum»	***ScopingMethod***	*ScopingMethodEnum*	0:1	Indication of how member stops of a FARE ZONE are specified; The default value is '*explicitStops'.* • *explicit* • *implicitSpatialProjection* • *explicitPeripheryStops* • *other*
«FK»	***Transport-OranisationRef***	*(TransportOrganisationRef) OperatorRef | AuthorityRef*	0:1	Reference to OPERATOR of FARE ZONE.
«cntd»	***GroupOf-Operators***	*GroupOfOperators*	0:*	Reference to GROUP OF OPERATORs (also AUTHORITies) for FARE ZONE.
«cntd»	***fareSections***	*FareSection*	0:*	FARE SECTIONs in FARE ZONE.
«cntd»	***neighbours***	*FareZoneRef*	0:*	Adjacent FARE ZONEs.

Table 30 – *FareZone* – XML Element

9.5 FareSection

A subdivision of a JOURNEY PATTERN consisting of consecutive POINTs IN JOURNEY PATTERN, used to define an element of the fare structure.

Classifi-cation	Name	Type	Cardin-ality	Description
::>	::>	*CommonSection*	::>	FARE SECTION inherits from COMMON SECTION. See NeTEx Part1.
«PK»	***id***	*FareSectionIdType*	1:1	Identifier of FARE SECTION.

	Name	*MultilingualString*	0:1	Name of FARE SECTION.
«FK»	***JourneyPatternRef***	*JourneyPatternRef+*	0:1	Reference to a JOURNEY PATTERN that FARE SECTION follows.
«FK»	***FromFarePointRef***	*FarePointInPatternRef*	0:1	Reference to FARE POINT IN PATTERN at which FARE SECTION starts.
«FK»	***ToFarePointRef***	*FarePointInPatternRef*	0:1	Reference to FARE POINT IN PATTERN at which FARE SECTION ends.

Table 31 – *FareSection* – XML Element

9.6 FarePointInPattern (Subcomponent of FareSection)

A POINT IN PATTERN which represents the start or end of a FARE SECTION.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*PointInJourneyPattern*	::>	FARE POINT IN PATTERN inherits from POINT IN JOURNEY PATTERN. See NeTEx Part1.
«PK»	***id***	*FaresPointInPattern-IdType*	1:1	Identifier of a FARE POINT IN PATTERN.
	ScheduledStop-PointView	*ScheduledStopPointView*	0:1	Derived information about the SCHEDULED STOP POINT, such as its name – see NeTEx Part1.
	Abridgment-Ranking	*xsd:integer*	0:1	Relative ranking for omitting this FARE POINT IN PATTERN when presenting an abridged version of the series as an itinerary. 1=High, i.e. Omit first.
«enum»	***Presentation-Position***	*SeriesPresentationEnum*	0:1	Relative position for showing this FARE POINT IN PATTERN in an itinerary when there is a choice according to rail conventions. For example, *(A / B) * C* versus *(B/A) * C.*

				• *noStation; requiredStation; optionalLeft; optionalRight*
	IsForbidden	*xsd:boolean*	0:1	Whether use of fare point is forbidden- can be used to explicitly exclude certain routings. The default value is '*false*'.
	Interchange-Allowed	*xsd:boolean*	0:1	Whether interchange to another service is allowed at this STOP POINT.
	IsFareStage	*xsd:boolean*	0:1	Whether stop is considered to be a fare stage. +v1.1

Table 32 – *FarePointInPattern* – XML Element

9.7 SeriesConstraint

A specialization of TARIFF ZONE to include designated FARE SECTIONs.

A FARE ONE is used just as if it were a SCHEDULED STOP POINT, but has some extra attributes.

An extension of a DISTANCE MATRIX ELEMENT, a cell of an origin-destination matrix for TARIFF ZONEs or STOP POINTs, expressing a fare distance for the corresponding trip (as a value in km, number of fare units etc.) and possibly a constraint to allow travel only on specific routes.

Table 1 — Classification	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	SERIES CONSTRAINT inherits from PRICEABLE OBJECT.
«PK»	***id***	*SeriesConstraintIdType*	1:1	Identifier of SERIES CONSTRAINT.
	order	*xsd:integer*	0:1	Relative priority when there are multiple SERIES CONSTRAINTs between the same points. (Equivalent to route number on TAP TSI Series Number).
	PrivateCode	*PrivateCodeType*	0:1	Private Code associated with element.
	Itinerary	*xsd:normalizedString*	0:1	Stylised text description of SERIES CONSTRAINT. See Tap TSI 5.1. and above
	SymbolMarking-UsualRoute	*xsd:normalizedString*	0:1	Symbol to use to denote the usual route.
	SeriesType	*SeriesTypeEnum*	0:1	Classification of SERIES CONSTRAINT. Default is station to statistationToStation Series runs from a station to a station. originToBorder Series runs from origin to border point. borderToDestination Series runs from border point to destination. transit Series is a transit series running between two border points.on.

	RoutingType	*RoutingTypeEnum*	0:1	Whether this is a direct i.e. no changes required point to point or indirect. .direct No changes needed. indirect Changes needed. both Both direct and indirect routings.
	FareBasis	*FareBasisEnum*	0:1	Fare basis used to price Series. route distance
	Distance1stClass	*DistanceType*	0:1	Notional distance along SERIES CONSTRAINT for computation of First Class fares.
	Distance2ndClass	*DistanceType*	0:1	Notional distance along SERIES CONSTRAINT for computation of Second Class fares.
	Discrete	*xsd:boolean*	0:1	Whether SERIES CONSTRAINT can only be used by itself, or whether it can be used in a chain of series.
«FK»	***FromConnectionRef***	*ConnectionRef*	0:1	Reference to CONNECTION associated with origin end of SERIES CONSTRAINT.
«FK»	***ToConnectionRef***	*ConnectionRef*	0:1	Reference to CONNECTION associated with destination end of SERIES CONSTRAINT.
“cntd»	***farePointsInPattern***	*FarePointInPattern*	0:*	FARE POINTs IN PATTERN in SERIES CONSTRAINT.
“cntd»	***JourneyPatterns***	*JourneyPattern*Ref	0:*	References to JOURNEY PATTERN or patterns. equivalent to the SERIES CONSTRAINT.
“cntd»	***prices***	*SeriesConstraintPrice*	0:*	SERIES CONSTRAINT PRICES for the SERIES CONSTRAINT.
“cntd»	***replaces***	*SeriesConstraintRef*	0:*	Replaces the specified SERIES. (Needed for TAP TSI)

Table 33 – *SeriesConstraint* – XML Element

A SERIES CONSTRAINT allows a routing constraint to be specified concisely as a series of FARE POINTs in JOURNEY PATTERN – series are used in TAP TSI for many rail products. The TAP TSI B1 document defines by a notation for describing the constraints to staff and passengers as a “route description” or itinerary on a ticket.

For example: the following set of diagrams show four paths through a simple network from an origin station A to a destination station D, with two designated via stations B and C. Use of the network could be restricted to a specific path, as in (i) *[A * B * D]* (Figure 22) or (ii) *[A * C * D]* (Figure 22), or it could allow any path within an envelope delineated by the intermediate stops, as in *[A * (B/C) * D]*. In this case, the route description (B/C) does not mean that the passenger must take a train passing exactly by either B or C, it means that the passenger has a choice of all routes between a “tariff envelope” of a station on the left and a station on the right (route range), as shown in (i), (ii), (iii) (Figure 23), and (iv) (Figure 24), . Thus a ticket [A * D] with SERIES CONSTRAINT (B/C) would allow travel along any of the following four routes:

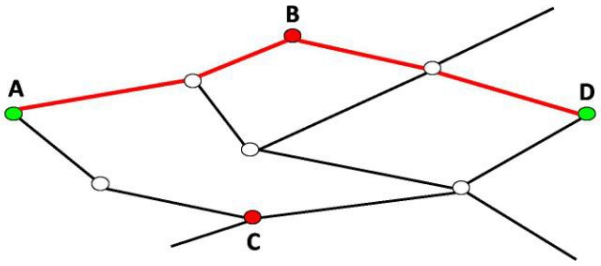


Figure 21 — Series Example (i) [A * B * D]

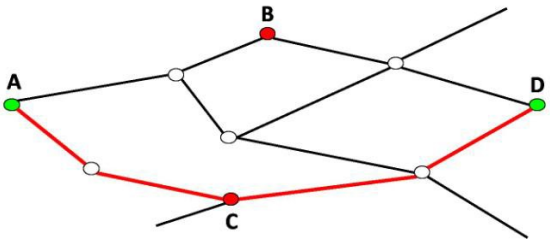


Figure 22 — Series Example (ii) [A * C * D]

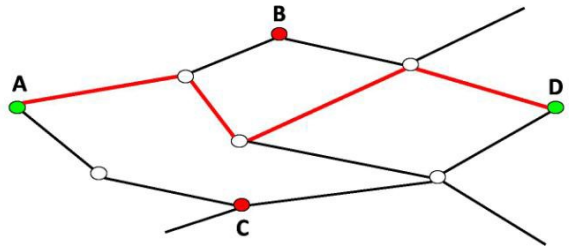


Figure 23 — Series Example (iii) [A * (B/C) * D]

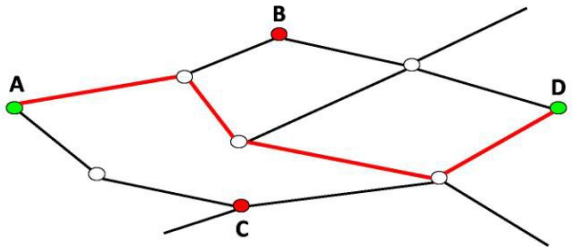


Figure 24 — Series Example (iv) [A * (B/C) * D]

9.7.1 *SeriesConstraint Notation for printed itineraries*

A sequence can be indicated on a ticket or other media as a sequence of station names in an itinerary or route description.

Example	**Notation**		**Note**
A * B * C * D	Sequence	User must go via B and C	
A *(B/C) * D	Choice	User may travel within envelope of B/C	The order to show B or C is indicated by a Presentation Position
P *[* Y]* * ***X*** *[* Z *]* ***Q*** *Shown as*	Abridgement	Intermediate stations may be omitted if there is insufficient space	The relative priority for omitting stations is indicated by an AbridgementRanking

P * ***X*** * ***Q***			
	Presentation Position	The ***PresentationPosition*** attribute defines where a station appears in a printed route description.	Relative positiion to be given to name of point when presenting it in a list of stops (Left, right etc). Allows TAP itinerary to be constructed.

Table 34 – SeriesConstraint Notation

10 EFIP: Pricing

10.1 General

The FARE PRICE Model allows prices to be specified for any element of the tariff structure which may hold a price value used in the overall price calculation.

An important principle of Transmodel is that prices should be separate from the things that they price. This allows for alternative price sets with different validities to be exchanged without revising the rest of the fare model.

10.2 Fare Prices

A FARE PRICE represents an individual price (both value and pricing related parameters) that can be associated with a PRICEABLE OBJECT. Its value is in a stated PRICE UNIT (currency or otherwise).

A FARE PRICE may be given as an absolute amount (e.g. €23.00) or be derived using a PRICING RULE from another base price (e.g. the child fare is 50% of adult fare). When it is computed, a derived price is resolved into an absolute value (which may in turn be used as a base price for a further derivation); it may however also retain references to the base price and rule from which it is derived as well as the resulting amount, thus providing a justification.

EFIP1F In the Basic profile all prices are absolute

EFIP2F In the Advanced profile all prices may also be derived

Note: In NeTEx, FARE PRICEs may also be dynamic (rather than static), that is provided by a named PRICING SERVICE at run time .

Transmodel/NeTEx provide two ways of specifying fare prices

- As CELL PRICE fare prices fares associated with a CELL that refernces all the relevant factors (i.e. PRICEABLE OBJECTS, and other elements) for which it represents the specifc price
- As specifc prices directly associated with each different type of PRICEABLE OBJECT that they price.
- CONTROLLABLE ELEMENT PRICE;
- FARE STRUCTURE ELEMENT PRICE;
- GEOGRAPHICAL INTERVAL PRICE;
- DISTANCE MATRIX PRICE;
- TIME INTERVAL PRICE;
- VALIDABLE ELEMENT PRICE;

- USAGE PARAMETER PRICE;
- FARE PRODUCT PRICE;
- SALES OFFER PACKAGE PRICE.
- FULFILMENT METHOD PRICE.
-

EFIP1F The EFIP uses CELL PRICEs

EFIP1F PRICEs and pricing parameters are placed in a separate frames.

Note: In NeTEx, FARE PRICEs may also be **dynamic (***rather than* **static)**, that is provided by a named PRICING SERVICE at run time. Dynamic prices are excluded from the EFIP.

It may be necessary to group price entities into PRICE GROUPs, in order:

- to group all possible access rights or products into a few categories , each of them having a price reference (products of price 'A', 'B', etc.);
- to group prices into categories to which the same increase, in value or percentage, may be applied;
- To organize large numbers of prices.

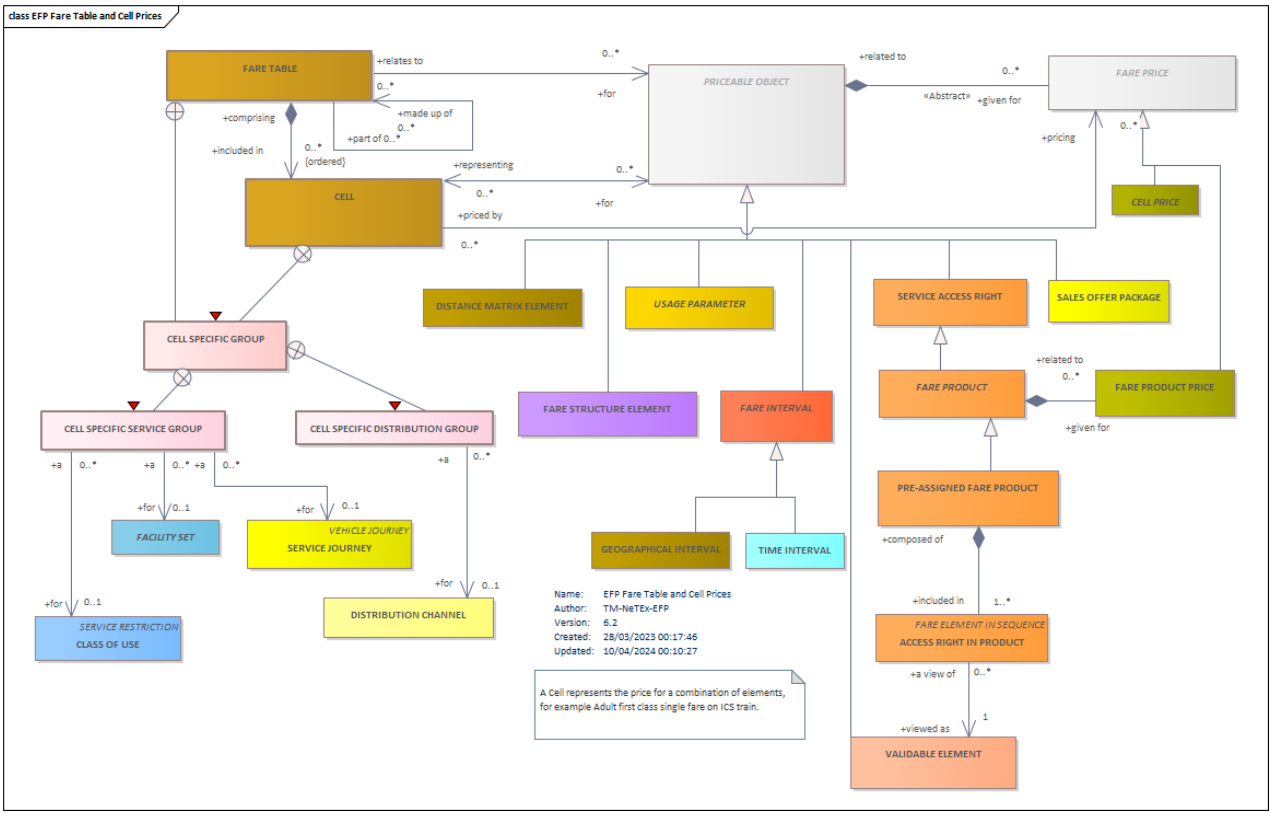


Figure 26 — CellPrices (UML)

10.3 FarePrice

A set of all possible price features for a Fare element.

Classification	Name	Type	Cardinality	Description
::>	::>	*VersionedChild*	::>	FARE PRICE inherits from VERSIONED CHILD
«PK»	***id***	*FarePriceIdType*	1:1	Identifier of FARE PRICE.
	Name	*MultilingualString*	0:1	Name of PRICE.
	Description	*MultilingualString*	0:1	Description of PRICE. +v1.1

	PrivateCode	*PrivateCode*	0:1	External identifier of PRICE. +v1.1
	StartDate	*xsd:date*	0:1	Start date for PRICE validity.
	EndDate	*xsd:date*	0:1	End date for PRICE validity.
	Amount	*AmountType*	0:1	Price in a specified currency.
	Currency	*CurrencyType*	0:1	Currency ISO 4217 code (This in an optimization to allow PRICE UNITs to be omitted).
«FK»	***PriceUnitRef***	*PriceUnitRef*	0:1	Reference to a PRICE UNIT; may be a currency.
	Units	*xsd:decimal*	0:1	Amount in designated unit.
«cntd»	***ruleStepResults***	*RuleStepResult*	0:*	RULE STEP RESULTS describing derivation of price.
	IsAllowed	*xsd:boolean*	0:1	Whether the FARE PRICE is allowed. The default is '*true*'.
«FK»	***PricingServiceRef***	*PricingServiceRef*	0:1	Reference to a PRICE SERVICE which can provide / provided price.
«FK»	***FarePriceRef***	*FarePriceRef+*	0:1	Reference to a FARE PRICE from which this fare price is derived using a PRICING RULE.
«FK»	***a*** ***PricingRuleRef***	*PricingRuleRef+*	0:1	Reference to a PRICING RULE used to derive price.
«cntd»	b ***PricingRule***	*PricingRule*	0:1	PRICING RULE used to derive price.
	CanBeCumulative	*xsd:boolean*	0:1	Whether discount can be used cumulatively in combination with other discounts.
«FK»	***RoundingRef***	*RoundingRef*	0:1	Rounding to use on calculation.

	Ranking	*xsd:integer*	0:1	Relative ranking of price relative to other prices.

Table 36 – FarePrice – XML Element

10.4 RuleStepResult (Subcomponent of FARE PRICE)

The ***RuleStepResult*** holds one or more intermediate steps relating to the calculation of a price from another price, for example the tax or a discount that has been applied.

Classifi-cation	Name	Type	Cardinality	Description
«PK»	***id***	*RuleStepResultIdType*	1:1	Identifier of PRICE RULE STEP RESULT.
«atr»	***order***	*xsd:integer*	1:1	Order in which step was done.
«FK»	***FarePriceRef***	*FarePriceRef+*	0:1	Reference to a FARE PRICE from which this fare price is derived using a PRICING RULE
	Amount	*AmountType*	0:1	Price in a specified currency.
	Currency	*CurrencyType*	0:1	Currency ISO 4217 code (This in an optimization to allow PRICE UNITs to be omitted).
«FK»	***PriceUnitRef***	*PriceUnitRef*	0:1	Reference to a PRICE UNIT.
	Units	*xsd:decimal*	0:1	Amount in designated unit.
«FK»	***PriceUnitRef***	*PriceUnitRef*	0:1	Reference to a PRICE UNIT; may be a currency.
	RateUsed	*xsd:decimal*	*0:1*	Discount rate used.
	Adjustment-Amount	*CurrencyType*	0:1	Step calculation amount, in same currency as STEP RESULT Amount. (i.e. difference between Base Amount and result Amount. PRICE) +v1.1

	AdjustmentUnits	*xsd:decimal*	*0:1*	Step calculation Units, in same PRICE UNIT as STEP RESULT Amount.(i.e. difference between base Units and Result Units. +v1.1
«FK»	***PricingRuleRef***	*PricingRuleRef+*	0:1	Reference to a PRICING RULE used in calculation step to derive price.
«FK»	***RoundingRef***	*RoundingRef*	0:1	Reference to a ROUNDING method used in calculation step to derive price.
«FK»	***RoundingStep-Ref***	*RoundingStepRef*	0:1	Reference to a ROUNDING STEP used in calculation step to derive price.
	Narrative	*MultilingualString*	0:1	Textual explanation of calculation

Table 35 – *RuleStepResult* – XML Element

10.5 PriceGroup

A grouping of prices, allowing the grouping of numerous possible consumption elements into a limited number of price references, or to apply grouped increases, in value or percentage. (“A price band”)

PRICE GROUPS provide a further level of indirection describing prices – thus a price can be given as a PRICE GROUP rather than an absolte value.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*GroupOfEntities*	::>	PRICE GROUP inherits from GROUP OF ENTITies.
«PK»	***id***	*PriceGroupIdType*	1:1	Identifier of PRICE GROUP.
	StartDate	*xsd:date*	0:1	Start date for PRICE GROUP.
	EndDate	*xsd:date*	0:1	End date for PRICE GROUP.
«FK»	***RoundingRef***	*RoundingRef*	0:1	Rounding to use on calculation.

«FK»	***members***	*FarePrice* | *FarePriceRef+*	0:*	PRICEs in PRICE GROUP. Each price can represents a priceable combination that shares the common price.
«FK»	***FarePriceRef***	*FarePriceRef+*	0:1	Reference to a FARE PRICE to be used as the price for the PRICE GROUP.

Table 36 – *PriceGroup* – XML Element

10.6 Pricing Parameters

A set of parameters may be associated with a price to use when computing prices.

- A PRICING RULE names a method of calculation that a consumer system must understand and interpret. There are two types of common PRICING RULES that have specific parameters associated with them:
- A DISCOUNTING RULE specifies parameters relating to discounting; discounts may be specified as either a percentage (e.g. 10%) or an absolute amount (e.g. 5 €).
- A LIMITING RULE is a DISCOUNTING RULE with further parameters to set limits on the results, for example to set a minimum and maximum price.

More than one PRICING RULE may be applied successively to derive a price; PRICING RULESs may be chained together. (e.g.Adult fare is distance GEOGRAPHIC INTERVAL PRICE à Child fare is 50% of adult fare à Final price is price without tax + Vat at 15%). Where a price is derived from another price, the intermediate derivation steps may be recorded using a RULE STEP RESULT. For example, a concessionary price might be calculated from a full fare by applying a discount, then deducting tax.

ROUNDING conditions may se specifed for use when computing a derived price.

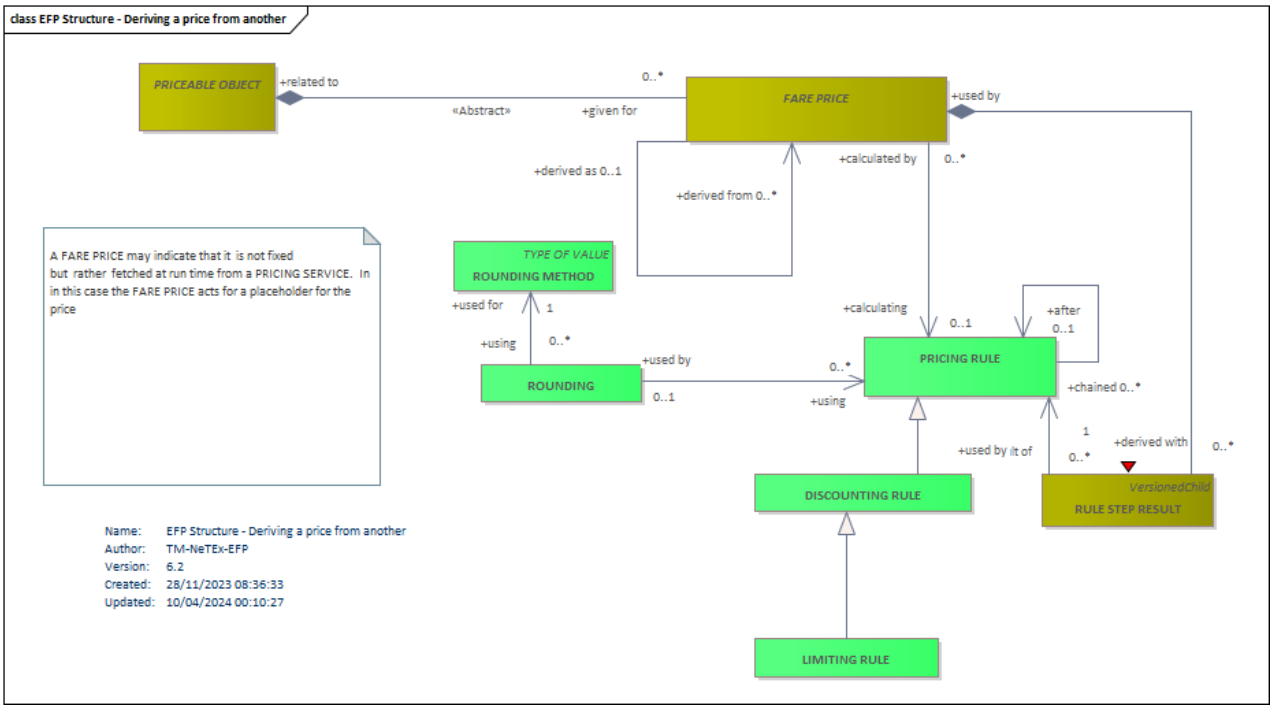


Figure 27 — Pricing Parameters (UML)

10.7 PricingParameterSet

A set of reusable pricing Parameters directing the rounding of values that are the result of calculations.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*DataManagedObject*	::>	PRICING PARAMETER SET inherits from DATA MANAGED OBJECT. See NeTEx Part1.
	id	*PricingParameterSet-IdType*	1:1	Identifier of PRICING PARAMETER SET.
	Name	*MultilingualString*	0:1	Name of PRICING PARAMETER SET.
«FK»	***PriceUnitRef***	*PriceUnitRef*	0:1	Reference to a default PRICE UNIT.

«cntd»	***priceUnits***	*PriceUnit*	0:*	Available PRICE UNITs.
«cntd»	***pricingRules***	*PricingRule*	0:1	PRICING RULEs available to use in pricing.
	AllowCumulative Discounts	*xsd:boolean*	0:1	Whether cumulative discounts are allowed.
«FK»	***RoundingRef***	*RoundingRef*	0:1	Reference to a default ROUNDING.
«cntd»	***roundings***	*Rounding*	0:1	ROUNDINGs available to use in pricing.
«FK»	***DayTypeRef***	*DayTypeRef*	0:1	Default FARE DAY.
«cntd»	***monthValidity-Offsets***	*MonthValidityOffset*	0:12	Day offsets for each month in year to use to decide activation date of certain products.
«cntd»	***pricingServices***	*PricingService*	0:*	PRICING SERVICEs available to use.

Table 37 – *PricingParameterSet* – XML Element

10.8 Rounding

Parameters directing the rounding of values that are the result of calculations.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*DataManagedObject*	::>	ROUNDING inherits from DATA MANAGED OBJECT. See NeTEx Part1.
	id	*RoundingIdType*	1:1	Identifier of ROUNDING.
	Name	*MultilingualString*	0:1	Name of ROUNDING.

«enum»	***RoundingMethod***	*RoundingMethodEnum*	0:1	Method to use to *round: down, up, split, none.* See allowed values below. • *none; down; up; split* • *stepTable*
	Rounding-Modulus	*decimal*	0:1	Amount by which rounding is to be quantised, i.e. results should be quantised to nearest whole multiple of this value, for example, 0.10, 0.20, 0.30 cents, or 1.00 Euro, 1.6 Euro, etc.
«cntd»	***roundingSteps***	*RoundingStep*	0:*	Table of explicit ROUNDING STEPs.

Table 38 – *Rounding* – XML Element

10.9 PriceUnit

A unit to express prices: amount of currency, abstract fare unit, ticket unit or token etc.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*TypeOfValue*	::>	PRICE UNIT inherits from TYPE OF VALUE.
«PK»	***id***	*PriceUnitIdType*	1:1	Identifier of PRICE UNIT.
	Precision	*xsd:integer*	0:1	Precision of PRICE UNIT.

Table 39 – *PriceUnit* – XML Element

10.10 PricingRule (Abstract in EFIP)

A rule used for the calculation of FARE PRICE, determined either by a set of parameters to be applied to a reference price or by a more complex algorithm.
NOTE: PRICING RULEs may be chained together.

Classifi-cation	Name	Type	Cardin-ality	Description

::>	::>	*DataManagedObject*	::>	PRICING RULE inherits from DATA MANAGED OBJECT.
«PK»	***id***	*PricingRuleIdType*	1:1	Identifier of PRICING RULE.
	Name	*MultilingualString*	0:1	Name of PRICING RULE.
	Description	*MultilingualString*	0:1	Description of PRICING RULE.
	MethodName	*xsd:NCNAME*	0:1	Calculation method associated with PRICING RULE.
«FK»	***TypeOfPricing-RuleRef***	*TypeOfPricing-RuleRef*	0:1	Classification of PRICING RULE. +v1.1
«FK»	***PricingRuleRef***	*PricingRuleRef+*	0:1	PRICING RULE to chain to from this one.
	Factor	*xsd:decimal*	0:1	Numeric factor associated with PRICING RULE.
	Currency	*CurrencyType*	0:1	Currency associated with PRICING RULE. +v1.1
«FK»	***PriceUnitRef***	*PriceUnitRef*	0:1	PRICE UNIT for PRICING RULE. +v1.1
	url	*xsd:anyURI*	0:1	URL associated with PRICING RULE.

Table 40 – *PricingRule* – XML Element

10.10.1 *DiscountingRule*

Parameters of a rule for computing a discounted price from another price.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*PricingRule*	::>	DISCOUNTING RULE inherits from PRICING RULE.
«PK»	***id***	*DiscountingRuleIdType*	1:1	Identifier of DISCOUNTING RULE.

	Discount-AsPercentage	*PercentageType*	0:1	Discount of PRICE as a percentage.
	DiscountAsValue	*AmountType*	0:1	Discount of PRICE as a value.
	CanBe-Cumulative	*xsd:boolean*	0:1	Whether discount can be used cumulatively in combination with other discounts.

Table 41 – *DiscountingRule* – XML Element

10.10.2 *LimitingRule*

Parameters of a rule for computing a FARE PRICE from another FARE PRICE, subject to minima or maxima.

Classifi-cation	Name	Type	Cardin-ality	Description
::>	::>	*DiscountingRule*	::>	LIMITING RULE from DISCOUNTING RULE.
«PK»	***id***	*LimitingRuleIdType*	1:1	Identifier of LIMITING RULE.
	MinimumPrice	*AmountType*	0:1	Minimum amount at which to cap discounted fare.
	MinimumPrice-AsPercentage	*PercentageType*	0:1	Minimum PRICE expressed as a percentage of the total price.
	MinimumPrice-AsMultiple	*PercentageType*	0:1	Minimum PRICE expressed as a multiple of a unit fare.
	MaximumPrice	*AmountType*	0:1	Maximum amount at which to cap discounted fare.
	MaximumPrice-AsPercentage	*PercentageType*	0:1	Maximum PRICE expressed as a percentage of the total price.

	MaximumPrice-AsMultiple	*PercentageType*	0:1	Maximum PRICE expressed as a multiple of a unit fare.
	MinimumLimitPrice	*AmountType*	0:1	Limiting amount below which resulting fare may not be sold.
	MinimumLimit-AsPercentage	*PercentageType*	0:1	Minimum limit expressed as a percentage of the total price.
	MaximumLimitPrice	*AmountType*	0:1	Limiting amount above which resulting fare may not be sold.
	MaximumLimit-AsPercentage	*PercentageType*	0:1	Maximum limit expressed as a percentage of the total price.

Table 44 – LimitingRule – XML Element

10.10.3 *Dynamic and Yield managed pricing*

A web service used to provide prices dynamically at time of booking or purchase.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*DataManagedObject*	::>	PRICING SERVICE inherits from DATA MANAGED OBJECT.
«PK»	***id***	*PricingServiceIdType*	1:1	Identifier of PRICING SERVICE.
	Name	*MultilingualString*	0:1	Name of PRICING SERVICE.
	Description	*MultilingualString*	0:1	Description of PRICING SERVICE.
	Url	*xsd:anyURI*	0:1	Url at which service is available.
«FK»	***OrganisationRef***	*(OrganisationRef)*	0:1	ORGANISATION that provides service. has contact details etc.

Table 45 – PricingService – Element

10.11 Prices for fare structure elements

An ENTITY which can have a price is a specialisation of PRICEABLE OBJECT; both individual structural elements of a TARIFF (DISTANCE MATRIX ELEMENT, GEOGRAPHCIAL INTERVAL, TIME INTERVAL, FARE STRUCTURE ELEMENT, etc.; product and product offer components (FARE PRODUCT, SALES OFFER PACKAGE, etc); and various other parameters (USAGE PARAMETER FULFILMENT METHOD, etc) are PRICEABLE OBJECTS and may be assigned FARE PRICEs. A pricing algorithm will use these variously to compute an actual price for a specific set of choices.

There are different specialisations of FARE PRICE for different types of PRICEABLE OBJECTs, for example a DISTANCE MATRIX ELEMENT PRICE, a FARE PRODUCT PRICE, etc., etc. For brevity, in the EFIP specification the separate fare is not used - CELL PRICEs are used instead.

FARE PRICEs can be in any PRICE UNIT (currency or otherwise) and can have a start date.

EFIP1>> In the EFIP, all prices must be placed in a separate FARE FRAME formed of the fare descriptions.

10.12 PriceableObject

An element which may have a FARE PRICE.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*DataManagedObject*	::>	PRICEABLE OBJECT inherits from DATA MANAGED OBJECT.
«PK»	***id***	*PriceableObjectIdType*	1:1	Identifier of PRICEABLE OBJECT.
	Name	*MultilingualString*	0:1	Name of PRICEABLE OBJECT.
	Description	*MultilingualString*	0:1	Description of PRICEABLE OBJECT.
	Url	*xsd:AnyURI*	0:1	URL to web page with information about PRICEABLE OBJECT. +v1.1

«cntd»	***infoLinks***	*InfoLink*	0:*	Additional hyperlinks for +v1.1
«cntd»	***alternative-Names***	*AlternativeName*	0:*	ALTERNATIVE Name for element.
«cntd»	***notice-Assignments***	*NoticeAssignment*	0:*	NOTICE ASSIGNMENTs associated with the element.
«FK»	***PricingService-Ref***	*PricingServiceRef*	0:1	PRICING SERVICE to use to fetch prices dynamically.
«FK»	***PricingRuleRef***	*PricingRuleRef+*	0:1	Default PRICING RULE to use to derive prices from this element.
«cntd»	***priceGroups***	*PriceGroup*	0:*	PRICE GROUPs associated with PRICEABLE OBJECT.
«cntd»	***fareTables***	*FareTable*	0:*	FARE TABLEs associated with PRICEABLE OBJECT.

Table 46 – PriceableObject – XML Element

10.13 Organising prices for exchange

Even a relatively simple tariff structure may have a large number of prices; depending not only on the size of the network, but also combinatorically on the number of different features in the tariff structure. Certain tariff structures, notably point-to-point fares, may have very large price matrices.

NeTEx uses FARE TABLEs to organise fares into nested groups so that common properties of prices do not need to be repeated. FARE TABLEs allow any order to be chosen for the hierarchy of nesting used.

Each FARE TABLE contains one or more FARE TABLE CELLs. Each cell may hold a FARE PRICE for a specifc PRICEABLE OBJECT. It may also hold reference for one or more PRICEABLE OBJECTs as well as for other pricing factors that further describe the meaning of the price (see CELL PRICE ASSIGNMENT below). A FARE TABLE may include other FARE TABLEs (but not include itself, either directly or indirectly).

10.14 FareTable

A FARE TABLE is a grouping of prices that may be associated with various combinations of fare elements such as the DISTANCE MATRIX ELEMENT, FARE STRUCTURE ELEMENT, GEOGRAPHICAL INTERVAL, TIME INTERVAL, USAGE PARAMETER, etc.

Classification	Name	Type	Cardinality	Description
::>	::>	*PriceGroup*	::>	FARE TABLE inherits from PRICE GROUP.
«PK»	***id***	*FareTableIdType*	1:1	Identifier of FARE TABLE.
	StartDate	*xsd:date*	0:1	Start date for PRICE validity.
	EndDate	*xsd:date*	0:1	End date for PRICE validity.
«FK»	***RoundingRef***	*RoundingRef*	0:1	Rounding to use on calculation.
«FK»	***TypeOfFare-TableRef***	*TypeOfFareTableRef*	0:1	Classification of FARE TABLE to apply.
«cntd»	***pricesFor***	*PriceableObjectRef+*	0:*	PRICEABLE OBJECT elements which may be given a price and so associated with this CELL.
«cntd»	***usedIn***	*Choice*	0:1	A fare element associated with the FARE TABLE.
«FK»	***a*** ***TariffRef***	*TariffRef*	1:*	TARIFF to which PRICEs of FARE TABLE apply.
«FK»	***b*** ***GroupOf-Distance-Matrix-ElementsRef***	*GroupOfDistanceMatrix-ElementsRef*	1:*	GROUP OF DISTANCE MATRIX ELEMENTs associated with a FARE TABLE.
«FK»	***c*** ***GroupOf-SalesOffer-PackagesRef***	*GroupOf-SalesOfferPackagesRef*	1:*	GROUP OF SALES OFFER PACKAGEs associated with a FARE TABLE.
«FK»	***OrganisationRef***	*(OrganisationRef)*	0:1	OPERATOR or AUTHORITY to which FARE

				PRICEs apply.
«cntd»	***limitations***	*UsageParameterRef+*	0:*	USAGE PARAMETER or PARAMETERs to which the CELL PRICE applies.
«cntd»	***specifics***	*As for Cell*	0:*	PARAMETERs to which the CELL PRICE applies.
«cntd»	***columns***	*FareTableColumnHeading*	0:*	Column headings to use when presenting table.
«cntd»	***rows***	*FareTableRowHeading*	0:*	Row headings to use when presenting table.
«cntd»	***includes***	*FareTable*	0:*	FARE TABLEs nested within this table. Can be recursive. +v1.1
	EmbargoUntil	*xsd:dateTime*	0:1	Prices must not be released until this date.
«cntd»	***prices***	*FarePrice*	0:*	An optimization – A TABLE may hold prices directy withouthe ue of cells.
«cntd»	***cells***	*Cell | CellInContext*	0:*	A tuple within a FARE TABLE that associates one or more fare entities with a price.
«cntd»	***notice-Assignments***	*NoticeAssignment*	0:*	NOTICEs that apply to whole FARE TABLE

Table 47 – FareTable – XML Element

10.14.1 *Cell (Subcomponent of FareTable)*

A unique individual combination of features within a FARE TABLE, used to associate a FARE PRICE with a fare element.

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*VersionedChild*	::>	CELL inherits from VERSIONED CHILD
«PK»	***id***	*CellIdType*	1:1	Identifier of CELL.

	Name	*MultilingualString*	0:1	Name of CELL.
	Description	*MultilingualString*	0:1	Description of CELL.
	price	*Choice*	1:1	One of the following three
«cntd»	***a*** ***CellPrice***	*FarePriceStructure*	1:1	Fare Price held inline in CELL. See FARE PRICE for description.
«FK»	***b*** ***FarePriceRef***	*FarePriceRef+*	1:1	Reference to another FARE PRICE providing price for CELL
«FK»	***c*** ***Price-GroupRef***	*PriceGroupRef*	1:1	Reference to a FARE GROUP providing price for CELL via its lead price.
«cntd»	***Priceable-ObjectRef***	*PriceableObjectRef+*	0:*	Fare structure elements which may be given a price and so associated with this CELL.
«FK»	***GroupOf-DistanceMatrix-ElementsRef***	*GroupOfDistanceMatrix-ElementsRef*	0:1	Reference to a GROUP OF DISTANCE MATRIX ELEMENTS) associated with an individual CELL or FARE TABLE.
«FK»	***GroupOfLines-Ref***	*GroupOfLinesRef*	0:1	A GROUP OF LINEs for which the CELL provides a price.
«FK»	***LineRef***	*LineRef*	0:1	A LINE for which the CELL provides a price.
«FK»	***SiteRef***	*SiteRef*	0:1	A SITE for which the CELL provides a price.
«FK»	***TariffZoneRef***	*TariffZoneRef*	0:1	A TARIFF ZONE for which the CELL provides a price.
«FK»	***FareSectionRef***	*FareSectionRef*	0:1	A FARE SECTION for which CELL provides a price.
«enum»	***DirectionType***	*RelativeDirectionEnum*	0:1	For fares for DISTANCE MATRIX ELEMENTs, DIRECTION in which price applies. See Part1 for

				allowed values.
«enum»	***RoutingType***	*RoutingTypeEnum*	0:1	Whether fare is for direct (i.e. no changes required point-to-point fare) or indirect routing. See allowed values below.
«enum»	***FareClass***	*FareClassEnum*	0:1	A FARE CLASS for which the CELL provides a price. See Part1.
«FK»	***ClassOfUseRef***	*ClassOfUseRef*	0:1	A CLASS OF USE (Seat Class) for which the CELL provides a price.
«FK»	***FacilitySetRef***	*FacilitySetRef*	0:1	A FACILITY SET for which the CELL provides a price.
«FK»	***TypeOfProduct-CategoryRef***	*TypeOfProductCategoryRef*	0:1	A TYPE OF PRODUCT CATEGORY for which the CELL provides a price.
«FK»	***TypeOfService-Ref***	*TypeOfServiceRef*	0:1	A TYPE OF SERVICE for which the CELL provides a price.
«FK»	***ServiceJourney-Ref***	*ServiceJourneyRef*	0:1	A SERVICE JOURNEY for which the CELL provides a price.
«FK»	***TrainNumberRef***	*TrainNumberRef*	0:1	A TRAIN NUMBER for which the CELL provides a price.
«FK»	***GroupOfServicesRef***	*GroupOfServicesRef*	0:1	A GROUP OF SERVICEs for which the CELL provides a price.
«FK»	***TypeOfFare-ProductRef***	*TypeOfFareProductRef*	0:1	TYPE OF FARE PRODUCT for which the CELL provides a price.
«FK»	***Distribution-ChannelRef***	*DistributionChannelRef*	0:1	DISTRIBUTION CHANNEL for which the CELL provides a price.
«FK»	***GroupOf-***	*GroupOfDistribution-*	0:1	GROUP OF DISTRIBUTION CHANNELs for which

	Distribution-ChannelsRef	*ChannelsRef*		the CELL provides a price.
«enum»	***Payment-Methods***	*PaymentMethodEnum*	0:1	PaymentMethod standard value for which the CELL provides a price. See Part1.
«FK»	***TypeOfPayment-MethodRef***	*TypeOfPaymentMethodRef*	0:1	TYPE OF PAYMENT METHOD for which the CELL provides a price.
«FK»	***TypeOfTravel-DocumentRef***	*TypeOfTravelDocumentRef*	0:1	TYPE OF TRAVEL DOCUMENT for which the CELL provides a price.
«FK»	***ColumnRef***	*ColumnRef*	0:1	Reference to a column in the FARE TABLE to which this CELL should be assigned.
«FK»	***RowRef***	*RowRef*	0:1	Reference to a row in the FARE TABLE to which this CELL should be assigned.

Table 42 – *Cell* – XML Element

10.14.2 *Cell price assignment*

Since prices may exist and be exchanged independently of the things they price, each price must be able to describe what it represents in terms of references to specific model elements.

Each CELL represents a FARE PRICE; the individual FARE PRICE may represent the combination of one or more PRICEABLE OBJECTs such as a FARE STRUCTURE ELEMENT, FARE PRODUCT, USAGE PARAMETER, DISTANCE MATRIX ELEMENT, SERIES CONSTRAINT, TIME INTERVAL, etc.

In a product definition, some of these elements may themselves have been defined as lists of choices of further pricing factors of a FARE STRUCTURE ELEMENT, such as TARIFF ZONEs, CLASSes OF USE, PRODUCT CATEGORies, etc.; there may be a separate price for each possible combination of these respective factors. The CELL allows each combination of elements to be specified directly along with an amount in currency or other PRICE UNIT.

The CELL can be considered as an optimisation of a VALIDITY PARAMETER ASSIGNMENT that states the selection of parameters that apply to an individual price.

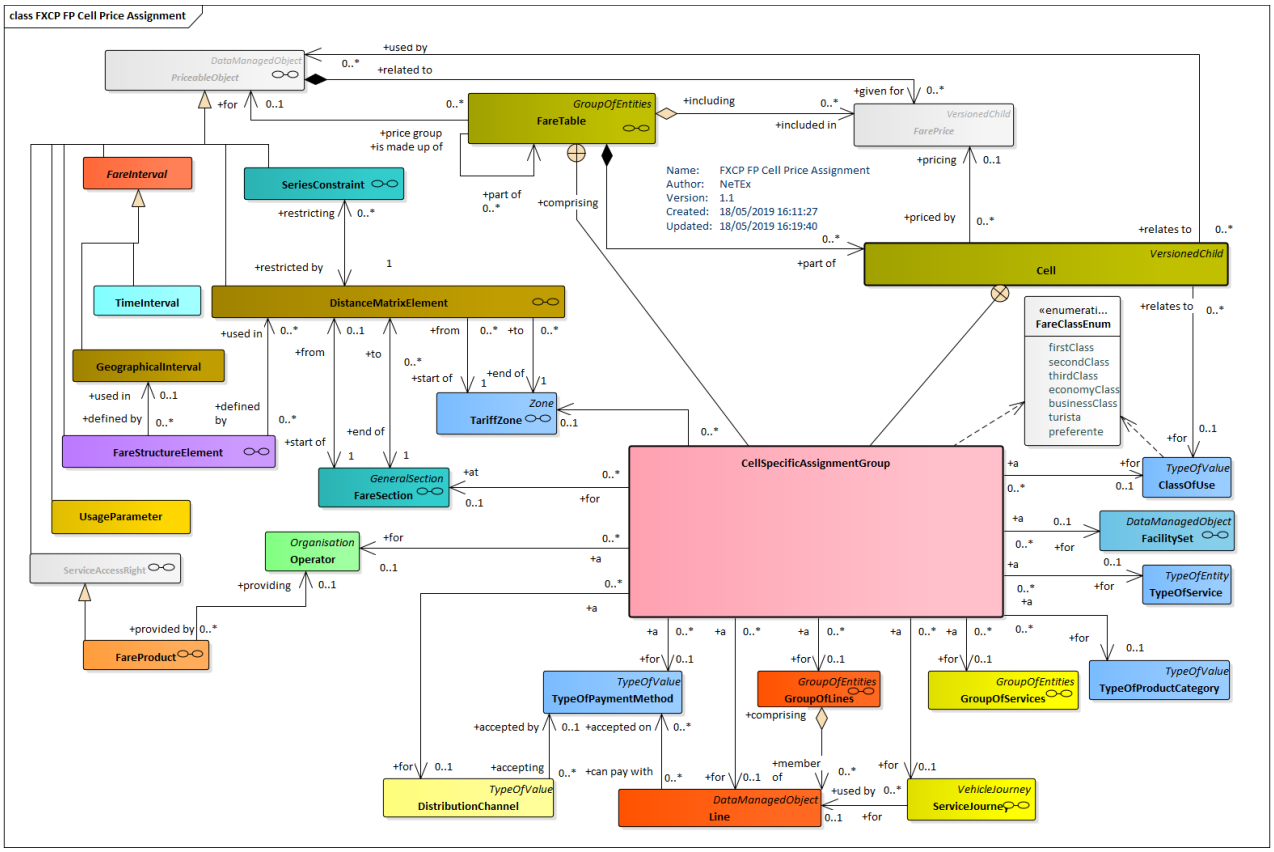


Figure 28 — Cell Price Assignment Model (UML)

10.15 Pricing factors and data set size

As noted above in XXXX the most common tariff structures have prices corresponding to combinations of the following price factors:

Point-to-point Trip

- **DISTANCE MATRIX ELEMENTs** x USER PROFILE x (SALES OFFER PACKAGE).

Stage fare trip

- **GEOGRAPHICAL INTERVAL** x USER PROFILE x (SALES OFFER PACKAGE).

Zonal Fare, trip or pass

- **FARE ZONE x TIME INTERVAL** x USER PROFILE x (SALES OFFER PACKAGE).

The SALES OFFER PACKAGE implies a specific media (TYPE OF TRAVEL DOCUMENT and FAREPRODUCT trip, pass, etc)

In addition, a number of additional factors may also apply combinatorially. for example FARE DEMAND FACTOR (Peak/off peak uses), Number of units (e.g. for a Carnet FARE QUALITY FACTOR), PAYMENT METHOD (epayment may be cheaper – can be a FARE STRUCTURE ELEMENT), DISTRIBUTION CHANNEL (online may be cheaper), FULFILMENT METHOD (post may be more expensive).

The following rough orders of magnitude for the different basic fare pricing factors result:

- DISTANCE MATRIX ELEMENTs; assuming prices are the same both ways if there are between n = 50 and a 1000 stops in the network, then there will be $(N^2-N)/2$. i.e. between ~1200 < ~500,000 O/D pairs.
- FARE ZONEs ;say between ~ 3 < ~50 zones.
- GEOGRAPHICAL INTERVALs; say between ~ 1 < ~ 10 fare stages.
- TIME INTERVALs say a pass is available for 1D, 2D, 3D,1W, 1M, 2M, 1Y) between ~ 1 < ~8 up to eight available time periods for a pass.
- USER PROFILEs (Adult, Child, Student, Senior, Disabled): ~ 2 < ~5 up to five different user types.

- SALE OFFER PACKAGEs (paper, smartcard, mobile app) ~ 1 < ~3 up to three media options.

Table 43 – *Price combinations for different fare structures*

	Min Bound	Prices	Max bound	Prices
Point-to-point Trip	1200 x 2 x 1	**~ 2400**	500000 x 5 x 3	**~ 7.5 million**
Stage fare trip	1 x 2 x 1	**~2**	10 x 3 x 5	**~ 150**
Zonal Fare, trip or pass	3 x 1 x 2 x 1	**~6**	50 x 8 x 5 x3	**~ 6000**

10.16 Pricing factors and fare table nesting

The above discussion indicates that price data sets may be large

The hierarchy used to present prices to passengers is therefore arbitrary and depends on the marketing decisions of an operator. The presentation of prices may thus be organised by any of product, user type, zone /origin-destination, media, etc.

For example, consider the following set of prices giving point-to-point fare for two destinations..

Table 44 – *Example fare table data*

FARE PRICE	DISTANCE MATRIX ELEMENT	USER PROFILE	SALES OFFER PACKAGE		AMOUNT
			FARE PRODUCT	TYPE OF TRAVEL DOCUMENT	
001	A+B	Adult	Single Trip	Paper Ticket	10.00
002	A+B	Adult	Single Trip	Mobile APp	9.00
003	A+B	Child	Single Trip	Paper Ticket	5.00
004	A+B	Child	Single Trip	Mobile APp	4.50
005	A+C	Adult	Single Trip	Paper Ticket	20.00

006	A+C	Adult	Single Trip	Mobile APp	18.00
007	A+C	Child	Single Trip	Paper Ticket	10.00
008	A+C	Child	Single Trip	Mobile APp	9.00
009	A+B	Adult	Return Trip	Paper Ticket	18.00
010	A+B	Adult	Return Trip	Mobile APp	17.00
011	A+B	Child	Return Trip	Paper Ticket	9.00
012	A+B	Child	Return Trip	Mobile APp	8.00
013	A+C	Adult	Return Trip	Paper Ticket	36.00
014	A+C	Adult	Return Trip	Mobile APp	24.00
015	A+C	Child	Return Trip	Paper Ticket	18.00
016	A+C	Child	Return Trip	Mobile APp	16.00

A nesting according to the order of elements in the table above (that is, DISTANCE MATRIX ELEMENT, USER PROFILE, SALE OFFER PACKAGE) would be as follows:

FARE TABLE 1.

èDISTANCE MATRIX ELEMENT: *"A+B"*

èFARE TABLE 1.1

àUSER PROFILE: *"Adult"*

èFARE TABLE 1.1.1 *"Single trip"*.

è*SALES OFFER PACKAGE PRICE*

à*SALES OFFER PACKAGE "Single trip, paper"*

è*SALES OFFER PACKAGE PRICE*

à*SALES OFFER PACKAGE "Single trip, mobile app"*

èFARE TABLE 1.1.2 *"Return trip"*.

èSALES OFFER PACKAGE PRICE

àSALES OFFER PACKAGE *"Return trip, paper"*

èSALES OFFER PACKAGE PRICE

àSALES OFFER PACKAGE *"Return trip, mobile app"*

èFARE TABLE 1.2

àUSER PROFILE: *"Child"*

èFARE TABLE 1.2.1 *"Single trip"*.

èSALES OFFER PACKAGE PRICE

àSALES OFFER PACKAGE *"Single trip, paper"*

èSALES OFFER PACKAGE PRICE

->SALES OFFER PACKAGE *"Single trip, mobile app"*

èFARE TABLE 1.2.2 *"Return trip"*.

èSALES OFFER PACKAGE PRICE

àSALES OFFER PACKAGE *"Return trip, paper"*

èSALES OFFER PACKAGE PRICE

àSALES OFFER PACKAGE *"Return trip, mobile app"*

etc

But other nestings are equally valid for presentation, for example

SALES OFFER PACKAGE à DESTINATION MATRIX ELEMENT à USER PROFILE è USAGE PARAMETER PRICE

Or even

FARE PRODUCT à DESTINATION MATRIX ELEMENT à USER PROFILE à TYPE OF TRAVEL DOCUMENT èSALES OFFER PACKAGE PRICE

10.16.1 *Fare table nesting rules*

When creating FARE TABLEs for the EFIP, the following principles should be followed

a. FARE PRICEs shall be nested in FARE TABLEs for clarity and to reduce data volumes
b. Each NESTING LAYER shall correspond to one further pricing factor
c. The type of fare price used shall be for that of lowest level elemen. For example if the lowest level is the USER PROFILE (a type of USAGE PARAMETER), such as 'Adult' or 'Child', then it shall be a USAGE PARAMETER PRICE, if it is a TYPE OF TRAVEL DOCUMENT (as associated with a SALES OFFER PACKAGE) it shall be a SALES OFFER PACKAGE PRICE, etc
d. Where the price is for a specific fare element such as a DISTANCE MATRIX ELEMENT, TIME INTERVAL, GEOGRAPHICAL INTERVAL etc, the price shall be embedded in line and must include a reference to the priced element.
e. Where the price is for an additional option out of several specified for a single FARE STRUCTURE ELEMENT (as say may be the case for a FARE ZONE or a CLASS OF USE, a reference to the specific option should be), should be included in the CELL, for example: a FareZoneRef, or ClassOfUseRef, so that a consuming system can determine its meaning,
f. The preferred order for nesting tables is

   SALES OFFER PACKAGE / {spatial Factor] / (TIME INTERVAL) / USER PROFILE / CLASS OF USE /

   Where Spatial factor is either

   (1) DISTANCE MATRIX ELEMENT or

   (2) FARE STRUCTURE ELEMENT + ZONE

   (3) GEOGRAPHiCAL INTERVAL

Table 45 – *Common fare pricing factors*

	Pricing factor	**Tariff element**	**FARE PRICE**	**Cell Assigment**
Point-to-point	*DISTANCE MATRIX*	*DISTANCE MATRIX*	*DISTANCE MATRIX*	***[DistanceMatrixElementRef)***

fare	*ELEMENT*	*ELEMENT*	*ELEMENT PRICE*	
Zonal fare	*FARE ZONE*	*FARE STRUCTURE ELEMENT*	*FARE STRUCTURE ELEMENT PRICE*	***[FareStructureElementRef)*** ***FareZoneRef***
Different media	*TYPE OF TRAVEL DOCUMENT*	*SALES OFFER PACKAGE ELEMENT*	*SALES OFFER PACKAGE PRICE*	***(SalesOfferPackageRef)*** ***TypeOfTravelDocumentRef*** ***FareProductRef***
Pass	*TIME INTERVAL*	*TIME INTERVAL*	*TIME INTERVAL PRICE*	***(TimeIntervalRef)***
Stage fare	*GEOGRAPHICAL INTERVAL*	*GEOGRAPHICAL INTERVAL*	*GEOGRAPHICAL INTERVAL PRICE*	***(GeographicIntervalRef)***
Carnet	*FARE QUALITY FACTOR*	*FARE QUALITY FACTOR*	*FARE STRUCTURE FACTOR PRICE*	***(FareQualityFactorRef)***
Peak/OffPeak	*FARE DEMAND FACTOR*	*FARE DEMAND FACTOR*	*FARE STRUCTURE FACTOR PRICE*	***(FareDemandFactorRef)***
Class of Use	*CLASS OF USE*	*FARE STRUCTURE ELEMENT*	*FARE STRUCTURE ELEMENT PRICE*	***(SalesOfferPackageRef)*** ***ClassOfUsetRef***
Delivery options	*FULFILMENT METHOD*	*FULFILMENT METHOD*	*FULFILMENT METHOD PRICE*	
	SUPPLEMENT FARE PRODUCT	*SUPPLEMENT FARE PRODUCT*	*FARE PRODUCT PRICE*	***(SupplementFareProductRef)***

10.16.2 *Additional pricing factors*

In addition, a number of additional factors may also apply combinatorially to the pricing table. for example FARE DEMAND FACTORs (Peak/off peak times), NUMBER OF UNITs (e.g. for a Carnet), PAYMENT METHODs (epayment may be cheaper), DISTRIBUTION CHANNELs (online may be cheaper), FULFILMENT METHODs (post may be more expensive).

10.17 Presenting prices in tabular format

A FARE TABLLE may also include elements to describe how the price in the table are to be presented in tabular form. Each CELL may be assigned to a ROW and COLUMN, and each ROW and COLUMN may have a heading.

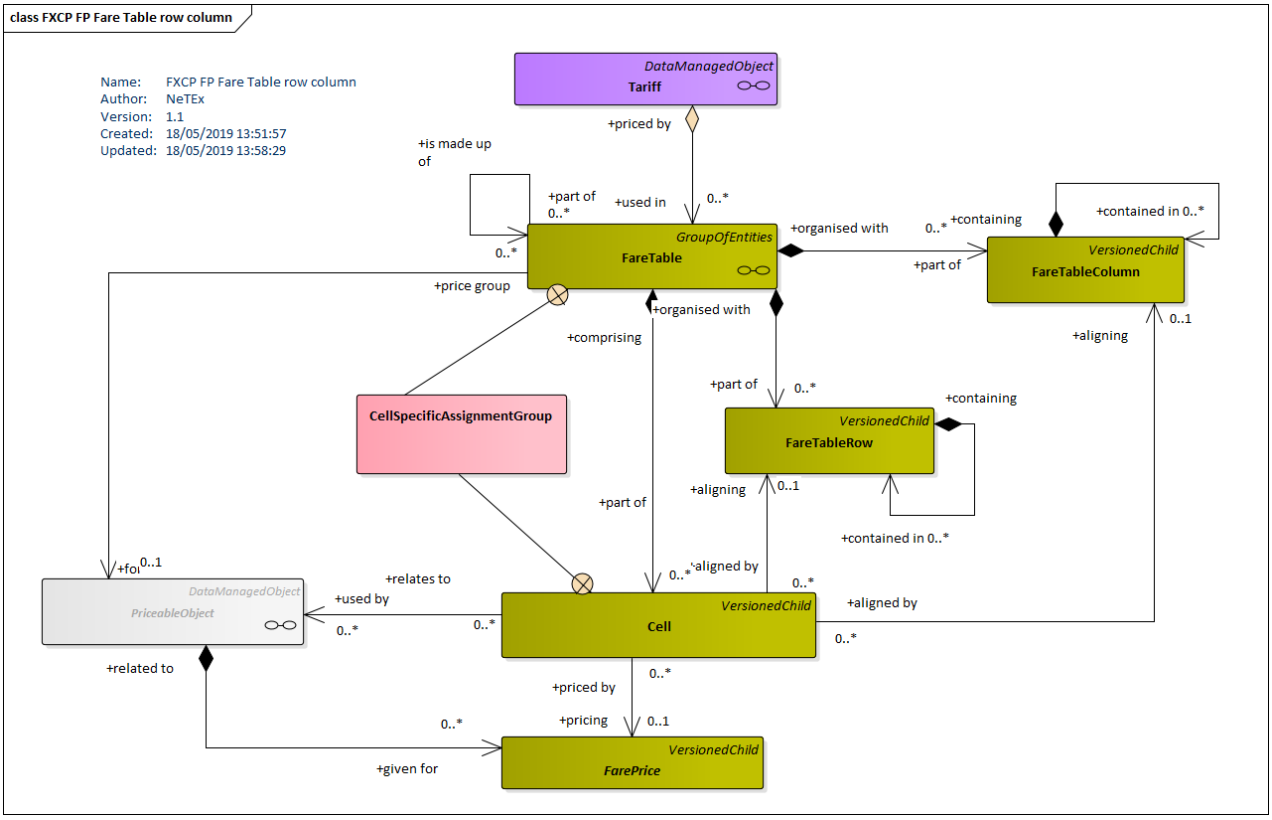


Figure 29 — Fare Table Row & Column Headings (UML)

10.17.1 *FareTableColumn (Subcomponent of FareTable)*

A Column heading for a FARE TABLE.

Table 46 – *Column* – XML Element

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*VersionedChild*	::>	FARE TABLE COLUMN inherits from VERSIONED CHILD
«PK»	***id***	*FareTableColumnIdType*	1:1	Identifier of FARE TABLE COLUMN.
	Name	*MultilingualString*	0:1	Name of FARE TABLE COLUMN.
	Description	*MultilingualString*	0:1	Description of FARE TABLE COLUMN.
«FK»	***FareTableRef***	*FareTableRef*	1:1	Reference to a FARE TABLE containing COLUMN
«cntd»	**n*otice-Assignments***	*NoticeAssignments*	0:*	NOTICEs that apply to whole FARE TABLE COLUMN.
«cntd»	***representing***	*(VersionOfObjectRef)*	0:*	ENTITIES that column represents. +v1.1
«cntd»	***columns***	*FareTable-ColumnHeading*	0:*	Nested FARE TABLE COLUMN headings to use when presenting table. Recursive. +v1.1

10.17.2 *FareTableRow (Subcomponent of FareTable)*

A Row heading for a FARE TABLE.

Table 47 – *Column* – XML Element

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*VersionedChild*	::>	FARE TABLE ROW inherits from VERSIONED CHILD.
«PK»	***id***	*FareTableRowIdType*	1:1	Identifier of FARE TABLE ROW.
	Name	*MultilingualString*	0:1	Name of FARE TABLE ROW.
	Description	*MultilingualString*	0:1	Description of FARE TABLE ROW.
«FK»	***FareTableRef***	*FareTableRef*	1:1	Reference to a FARE TABLE containing FARE TABLE ROW.
«cntd»	***notice-Assignments***	*NoticeAssignments*	0:*	NOTICEs that apply to whole FARE TABLE ROW.
«cntd»	***representing***	*(VersionOfObjectRef)*	0:*	ENTITIES that FARE TABLE ROW represents. +v1.1.
«cntd»	***rows***	*FareTableRowHeading*	0:*	Nested FARE TABLE ROW headings to use when presenting table. Recursive.

11 EFIP: Fare Structures

11.1 Fare Structures

The FARE STRUCTURE ELEMENT model describes the core elements – in particular, FARE STRUCTURE ELEMENT, FARE STRUCTURE ELEMENT IN SEQUENCE – of the fare structure. These can then be combined with other spatial, temporal and quality factors to specify the overall fare structure, as described later below.

A fare structure consists of generic quantitative rules for the limitation of access rights, allowing the calculation of prices. These rules are applied by a pricing engine to FARE STRUCTURE ELEMENTs, each providing parameters relevant to the calculation.

A FARE STRUCTURE ELEMENT is a service consumption right determined by a set of quantitative (spatial, temporal) rules (i.e. determined by fare structure parameters, such as zones, sections, hours, etc);

In some cases, a FARE STRUCTURE ELEMENT may involve several access rights that must be consumed in a particular order. This may be captured by a FARE STRUCTURE ELEMENT IN SEQUENCE.

A FARE STRUCTURE ELEMENT is a thus a sequence or a set of (one or several) of the most basic service elements, for all of which the values of the parameters characterizing it are of similar dimension.

A FARE STRUCTURE ELEMENT can be further characterised by specifying additional parameters by means of an ACCESS RIGHT PARAMETER ASSIGNMENT. See later chapter below.

A set of distinct FARE STRUCTURE ELEMENTs are grouped as a TARIFF.

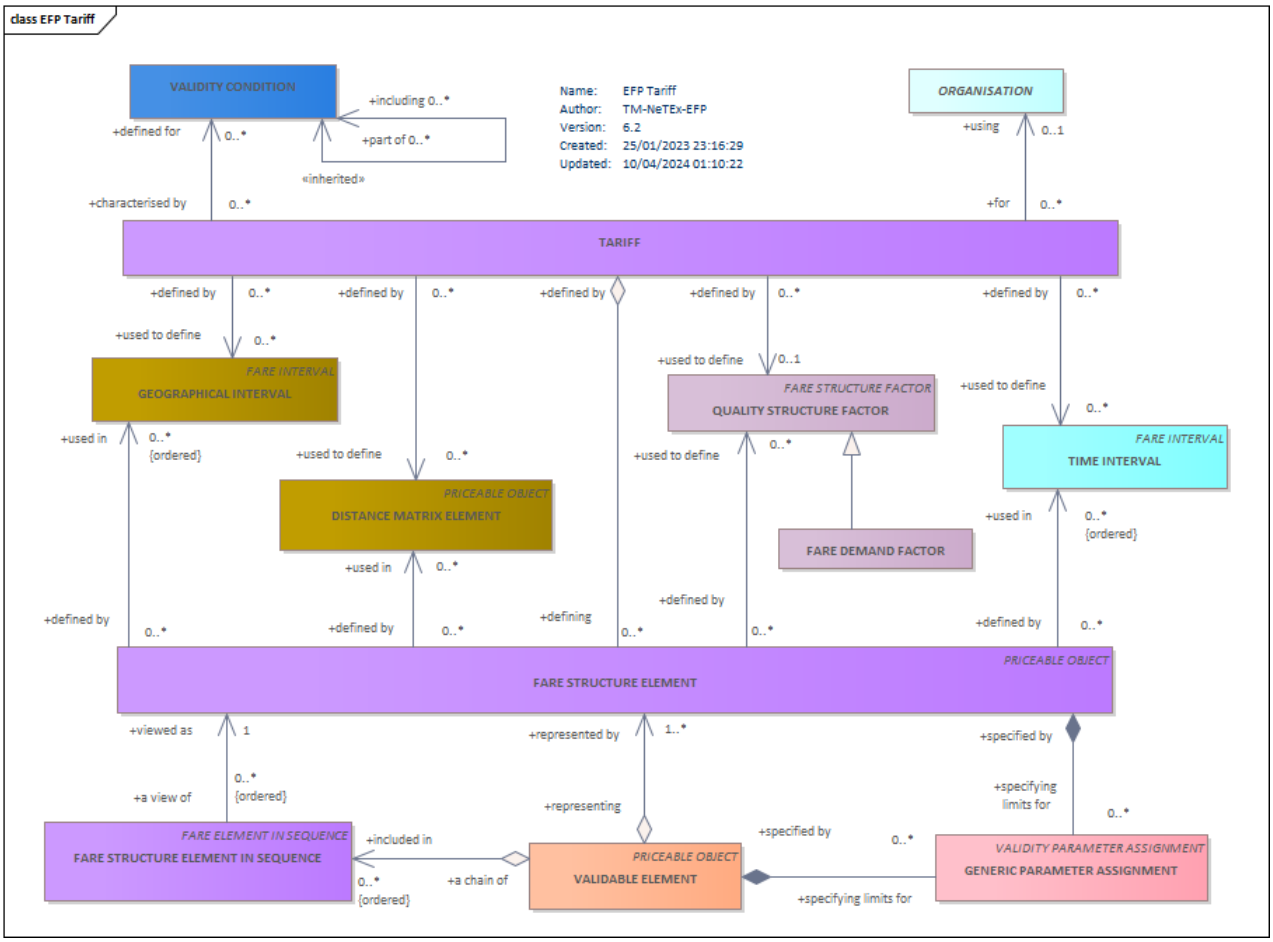


Figure 25 – EFP Tariff

11.2 Common tariff elements

Regardless of their specific structure (point-to-point, zonal, stage, trip, pass, etc), all TARIFFs are described by a TARIFF made up of FARE STRUCTURE ELEMENTs and other parameters.

11.2.1 *Tariff*

A TARIFF describes all parameters composing a particular fare structure. When applying fare structure rules, an algorithm will choose the parameters (e.g. a TIME INTERVAL) according to the valid TARIFF.

In an implementation, constraints should be set to ensure that, for a given TARIFF, only one factor of each type (e.g. GEOGRAPHICAL STRUCTURE FACTOR) is attached to any valid FARE STRUCTURE ELEMENT.

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*DataManagedObject*	::>	TARIFF inherits from DATA MANAGED OBJECT. See NeTEx Part1.
«PK»	***id***	*TariffIdType*	1:1	Identifier of TARIFF.
	Name	***MultilingualString***	0:1	Name of TARIFF.
«cntd»	***alternativeNames***	***AlternativeName***	0:*	Alternative names for TARIFF.
	Description	***MultilingualString***	0:1	Description of TARIFF.
«cntd»	***noticeAssignments***	***NoticeAssignment***	0:*	NOTICE ASSIGNMENTs for TARIFF.
«cntd»	***documentLinks***	***InfoLink***	0:*	Links for documents associated with TARIFF.
	PrivateCode	*PrivateCodeType*	0:1	Alternative identifier of an entity; can be used to associate with legacy systems.
«FK»	***OrganisationRef***	***(OrganisationRef)***	0:1	ORGANISATION to which TARIFF applies.
«FK»	***GroupOf-OrganisationsRef***	***GroupOf-OrganisationsRef***	0:1	GROUP OF ORGANISATIONs to which TARIFF applies.
«FK»	***LineRef***	***LineRef***	0:1	LINE to which TARIFF applies.
«FK»	***GroupOfLinesRef***	***GroupOfLinesRef***	0:1	GROUP OF LINEs to which TARIFF applies.
«FK»	***TypeOfTariffRef***	*TypeOfTariffRef*	0:1	Reference to a TYPE OF TARIFF.
«enum»	***TariffBasis***	***TariffBasisEnum***	0:1	Classification of how Tariff is priced. See allowed values below
	ReturnFareTwiceSingle	***xsd:boolean***	0:1	Whether the return ticket is the double as the fare for a single ticket.

«FK»	***GeographicalUnitRef***	***GeographicalUnitRef***	0:1	Reference to GEOGRAPHICAL UNIT for TARIFF.
«cntd»	***geographicalIntervals***	***GeographicalInterval***	0:*	GEOGRAPHICAL INTERVALs associated with TARIFF.
«cntd»	***geographical-StructureFactors***	***Geographical-StructureFactor***	0:*	GEOGRAPHICAL STRUCTURE FACTORs associated with TARIFF.
«FK»	***TimeUnitRef***	***TimeUnitRef***	0:1	Reference to TIME UNIT for TARIFF.
«cntd»	***timeIntervals***	***TimeInterval***	0:*	TIME INTERVALs associated with TARIFF.
«cntd»	***timeStructureFactors***	***TimeStructureFactor***	0:*	TIME STRUCTURE FACTORs associated with TARIFF.
«cntd»	***qualityStructureFactors***	***QualityStructureFactor***	0:*	QUALITY STRUCTURE FACTORs associated with TARIFF.
«cntd»	***fareStructureElements***	***FareStructureElement***	0:*	FARE STRUCTURE ELEMENTs associated with TARIFF.
«cntd»	***distanceMatrixElements***	***DistanceMatrix-Element***	0:*	DISTANCE MATRIX ELEMENTs associated with TARIFF.
«cntd»	***groupsOfDistanceMatrix Elements***	***GroupOfDistance-MatrixElements***	0:*	GROUPs OF DISTANCE MATRIX ELEMENTS associated with TARIFF.
«cntd»	***priceGroups***	***PriceGroup***	0:*	PRICE GROUPs for the TARIFF.
«cntd»	***fareTables***	***FareTable***	0:*	FARE TABLEs for the TARIFF.

Table 48 – Tariff – XML Element

11.2.2 *FareStructureElement*

A FARE STRUCTURE ELEMENT is a service consumption right determined by a set of quantitative (spatial, temporal) rules (i.e. determined by fare structure parameters, such as zones, sections, hours, etc); in effect each FARE STRUCTURE ELEMENT specifies a set of options affecting the pricing and use of a fare product.

In most cases, only one dimension, either a GEOGRAPHICAL, TIME TIME or QUALITY) STRUCTURE FACTOR is attached to each FARE STRUCTURE ELEMENT. There may be multiple dimensions (e.g. space, time duration, time of travel, etc), so that by picking one of each a choice of product options is specifed for purchase.In rare cases, different factors may apply to the same element, chosen by a rule depending on specific validity conditions. For example if fare prices depended on specific combinations of zone and durations of travel, rather than being priced separately for choice for zone and duration.

Note: In the full Transmodel representation, which also models the operational control and checking of fares as passengers travel, a FARE STRUCTURE element is further considered to be a sequence or set of CONTROLLABLE ELEMENTs to which rules for limitation of access rights and calculation of prices (fare structure) are applied. Thus for example, a single trip on the a metro with barriers on entry and exit may be made up of several separate rides on different lines, each a CONTROLLABLE ELEMENT, but can be treated as a single VALIDABLE ELEMENT (indicating whether the user has the rights to travel in the specified ones and can entry and/or exit) for fare definition purposes; and so can be represented as a FARE STRUCTURE ELEMENT offering a choice of just the permitted zone combinations.

In practice the parameters of a given FARE STRUCTURE ELEMENT need to be interpreted by a consumer of the data according to the tariff structure being used. A specific meaning can be indicated by means of the ***TariffBasis*** and the ***TypeOfFareStructureElement*** attributes.

EFIP1>> A ***TariffBasis*** must be indicated on a ***FareStructureElement***.

EFIP1>> A ***TypeOfFareStructureElemen t*** must be indicated on a ***FareStructureElement***

See later for the required values for Tariff Basis & TypeOfFareStructureElement.

Classification	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	FARE STRUCTURE ELEMENT inherits from PRICEABLE OBJECT**.**
«PK»	***id***	*FareStructureElementIdType*	1:1	Identifier of FARE STRUCTURE ELEMENT.
«enum»	***TariffBasis***	*TariffBasisEnum*	0:1	TARIFF BASIS to be used for the element. See allowed values below. • *flat, distance; unitSection, zone, zoneToZone; pointToPoint; route; tour; group; discount; period; free; other*

				EFIP1>> A ***TariffBasis*** must be indicated
«FK»	***TypeOfFare-Structure-ElementRef***	*TypeOfFareStructureElement Ref*	0:1	Classifcation of FARE STRUCTURE ELEMENT. EFIP1>> A ***TypeOfFareStructureElement*** must be indicated
		CHOICE		
«FK»	***a*** ***Geographical-IntervalRef***	*GeographicalIntervalRef*	0:1	Reference to GEOGRAPHICAL INTERVAL associated with FARE STRUCTURE ELEMENT.
«cntd»	***b*** ***geographical-Intervals***	*GeographicalInterval* | *GeographicalIntervalRef*	0:*	GEOGRAPHICAL INTERVALs associated with FARE STRUCTURE ELEMENT.
«cntd»	***c*** ***geographical-Structure-Factors***	*GeographicalStructureFactor* | *GeographicalStructure-FactorRef*	0:*	GEOGRAPHICAL STRUCTURE FACTORs associated with the FARE STRUCTURE ELEMENT.
		Choice		
«FK»	***a*** ***TimeInterval-Ref***	*TimeIntervalRef*	0:1	Reference to TIME INTERVAL associated with FARE STRUCTURE ELEMENT.
«cntd»	***b*** ***timeIntervals***	*TimeInterval* | *TimeIntervalRef*	0:*	TIME STRUCTURE INTERVALs associated with the FARE STRUCTURE ELEMENT.
«cntd»	***c*** ***timeStructure-Factors***	*TimeStructureFactor* | *TimeStructureFactorRef*	0:*	TIME STRUCTURE FACTORs associated with the FARE STRUCTURE ELEMENT.
		CHOICE		
«FK»	***a*** ***Quality-Structure-FactorRef***	*QualityStructureFactor Ref*	0:1	Reference to QUALITY STRUCTURE FACTOR associated with FARE STRUCTURE ELEMENT.

«cntd»	***b***	***quality-Structure-Factors***	*QualityStructureFactor* | *QualityStructureFactor*	0:*	QUALITY STRUCTURE FACTORs associated with the FARE STRUCTURE ELEMENT.
			Choice		
«FK»	***a***	***Distance-Matrix-ElementRef***	*DistanceMatrixElementRef*	0:1	Reference to DISTANCE MATRIX ELEMENT associated with FARE STRUCTURE ELEMENT.
«FK»	***b***	***distanceMatrixElements***	*DistanceMatrixElement* | *DistanceMatrixElementRef*	0:*	DISTANCE MATRIX ELEMENTs associated with FARE STRUCTURE ELEMENT.
«FK»	***c***	***GroupOf-Distance-Matrix-ElementsRef***	*GroupOfDistanceMatrix-ElementsRef*	0:1	Reference to GROUP OF DISTANCE MATRIX ELEMENTs associated with FARE STRUCTURE ELEMENT.
«cntd»	***d***	***GroupOf-Distance-Matrix-Elements***	*GroupOfDistanceMatrix-Elements*	0:1	GROUP OF DISTANCE MATRIX ELEMENTs associated with FARE STRUCTURE ELEMENT.
«cntd»	***fareStructure-ElementsIn-Sequence***		*FareStructureElement-InSequence* | *Controllable-ElementInSequence*	0:*	Child FARE STRUCTURE ELEMENTs in SEQUENCE making up the FARE STRUCTURE ELEMENT.
			CHOICE		Either multiple parameters wrapped in a tag, or a single parameter (an optimisation).
«cntd»	***a***	***validity-Parameter-Assignments***	*AccessRightParameter-Assignment*	0:*	GENERIC PARAMETER ASSIGNMENTs associated with the FARE STRUCTURE ELEMENT.
«cntd»	***b***	***Generic-Parameter-Assignment***	*GenericParameter-Assignment*	0:1	A single GENERIC PARAMETER ASSIGNMENT associated with the FARE STRUCTURE ELEMENT.
«cntd»	***c***	***Generic-Parameter-AssignmentIn***	*GenericParameter-Assignment*	0:1	A single GENERIC PARAMETER ASSIGNMENT associated with the FARE STRUCTURE ELEMENT. No ID needs to be given – will be

		Context			inferred from the assignment values. (OPTIMISATION).
«cntd»	***prices***		*FareStructureElementPrice*	0:*	Prices for the FARE STRUCTURE ELEMENT.

Table 49 – *FareStructureElement* – XML Element

11.2.3 *FareElementInSequence – (Abstract)*

An abstract framework element representing an element that is a one of sequence of ELEMENT, including its possible order in the sequence.

Specialised as FARE ESTRCUTURE ELEMENT IN SEQUENCE and CONTROLLABLE ELEMENT IN SEQUENCE.

Classifi-cation	Name	Type	Cardin-ality	Description
::>	::>	*VersionedChild*	::>	FARE ELEMENT IN SEQUENCE inherits from VERSIONED CHILD. See NeTEx Part1.
«PK»	***id***	*FareElementInSequenceIdType*	1:1	Identifier of FARE ELEMENT IN SEQUENCE.
«PK»	***order***	*xsd:positiveInteger*	0:1	Order of element within SEQUENCE.
	Name	*MultilingualString*	0:1	Name of FARE ELEMENT IN SEQUENCE.
	Description	*MultilingualString*	0:1	Description of FARE ELEMENT IN SEQUENCE.
	IsFirstInSequence	*xsd:boolean*	0:1	Whether element is the first in the sequence.
	IsLastInSequence	*xsd:boolean*	0:1	Whether element is the last in the sequence.
	Access-NumberIsLimited	*xsd:boolean*	0:1	Whether access number is limited. +v1.1
	MinimumAccess	*xsd:nonNegativeInteger*	0:1	Minimum number of accesses. +v1.1
	MaximumAccess	*xsd:nonNegativeInteger*	0:1	Maximum number of accesses. +v1.1

	AccessNumber	*xsd:nonNegativeInteger*	0:1	Access number in sequence.

Table 50 – *FareElementInSequence* – XML Element

11.2.4 *FareStructureElementInSequence [ADVANCED PROFILE ONLY]*

A FARE STRUCTURE ELEMENT as a part of a VALIDABLE ELEMENT, including its possible order in the sequence of FARE STRUCTURE ELEMENTs forming that VALIDABLE ELEMENT, and its possible quantitative limitation.

Note: FARE STRUCTURE ELEMENTs IN SEQUENCE are only needed for certain types of more complex fare where access rights must be consumed in a certain order to achieve a particular price.

Classification	Name	Type	Cardinality	Description
::>	::>	*FareElementInSequence*	::>	FARE STRUCTURE ELEMENT IN SEQUENCE inherits from FARE ELEMENT IN SEQUENCE.
«PK»	***id***	*FareStructureElement-InSequenceIdType*	1:1	Identifier of FARE STRUCTURE ELEMENT IN SEQUENCE.
«FK»	***FareStructure-ElementRef***	*FareStructureElementRef*	0:1	Reference to a FARE STRUCTURE ELEMENT.
«FK»	***Validable-ElementRef***	*ValidableElementRef*	0:1	Reference to a VALIDABLE ELEMENT.
«cntd»	***validityParameter-Assignments***	*ValidityParameterAssignment*	0:*	VALIDITY PARAMETER ASSIGNMENTs associated with the ELEMENT IN SEQUENCE.

Table 51 – *FareStructureElementInSequence* – XML Element

11.3 Using Fare Structure elements to define a tariff

A NeTEx fare structure is based on quantitative rules that describe the access rights regulating the consumption of transport services in terms of their spatial (GEOGRAPHICAL), temporal (TIME) and other (QUALITY) aspects.

Most fare structures (other than flat fares) are fundamentally space-based, or more precisely, distance-based, for example being based on intervals, graduated distance, origin/destination pairs, or specific zones.

Some of these types may be combined together. Some fare structures also have a temporal component, for example the price of a season pass typically depends on its duration.

The various fare structure elements are grouped as a TARIFF. In practice, only certain types of TARIFFs (i.e. ways of combining of FARE STRUCTURE ELEMENTS) are found. For example:

Spatial

- Point-to-Point or Zone-to-Zone
- Zonal
- Fare Stage.

Temporal

- Specific use: Single, Return
- Timed Use
- Multiple use (Carnet)
- Day pass
- Period Pass.

In the NeTEx specification the fare structure elements are presented all together and so the model, though powerful, is somewhat difficult to understand. Here, for clarity, we present in turn just the elements needed for specifc types of fare structures that are found in the EFIP

Note that NeTEx also has capabilities to describe other more complex fare structuress (for example that depend or routing (See SERIES CONSTRAINT in the NeTEx document). These are outside of the scope of the EFIP.

11.4 Point-to-point fare structure

To describe a Point-to-point (P2P) or zone-to-zone (Z2Z) fare in its simplest form, a DISTANCE MATRIX element is used. Each DISTANCE MATRIX ELEMENT represents the access rights to travel between two stops or zones, and may have one or more DISTANCE MATRIX ELEMENT PRICEs associated directly with it in a designated PRICE UNIT.

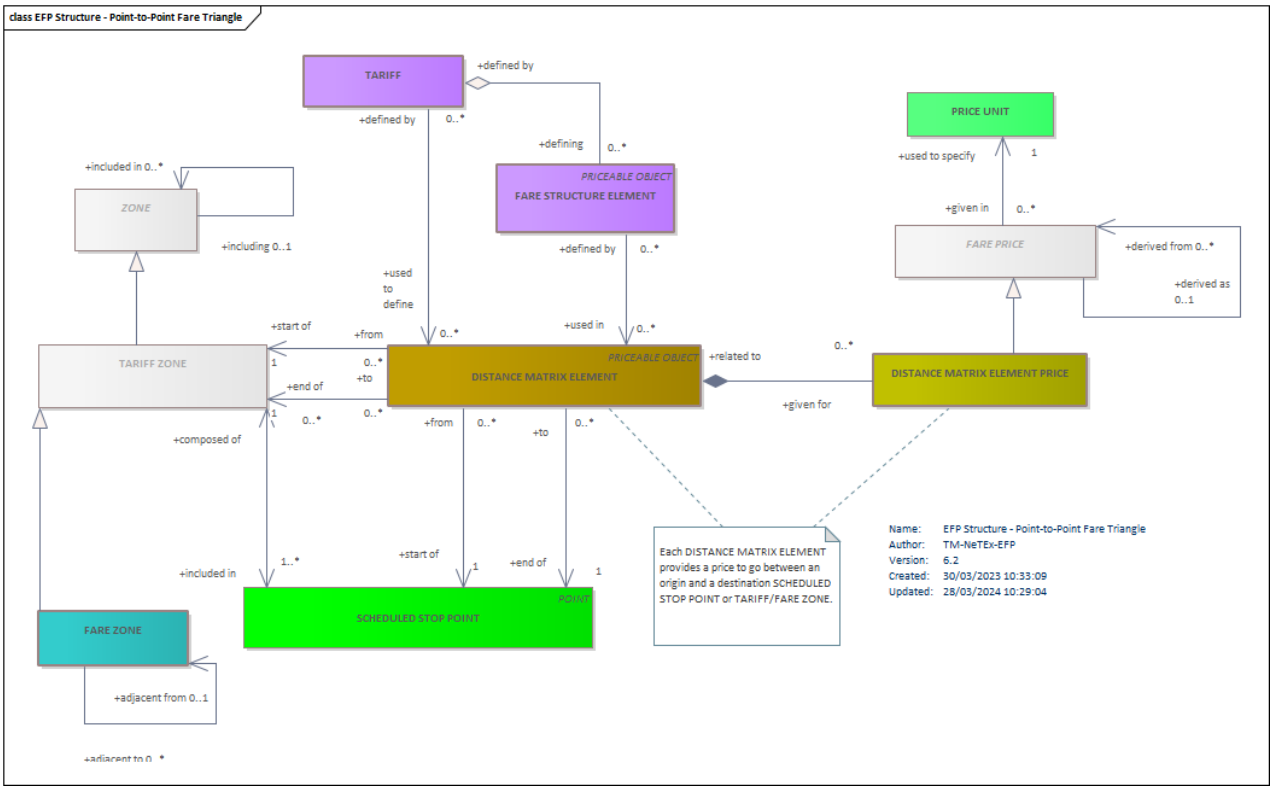


Figure 31 — Point-to-point Fare structure (UML)

The TARIFF is thus in effect a set of DISTANCE MATRIX elements, comprising a classic “fare triangle”

Outward (Absolute Fare Price)¶

Ask Ave					
Bath Pla	£0.40				
Cam Sq	£0.50	£0.40			
Dee Sta	£0.75	£0.75	£0.50		
Ely Rd	£1.00	£1.00	£0.75	£0.40	
	Ask Ave	Bath Pla	Cam Sq	Dee Sta	Ely Rd

Table 1-2 Example Triangular Fare Table with Absolute Prices¶

Figure 32 — Example: Distance Matrix Element – Triangular Fare table with absolute Prices (EXM)

11.4.1 *DistanceMatrix Element*

Each DISTANCE MATRIX ELEMENT represents the fare between an origin and a destination pair; either two SCHEDULED STOP POINTs or two TARIFF ZONEs, or two FARE SECTIONs. A GROUP OF DISTANCE MATRIX ELEMENTs specifies a set of DISTANCE MATRIX ELEMENTs, allowing a common set of prices for between different origin-destination pairs if required.

Classi fi- cation	Name	Type	Cardi nality	Description
::>	::>	*PriceableObject*	::>	DISTANCE MATRIX ELEMENT inherits from PRICEABLE OBJECT**.**
«PK»	***id***	*DistanceMatrix-ElementIdType*	1:1	Identifier of a DISTANCE MATRIX ELEMENT.
	Name	*MultilingualString*	0:1	Name of DISTANCE MATRIX ELEMENT.
	Distance	***DistanceType***	0:1	Distance between origin and destination of a DISTANCE MATRIX ELEMENT.
	RelativeRanking	***xsd:integer***	0:1	Relative preference assigned to this element if there are multiple entries between two points.
	IsDirect	***xsd:boolean***	0:1	Whether journey is direct or requires changes.
	InverseAllowed	***xsd:boolean***	0:1	Whether an inverse element in the opposite direction with the same prices may be assumed – optimisation to reduce data volumes.
		Choice	1:1	Origin of DISTANCE MATRIX ELEMENT
«FK»	***a*** ***StartStopPointRef***	*ScheduledStopPointRef*	0:1	Start SCHEDULED STOP POINT at which a DISTANCE MATRIX ELEMENT begins.
«FV»	***b*** ***StartStopPointView***	*ScheduledStopPoint-View*	0:1	Details of origin SCHEDULED STOP POINT.

«FK»	***c***	***StartTariffZoneRef***	*TariffZoneRef*	0:1	Start TARIFF ZONE at which a DISTANCE MATRIX ELEMENT begins.
«FV»	***d***	***StartTariffZoneView***	*TariffZoneView*	0:1	Details of origin TARIFF ZONE.
«FK»	***e***	***StartFareSectionRef***	*FareSectionRef*	0:1	Start FARE SECTION at which a DISTANCE MATRIX ELEMENT begins.
«FK»	***f***	***StartFarePoint-InPatternRef***	*FarePointInPatternRef*	0:1	Start FARE POINT IN PATTERN at which a DISTANCE MATRIX ELEMENT begins. (Handles case of repeated visits)
			Choice	1:1	Destination of DISTANCE MATRIX ELEMENT.
«FK»	***a***	***EndStopPointRef***	*ScheduledStopPointRef*	0:1	End SCHEDULED STOP POINT at which a DISTANCE MATRIX ELEMENT ends.
«FV»	***b***	***EndStopPointView***	*ScheduledStopPoint-View*	0:1	Details of destination SCHEDULED STOP POINT
«FK»	***c***	***EndTariffZoneRef***	*TariffZoneRef*	0:1	Final TARIFF ZONE at which a DISTANCE MATRIX ELEMENT ends.
«FV»	***d***	***EndTariffZoneView***	*TariffZoneView*	0:1	Details of origin TARIFF ZONE.
«FK»	***e***	***EndFareSectionRef***	*FareSectionRef*	0:1	End FARE SECTION at which a DISTANCE MATRIX ELEMENT ends.
«FK»	***f***	***EndFarePoint-InPatternRef***	*FarePointInPatternRef*	0:1	End FARE POINT IN PATTERN at which a DISTANCE MATRIX ELEMENT ends. (Handles case of repeated visits).
«cntd»	***seriesConstraints***		*SeriesConstraintRef*	0:*	SERIES CONSTRAINTs associated with this DISTANCE MATRIX ELEMENT.
«cntd»	***structureFactors***		*GeographicalStructureFactorRef*	0:*	STRUCTURE FACTORs associated with this DISTANCE MATRIX ELEMENT.
«cntd»	***tariffs***		*TariffRef*	0:*	TARIFFs for the DISTANCE MATRIX ELEMENT.

FK	***FareTableRef***	*FareTableRef+*	0:1	Primary FARE TABLE for the DISTANCE MATRIX ELEMENT.
«cntd»	***fareTables***	*FareTableRef*	0:*	FARE TABLEs for the DISTANCE MATRIX ELEMENT.

Table 52 – *DistanceMatrixElement* – XML Element

11.4.2 *GroupOfDistanceMatrixElements*

A grouping of DISTANCE MATRIX ELEMENTs. May be used to provide reusable Origin / Destination pairs.

Classifi-cation	**Name**	**Type**	**Cardin-ality**	**Description**
::>	::>	*GroupOfEntities*	::>	GROUP of DISTANCE MATRIX ELEMENTs inherits from GROUP OF ENTITies.
«PK»	***id***	*GroupOfDistance-MatrixElementsIdType*	1:1	Identifier of GROUP of DISTANCE MATRIX ELEMENTS.
	UseToExclude	*xsd:boolean*	0:1	Whether contents of Group should be used to exclude (true) from a larger list. The default value is '*false* (i.e. "include")
«cntd»	***priceGroups***	*PriceGroup*	0:*	PRICE GROUPs for the GROUP OF DISTANCE MATRIX ELEMENTs.
«cntd»	***fareTables***	*FareTable*	0:*	FARE TABLEs for the GROUP OF DISTANCE MATRIX ELEMENTs.
	Distance	*DistanceType*	0:1	Distance between origins and destinations of a GROUP OF DISTANCE MATRIX ELEMENTs.
«cntd»	***structureFactors***	*GeographicalStructure-FactorRef*	0:*	References to GEOGRAPHICAL STRUCTURE FACTORs.

«cntd»	***notice-Assignments***	*NoticeAsssignment*	0:*	NOTICE ASSIGNMENTs for GROUP OF DISTANCE MATRIX ELEMENTs.
«cntd»	***members***	*DistanceMatrixElements*	0:*	References to members of the GROUP OF DISTANCE MATRIX ELEMENTs.
«cntd»	***prices***	*DistanceMatrix-ElementPrice*	0:*	Prices for the GROUP OF DISTANCE MATRIX ELEMENTs.

Table 53 – *GroupOfDistanceMatrixElements* – XML Element

11.5 Geographical interval fare structures

In many cases, the values used for applying the fare structure rules will be derived from the description of the actual elements consumed when making a trip. This may be done in different ways

- **Actual distance:** For a true “kilometre distance” fare, the length of a trip in km will be derived from the JOURNEY PATTERN description (using the length of the LINKs composing the JOURNEY PATTERN).

- **Zone-to-zone:** Some fare structure systems will use arbitrary fare distances between the origin and the destination of a FARE STRUCTURE ELEMENT. This is typically the case when a zone-matrix fare system is used. Some TARIFF ZONEs (usually a few) are defined and a specific fare distance parameter is defined for each possible origin/destination pair of TARIFF ZONEs. Such parameter values are likely to differ from an exact calculation based on the covered distance. These values are stored in as DISTANCE MATRIX ELEMENTs. In the simplest form, as introduced in the previous section, the resulting price is given directly for each element. (so the distance need not be exposed to the passenger at all). In a more elaborated form, a GEOGRAPHICAL UNIT with a UNIT price is associated with eeach DISTANCE MATRIX ELMENT and is used to compute a price from the element’s distance. In other words the allocation of prices may be as ***arbitrary amounts*** or a ***distance function***

- **Point-to-point:** In a similar way to a zone-to-zone tariff, the fare distances between SCHEDULED STOP POINTs may not necessariiy be taken from the actual trip distance, but be stored as specific values for each possible trip. In such a case, a DISTANCE MATRIX ELEMENT will store the chosen distance value between each origin/destination pair of SCHEDULED STOP POINTs and a corresponding price may be allocated or computed using a GEOGRAPHICAL UNIT PRICE.

- **Unit zone or section distance:** Many graduated fare structures will use a count of TARIFF ZONEs or FARE SECTIONs, such that the fare depends on the number of ones or sections consumed (rather than any specific zone or section); thus the GEOGRAPHICAL UNIT is a zone or section. A projection

of the actual TARIFF ZONEs or FARE SECTIONs on the JOURNEY PATTERN needed to make trip by a passenger will allow a pricing engine to derive the number of zones or sections crossed during the trip.

- ***Geographical interval fare***, a price is associated with two more intervals each applying to a range of distances; thus the pricing is stepped and independent of any particular stop or zone. To price a journey a distance between an origin and a destination, taken from a DISTANCE MATRIX ELEMENT (or computed directly from the underlying SERVICE PATTERN) is used to lookup the appropriate interval price that applies to use.

Note that it is possible to present any distance based fare as a Point-to-point fare by using the distances and unitis to computing an origin/destination price.

11.5.1 *Simple Geographic Intervals*

In a simple interval distance, fare, a price is associated with one more GEOGRAPHICAL INTERVALs, each applying to a range of distances (so it can be progressive, getting cheaper for longer distances). To price a journey a distance (which may be stated in arbitray units as well as actual physical distances) is used to lookup the appropriate GEOGRAPHICAL INTERVAL to use.

The most basic usage is for a Unit Zone tariffs, where there is a unit cost per zone, regardless of which zone is used. In this case the pricing is not progressive, i.e. does not get cheaper if more ones are consumed.

The TARIFF thus includes a set of GEOGRAPHICAL INTERVALs.

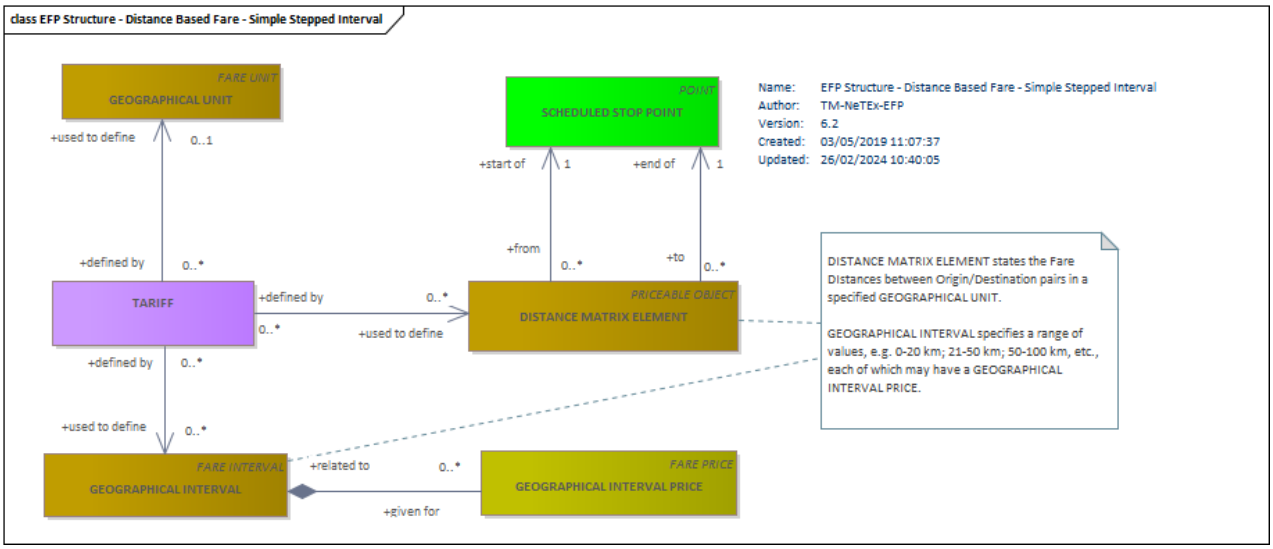


Figure 33 — Simple Interval Distance Fare Structure (UML)

11.5.2 *Geographic factors*

The simple spaced-based fare structures described above may be combined in more complex structures.

In most cases of fare structures using GEOGRAPHICAL INTERVALs, the fare will be flat within the range of each interval, which means that the fare is the same all along the interval. However, the fares may vary with each interval, depending on a graduation based on a GEOGRAPHICAL UNIT. Such a unit is not necessarily the same as the unit describing the interval. For instance, the fares may be graduated, the price per km differing according to the number of zones crossed (e.g. to allow lower prices for long trips).

Similarly, a graduated fare structure may be influenced by the type of trip, as regards the geography of the network. If the fare is based on the number of fare sections crossed, it may vary, for instance, depending on whether the trip is from a suburb to the city centre or between two suburbs. This structure will associate GEOGRAPHICAL INTERVALs (fare sections) and DISTANCE MATRIX ELEMENTs (using a set of TARIFF ZONEs, e.g. “centre” and “suburbs”).

The entity GEOGRAPHICAL STRUCTURE FACTOR allows the combinination of two simple structures in a complex factor. It is identified by a GEOGRAPHICAL UNIT, describing the used graduation unit, and by either a GEOGRAPHICAL INTERVAL or a DISTANCE MATRIX ELEMENT.

In an implementations, GEOGRAPHICAL STRUCTURE FACTORs would probably be associated in sets related to one fare calculation rule, in order to allow an algorithm to choose the appropriate rule.

A GEOGRAPHICAL STRUCTURE FACTOR is thus used for more elaborate interval fares, each associaties a GEOGRAPHICAL UNIT PRICE (e.g price per kilometer) with a GEOGRAPHICAL INTERVAL. As for a simple interval, this may be used with a distance to arrive at a price.

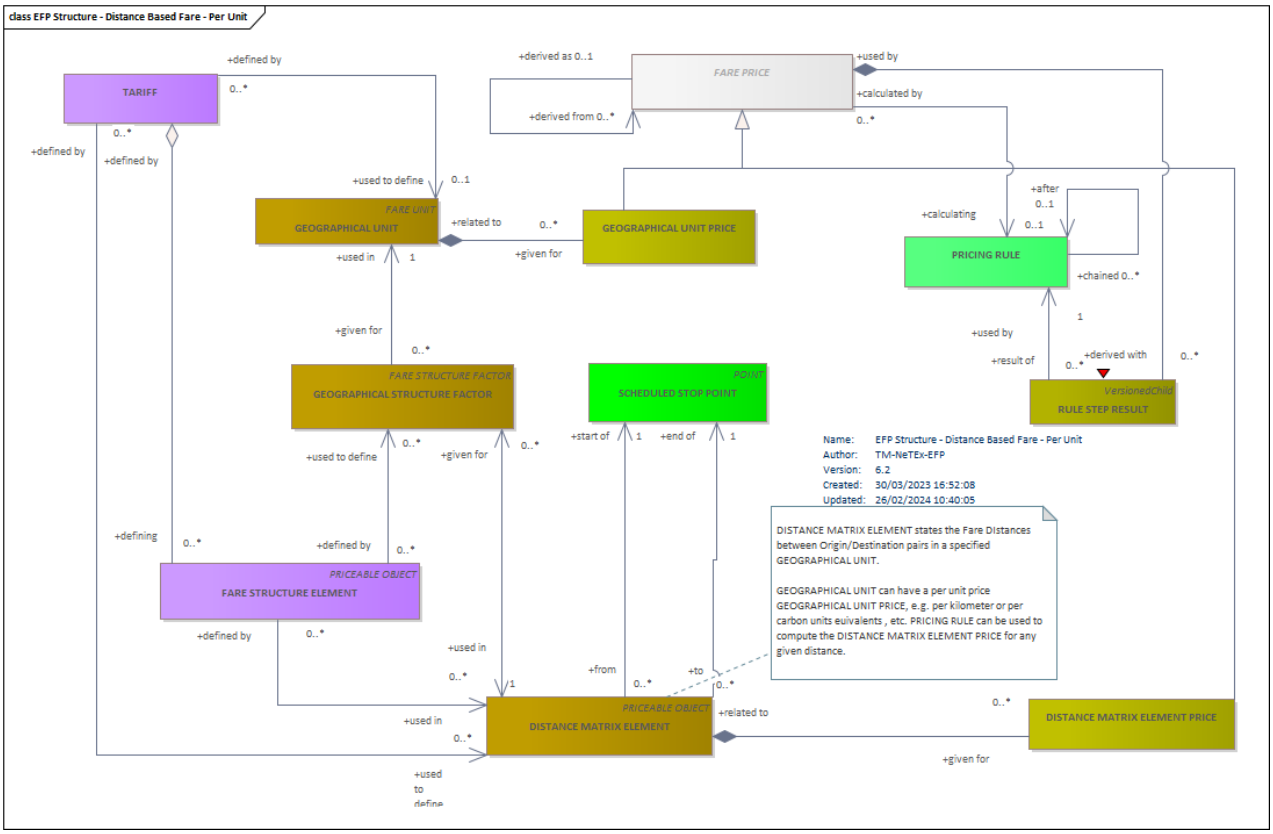


Figure 34 — Simple Interval Distance Fare Structure

11.5.3 *Zonal fares*

In a Zonal tariff, a price is assigned to each zone on an arbitrary basis. Each zone or combination of zones is defined as a FARE ZONE. A NETWORK may compromise a number of ZONEs. A FARE STRUCTURE ELEMENT will assign a list of zones as its parameters, one of which may be selected when the product is purchased. In order to be able to associate a separate price with each zone, a DISTANCE MATRIX ELEMENT is used with the start and end zone the same.

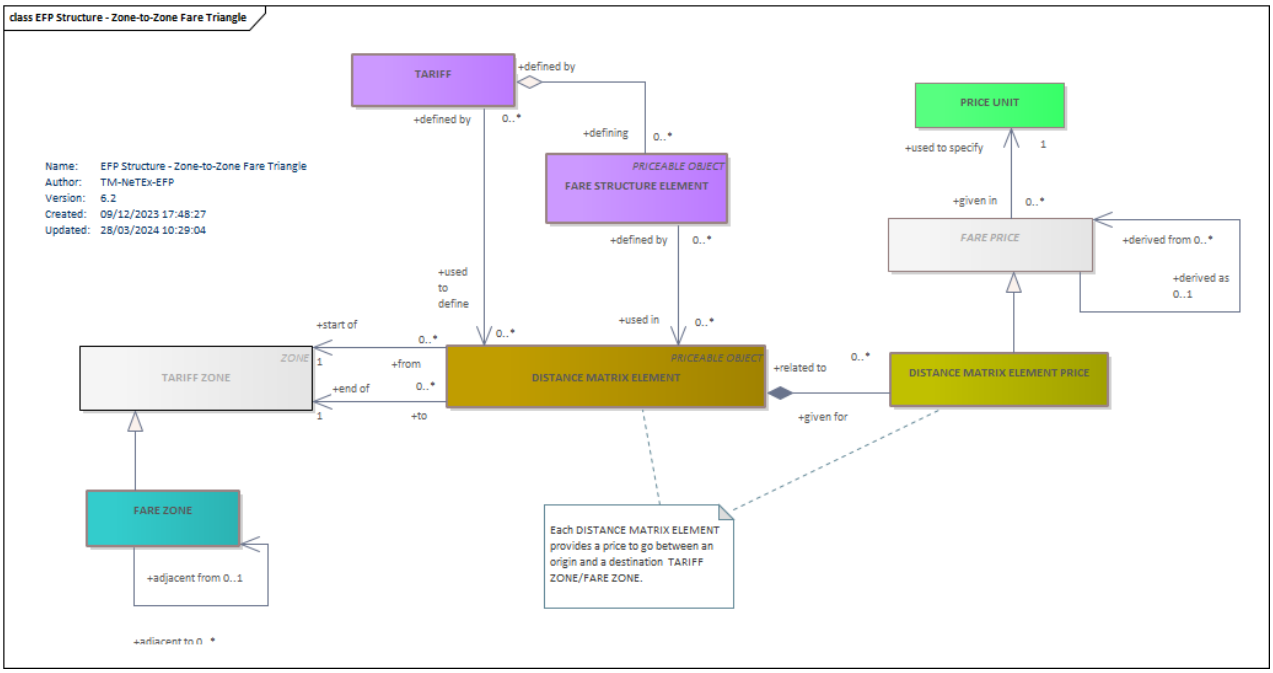


Figure 35 — Zonal Fare Structure

Routing constraints may be applied to specify the order that the zones are traversed.

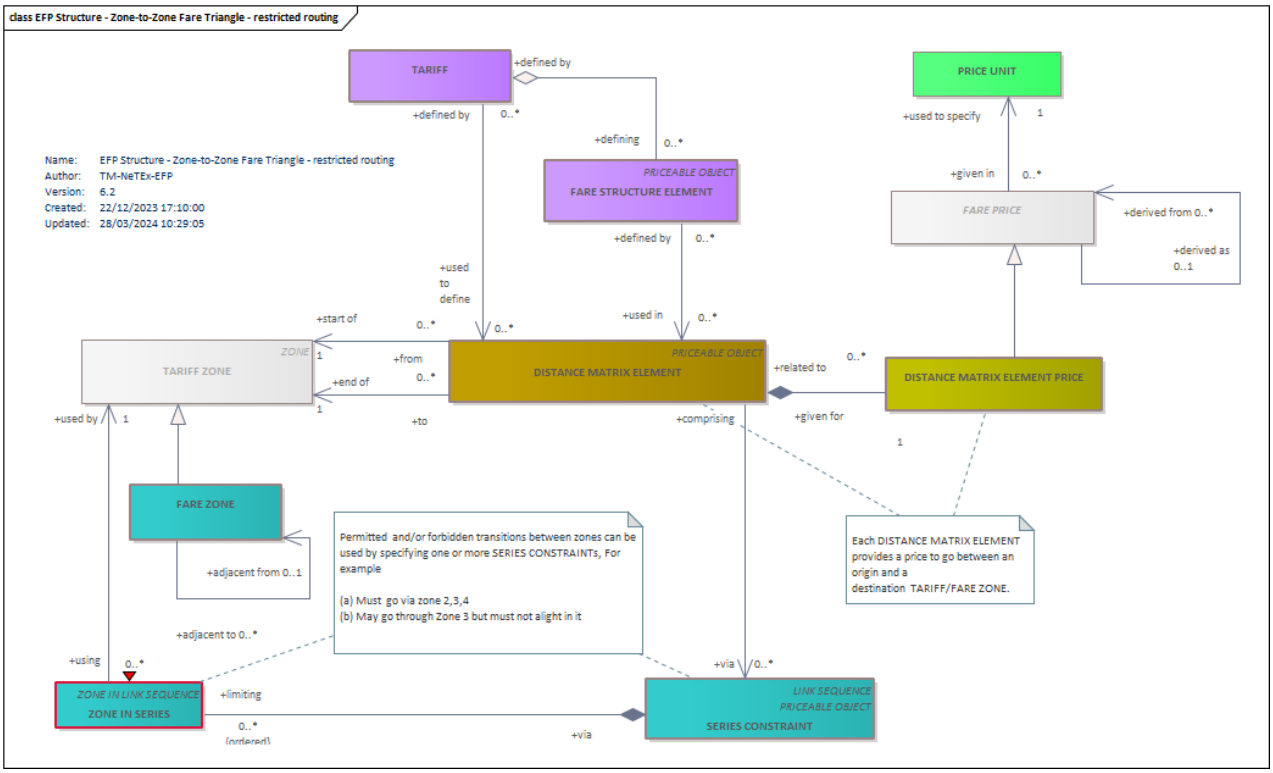


Figure 36 — Zonal Fare Structure with restricted routing

It may also be that the use is restricted to certain lines in the zone.

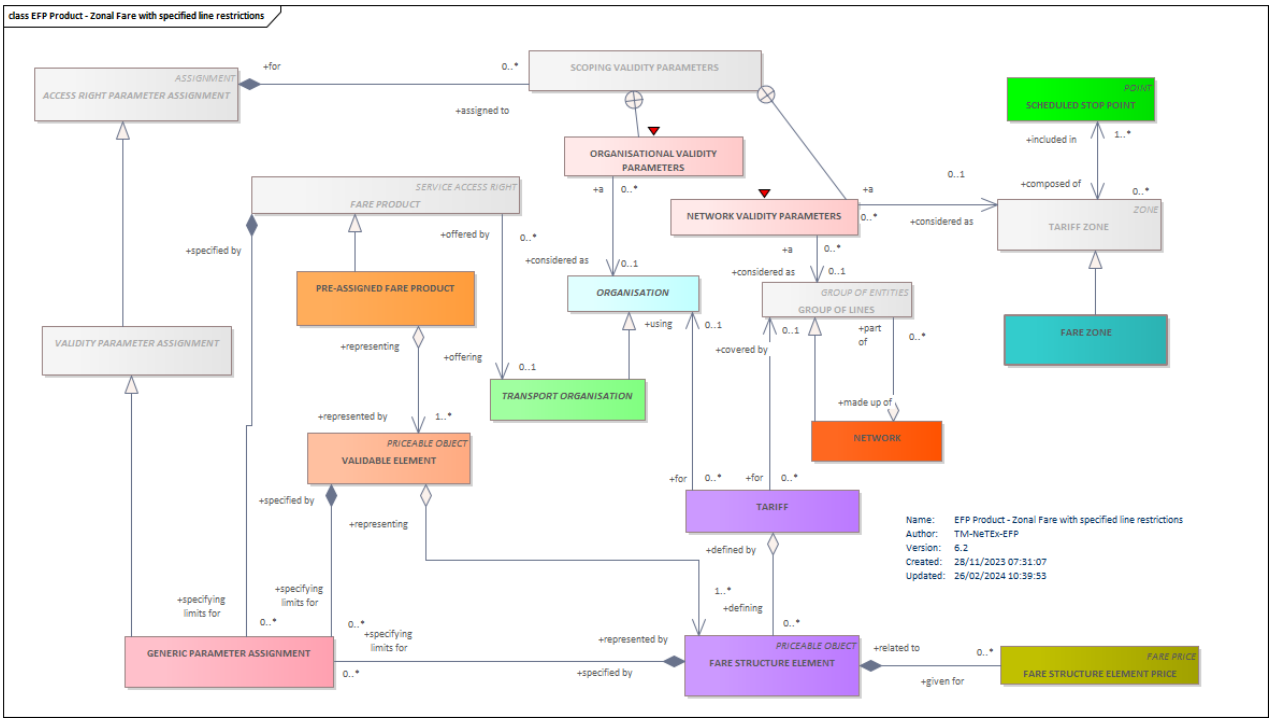


Figure 37 — Zonal Fare Structure with line use restrictions

11.5.4 *FareInterval (Abstract)*

An interval-based aspect of the fare structure.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	FARE INTERVAL inherits from PRICEABLE OBJECT.
«PK»	***id***	*FareIntervalIdType*	1:1	Identifier of FARE INTERVAL.

Table 54 – *FareInterval* – XML Element

11.5.5 *GeographicalInterval*

A geographical interval specifies the access rights for the FARE STRUCTURE ELEMENTs within the range of this interval: “20-5 km”, “4-6 zones”, etc.

Each GEOGRAPHICAL INTERVAL will store the minimum and the maximum value describing the corresponding distance interval, on which a certain fare will be applied.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*FareInterval*	::>	GEOGRAPHICAL INTERVAL inherits from FARE INTERVAL.
«PK»	***id***	*GeographicalInterval-IdType*	1:1	Identifier of GEOGRAPHICAL INTERVAL.
	Start-Geographical-Value	*xsd:decimal*	0:1	Start value for GEOGRAPHICAL INTERVAL.
	End-Geographical-Value	*xsd:decimal*	0:1	End value for GEOGRAPHICAL INTERVAL.
	NumberOfUnits	*xsd:integer*	0:1	Number of units in GEOGRAPHICAL INTERVAL.
«enum»	***IntervalType***	*IntervalTypeEnum*	0:1	Classification of interval type. See allowed values below. • *stop* • *tariffZone* • *distance* • *section* • *coupon* • *other*

«FK»	***Geographical-UnitRef***	*GeographicalUnitRef*	0:1	GEOGRAPHICAL UNIT for interval.
«cntd»	***prices***	*Geographical-IntervalPrice*	0:*	Prices for the GEOGRAPHIC INTERVAL.

Table 55 – *GeographicalInterval* – XML Element

11.5.6 *FareUnit (Abtract)*

A unit associated with a FARE STRUCTURE FACTOR.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	FARE UNIT inherits from PRICEABLE OBJECT.
«PK»	***id***	*FareUnitIdType*	1:1	Identifier of FARE UNIT.
	nameOfClassOfUnit	*NameOfClass*	0:1	Type of Class used for zone; ***DistanceType***, etc e.g. ***TariffZone***. This is metadata to facilitate programming.

Table 56 – *FareUnit* – Element

11.5.6.1 *GeographicalUnit*

A unit for calculating geographical graduated fares.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*FareUnit*	::>	GEOGRAPHICAL UNIT inherits from FARE UNIT.
«PK»	***id***	*GeographicalUnitIdType*	1:1	Identifier of GEOGRAPHICAL UNIT.
	Distance	*DistanceType*	0:1	If distance-based unit, length of unit.

«cntd»	***prices***	*GeographicalUnitPrice*	0:*	Prices associated with GEOGRAPHICAL UNIT

Table 57 – *GeographicalUnit* – Element

11.5.7 *FareStructureFactor (Abstract)*

A factor influencing access rights definition or calculation of prices (abstract framework element).

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	FARE STRUCTURE FACTOR. inherits from PRICEABLE OBJECT.
«PK»	***id***	*FareStructureFactorIdType*	1:1	Identifier of FARE STRUCTURE FACTOR.
	PrivateCode	*PrivateCodeStructure*	0:1	External code associated with factor. +v1.1
	Factor	*xsd:anyType*	0:1	Arbitrary values associated with factor.
«PK»	***TypeOfFare-Structure-FactorRef***	*TypeOfFareStructure-FactorRef*	1:1	Reference to a classification of the FARE STRUCTURE FACTOR.

Table 58 – *FareStructureFactor* – XML Element

11.5.7.1 *GeographicalStructureFactor*

The value of a GEOGRAPHICAL INTERVAL or a DISTANCE MATRIX ELEMENT expressed by a GEOGRAPHICAL UNIT.

Geographical fare structures can be used with many different types of units. For example:

- True Distance based fares – e.g. Kilometres.
- Fares based on the number of zones traversed.
- Fares based on the number of fare stages traversed.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*FareStructureFactor*	::>	GEOGRAPHICAL STRUCTURE FACTOR inherits from FARE STRUCTURE FACTOR.
«PK»	***id***	*GeographicalStructure-FactorRef*	1:1	Identifier of GEOGRAPHICAL STRUCTURE FACTOR.
«FK»	***TariffRef***	*TariffRef*	0:1	Reference to a TARIFF. EFP1>>Should be embedded within a TARIFF.
«FK»	***DistanceMatrix-ElementRef***	*DistanceMatrix-ElementRef*	0:1	Reference to a DISTANCE MATRIX ELEMENT.
«FK»	***Geographical-IntervalRef***	*Geographical-IntervalIdType*	0:1	Reference to a GEOGRAPHICAL INTERVAL.
«FK»	***Geographical-UnitRef***	*GeographicalUnitRef*	0:1	Reference to GEOGRAPHICAL UNIT.
	NumberOfUnits	*NumberOfUnits*	0:1	Quantity of units.
	AmountFactor	*xsd:decimal*	0.1	Arbitrary amount factor associated with GEOGRAPHICAL STRUCTURE FACTOR.

Table 59 – *GeographicalStructureFactor* – XML Element

11.6 Time interval fare structure

The time-based fare structures are described in a similar way to the space-based structures. A TIME INTERVAL describes intervals of time (0-1 hour, 1-3 hours, etc.) during which a certain fare is applied to FARE STRUCTURE ELEMENTs. A graduated time-based structure will be defined using a TIME UNIT (e.g. days, hours or minutes).

TIME INTERVALS and TIME UNIT may be combined into TIME STRUCTURE FACTORs. This allows for instance to specify a fare per hour spent, which varies depending on the range of days spent.

TIME INTERVALs are used to define day and season passes,

Note: The concept of TIME INTERVAL should be carefully distinguished from that of a USAGE VALIDITY PERIOD;

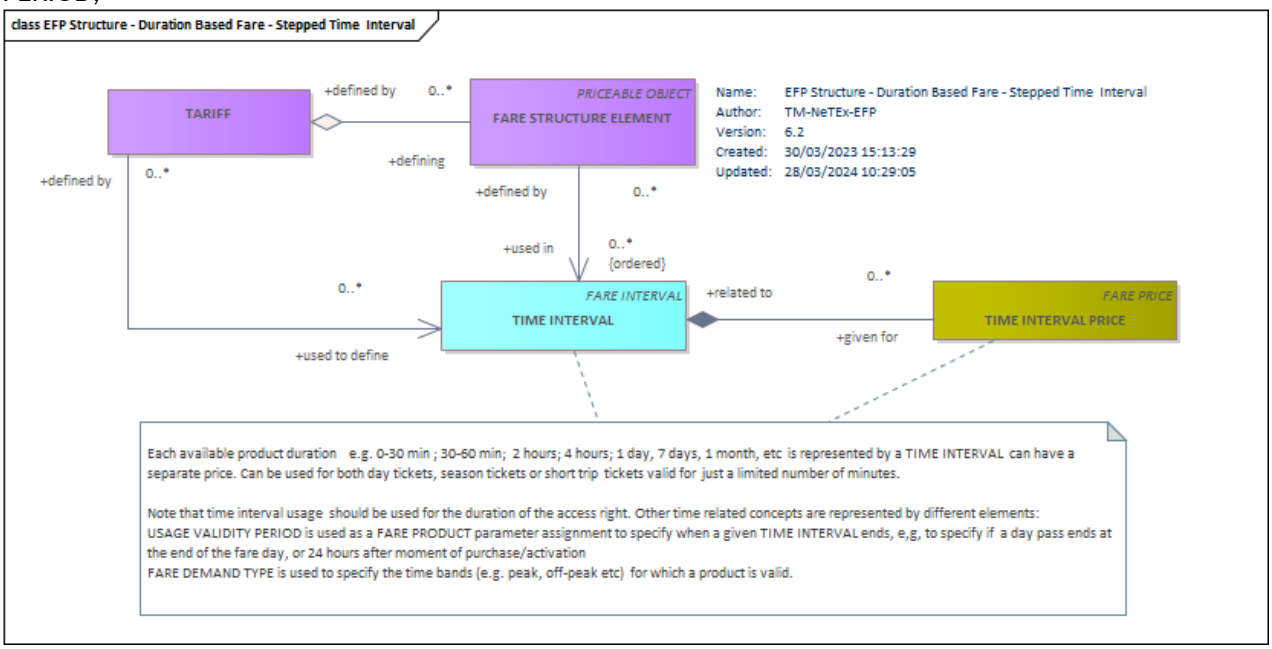


Figure 38 — Time Interval Fare Structure (UML)

11.6.1 *TimeInterval*

A time-based interval specifying access rights for the FARE STRUCTURE ELEMENTs within the range of this interval: “0-1 hours,” “1-3 days”, etc.

EFP1>>For the FXC, any ***TimeInterval*** instances should be nested within a ***Tariff***.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*FareInterval*	::>	TIME INTERVAL inherits from FARE INTERVAL.
«PK»	***id***	*TimeIntervalIdType*	1:1	Identifier of TIME INTERVAL.
	StartTime	*xsd:time*	0:1	Start of TIME INTERVAL.

	EndTime	*xsd:time*	0:1	End of TIME INTERVAL.
	DayOffset	*DayOffsetType*	0:1	Day offset of end time from start time.
	Duration	*xsd:duration*	0:1	Interval expressed as duration.
	Minimum-Duration	*xsd:duration*	0:1	Minimum Duration for TIME INTERVAL. +v1.1
«cntd»	***prices***	*TimeIntervalPrice*	0:*	Prices for the TIME INTERVAL.
«cntd»	***timeStructure-Factors***	*TimeStructureFactor*	0:*	TIME STRUCTURE FACTORs using the TIME INTERVAL.

Table 60 – *TimeInterval* – XML Element

11.6.2 *TimeUnit*

A unit for calculating time-based graduated fares.

EFP1>>For the FXC, any ***TimeUnit*** instances should be nested within a ***Tariff***.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*FareUnit*	::>	TIME UNIT inherits from FARE UNIT.
«PK»	***id***	*TimeUnitIdType*	1:1	Identifier of TIME UNIT.
	Type	*xsd:NCName*	0:1	Name of XML class associated with unit e.g. gday, gMonth. This is metadata.
	Duration	*xsd:duration*	0:1	Duration associated with unit, e.g. P1D, PT1S.
	prices	*TimeUnitPrice*	0:*	Prices associated with TIME UNIT.

Table 61 – *TimeUnit* – Element

11.6.3 *TimeStructureFactor [ADVANCED PROFILE ONLY]*

The value of a TIME INTERVAL expressed by a TIME UNIT.

A TIME STRUCTURE FACTOR PROVIDES a more elaborate form of TIME INTERVAL – it can be in any arbitrary unit – e.g. Scholastic term, and can also be combined with a QUALITY STRUCTURE FACTOR.

EFP1>>For the FXC, any ***TimeStructureFactor*** instances should be nested within a ***Tariff***.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*FareStructureFactor*	::>	TIME STRUCTURE FACTOR inherits from FARE STRUCTURE FACTOR.
«PK»	***id***	*TimeStructureFactor-IdType*	1:1	Identifier of TIME STRUCTURE FACTOR.
«FK	***TariffRef***	*TariffRef*	0:1	Reference to TARIFF associated with TIME STRUCTURE FACTOR.
«FK»	***TimeIntervalRef***	*TimeIntervalRef*	0:1	Reference to TIME INTERVAL associated with factor.
«FK»	***TimeUnitRef***	*TimeUnitRef*	0:1	Reference to TIME UNIT associated with factor.
«FK»	***QualityStructure-FactorRef***	*QualityStructure-FactorRef*	0:*	QUALITY FACTOR associated with the TIME STRUCTURE FACTOR.

Table 62 – *TimeStructureFactor* – XML Element

11.7 Quality fare structure

QUALITY FARE STRUCTURE can be used to define arbitrary fare structure qualities.

A particular use is to specify the number of units (of trips or passes) for a carnet that is available in different multiples.

Two specialisations can be used for specific aspects: A FARE DEMAND FACTOR defines a 'time band' for travel, e.g. *peak* or *off-peak*, and a FARE QUOTA FACTOR defines a limited allocation of seats available at a particular price (The latter is not in scope for the EFIP).

A FARE DEMAND FACTOR specifies a named period for travelling during which specific tariff rates may apply. The standard NeTEx VALIDITY CONDITION elements (See NeTEx Part1) can be used to specify the day types and timebands of the period. In large networks this furthermore may have a complex definition whereby the start and end are not uniform over the whole network but vary from stop to stop. This can be represented by the START TIME AT STOP element.

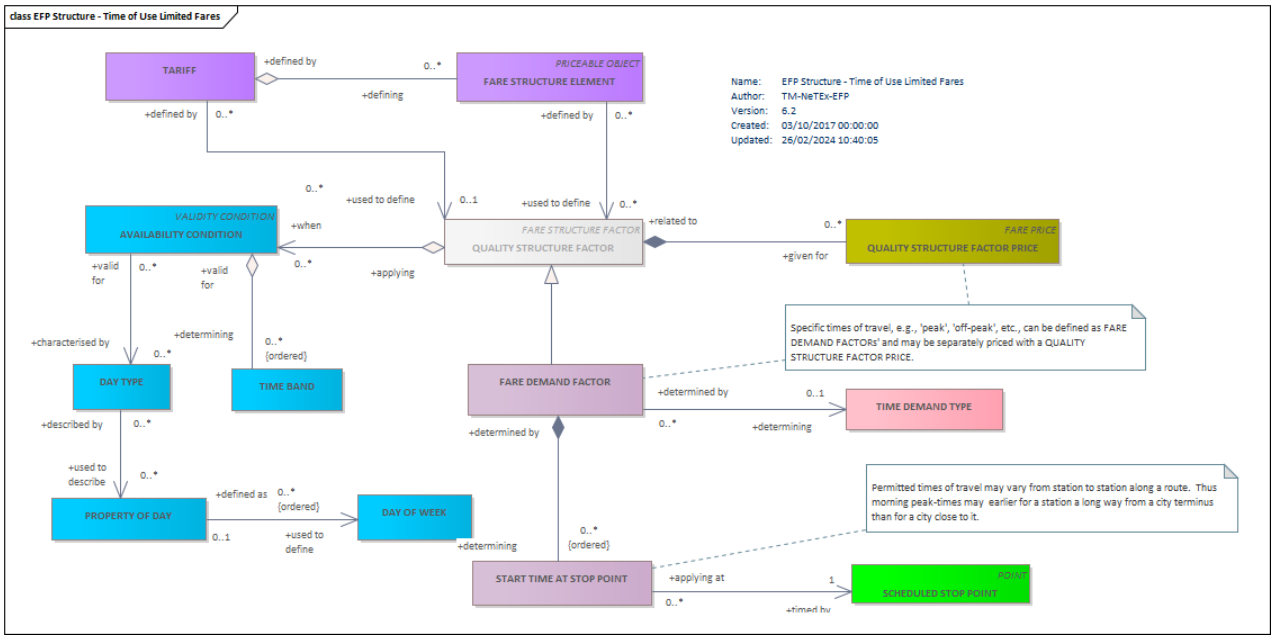


Figure 39 — Quality Fare Structure – Conceptual MODEL (UML)

11.7.1 *QualityStructureFactor*

A factor influencing access rights definition or calculation of prices, based on the quality: traffic congestion threshold, early/late reservation etc.

EFP1>>For the FXC, any ***QualityStructureFactors*** instances should be nested within a ***Tariff***.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*FareStructureFactor*	::>	QUALITY STRUCTURE FACTOR inherits from FARE STRUCTURE FACTOR.
«PK»	***id***	*QualityStructure-FactorIdType*	1:1	Identifier of QUALITY STRUCTURE FACTOR.
	Factor	*xsd:anyType*	0:1	General Factor amount.
	Value	*xsd:anyType*	0:1	Quantitative quality value.
	prices	*QualityStructureFactor-Price*	0:*	Price for QUALITY STRUCTURE FACTOR.

Table 63 – *QualityStructureFactor* – XML Element

11.7.2 *FareDemandFactor*

A FARE DEMAND FACTOR is a specalisation of QUALITY STRUCTURE FACTOR and is named set of parameters defining a period of travel with a given price, for example; "*peak", "off-peak", "super off-peak",* etc.

EFP1>>For the FXC, any ***FareDemandFactor*** instances should be nested within a ***Tariff***.

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*QualityStructureFactor*	::>	FARE DEMAND FACTOR inherits from QUALITY STRUCTURE FACTOR.
«PK»	***id***	*FareDemandFactorIdType*	1:1	Identifier of a FARE DEMAND FACTOR.
«enum»	***FareDemand-Type***	*FareDemandTypeEnum*	0:1	TIME DEMAND TYPE corresponding to FARE DEMAND FACTOR. See allowed values below.

				• peak ;middle; offPeak; superOffPeak; night; specialEvent
«FK»	***TimeDemand-TypeRef***	*TimeDemandTypeRef*	0:1	TIME DEMAND TYPE corresponding to FARE DEMAND FACTOR. See NeTEx Part2.
«enum»	***StopUse-Constraint***	*StopUseConstraintEnum*	0:1	Nature of constraint on uses of stop. See allowed values. +v1.1. • *arriving; departing; passingThrough;night*
«cntd»	***startTimesAt-StopPoints***	*StartTimeAtStopPoint*	0:*	Start times at SCHEDULED STOP POINTs for FARE DEMAND TYPE.

Table 64 – *FareDemandFactor* – XML Element

11.7.3 *StartTimeAtStopPoint (Subcomonent of FareDemandFactor)*

A time at which a fare time band (time band peak, off peak) is deemed to begin for trips starting at a particular station.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*VersionedChild*	::>	START TIME AT STOP POINT inherits from VERSIONED CHILD. See NeTEx Part1.
«PK»	***id***	*StartTimeAtStop-PointIdType*	1:1	Identifier of START TIME AT STOP POINT
«FK»	***FareDemand-FactorRef***	*FareDemandFactorRef*	0:1	FARE DEMAND FACTOR for which start time applies.
«FK»	***ScheduledStop-PointRef***	*ScheduledStopPointRef*	1:1	SCHEDULED STOP POINT at which time band start applies.
	StartTime	*xsd:time*	0:1	Time at which time band starts at station.

	EndTime	*xsd:time*	0:1	Time at which time band ends at station.
	DayOffset	*DayOffsetType*	0:1	Day offset of end time from start time. Zero is same day.

Table 65 – *StartTimeAtStopPoint* – XML Element

11.8 Validable Elements

11.8.1 *General*

The control system of a Public Transport organisation is organised in order to regularly "validate" the consumption of access rights, i.e. to check that the passengers have the right ticket for the transport on which they are travelling. The validation process is aimed at specifying that an access right is valid, has been consumed and that this consumption was allowed. It uses the results of one or several consecutive controls.

Such a validated access right may include several components for which the fare structure is different. For instance, a fare product may include a discount for travellers using a car park and then public transport. If the fare structure of these two components is different (e.g. flat fares for public transport and price based on duration of stay for car parking), they will be described by two different FARE STRUCTURE ELEMENTs. The discount is granted only when the validation process recognises that both have been consumed in sequence.

Therefore, a VALIDABLE ELEMENT is defined as a sequence or a set of FARE STRUCTURE ELEMENTs, to be consumed as a whole (or validated in one go) i.e. it is not foreseen to use the different elements of the sequence separately in the sense that if one of the elements is consumed separately, then the whole access right is considered as consumed.

Examples of VALIDABLE ELEMENTs are the following:

- A **simple** FARE STRUCTURE ELEMENT to be validated in itself (e.g. a point-to-point trip on a rail network). In such a case, the VALIDABLE ELEMENT will be identical to the FARE STRUCTURE ELEMENT;

- **Chained** FARE STRUCTURE ELEMENTs of which the successive consumption allows a discount, as in the park and ride above example. Such a discount may be applied with a discounted joint ticket, or by a discount on the latest consumed element, or by a discount to both elements with a post-payment system;

- **Composite** access rights (e.g. trips or rides) where the fare structure changes during consumption, for instance on a train link composed of two sections, each operated by an operator applying a different fare structure to that the other.

In other words, the VALIDABLE ELEMENT provides a functional grouping (e.g. "*Metro trip*", *"rail trip" "rail return trip")* with which to relate fine grained access-right components to a FARE PRODUCT.

A VALIDABLE ELEMENT can be limited to a particular scope (e.g. MODE, OPERATOR, LINE etc) that is more specific than that of the Tariff / FareStructure elements on which it is based. via an associated VALIDITY PARAMETER ASSIGNMENT.

VALIDABLE ELEMENTs are used to indicate the consumption rights of a PREASSIGNED FARE PRODUCT or AMOUNT OF PRICE UNIT product and/or the allowed discount rights (USAGE DISCOUNT RIGHT)

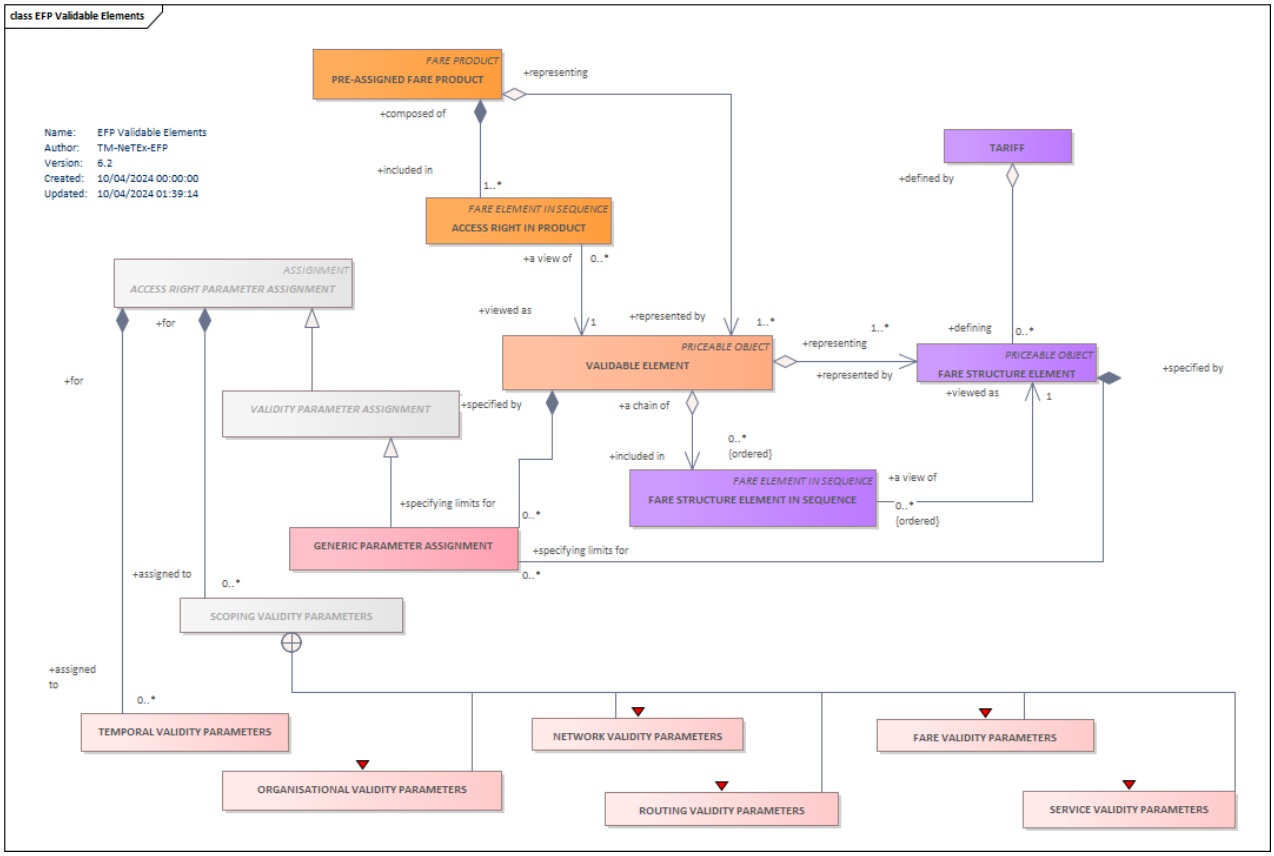


Figure 40 — Validable Element (UML)

11.8.2 *ValidableElement*

A sequence or set of FARE STRUCTURE ELEMENTs, grouped together to be validated in one go.

Classification	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	VALIDABLE ELEMENT inherits from PRICEABLE OBJECT**.**
«PK»	***id***	*ValidableElementIdType*	1:1	Identifier of VALIDABLE ELEMENT.
XGRP	***ValidableElement-StructureGroup***	***xmlGroup***	1:1	Structure elements making up VALIDABLE ELEMENT.
XGRP	***ValidableElement-ProductGroup***	***xmlGroup***	1:1	Product elements making up VALIDABLE ELEMENT.
«cntd»	***fareStructure-Elements***	*FareStructureElement*	0:*	FARE STRUCTURE ELEMENTs making up VALIDABLE ELEMENT.
«cntd»	***elementsInSequence***	*FareStructureElement-InSequence*	0:*	FARE STRUCTURE ELEMENTs IN SEQUENCE making up VALIDABLE ELEMENT.
«cntd»	***accessRights-InProduct***	*AccessRightInProduct*	0:*	ACCESS RIGHT IN PRODUCT making up VALIDABLE ELEMENT.
«cntd»	***discountRights***	*FareProductRef+*	0:*	Discount rights in Product making up VALIDABLE ELEMENT.
«cntd»	***amountOfPriceUnit***s	*AmountOfPriceUnitRef*	0:*	AMOUNTs OF PRICE UNIT making up VALIDABLE ELEMENT.
«cntd»	***thirdPartyProducts***	*ThirdPartyProductRef*	0:*	THIRD PARTY PRODUCTs for VALIDABLE ELEMENT.
«cntd»	***validityParameter-***	*ValidityParameter-*	0:*	VALIDITY PARAMETER ASSIGNMENTs for

	Assignments	*Assignment*		VALIDABLE ELEMENT.
*«cntd»	***prices***	*ValidableElementPrice*	0:*	VALIDABLE ELEMENT PRICEs for element.

Table 66 – *ValidableElement* – XML Element

12 EFIP: Fare Products

12.1 Fare Product

A FARE PRODUCT is a marketable element made available to the public. It can be purchased and enables the owner to consume public transport or other services under specific conditions. FARE PRODUCT itself is abstract; it has a number of different specialisations, each representing a different sort of FARE PRODUCT. For example specified access rights to travel (PRE-ASSIGNED FARE PRODUCT). The different specialisations are described later below.

A FARE PRODUCT is immaterial, that is, it is independent of any physical representation but can be materialised on various TRAVEL DOCUMENTs by means of a SALES OFFER PACKAGE. The same FARE PRODUCT might have different materialisations on different media, for instance, a “monthly pass” FARE PRODUCT may be variously incorporated on a specific paper ticket or stored on an electronic card.

The same FARE PRODUCT can be used in one or more SALES OFFER PACKAGEs (see later) to described a marketable product that the user can actually buy materialised onto a TYPE OF TRAVEL DOCUMENT, for example a metro trip might be available as both a paper ticket and as a smartcard transaction.

A FARE PRODUCT is specific to a particular CHARGING MOMENT, which is a combination of: payment method (pre-payment or post-payment and account location (account stored on a TRAVEL DOCUMENT or in a central account). The CHARGING MOMENT is significant in particular for distinguishing Pay as You GO and capped products where payment is not a simple prepayment. The same access rights when presented to the public (i.e. when they become FARE PRODUCTs) may differ, for instance, the “access right to the metro network” may be advertised as two products: one as prepaid (materialised as a simple ticket), another as post-paid (materialised on an electronic card).

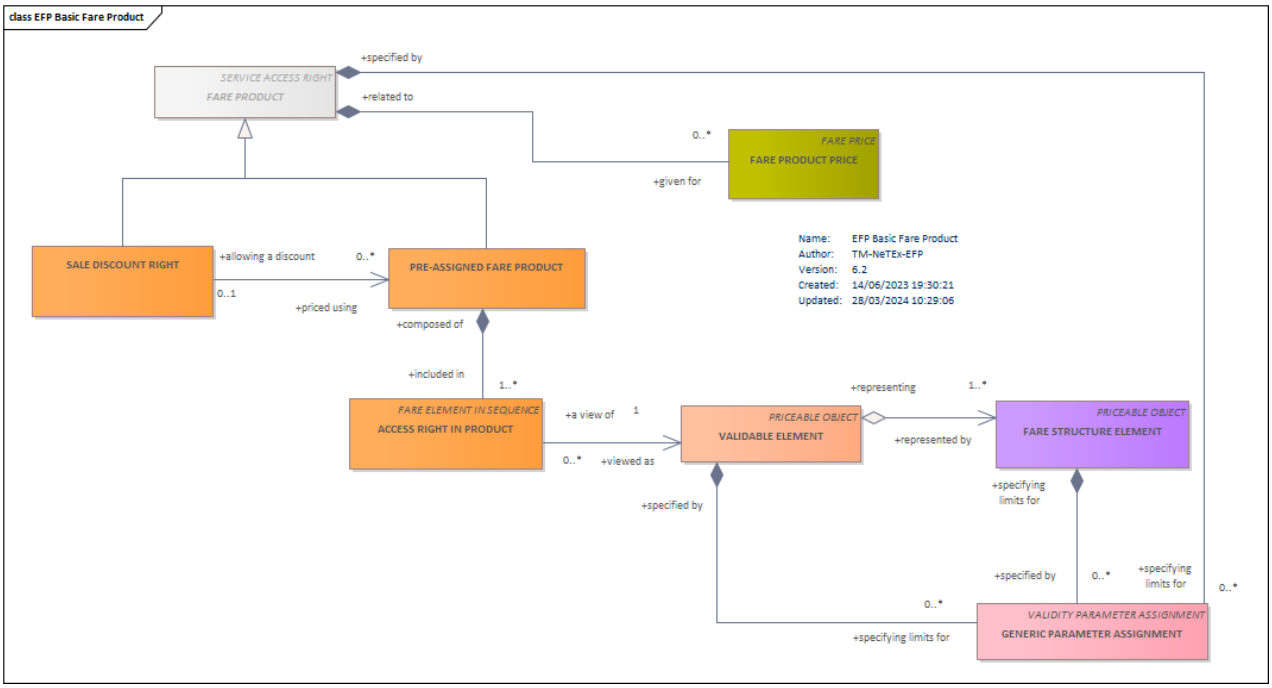


Figure 41 — EFIP – Fare Product Overview (UML)

The most classical FARE PRODUCTs are combinations of specified access rights (single ticket, commuter week ticket, monthly pass, etc.). Such a PRE-ASSIGNED FARE PRODUCT is defined as a FARE PRODUCT consisting of one or several VALIDABLE ELEMENTs.

Typical examples of PRE-ASSIGNED FARE PRODUCTs are the following:

- any VALIDABLE ELEMENT that is directly marketable, e.g. access right granted by a single ticket, access right granted by a park and ride discount ticket, etc. In such a case, the PRE-ASSIGNED FARE PRODUCT is identical to the VALIDABLE ELEMENT;

- a week card allowing one or two specified trips for each day of the week, each trip being defined as a VALIDABLE ELEMENT that should be consumed in one go during a specified time band of the considered day;

- a monthly pass allowing the unlimited consumption of several specified trips, each being defined as a VALIDABLE ELEMENT, for example travel within a specified zone.

The VALIDABLE ELEMENT V

- A given FARE PRODUCT (and subsequent SALES OFFER PACKAGEs) may comprise a number of different values for each feature of the fare structure. For example, a FARE PRODUCT for a set of point-to-point journeys (each represented by a DISTANCE MATRIX ELEMENT) might include parameters for *first class*, *second class*, *and for peak* or *off-peak*); each allowed for different USER PROFILEs such as *adult*, *child*, *senior* and *student* – *and* every separate selectable combination having a separate price. Thus, there is not normally a separate FARE PRODUCT for each combination of features that a user may buy and it is possible to represent a large set of offerings by a single FARE PRODUCT.

- The user's actual purchase will be described by a TRAVEL SPECIFICATION which indicates which specific features have been selected, for example *an adult single second class ticket between Lille and Valenciennes*.

12.1.1 *ServiceAccessRight (Abstract)*

A SERVICE ACCESS RIGHT is an immaterial marketable element (access rights, discount rights etc) that may be specialised to make various types of product.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	SERVICE ACCESS RIGHT inherits from PRICEABLE OBJECT**.**
«PK»	***id***	*ServiceAccessRightIdType*	1:1	Identifier of SERVICE ACCESS RIGHT.
«AK»	***PrivateCode***	*PrivateCodeType*	0:1	Alternative identifier of an entity; can be used to associate with legacy systems.
	InfoUrl	*xsd:anyURI*	0:1	Link for product information.
«cntd»	***documentLinks***	*InfoLink*	0:*	InfoLinks for external links. For PFDs, etc =V1.1

Table 67 – *ServiceAccessRight* – XML Element

12.1.2 *FareProduct (Abstract)*

An immaterial marketable element (access rights, discount rights etc), specific to a CHARGING MOMENT.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	FARE PRODUCT inherits from SERVICE ACCESS RIGHT**.**
«PK»	***id***	*FareProductIdType*	0:1	Identifier of FARE PRODUCT.
«FK»	***Charging-MomentRef***	*ChargingMomentRef*	0:1	Reference to a CHARGING MOMENT for product +v1.1.
«enum»	***Charging-MomentType***	*Charging-MomentTypeEnum*	0:1	Enumeration of standardised Charging moment values. See allowed values. _v1.1. • *beforeTravel* • *onStartOfTravel* • *beforeEndOfTravel* • *onStartThenAdjustAtEndOfTravel* • *onStartThenAdjustAtEndOfFareDay* • *onStartThenAdjustAtEndOfChargePeriod* • *atEndOfTravel* • *atEndOfFareDay* • *atEndOfChargePeriod* • *free* • *anyTime* • *other*

«FK»	***typesOfFare-ProductRef***	*TypeOfFareProductRef*	0:*	Classifications of FARE PRODUCT. (made *:* in v1.1)
«FK»	***Transport-OrganisationRef***	*(TransportOrganisationRef) OperatorRef | AuthrityRef*	0:1	OPERATOR or AUTHORITY in charge of the FARE PRODUCT.
«cntd»	***Condition-Summary***	*ConditionSummary*	0:1	Summary description of conditions on FARE PRODUCT.
«FK»	***BaseFare-ProductRef***	***FareProductRef+***	0:1	Another FARE PRODUCT which this product extends. Will assume all properties of base product unless specifically overridden.
«cntd»	***validityParameter-Assignments***	***AccessRightParameter-Assignment***	0:*	VALIDITY PARAMETER ASSIGNMENTs relating to FARE PRODUCT.
«cntd»	***validableElements***	***ValidableElement***	0:*	VALIDABLE ELEMENTs for FAR PRODUCT.
«cntd»	***accessRights-InProduct***	***AccessRightInProduct***	0:*	ACCESS RIGHTs in PRODUCT for FAR PRODUCT.
«cntd»	***tariffs***	***TariffRef***	0:*	TARIFFs used by FARE PRODUCT.
«cntd»	***prices***	***FareProductPrice***	0:*	FARE PRODUCT PRICEs in PRICE GROUP.

Table 68 – *FareProduct* – XML Element

12.1.3 *AccessRightInProduct (Subcomponent of FareProduct)*

ACCESS RIGHT in PRODUCT links a FARE PRODuCT to a VALIDABLE ELEMENT to specify the access righst of the product.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*FareElementInSequence*	::>	ACCESS RIGHT IN PRODUCT inherits from FARE ELEMENT IN SEQUENCE.
«PK»	***id***	*AccessRightInProduct-*	1:1	Identifier of ACCESS RIGHT IN PRODUCT.

		IdType		
«FK»	***Validable-ElementRef***	*ValidableElementRef*	0:1	Reference to a VALIDABLE ELEMENT for which access rights are specified.
«FK»	***Preassigned-FareProducRef***	*PreassignedFareProducRef*	0:1	Reference to a PRE ASSIGNED FARE PRODUCT for which access rights are specified.

Table 69 – *AccessRightInProduct* – XML Element

12.1.4 *ChargingMoment*

A classification of FARE PRODUCTs according to the CHARGING MOMENT and the account location: pre-payment with cancellation (throw-away), pre-payment with debit on a value card, pre-payment without consumption registration (pass), post-payment etc.

The classical examples of CHARGING MOMENT are the following:

- pre-payment with cancellation (throw-away tickets);
- pre-payment with debit on a TRAVEL DOCUMENT (value card);
- pre-payment without registration of the consumption (unlimited pass);
- post-payment (electronic card with central account and monthly debiting);
- free of charge.

These main categories may be subdivided according to the operator specific requirements.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*TypeOfValue*	::>	TYPE OF CHARGING MOMENT inherits from TYPE OF VALUE. See NeTEx Part1.
«PK»	***id***	*ChargingMomentIdType*	1:1	Identifier of TYPE OF CHARGING MOMENT.

Table 70 – *ChargingMoment* – Element

12.1.5 *ConditionSummary (Subcomponent of FareProduct)*

A summary of the properties of a FARE PRODUCT or PACKAGE that can be used to generate passenger information.

Classifi-cation	Name	Type	Cardinality	Description
«enum»	***FareStructure-Type***	*FareStructureTypeEnum*	1:1	Classification of fare type. • *networkFlatFare; lineFlatFare; zonalFare; zoneToZoneFare; zoneSequenceFare; pointToPointFare; stageFare* • *cappedZonalFare; cappedFlatFare* • *store; other*
«enum»	***TariffBasis***	*TariffBasisEnum*	0:1	Basis for Tariff. • *flat; distance; route; zoneToZone; pointToPoint; tour; group; discount; other*
	HasNotices	*xsd:boolean*	0:1	Whether there are notices associated with the product.
	ProvidesCard	*xsd:boolean*	0:1	Whether a card is provided with the product.
	GoesOnCard	*xsd:boolean*	0:1	Whether the product goes on a card.
	IsPersonal	*xsd:boolean*	0:1	Whether the product is a sold anonymously or to an identified person.
	RequiresPhoto	*xsd:boolean*	0:1	Whether use of the product requires a photo to be provided.
	MustCarry	*xsd:boolean*	0:1	Whether use of the card must be carried in order to use product.

	Requires-Account	*xsd:boolean*	0:1	Whether the product requires the user to register for an account for billing. +v1.1
	IsSupplement	*xsd:boolean*	0:1	Whether the package is a supplement to another product
	Requires-Entitlement	*xsd:boolean*	0:1	Whether the product requires entitlement to other products.
	Gives-Entitlement	*xsd:boolean*	0:1	Whether the product grants entitlements to other products.
«enum»	***HasOperator-Restrictions***	*Operator RestrictionEnum*	0:1	Limitations as to which OPERATOR's services may be used. See allowed values below. • *anyTrain* • *restricted* • *specifiedOperatorOnly*
	HasTravelTime-Restrictions	*xsd:boolean*	0:1	Whether limitations apply as to when travel may take place.
	HasRoute-Restrictions	*xsd:boolean*	0:1	Whether limitations apply as to the route that may be used.
«enum»	***HasTrain-Restrictions***	*TrainRestrictionEnum*	0:1	Limitations as to which trains may be used. See allowed values below. • anyTrain • restricted • specifiedTrainOnly • specifiedTrainsOnly • specifiedTrainAndConnections
	HasZone-	*xsd:boolean*	0:1	Whether limitations apply as to the area in which

	Restrictions			travel may take place.
	CanBreak-Journey	*xsd:boolean*	0:1	Whether the user is allowed to break journey, i.e. leave transport network, at an intermediate point.
	ReturnTripsOnly	*xsd:boolean*	0:1	Whether must buy a return trip.
	CanChange-Class	*xsd:boolean*	0:1	Whether user can change class
	IsRefundable	*xsd:boolean*	0:1	Whether the ticket is refundable
	IsExchangable	*xsd:boolean*	0:1	Whether the ticket is exchangeable
	HasExchange-Fee	*xsd:boolean*	0:1	Whether there is a fee for exchanges.
	HasDiscounted-Fares	*xsd:boolean*	0:1	Whether discounted Fares are allowed.
	AllowAdditional-Discounts	*xsd:boolean*	0:1	Whether more than one discount may be applied, e.g. Child + Companion.
	Allow-Companion-Discount	*xsd:boolean*	0:1	Whether there is a companion discount.
	HasMinimum-Price	*xsd:boolean*	0:1	Whether there is a minimum price when combining elements.
	Requires-PositiveBalance	*xsd:boolean*	0:1	Whether the product requires a positive stored balance to be used.
	PenaltyWIthout-Ticket	*xsd:boolean*	0:1	Whether there is a penalty for travelling without a ticket, i.e. tickets cannot be bought on-board. +v1.1
	AvailableOn-	*xsd:boolean*	0:1	Whether the product is available on subscription.

	Subscription			+v1.1
	HasPurchase-Conditions	*xsd:boolean*	0:1	Whether purchase conditions apply to the sale of the product, e.g. when must be bought or who may purchase.
	HasDynamic-Pricing	*xsd:boolean*	0:1	Whether product has dynamic pricing.
	Requires-Reservation	*xsd:boolean*	0:1	Whether a Reservation is required.
	HasReservation-Fee	*xsd:boolean*	0:1	Whether there is a fee for Reservations.
	HasQuota	*xsd:boolean*	0:1	Whether limited quota for the offer or it can be sold in unlimited numbers.

Table 71 – *ConditionSummary* – XML Element

12.2 Types of Fare Product

FARE PRODUCT itself is abstract – it is specialised to make a number of concrete fare types.

There are four fundamental types of FARE PRODUCTs that are found most commonly;

- PRE-ASSIGNED FARE PRODUCT is a marketable combination of specified VALIDABLE ELEMENTs. It is the most common FARE PRODUCT in public transport (materialised e.g. as single ticket, monthly pass etc.);
- AMOUNT OF PRICE UNIT is a FARE PRODUCT expressed by a specified number of PRICE UNITs (currency unit, token, etc.). It is not pre-assigned, which means that it gives the right to consume any VALIDABLE ELEMENT from a specified list. The main types of AMOUNT OF PRICE UNIT are value cards or electronic purses, which are debited for each transaction. In some cases, single tickets should be considered as AMOUNT OF PRICE unit, when it is required to punch a variable number of tickets according to the length of the intended trip;
- SALE DISCOUNT RIGHT is a FARE PRODUCT allowing its holder to benefit from discounts when purchasing specific SALES OFFER PACKAGEs. Train companies for instance usually propose such discounts (e.g. 30 % discount card);

- SUPPLEMENT PRODUCT: An ancillary product, such as a seat class upgrade or a meal, that can only be purchased in addition to another product.

Two further types of FARE PRODUCTs are used in: complex products

- USAGE DISCOUNT RIGHT is a FARE PRODUCT allowing its holder to benefit from discounts when consuming specified VALIDABLE ELEMENTs. For instance, such a product grants to its holder a discount when consuming park and ride sequences, whereas parking or PT rides consumed alone are charged at the normal fare. This kind of discount is particularly meaningful with post-payment methods.
- CAPPED DISCOUNT RIGHT a refinement of a SALE DISCOUNT RIGHT used for advanced electronic pay as you go fares, where once a certain amount of consumption has been achieved within a certain interval, a cap (as specified by one or more CAPPING RULES) is applied, for example limiting the daily use to no more than the cost of a day pass.

In addition, two other types of non-travel “product”, can be declared and referenced:

- an ENTITLEMENT PRODUCT: may also be used to represent non-transport related qualifications such as disability cards, military cards or pensioner passes that are pre-requisites for the purchase or consumption of travel products.
- a THIRD-PARTY PRODUCT: A FARE PRODUCT that is marketed together with a Public Transport FARE PRODUCT. It is a product not fully described by the system. Out of scope of the EFIP.

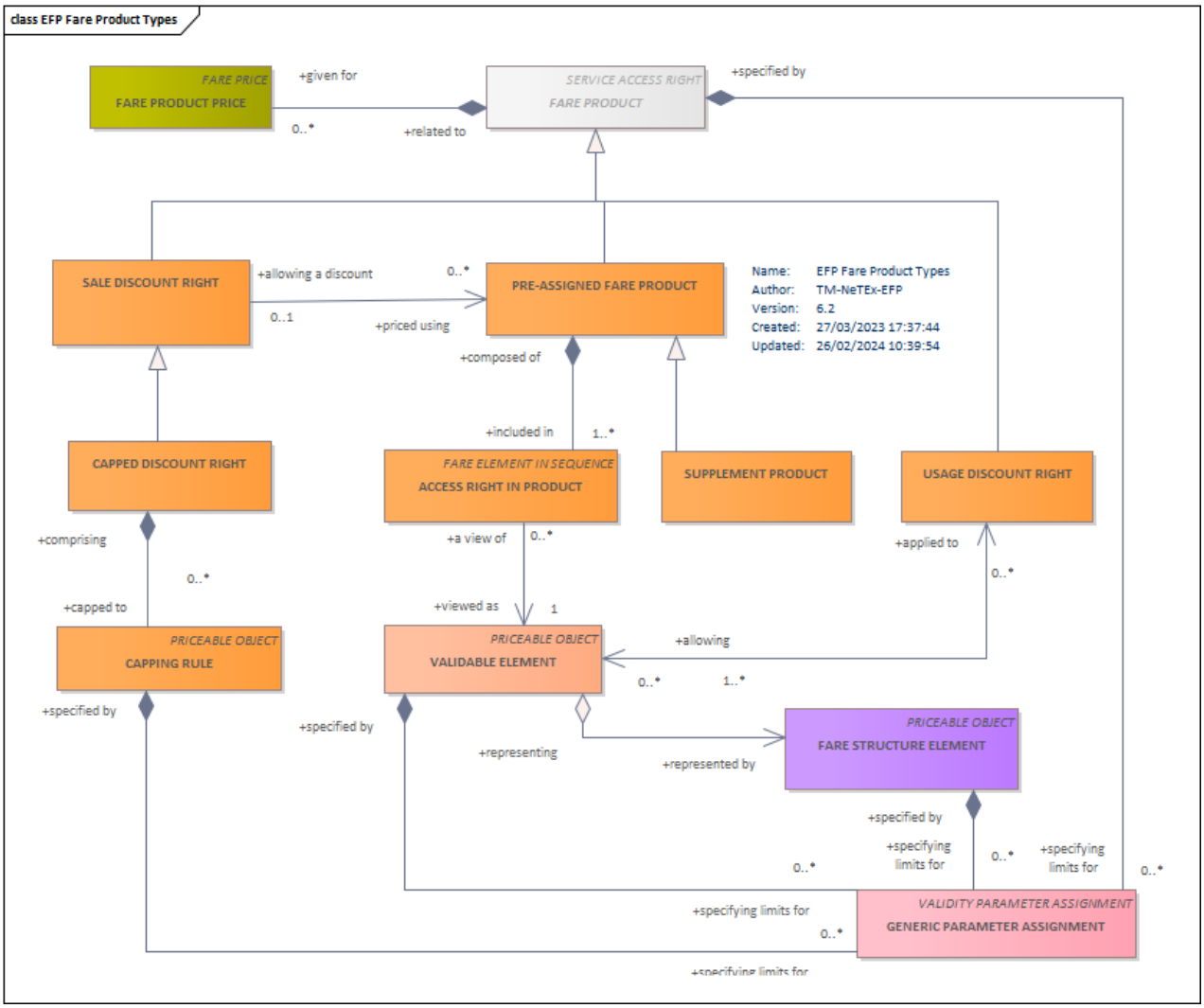


Figure 42 — Types of Fare Product (UML)

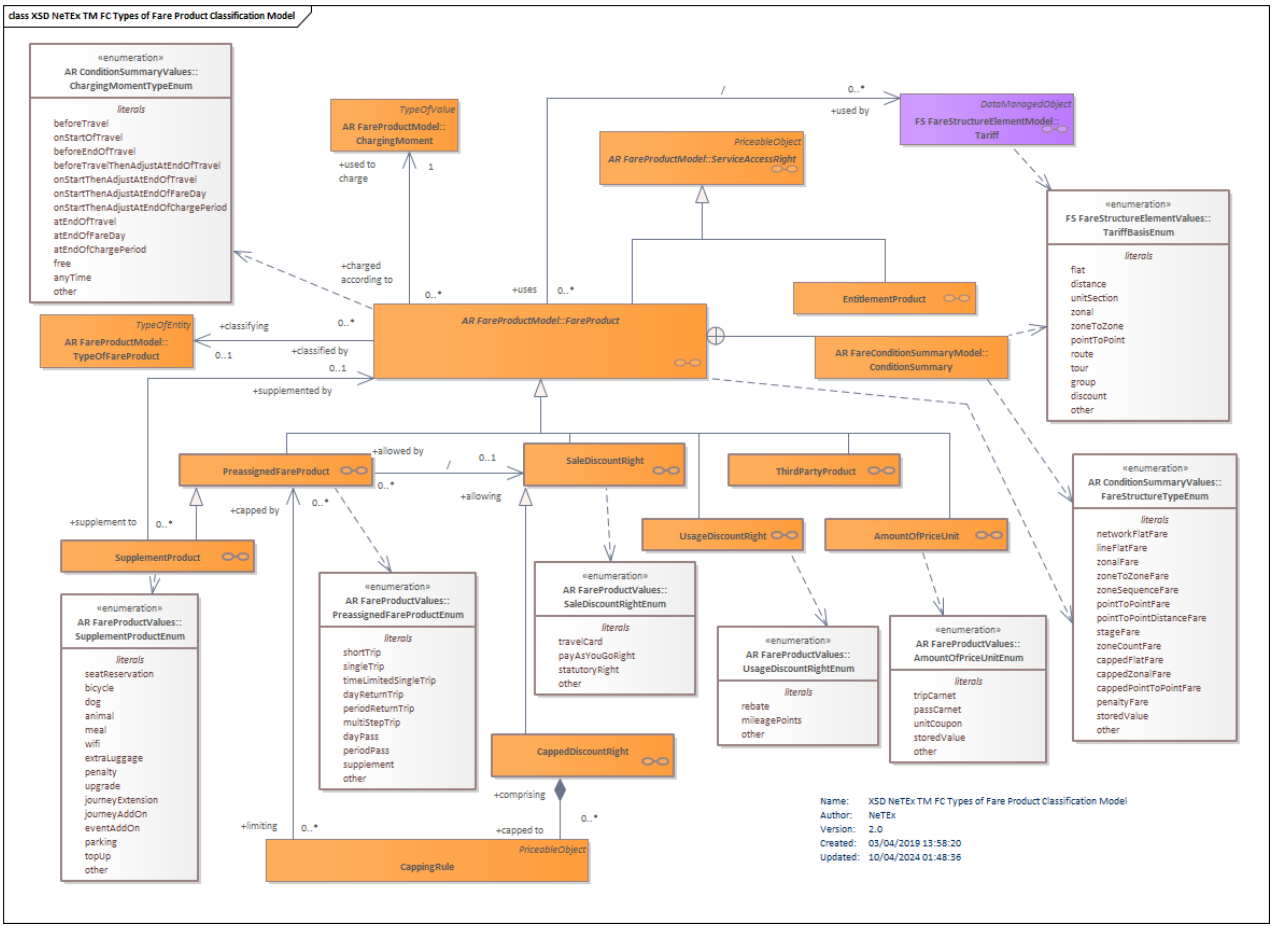


Figure 43 — NeTEx Types of Fare Product - Classifications (UML)

12.2.1 *PreassignedFareProduct*

A FARE PRODUCT consisting of one or several VALIDABLE ELEMENTs, specific to a CHARGING MOMENT.

Classifi-cation	Name	Type	Cardin-ality	Description
::>	::>	*FareProduct*	::>	PREASSIGNED FARE PRODUCT inherits from FARE PRODUCT.

«PK»	***id***	*Preassigned-FareProductIdType*	1:1	Identifier of PREASSIGNED FARE PRODUCT.
«enum»	***ProductType***	*PreassignedFareProduct-Enum*	1:1	Classification of PREASSIGNED FARE PRODUCT. See allowed values. +v1.1 • *singleTrip; ShortTrip; timeLimitedSingleTrip* • *dayReturnTrip; periodReturnTrip* • *multistepTrip* • *dayPass; periodPass* • *supplement* • *other*

Table 72 – *PreassignedFareProduct* – XML Element

12.2.2 *SupplementProduct*

An additional FARE PRODUCT that may be used to describe additional purchases entitled by another product.

A SUPPLEMENT PRODUCT is usually constrained by some or all of the parameters of the supplemented product, e.g. same service, same route, etc. For example, a *PlusBus* ticket can only be bought for the origin or destination of a trip, or a first-class upgrade must be for the same trip as the original second-class ticket.

It may be part of passenger information to indicate which supplements are available:

- A SUPPLEMENT PRODUCT can indicate one or more prerequisite PREASSIGNED FARE PRODUCTs. Further details about the dependency can be specified using an ENTITLEMENT REQUIRED parameter. The ENTITLEMENT CONSTRAINT parameter can place restrictions on the supplement's parameters to align the with the base product.

- A PREASSIGNED FARE PRODUCT that has supplements may use an ENTITLEMENTs GIVEN usage parameter to declare the other products that are supplements for it.

Classifi-cation	Name	Type	Cardi	Description

				nality	
::>	::>		*PreassignedFare-Product*	::>	SUPPLEMENT PRODUCT inherits from PREASSIGNED FARE PRODUCT.
«PK»	***id***		*SupplementProductIdType*	1:1	Identifier of SUPPLEMENT PRODUCT.
«enum»	***Supplement-ProductType***		*SupplementProduct-TypeEnum*	0:1	Classification of SUPPLEMENT PRODUCT. See allowed values. +v1.1 • *bcycle; dog; animal* • *meal; wifi* • *extraLuggage* • *upgrade; journeyExtension; journeyAddOn; eventAddOn* • *topUp* • *parking* • *other*
	Choice				
«FK»	*a*	***SupplementToFareProduct-Ref***	*FareProductRef+*	0:1	Reference to base PRE ASSIGNED FARE PRODUCT OFFER for which this is a supplement.
«cntd»	*b*	***SupplementTo***	*FareProductRef+*	0:*	Reference to base PRE ASSIGNED FARE PRODUCT OFFER for which this is a supplement.

Table 73 – *SupplementProduct* – XML Element

12.2.3 *AmountOfPriceUnit*

A FARE PRODUCT consisting of a stored value of PRICE UNITs: an amount of money on an electronic purse, amount of units on a value card etc.

Classifi-cation	Name	Type	Card in-ality	Description
::>	::>	*FareProduct*	::>	AMOUNT OF PRICE UNIT inherits from FARE PRODUCT.
«PK»	***id***	*AmountOfPriceUnitIdType*	1:1	Identifier of AMOUNT OF PRICE UNIT.
«enum»	***ProductType***	*AmountOfPriceUnitEnum*	1:1	Classification of AMOUNT OF PRICE UNIT. See allowed values. +v1.1 • *tripCarnet; passCarnet* • *unitCoupon; storedValue; other*
«FK»	***PriceUnitRef***	*PriceUnitRef*	0:1	Reference to a PRICE UNIT.
	Amount	*xsd:decimal*	0:1	Number of units.

Table 74 – *AmountOfPriceUnit* – XML Element

12.2.4 *UsageDiscountRight*

A FARE PRODUCT allowing a customer to benefit from discounts when consuming VALIDABLE ELEMENTs.

Classifi-cation	Name	Type	Card in-ality	Description
::>	::>	*FareProduct*	::>	USAGE DISCOUNT RIGHT inherits from FARE PRODUCT.
«PK»	***id***	*UsageDiscounRightIdType*	1:1	Identifier of USAGE DISCOUNT RIGHT.
«enum»	***ProductType***	*UsageDiscountRightEnum*	1:1	Classification of USAGE DISCOUNT RIGHT. See allowed values. +v1.1

				• *mileagePoints* • *usageRebate* • *other*

Table 75 – *UsageDiscountRight* – XML Element

12.2.5 *SaleDiscountRight*

A FARE PRODUCT allowing a customer to benefit from discounts when purchasing SALES OFFER PACKAGEs.

Classification	Name	Type	Cardinality	Description
::>	::>	*FareProduct*	::>	SALE DISCOUNT RIGHT inherits from FARE PRODUCT.
«PK»	***id***	*SaleDiscountRightIdType*	1:1	Identifier of SALE DISCOUNT RIGHT.
«enum»	***ProductType***	*SaleDiscountRightEnum*	1:1	Classification of SALE DISCOUNT RIGHT. See allowed values. +v1.1 • *travelCard* • *payAsYouGoRight* • *other*

Table 76 – Sale Discount Right – XML Element

12.2.6 *Pay as you go fare and capping*

Account based electronic fare systems enable an effective way of simplying the choice of fares for the passenger – capping. With capping, if users nerd not decide in advance whether they should buy a period pass but simply make their trips.

A CAPPED DISCOUNT PRODUCT is a specialisation of a SALES DISCOUNT RIGH and can be used to represent a pay as you go fare product, such that once a certain amount of consumption has been achieved within a certain interval, a cap (as specified by one or more CAPPING RULEs) is applied, for example limiting the daily use to no more than the cost of a day pass, or weekly use to no more than a week pass.

The CAPPED DISCOUNT RIGHT does not of itself give access rights; merely the right to purchase other products at a discount. The usage is controlled by recording the VALIDABLE ELEMENTs of the other products purchased (usually trips) against the account of the passenger and then performing an overall calculation for the customer at the end of the period. PAYG products are characterised by a different CHARGIMG MOMENT from classical prodcuts; the customer may be charged some of the amount atstrat of travel with an adjustment later, or billing may even be deferred entirely to the end of period.

The same CAPPED DISCOUNT PRODUCT may set different caps for different VALIDABLE ELEMENTs, for exampelf or different modes (a metro trip, bus trip, river trip, etc.), each specified by a different CAPPING RULE – as say is the case for TfL's OYSTER profuct

In some cases, the start of the capping period is fixed (e.g. Monday, 1st of the moth etc) in other cases it is variable. This can be specified using attributes on a USAGE VALIDITY PERIOD parameter.

In practice, only certain combinations of tariff and product type are found in common use, with some types being more common on specific modes. NeTEx allows arbitrary user defined classifications to be made using a TYPE OF FARE PRODUCT ELEMENT. It also provides a number of enumerated values for the most common product types.

12.2.7 *CappedDiscountRight [ADVANCED PROFILE ONLY]*

A specialisation of SALE DISCOUNT RIGHT where the discount is expressed as a capping limit for a given time interval. For example, the London Oyster card fare, which charges for each journey at a reduced price until travel equivalent to a day pass has been consumed.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*SaleDiscountRight*	::>	CAPPED DISCOUNT RIGHT inherits from SALE DISCOUNT RIGHT**.**
«PK»	***id***	*CappedDiscountRight-IdType*	1:1	Identifier of CAPPED DISCOUNT RIGHT.
«cntd»	***cappingRules***	*CappingRule*	0:*	A set of parameters set a price cap on a product.

Table 77 – *CappedDiscountRight* – XML Element

12.2.8 *CappingRule (Subcomponent of CappedDiscountRight)*

A capping limit for a given time interval, where the capping is expressed by another product. For example, the London Oyster card fare, which charges for each journey at a reduced price until travel equivalent to a day pass for the mode of travel has been consumed. A CAPPING RULE is a PRICEABLE OBJECT and may have USAGE PARAMETERS such as a USAGE VALIDITY PERIOD to specify how long the capping period is and a CHARGING POLICY to specify rules about travelling under credit.

CAPPING may also be based on simply on distance.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	CAPPING RULE inherits from PRICEABLE OBJECT.
«PK»	***id***	*CappingRuleIdType*	1:1	Identifier of CAPPING RULE.
«cntd»	***Maximum-Distance***	*LengthType*	0:*	Capping distance if distance-based cap.
«enum»	***CappingPeriod***	*CappingPeriodEnum*	0:1	Period over which capping applies, e.g. daily. See allowed values below. A quantitative value can be set with a USAGE VALIDITY PERIOD, along with a more detailed definition of the start and end times. • *day; week; month*
«FK»	***CappedDiscount-RightRef***	*CappedDiscount-RightRef*	0:1	CAPPED DISCOUNT RIGHT for which this rule applies.
«FK»	***PreassignedFareProductRef***	*PreassignedFare-ProductRef*	0:1	PREASSIGNED FARE PRODUCT whose prices set cap the for this product. Usually a Pass
«FK»	***Validable-ElementRef***	*ValidableElementRef*	0:1	VALIDABLE ELEMENT of anotherproduct, usually a single trip, for which capping applies.
«cntd»	***validityParameteAssignments***	*ValidityParameterAssignment+*	0:*	VALIDITY PARAMETER ASSIGNMENTS for this rule.

«cntd»	***prices***	*CappIngRulePrice*	0:*	Capping FARE PRICEs for this rule.

Table 78 – *CappingRule* – XML Element

12.3 Validity parameters and their assignment

Apart from the quantitative parameters used in the fare structure. such as time intervals and distance, other parameters may be used by a fare system in order to limit the validity of particular access rights. For example, which OPERATORs accept a fare product, on which LINEs may it be used, what CLASSES OF USE are available?

NeTEx provides a general purpose mechanism for stating how NeTEx elements may be used as parameters to limit or scope various aspects of fare definition; the ACCESS RIGHT PARAMETER ASSIGNMENT.

Parameter assignments may be made of a number of different NeTEx elements, including a FARE STRUCTURE ELEMENT, DISTANCE MATRIX ELEMENT, GROUP OF DISTANCE MATRIX ELEMENTS, FARE PRODUCT, SALES OFFER PACKAGE, VALIDABLE ELEMENT, or CONTROLLABLE ELEMENT.

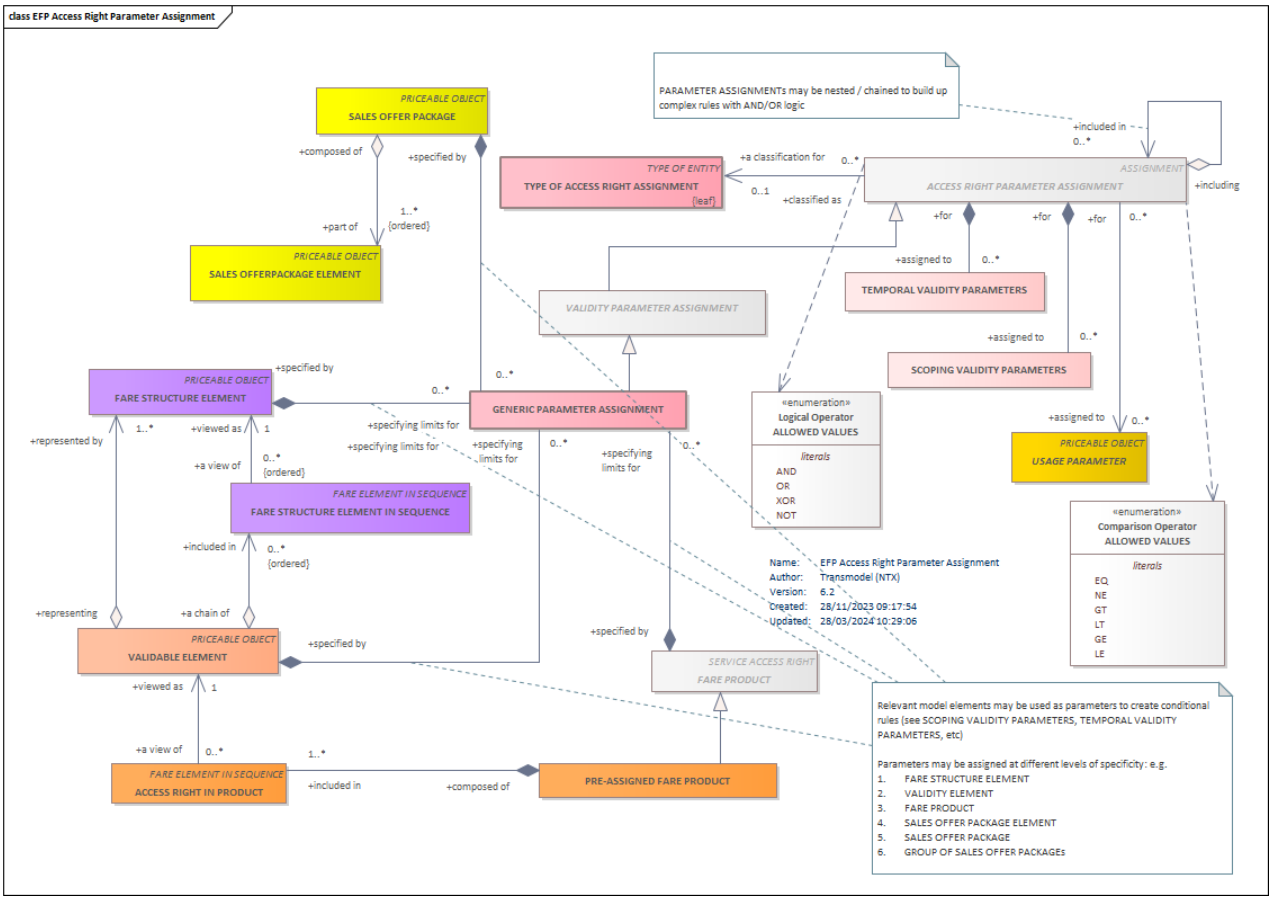


Figure 44 — EFIP – Access ight Parameter Assignments

12.3.1 *Generic and specific assignments*

Parameter assignment can be used both to define the sets of alternative possibilities right making up a fare structure, and to describe the actual choices made by a passenger when they select specific access rights out of the set of theoretical possibilities in order to specify the actual access rights that will be consumed by a particular trip.

The processes that consist of assigning a fare parameter to either a theoretical or consumed access right are very similar. The assignment of such parameters to an element of the fare system is therefore described using a generic entity ACCESS RIGHT PARAMETER ASSIGNMENT, each of which may assign one or more elements as parameters. It has two specializations:

- A GENERIC PARAMETER ASSIGNMENT, which attaches a fixed parameter to a certain class of rights, denoting it as being theoretically allowed (possibly with multiple alternatives) within a given fare product and
- A SPECIFIC PARAMETER ASSIGNMENT, which assigns a limiting parameter to a particular right, within a certain fare structure, thus representing the choice of a specific set of parameters for consumption on an individual trip. (Note that SPECIFIC PARAMETER ASSIGNMENTs are out of scope of the EFIP, which only covers what is offered, not what is purchased).

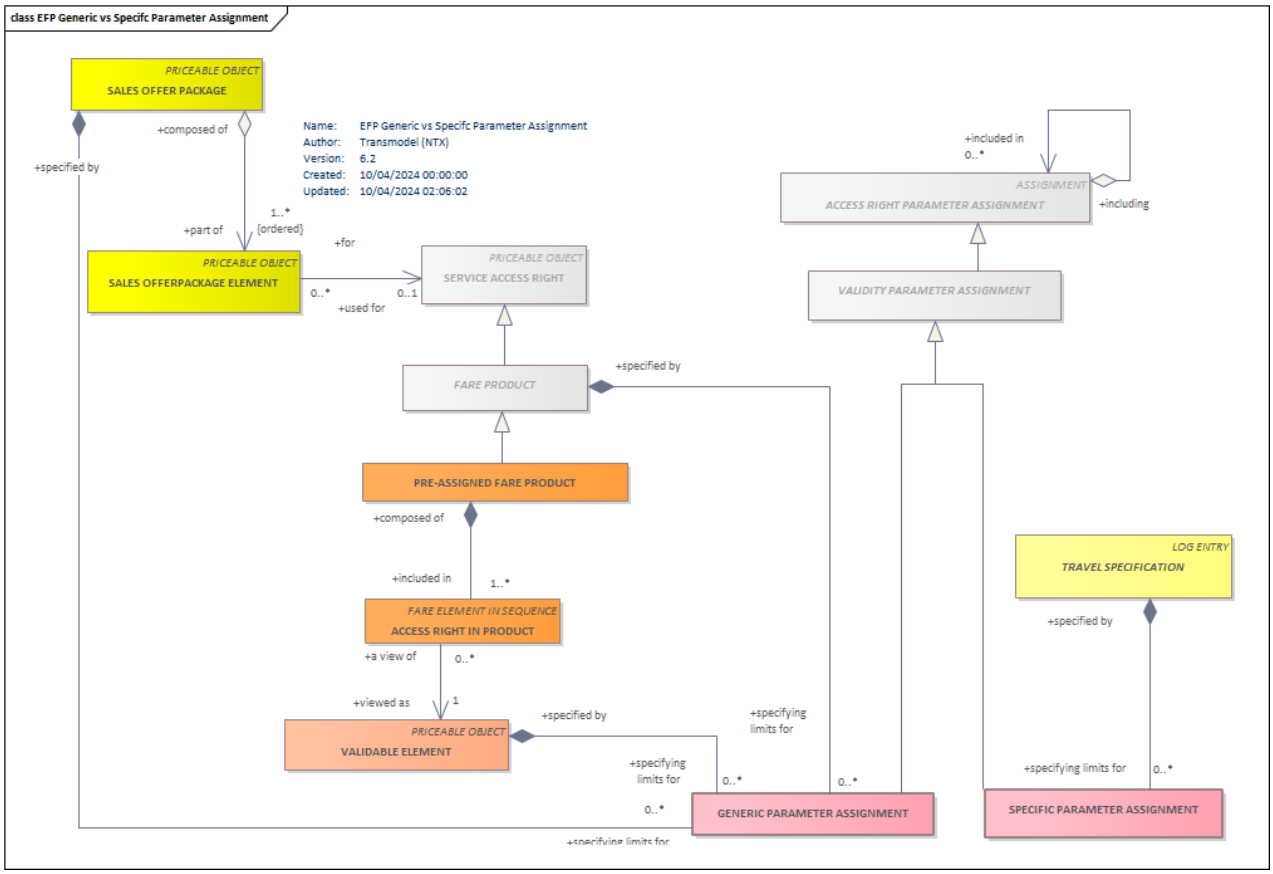


Figure 45 — EFIP Generic and specific Parameter Assignments (UML)

12.3.2 *Using parameter assignments – basic use*

In most cases the use of ACCESS RIGHT PARAMETER ASSIGNMENT is simple - they simply reference the NeTEx elements that are to be specified as restricting the validity. There are however a large number of different NeTEx element types that can be referenced as parameters, so these are organised into functionally related groups.

More than one element may be referenced as a parameter by a single assignment. The default interpretation is that all the values are logically ANDed together, but other types of ***LogicalOperator*** may be used (OR, XOR).

It should be remembered that a GENERIC PARAMETER ASSIGNMENT is presenting the possible list of of theoretical choices that a SPECIFIC PARAMETER ASSIGNMENT can choose from. Some parameters are themselves lists, for example GROUP OF LINEs, or GROUP OF OPERATOR), so in certain complex cases, i.e. where there is a list of lists a GENERIC PARAMETER ASSIGNMENT may need to indicate whether a SPECIFIC PARAMETER ASSIGNMENT may choose any or all of the members of a list. This can be specified by a ***SetSelectionOperator***.

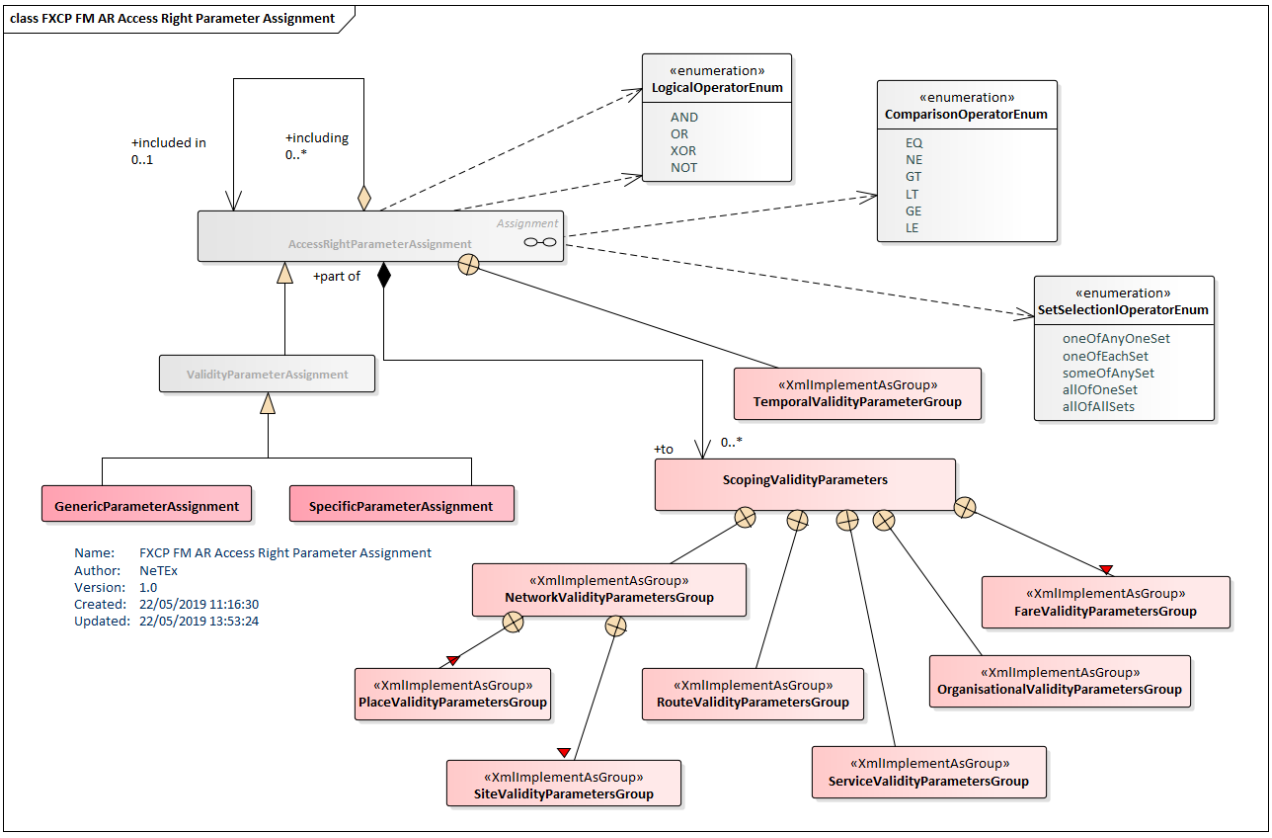


Figure 46 — EFIP – Combining Access right Parameter Assignments (UML)

The following figure shows some examples of validity parameters that are commonly assigned with a ACCESS RIGHT PARAMETER ASSIGNMENT to define products. The parameters are shown with the scope of the basic profile.

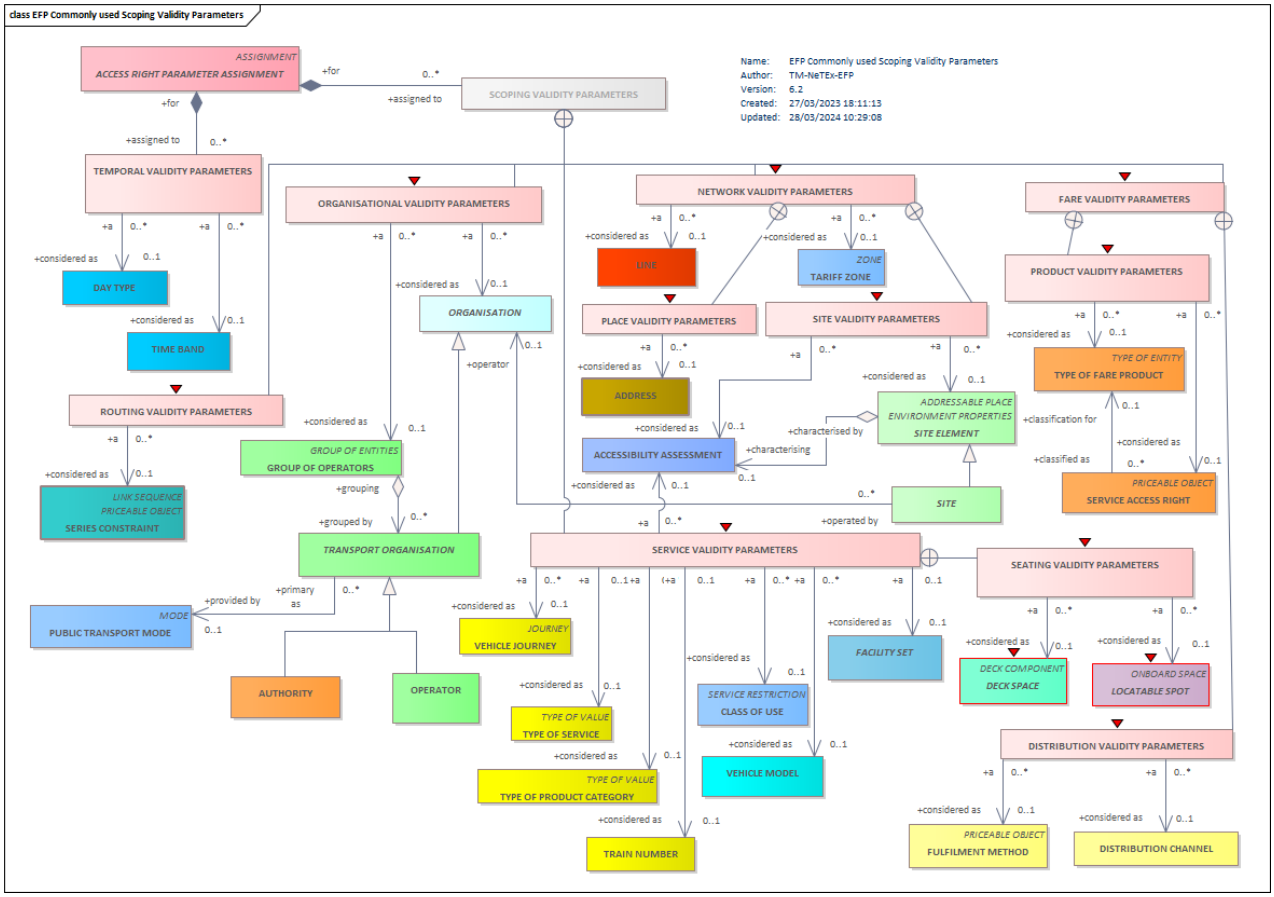


Figure 47 — EFIP - Commonly used Validity Parameters (UML)

12.3.3 *Using parameter assignments – advanced use*

More elaborate access right limitation rules can be specified

(1) By building ***compound conditions***, that include one or more ACCESS RIGHT PARAMETER ASSIGNMENTs within another assignment, the way the individual assignments should be combined together is indicated by by a ***TypeOfGrouping*** Logical operator (AND, OR OR) on the enclosing assignment.

(2) By using a ***ComparisonOperator*** to compare o the values of an element.

An ACCESS RIGHT PARAMETER ASSIGNMENT typically compares a parameter value to a characteristic of the related object. The A***ssignmentType*** attribute uses a ***ComparisonOperator*** that allows for such a comparison. There are different types of possible comparisons, specified by the ***ComparisonOperator***, ('GT', 'EQ', 'LT', etc)., e.g.:

- 'EQ': limits the assignment to be stricty the same as the identity of the reference of the parameter, e.g. the access right is limited to services provided by the specified OPERATOR, or the consumption must occur only on LINE "27". This is the defaut interpreration and covers tha majority of cases.

- 'NE': limits the assignment to exclude the given reference, e.g. in order to represent the rule 'the access right is valid on all bus network LINEs except for LINE 278 and LINE 66' or 'the access right to zone 4 is not valid between "2 a.m. – 4 a.m.'

- 'GE:' is greater than or equal to some inherent value of the reference of the parameter, e.g. in order to indicate that the consumption has to end after "11.00 p.m." a 'GE' reference to a ***Timeband*** with an ***EndTime*** of 23:00, could be used. For such a comparison to be meaningful, the referenced parameter must generally have some inherent primary attribute with a monotonic range., or a subcomponent that is ordered. For an example of the latter case, consider a VEHICLE JOURNEY following a specific SERVICE PATTERN. To indicate that consumption must start only after a give stop it would be meaningful to specify a compound condition that ANDs a reference to the overall SERVICE JOURNEY with a reference using a 'GE' ***LogicalOperator*** to a specific SCHEDULED STOP POINT in the JOURNEY PATTERN.

- 'LE' is equal or smaller than some inherent value of the reference of the parameter, e.g.For example to indicate that the consumption has to end before "11.00 p.m.". a reference to a ***Timeband*** with an ***EndTime*** of 23:00, could be used.

EFP1>> The UK basic profile uses only the 'EQ' logical operator.

12.3.4 *Assigning access right parameters*

There are many different tyes of parameter that can be selected to express the many and various conditions found in the definitions of fare products.

Note that many of the condituons are only needed for the compelx sales and after sales conditions attaching to prioducts; a such they are relevant for passenger information but are not needed to describe the basic availability and prices of fares.

The validity parameters are considered as being of two main types:

- TEMPORAL VALIDITY PARAMETERS, reflecting temporal limitations and
- SCOPING VALIDITY PARAMETERS, reflecting spatial and consumption limitations.

The SCOPING PARAMETERS in their turn may be further grouped as follows:

- ORGANISATIONAL VALIDITY PARAMETERS
- NETWORK VALIDITY PARAMETERS
  - SITE VALIDITY PARAMETERS
  - PLACE VALIDITY PARAMETERS
- ROUTING VALIDITY PARAMETERS
- SERVICE VALIDITY PARAMETERS
  - SEATING VALIDITY PARAMETERS
- FARE VALIDITY PARAMETERS.
  - TARIFF VALIDITY PARAMETERS.
  - PRODUCT VALIDITY PARAMETERS.
  - DISTRIBUTION VALIDITY PARAMETERS.

12.3.5 *Access right parameters: Temporal*

The temporal validity parameters of an ACCESS RIGHT PARAMETER ASSIGNMENT which can be used to restrict when an assignment applies.:

- The DAY TYPE or OPERATING DAY on which the assignment applies.
- The TIMEBANDs during which the assignment applies.
- The OPERATING PERIODs during which the assignment applies.
- The VALIDITY CONDITION or AVAILABILITY CONDITION restricting the assignment.

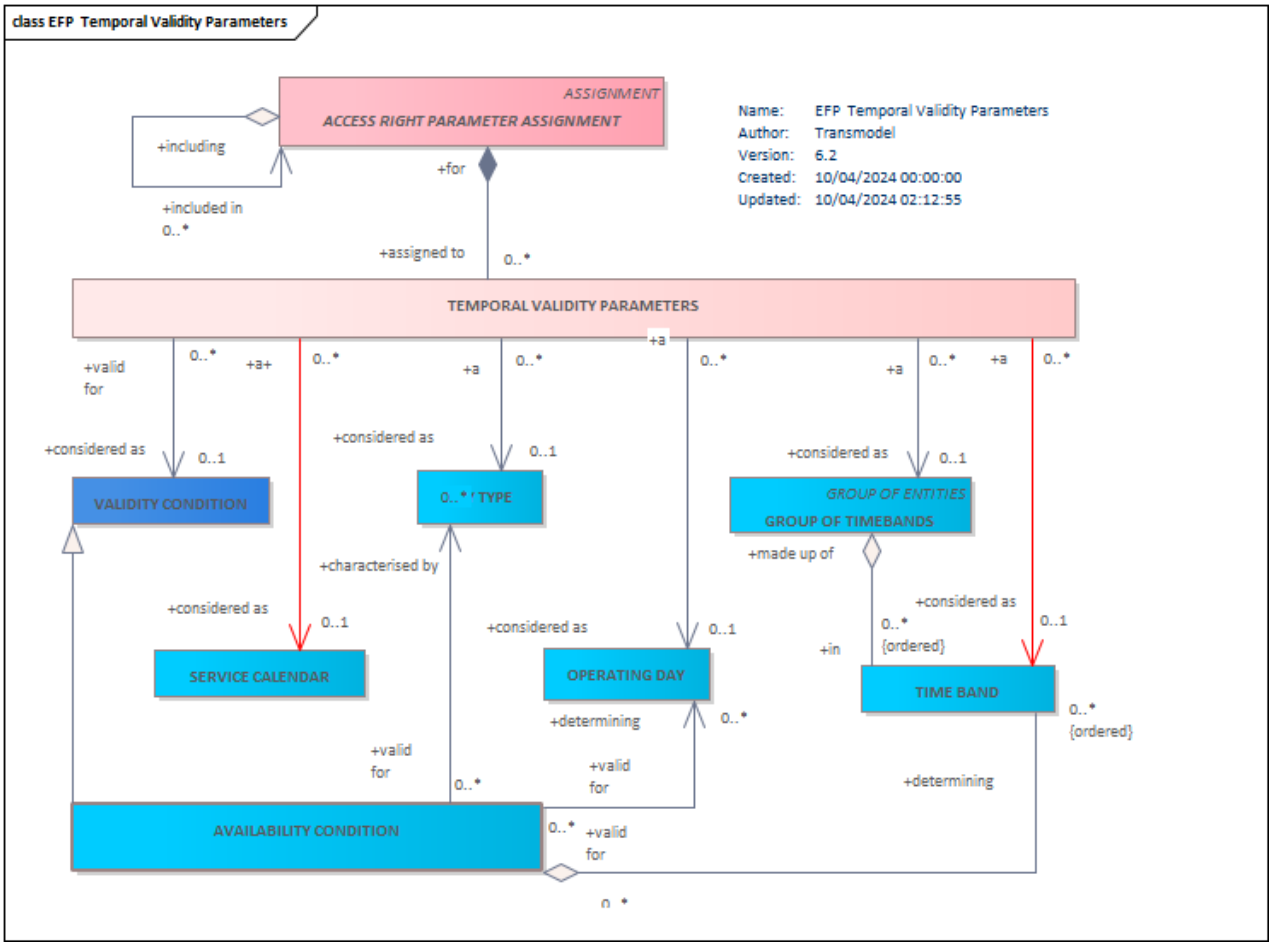


Figure 48 — Access Right Parameters: Temporal Validity Parameters (UML)

12.3.6 *Access right parameters: Network & service*

Network scoping parameters cover network and services and can include:

PLACE VALIDITY PARAMETERS

- The ADDRESS to which the assignment applies.
- The TOPOGRAPHIC PLACE to which the assignment applies.

SITE VALIDITY PARAMETERS

- The STOP PLACE, PARKING or POINT OF INTEREST to which the assignment applies.

NETWORK VALIDITY PARAMETERS

- The SCHEDULED STOP POINT to which the assignment applies.
- The TARIFF ZONE to which the assignment applies.
- The LINE or GROUP OF LINEs to which the assignment applies.
- The TYPE OF LINE to which the assignment applies.
- The ERIES CONSTRAINT to which the assignment applies.

ORGANISATION VALIDITY PARAMETERS

- The OPERATOR, AUTHORITY or GROUP OF AUTHORITIES to which the assignment applies.
- The TARIFF ZONE to which the assignment applies.
- The LINE or GROUP OF LINEsZONE to which the assignment applies.

SERVICE VALIDITY PARAMETERS

- The SERVICE JOURNEY GROUP OF SERVICEs, or TRAIN NUMBER to which the assignment applies.
- The TYOE OF ERVICE or TYPE OF PRODUCT CATEGORY to which the assignment applies.
- The TOPOGRAPHIC PLACE to which the assignment applies.

SEATING VALIDITY PARAMETERS

- The PASENGER SEAT to which the assignment applies.

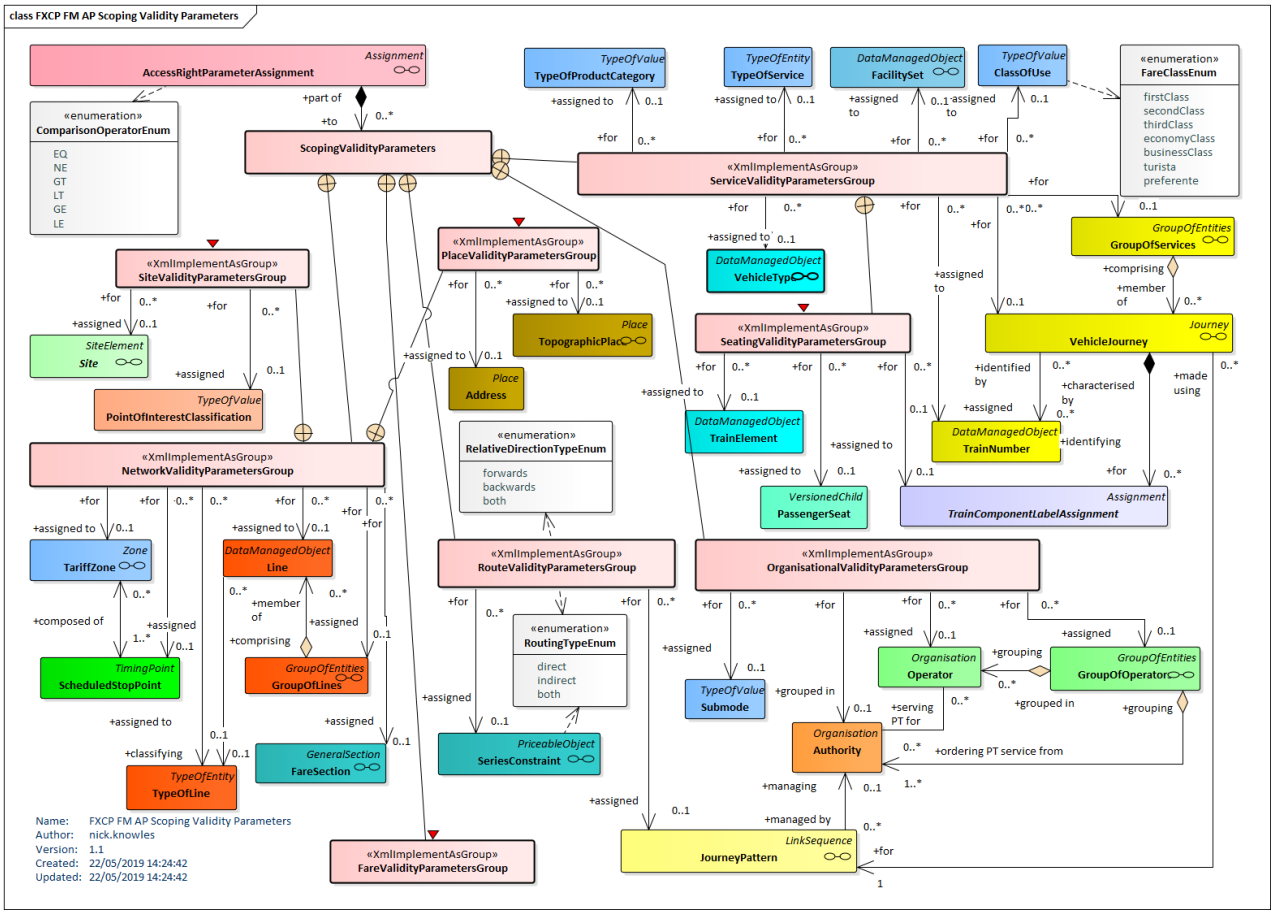


Figure 49 — Network Validity Parameters – (UML)

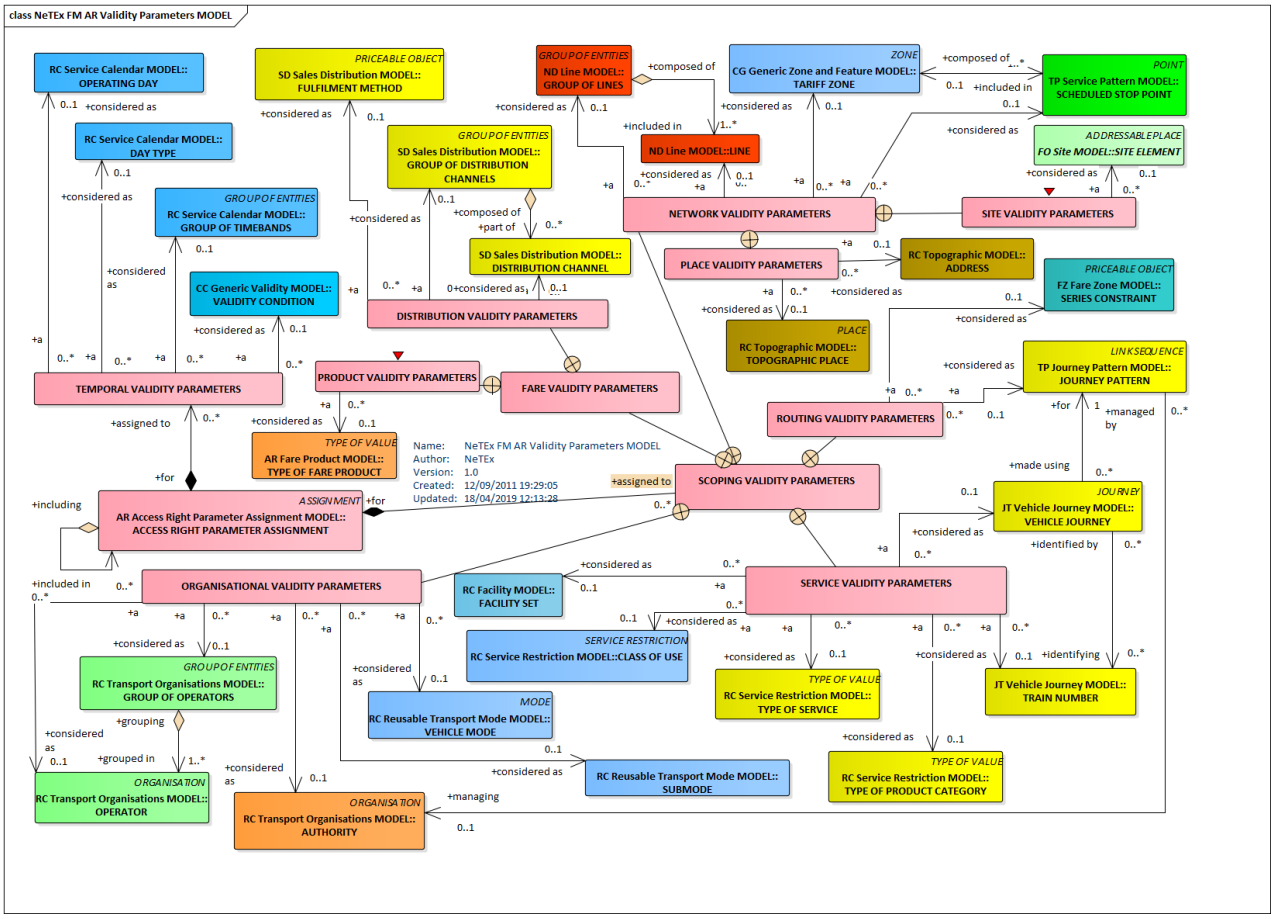


Figure 50 — Network Validity Parameters – (UML)

12.3.7 *Access right parameters: Fare validity*

Fare scoping parameters cover fare elements – other than the elemenst to which the assignments are made (FARE PRODUCT, SALESOFFER PACKE, VALIDABLE Element, etcetc) and can include:

USAGE VALIDITY PARAMETERS

- Any of the NeTEx USAGE PARAMETERs which the assignment applies.

TARIFF VALIDITY PARAMETERS

- Classifiers of TARRIFs such as TYPE OF TARIFF, TYPE OF FARE STRUCTURE ELEMENT, to which the assignment applies.

PRODUCT VALIDITY PARAMETERS

- Classifiers of FARE PRODUCTs, such as TYPE OF FARE PRODUCT, CHARGING MOMENT, TYPE OF CONCESSION, to which the assignment applies.

SALESS OFFER VALIDITY PARAMETERS

- TYPEs of TRAVEL DOCUMENT to which the assignment applies.
- Classifiers of SALES OFFER PACKGEs, such as TYPE OF SALES OFFER PACKGE to which the assignment applies.

DISTRIBUTION VALIDITY PARAMETERS

- DISTRIBUTION CHANNELs and GROUPs of DISTRINUTION CHANNELs to which the assignment applies.
- FULFILMENT METHODs to which the assignment applies.

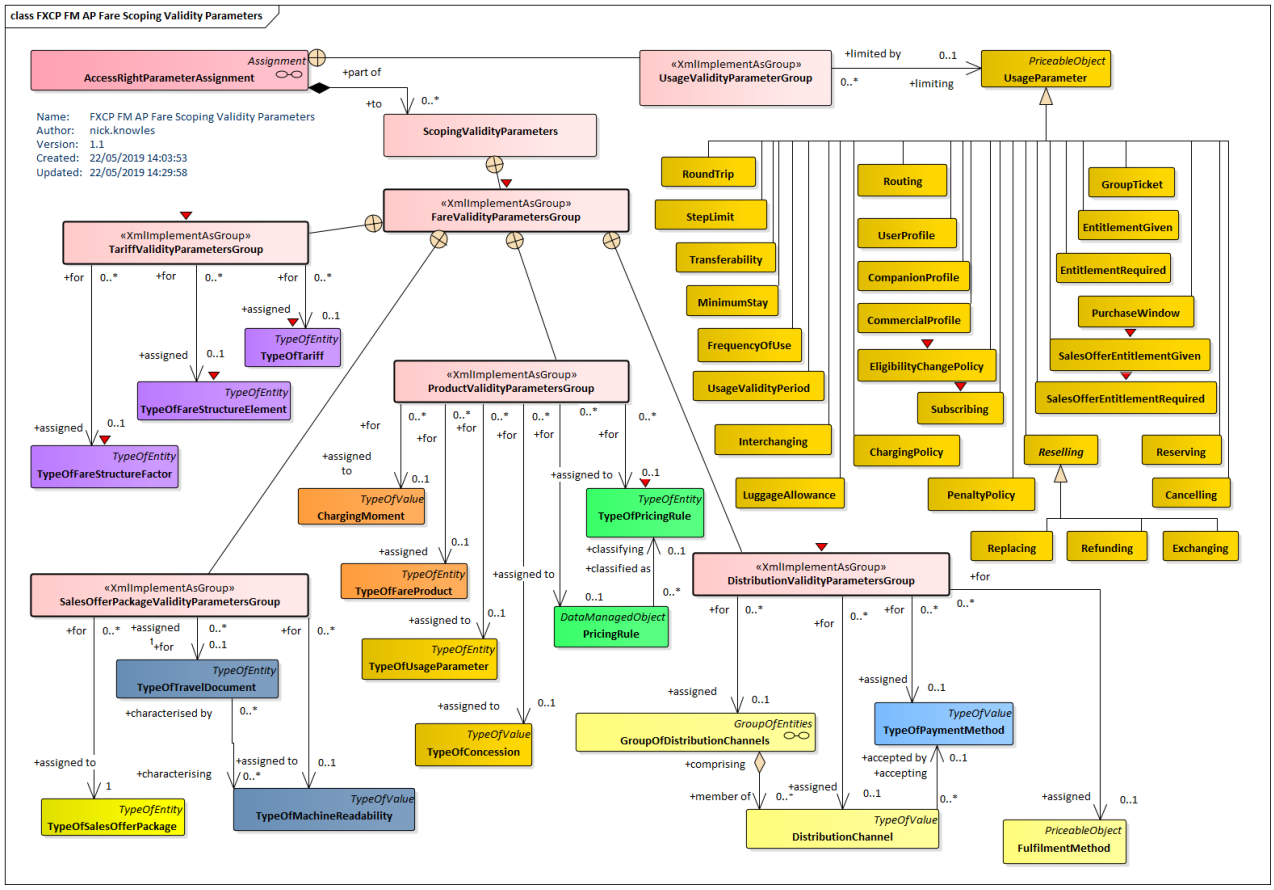


Figure 51 — Access Right Parameters: Fare Scoping Validity Parameters (UML)

12.4 Access Right Parameters

12.4.1 *AccessRightParameterAssignment (Abstract)*

The assignment of a fare collection parameter (referring to geography, time, quality or usage) to an element of a fare system (access right, validated access, control mean, etc.).

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*Assignment*	::>	ACCESS RIGHT PARAMETER ASSIGNMENT inherits from ASSIGNMENT.
«PK»	***id***	*AccessRight-Parameter-AssignmentIdType*	1:1	Identifier of ACCESS RIGHT PARAMETER ASSIGNMENT.
	IsAllowed	*xsd:boolean*	0:1	Whether the specified assignments are allowed (true) or not (false).
«FK»	***TypeOfAssignmentRef***	*TypeOAccessRight-AssignmentRef*	0:1	Classification of ACCESS RIGHT PARAMETER ASSIGNMENT.
«enum»	***ChargingBasis***	*ChargingBasisEnum*	0:1	Whether the specified assignment is for charged access, discounted access or free access. See allowed values below. • *free; discounted; normal; any*
«FK»	***ValidableElementRef***	*ValidableElementRef*	0:1	VALIDABLE ELEMENT to which assignment is made.
«FK»	***ControllableElementRef***	*Controllable-ElementRef*	0:1	CONTROLLABLE ELEMENT to which assignment is made.
«FK»	***FareProductRef***	*FareProductRef+*	0:1	FARE PRODUCT to which assignment is made.
«FK»	***TariffRef***	*TariffRef*	0:1	TARIFF to which assignment is made.
«FK»	***FareStructure-ElementRef***	*FareStructure-ElementRef*	0:1	FARE STRUCTURE ELEMENT to which assignment is made.
«FK»	***FareStructureElement-***	*FareStructure-Element-*	0:1	FARE STRUCTURE ELEMENT IN SEQUENCE

	InSequenceRef	*InSequenceRef*		to which assignment is made.
«FK»	***DistanceMatrixElement-Ref***	*DistanceMatrixRef*	0:1	DISTANCE MATRIX ELEMENT to which ACCESS RIGHT PARAMETER is assigned.
«FK»	***Distance-MatrixInverseRef***	*DistanceMatrixRef*	0:1	DISTANCE MATRIX ELEMENT to which ACCESS RIGHT PARAMETER is assigned; reference is I inverse sense to that of element.
«FK»	***DistanceMatrixInverse-View***	*DistanceMatrixView*	0:1	VIEW of DISTANCE MATRIX ELEMENT to which ACCESS RIGHT PARAMETER is assigned. View includes details of origin and destination
«FK»	***SalesOfferPackageRef***	*SalesOffer-PackageRef*	0:1	SALES OFFER PACKAGE to which assignment is made.
«FK»	***GroupOfDistanceMatrix-ElementsRef***	*GroupOf-DistanceMatrix-ElementsRef*	0:1	GROUP OF DISTANCE MATRIX ELEMENTs to which ACCESS RIGHT PARAMETER is assigned.
«FK»	***GroupOfSales-OfferPackages-Ref***	*GroupOfSalesOffer-PackagesRef*	0:1	GROUP OF SALES OFFER PACKAGEs to which assignment is made.
«enum»	***Limitations-GroupingType***	*BooleanOperator-Enum*	0:1	Logical operator for combining USAGE PARAMETERs elements. The default is ‘AND’. ‘OR’ and ‘XOR’ should only be used if parameters are all of the same type.
«enum»	***LimitationsSetSelection-Type***	*SetOperatorEnum*	0:1	Where one or more parameter is a group containing multiple elements, (GROUP OF xxx), set operator for distinguishing between whole set and item interpretation of elements which are sets of elements. See allowed values below.
«FK»	***limitations***	*UsageParameterRef* +	0:*	References to USAGE PARAMETERs defining limitations made by ACCESS RIGHT

				PARAMETER ASSIGNMENT.
«enum»	**ValidityParameter-AssignmentType**	*Comparison-OperatorEnum*	0:1	Comparison operator for matching validity parameter values. See allowed values below.
XGRP	***ScopingVaidity-Parameters***	***ScopingVaidity-Parameters***	1:1	Time related properties assigned by ACCESS RIGHT PARAMETER ASSIGNMENT.
«enum»	***ValidityParameter-GroupingType***	*BooleanOperator-Enum*	0:1	Logical operator for combining network validity parameters, e.g. ‘AND’, ‘OR’, ‘XOR’. See allowed values below.
«enum»	***ValiditySetSelection-Type***	*SetOperatorEnum*	0:1	Where one or more parameter is a group containing multiple elements, (GROUP OF xxx), set operator for distinguishing between whole set and item interpretation of elements which are sets of elements. oneOfEachSet someOfAnySet allOfOneSet allOfAllSets
«cntd»	***temporalValidity-Parameters***	***TemporalValidity-ParametersGroup***	0:*	Temporal validity parameters assigned by ACCESS RIGHT PARAMETER ASSIGNMENT.
«cntd»	***validityParameters***	***LimitingValidity-ParametersGroup***	0:*	Validity parameters assigned by ACCESS RIGHT PARAMETER ASSIGNMENT.
«FK»	***DayTypeRef***	***ValidityConditionRef***	0:1	DAY TYPE to which ACCESS RIGHT PARAMETER is assigned.
«FK»	***GroupOfTimebandsRef***	*GroupOfTimebands Ref*	0:1	GROUP OF TIME BANDs to which ACCESS RIGHT PARAMETER is assigned.

«FK»	***OperatingDayRef***	*OperatingDayRef*	0:1	OPERATING DAY to which ACCESS RIGHT PARAMETER is assigned.
«FK»	***OperatingPeriodRef***	*OperatingPeriod-Ref*	0:1	OPERATING PERIOD to which ACCESS RIGHT PARAMETER is assigned. +v1.1
«FK»	***ValidityConditionRef***	*ValidityConditionRef*	0:1	VALIDITY CONDITION to which ACCESS RIGHT PARAMETER is assigned.
«enum»	***IncludesGroupingType***	*BooleanOperatorEnum*	0:1	Logical operator for combining included elements. The default is 'OR'. . AND; OR; XOR; NOT
«cntd»	***includes***	***AccessRight-Parameter-Assignment+***	0:*	ACCESS RIGHT PARAMETER ASSIGNMENTs making up a composite ACCESS RIGHT PARAMETER ASSIGNMENT.

Table 79 – *AccessRightParameterAssignment* – XML Element

12.4.2 *ValidityParameterAssignment (Abstract)*

An ACCESS RIGHT PARAMETER ASSIGNMENT relating a fare collection parameter to a theoretical FARE PRODUCT (or one of its components) or a SALES OFFER PACKAGE.

A specialisation of ACCESS RIGHT PARAMETER ASSIGNMENT.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*AccessRight-ParameterAssignment*	::>	VALIDITY PARAMETER ASSIGNMENT inherits from ACCESS RIGHT PARAMETER ASSIGNMENT.
«PK»	***id***	*ValidityParameter-AssignmentIdType*	1:1	Identifier of VALIDITY PARAMETER ASSIGNMENT.

«FK»	***QualityStructure-FactorRef***	*QualityStructure-FactorRef*	0:1	Reference to a QUALITY STRUCTURE FACTOR to which the ACCESS RIGHT PARAMETER ASSIGNMENT applies.

Table 80 – *ValidityParameterAssignment* – XML Element

12.4.3 *GenericParameterAssignment*

A VALIDITY PARAMETER ASSIGNMENT specifying generic access rights for a class of products (e.g. a time band limit - 7 to 10 a.m. - for trips made with a student pass). May include alternatives from which a purchaser selects.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*ValidityParameter-Assignment*	::>	GENERIC PARAMETER ASSIGNMENT inherits from VALIDITY PARAMETER ASSIGNMENTs
«PK»	***id***	*GenericParameter-AssignmentIdType*	1:1	Identifier of GENERIC PARAMETER ASSIGNMENT.

Table 81 – *GenericParameterAssignment* – XML Element

12.4.4 *ScopingValidityParameters (Subcomponent of GenericParameterAssignment)*

The Scoping Validity Parameters allows one or more validity parameters to be assigned. The many different possible parameters are organized into five groups. (Organisation, Network, Route, Service, Product) Multiple values are combined using the logical operator (AND, OR) specified by the ***ValidityParameterGroupingType***. AND is the default. For example LINE "22" and SCHEDULED STOP POINT "4563" means that the assignment applies specifically to stop "4563" of LINE "22".

Classifi-cation	Name	Type	Cardi nality	Description
XGRP	***OrganisationValidity-ParametersGroup***	***xmlGroup***	1:1	ORGANISATION related validity parameters for assignment.
XGRP	***NetworkValidity-ParametersGroup***	***xmlGroup***	1:1	NETWORK related validity parameters for assignment.
XGRP	***RouteValidity-ParametersGroup***	***xmlGroup***	1:1	ROUTE related validity parameters for assignment.
XGRP	***ServiceValidity-ParametersGroup***	***xmlGroup***	1:1	SERVICE related validity parameters for assignment.
XGRP	***ProductValidity-ParametersGroup***	***xmlGroup***	1:1	PRODUCT related validity parameters for assignment.

Table 82 – ScopingValidityParameters – XML Element

12.4.5 *OrganisationValidity-ParametersGroup (Subcomponent of GenericParameterAssignment)*

The ***OrganisationValidityParametersGroup*** specifies validity parameters defining general access rights for MODE and ORGANISATION for an ACCESS RIGHT PARAMETER ASSIGNMENT.

Classifi-cation	Name	Type	Cardin-ality	Description

«enum»		***VehicleModes***	*TransportModeEnum*	0:*	TRANSPORT MODEs to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1 for allowed values. See NeTEx Part1.
«enum»		***Transport-Submodel***	*TransportSubmodel-Enum*	0:1	TRANSPORT SUBMODE to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1 for allowed values. See NeTEx Part1.
«FK»		***GroupOf-OperatorsRef***	*GroupOfOperatorsRef*	0:1	GROUP OF OPERATORs to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.
			CHOICE		
	a	***AllOperators-Ref***	*EmtyType*	0:1	ALL OPERATORS apply to ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.
«FK»	b	***OperatorRef***	*OperatorRef*	0:1	OPERATOR to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.
			CHOICE		
	a	***AllAuthoritiesRef***	*EmptyType*	0:1	ALL AUTHORITIies apply to ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.
«FK»	b	***AuthorityRef***	*AuthorityRef*	0:1	AUTHORITY to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.

Table 83 – *OrganisationValidityParametersGroup* – Group

12.4.6 *NetworkValidityParametersGroup (Subcomponent of GenericParameterAssignment)*

The ***NetworkValidityParametersGroup*** specifies validity parameters defining access rights to network elements such as LINE, TARIFF ZONE and SCHEDULED STOP POINT for an ACCESS RIGHT PARAMETER ASSIGNMENT.

Classifi-cation	Name	Type	Cardin-ality	Description
«FK»	***GroupOfLinesRef***	*GroupOfLinesRef*	0:1	GROUP OF LINEs to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.
«FK»	***LineRef***	*LineRef*	0:1	LINE to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.
«FK»	***TypeOfLineRef***	*TypeOfLineRef*	0:1	TYPE OF LINE to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.
«FK»	***TariffZoneRef***	*TariffZoneRef*	0:1	TARIFF ZONE to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.
«FK»	***FareZoneRef***	*FareZoneRef*	0:1	FARE ZONE to which ACCESS RIGHT PARAMETER is assigned.
«FK»	***FareSectionRef***	*FareSectionRef*	0:1	FARE SECTION to which ACCESS RIGHT PARAMETER is assigned.
«FK»	***Scheduled-StopPointRef***	*ScheduledStopPointRef*	0:1	SCHEDULED STOP POINT to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.
XGRP	***PlaceValidity-ParameterGroup***	**xmlGroup**	1:1	PLACE validity parameters for assignment.
XGRP	***SiteValidity-ParameterGroup***	**xmlGroup**	1:1	SITE validity parameters for assignment.

Table 84 – *NetworkValidityParametersGroup* – Group

12.4.7 *PlaceValidityParametersGroup (Subcomponent of GenericParameterAssignment)*

The ***PlaceValidityParametersGroup*** specifies validity parameters defining use of PLACE elements for an ACCESS RIGHT PARAMETER ASSIGNMENT. For flexible and private travel modes, the travel may be between an ADDRESS or a TOPOGRAPHIC PLACE.

Classification	Name	Type	Cardinality	Description
«enum»	***PlaceUse***	*PlaceUseENum*	0:1	Use of ADDRESS or TOPOGRAPHIC PLACE. See allowed values below.+v1.1 • startAt; endAt; via; restrictTo; other
«FK»	***TopographicPlaceRef***	*TopographicPlaceRef*	0:1	TOPOGRAPHIC PLACE to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1. +v1.1
«FK»	***AddressRef***	*PostalAddressRef* | *RoadAddressRef*	0:1	ADDRESS to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1. +v1.1

Table 85 – *PlaceValidityParametersGroup* – Group

12.4.8 *SiteValidityParametersGroup (Subcomponent of GenericParameterAssignment)*

The ***SiteValidityParametersGroup*** specifies validity parameters defining access rights to SITE elements for an ACCESS RIGHT PARAMETER ASSIGNMENT. SITEs can be used for example to associate fare structure elements with POINTs OF INTEREST as for a travel product that also allows entry to museums and other tourist attractions, or to all SITEs of a given type of point interest using a POINT OF INTEREST CLASSIFICATION.

Classification	Name	Type	Cardinality	Description
«FK»	***SiteElementRef***	*SiteElementRef*	0:1	SITE ELEMENT to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.

«FK»	***PointOfInterest-Classification-ElementRef***	*PointOfInterest-ClassificationElementRef*	0:1	POINT OF INTEREST CLASSIFICATION to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1.

Table 86 – *SiteValidityParametersGroup* – Group

12.4.9 *RouteValidityParametersGroup (Subcomponent of GenericParameterAssignment)*

The RouteValidityParametersGroup specifies validity parameters defining access rights to particular routes (as in effect specified by DISTANCE MATRIX and SERIES CONSTRAINT elements) for an ACCESS RIGHT PARAMETER ASSIGNMENT.

Classifi-cation	Name	Type	Cardinality	Description
«enum»	***RoutingType***	*RoutingTypeEnum*	1:1	Type of routing to which assignment applies. See allowed values earlier under FARE ZONE model.
«enum»	***Directions***	*RelativeDirectioeEnum*	0:1	Directions in which assignment applies. See NeTEx Part1.
«FK»	***SeriesConstraint Ref***	*SeriesConstraintRef*	0:1	SERIES CONSTRAINT to which ACCESS RIGHT PARAMETER is assigned.
«FK»	***ServiceJourney-PatternRef***	*ServiceJourney-PatternRef*	0:1	SERVICE JOURNEY PATTERN to which ACCESS RIGHT PARAMETER is assigned.

Table 87 – *RouteValidityParametersGroup* – Group

12.4.10 *ServiceValidityParametersGroup (Subcomponent of GenericParameterAssignment)*

The ***ServiceValidityParametersGroup*** specifies validity parameters defining access rights to particular services or types of service for an ACCESS RIGHT PARAMETER ASSIGNMENT.

Classifi-cation	Name	Type	Cardinality	Description
«FK»	***ClassOfUseRef***	*ClassOfUseRef*	1:1	Reference to a CLASS OF USE (Seat Class).
«enum»	***FareClass***	*FareClassEnum*	0:1	FARE CLASS to which ACCESS RIGHT PARAMETER is assigned. See NeTEx Part1 for allowed values.
«FK»	***FacilitySetRef***	*FacilitySetRef*	0:1	FACILITY SET provided or available for fare.
«FK»	***TypeOfProduct-CategoryRef***	*TypeOfProduct-CategoryRef*	0:1	Type of PRODUCT CATEGORY to which ACCESS RIGHT PARAMETER is assigned.
«FK»	***Service-JourneyRef***	*ServiceJourneyRef*	0:1	VEHICLE JOURNEY to which ACCESS RIGHT PARAMETER is assigned.
«FK»	***TrainNumberRef***	*TrainNumberRef*	0:1	TRAIN NUMBER to which ACCESS RIGHT PARAMETER is assigned.
«FK»	***GroupOfServices Ref***	*GroupOfServicesRef*	0:1	GROUP OF SERVICEs to which ACCESS RIGHT PARAMETER is assigned.
«FK»	***VehicleTypeRef***	*VehicleTypeRef*	0:1	VEHICLE TYPE to which assignment is made.
«FK»	***TypeOf-ServiceRef***	*TypeOfServiceRef*	0:1	TYPE OF SERVICE to which assignment is made, for example whether the assignment is a night train.

Table 88 – *ServiceValidityParametersGroup* – Group

12.4.11 *SeatingValidityParametersGroup (Subcomponent of GenericParameterAssignment)*

The ***SeatingValidityParametersGroup*** specifies conditions on seating for an ACCESS RIGHT PARAMETER ASSIGNMENT. For example, whether a purchase applies to a particular carriage or seat.

Classifi-cation	Name	Type	Cardinality	Description
«FK»	***TrainElementRef***	*TrainElementRef*	0:1	Reference to a TRAIN ELEMENT to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***TrainComponent Label-AssignmentRef***	*TrainComponentLabel-AssignmentIRef*	0:1	Reference to a TRAIN COMPONENT LABEL ASSIGNMENT to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***PassengerSeat-Ref***	*PassengerSeatRef*	0:1	Reference to a PASSENGER SEAT to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.

Table 89 – *SeatingValidityParametersGroup* – Group

12.4.12 *TariffValidityParametersGroup (Subcomponent of GenericParameterAssignment)*

The ***TariffValidityParametersGroup*** specifies conditions on based on tariff structure elements for an ACCESS RIGHT PARAMETER ASSIGNMENT. For example, where a condition applies to a specific FARE STRUCTURE ELEMENT.

Classifi-cation	Name	Type	Cardin-ality	Description
«FK»	***TypeOfFare-StructureFactorRef***	*TypeOfFare-StructureFactorRef*	0:1	Reference to a TYPE OF FARE STRUCTURE FACTOR to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***TypeOfFare-Structure-ElementRef***	*TypeOfFare-StructureElementRef*	0:1	Reference to a TYPE OF FARE STRUCTURE ELEMENT to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***TypeOfTariffRef***	*TypeOfTariffRef*	0:1	Reference to a TYPE OF TARIFF to which the ACCESS RIGHT PARAMETER ASSIGNMENTs

				applies.

Table 90 – *TariffValidityParametersGroup* – Group

12.4.13 *ProductValidityParametersGroup (Subcomponent of GenericParameterAssignment)*

The ***ProductValidityParametersGroup*** specifies conditions on purchase or fulfilment for an ACCESS RIGHT PARAMETER ASSIGNMENT. For example, where a ticket may be purchased or collected, or whether a commercial condition such as refunding is restricted with a particular DISTRIBUTION CHANNEL.

Classifi-cation	Name	Type	Cardinality	Description
«FK»	***PricingRuleRef***	*PricingRuleRef+*	0:1	Reference to a PRICING RULE to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***TypeOfPricing-RuleRef***	*TypeOfPricingRuleRef*	0:1	Reference to a TYPE OF PRICING RULE to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***TypeOfFare-ProductRef***	*TypeOfFareProductRef*	0:1	Reference to a TYPE OF FARE PRODUCT to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***TypeOfUsage-ParameterRef***	*TypeOfUsageParameterRef*	0:1	Reference to a TYPE OF USAGE PARAMETER to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***ChargingMomentRef***	*ChargingMomentRef*	0:1	Reference to a CHARGING MOMENT to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***TypeOf-ConcessiontRef***	*TypeOfConcessionRef*	0:1	Reference to a TYPE OF CONCESSION to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.

Table 91 – *ProductValidityParametersGroup* – Group

12.4.14 *SalesOfferValidityParametersGroup (Subcomponent of GenericParameterAssignment)*

The ***SalesOfferValidityParametersGroup*** specifies conditions on SALESS OFFER PACKAGE properties for an ACCESS RIGHT PARAMETER ASSIGNMENT. For example, where a ticket may be purchased or collected, or whether a commercial condition such as refunding is restricted with a particular DISTRIBUTION CHANNEL.

Classification	Name	Type	Cardinality	Description
«FK»	***TypeOfSales-OfferPackageRef***	*TypeOfSales-OfferPackageRef*	0:1	Reference to a TYPE OF SALES OFER to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***TypeOfTravel-DocumentRef***	*TypeOfTravelDocumentRef*	0:1	Reference to a TYPE OF TRAVEL DOCUMENT to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***TypeOMachine-ReadabilityRef***	*TypeOfMachine-ReadabilityRef*	0:1	Reference to a TYPE OF MACHINE READABILITY to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.

Table 92 – *SalesOfferValidityParametersGroup* – Group

12.4.15 *DistributionValidityParametersGroup (Subcomponent of GenericParameterAssignment)*

The ***DistributionValidityParametersGroup*** specifies conditions on purchase or fulfilment for an ACCESS RIGHT PARAMETER ASSIGNMENT relation to distribution. For example, where a ticket may be purchased or collected, or how it may be paid for.

Classification	Name	Type	Cardinality	Description
«FK»	***Distribution-ChannelRef***	*FareStructureElement-Ref*	0:1	Reference to a DISTRIBUTION CHANNEL to which the ACCESS RIGHT PARAMETER

				ASSIGNMENTs applies.
«FK»	***GroupOf-Distribution-ChannelsRef***	*GroupOfDistribution-ChannelsRef*	0:1	Reference to a GROUP OF DISTRIBUTION CHANNELs to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***FulfIment-MethodRef***	*FareStructureElement-Ref*	0:1	Reference to a FULFILMENT METHOD to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.
«FK»	***TypeOPayment-MethodRef***	*TypeOPayment-MethodRef*	0:1	Reference to a TYPE OF PAYMENT METHOD to which the ACCESS RIGHT PARAMETER ASSIGNMENTs applies.

Table 93 – *DistributionValidityParametersGroup* – Group

12.5 Usage Parameters Overview

The validity of an access right (or of a marketable combination) may be limited by parameters related to who may cosume them (user profile, group ticket, etc.) or how they may be consumed (frequency of use, interchanging, transferability, etc.). Such parameters express in general additional rules to those expressed by the fare structure CONTROLLABLE ELEMENTs or FARE STRUCTURE ELEMENTs. Such parameters are described by the generic entity USAGE PARAMETER.

USAGE PARAMETERS may also describe pre-sale and after sales commercial conditions, such, transferability, refunding, exchanging etc.

The Usage parameters can be grouped into a number of broad categories; travel, eligibility, after-sales, etc

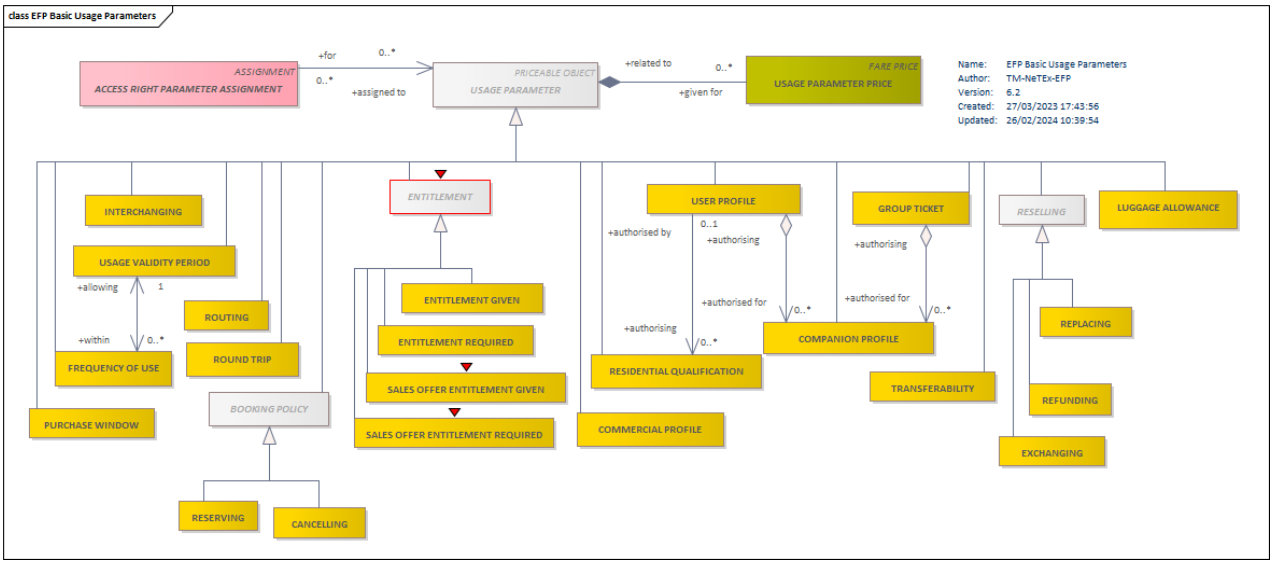


Figure 52 —Usage Parameters –(UML)

12.5.1 *Usage parameter overview: Travel*

Travel USAGE PARAMETERs specify limitations on travel such as.

Basic Fare Profile

- ROUND TRIP expressing the properties relating to single or return trip use of an access right.
- USAGE VALIDITY PERIOD describes a broad time limitation of access rights, especially passes. It may include a ‘standard duration' of validity (1-day, 1 month…), time limitations (‘start date’ and ‘end date’, ‘start time’ and ‘end time’), or a combination of both;
- FREQUENCY OF USE describes the limitation of an access right, depending on frequency of use during a VALIDITY PERIOD. For instance, a product is offered at a special fare if it is used more than 50 times in a month;
- INTERCHANGING expressing the limitations on making changes within a trip;

Advanced Fare Profile

- STEP LIMIT, a geographical parameter limiting the access rights by counts of stops, sections or zones;
- ROUTING, expressing the properties Limitations the limitations on routing of an access right.
- MINIMUM STAY, expressing the details of any minimum stay at the destination required to use the product;

12.5.2 *Usage parameter overview: Eligibility*

Eligibility: USAGE PARAMETERs specify limitations on who may buy or use prducts.

Basic Fare Profile

- USER PROFILE, which describes the social profile of a customer. It is generally used to allow discounts based on age groups (e.g. under 18), gender, profession, social status (e.g. student, retired, unemployed), etc.
- COMPANION PROFILE, indicating the number and characteristics of persons entitled to travel in a group or as companions to another USER PROFILE.
- GROUP TICKET describes the number and characteristics of persons possibly entitled to travel in addition to the holder of an access right.

Advanced Fare Profile

- RESIDENTIAL QUALIFICATION, categorising the users depending on their location of residence), often used for allowing discounts.
- COMMERCIAL PROFILE, which is used to describe customer categories depending on their commercial relations with the operator (frequent traveller, amount of purchase by a company, etc.). It is generally used to allow discounts.

12.5.3 *Usage parameter overview: Entitlement*

Entitlement USAGE PARAMETERs specify limitations on other product rights given or required by a product.

Basic Fare Profile

- ENTITLEMENT REQUIRED, indicating whether an ENTITLEMENT PRODUCT is required to use access right;
- ENTITLEMENT GIVEN, indicating whether a specific access right represents an ENTITLEMENT PRODUCT.

Advanced fare profile

- ENTITLEMENT CONSTRAINT:Where a product gives rights to another product, the dependent product may be constrained to particular properties of the prerequisite product. For example, a rail ticket might entitle the user to buy a local bus day pass at either end of the rail journey at a reduced rate; the allowed zones to purchased for the dependent product are restricted to the choices of origin and destination station made in the prerequisite product, and the day must be the same day of travel (or days of travel if it is a period return on different days).Such relative constraints can be expressed on an entitlement entity using an ENTITLEMENT constraint. Other examples might include a requirement to use the same media (TYPE OF TRAVEL DOCUMENT), to be the same type of user (i.e. USER PROFILE), or to use the same OPERATOR.
- Note that a SUPPLEMENT PRODUCT for a trip such as a seat reservation is normally assumed to be constrained to the same values as for the prerequisite product trip.

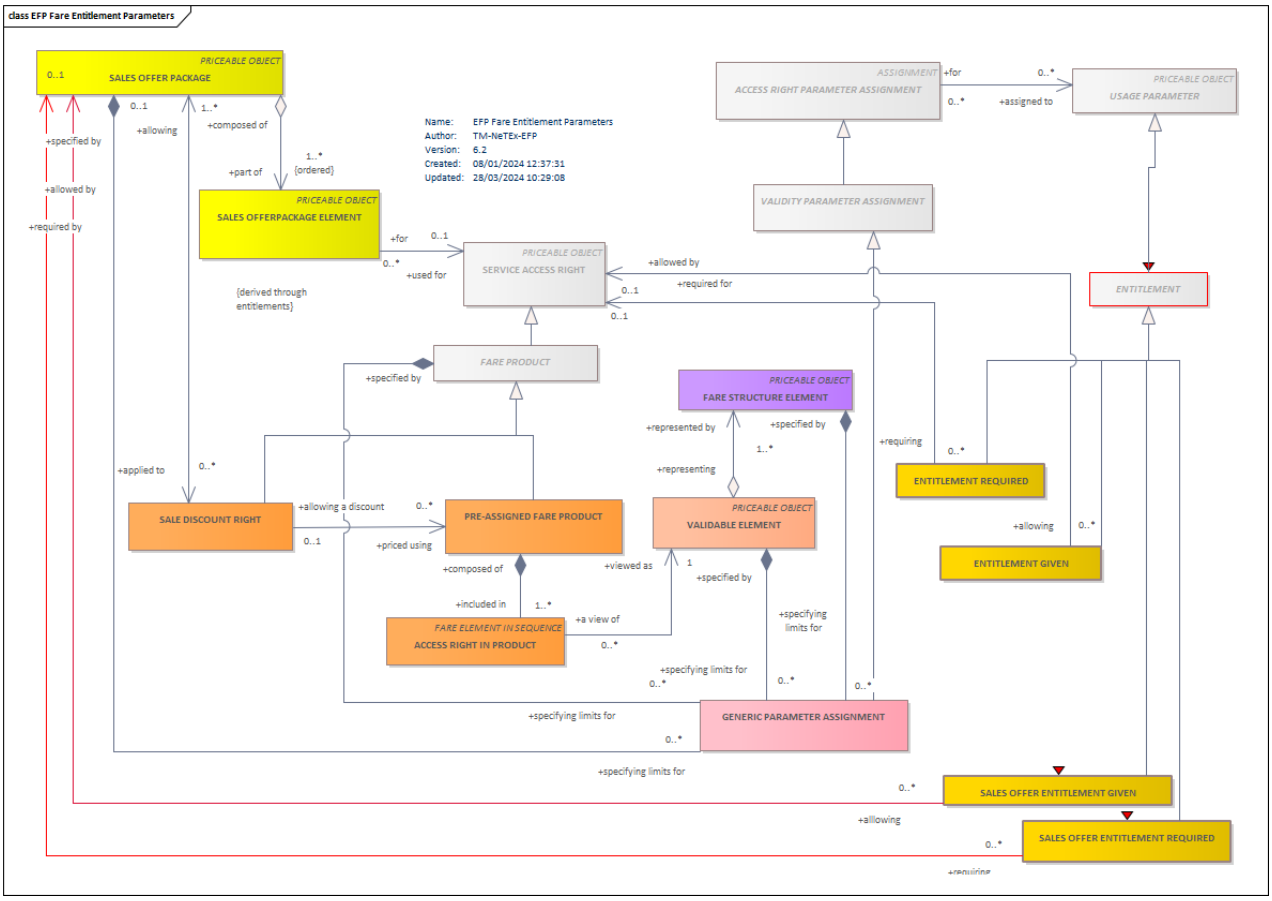


Figure 53 —Usage parameter overview: Product Entitlement (UML)

12.5.4 *Usage parameter overview: Luggage*

The LUGGAGE ALLOWANCE parameter specifies limitations on taking luggage.

12.5.5 *Usage parameter overview: Booking*

Booking USAGE PARAMETERs referring to sales and reservation (booking) specify limitations on booking transactions such as PURCHASE WINDOW, TRANSFERABILITY, RESERVING, EXCHANGING, REFUNDING.

Basic Fare Profile

- TRANSFERABILITY describes the right to transfer an access right to other persons than the original customer (number and characteristics of persons entitled to consume).
- PURCHASE WINDOW, indicating the period in which the product must be purchased.

Advanced fare profile

- RESERVING indicating whether the access right requires reservation.
- CANCELLING indicating whether and how a booking may be cancelled.
- BOOKING ARRANGEMENTS indicating how a booking can be made.

12.5.6 *Usage parameter overview: After-sales*

After-sales USAGE PARAMETERs describe commercial after sales conditions on products.

Basic Fare Profile

- The TRANSFERABILITY parameter specifies limitations on transferring a ticket to someone else.

Advanced fare profile

- The REPLACING parameter specifies whether the product can be replaced if lost or stolen.
- The REFUNDING parameter specifies limitations on refunds for a product and other resale properties.
- The EXCHANGING parameter specifies limitations on exchanging tickets for other tickets.
- The SUSPENDING parameter specifies conditions governing temporary suspension of a FARE PRODUCT, (i.e. period pass or subscription).

12.5.7 *Usage parameter overview: Charging*

Charging USAGE PARAMETERs describe commercial conditions relating to charging for products.

Advanced Fare Profile

- The CHARGING POLICY parameter specifies limitations on how a product may be charged. May be used to specify a minimum and maxim credit level.
- The PENALTY POLICY parameter specifies rules relating to penalty fares that may be incurred.
- The SUBSCRIBING parameter specifies rules relating to products bought on subscription, that is an agreement to make regular payments over a specified period in return for a discounted price.

12.5.8 *Prices for usage parameters*

Although all USAGE PARAMETERs have a PRICE the price has a different purpose in different cases.

Group	Usage Parameter	Comment on Price
Travel	ROUND TRIP	Pricing of single or return trip.
	USAGE VALIDITY PERIOD	Pricing for a trip of this length.
	FREQUENCY OF USE	Pricing for a product subject to this frequency of use constraint, if any
	INTERCHANGING	Price of making an interchange, if any
	MINIMUM STAY	Price for a minimum stay.
	STEP LIMIT	Pricing for a trip with the specified number of steps.
	ROUTING	Pricing for a route with these restrictions.
Eligibility	USER PROFILE	Price for a user of this type, if any
	COMMERCIAL PROFILE	Pricing for the commercial offer.
	GROUP TICKET	Pricing for a group ticket of this type.
	COMPANION PROFILE	Pricing for taking a companion along, if any.
	RESIDENTIAL QUALIFICATION	Pricing for someone who meets this residential criterion, if any.
Entitlement	ENTITLEMENT REQUIRED	Pricing for use for required product, if any
	ENTITLEMENT GIVEN,	Discount that gives to other product, if any
Luggage	LUGGAGE ALLOWANCE	Price of luggage carriage of the specified type, if any
Booking	PURCHASE WINDOW	Price of a ticket bought within the specified window, if any
	TRANSFERABILITY	Price of transferarig a ticket to someone else, if any
	EXCHANGING	Fee for exchanging ticket, if any

	REFUNDING	Fee for refunding ticket, if any.
	REPLACING	Fee for replacing a lost ticket, if any.
	RESERVING	Fee for reserving a ticket, if any.
Charging	CHARGING POLICY	Limits of credit associated with product.
	PENALTY POLICY	Fees for use deemed to incur penalty.
	SUBSCRIBING	Fee for setting up a subscription.
	SUSPENDING	Fee for suspending a perod pass or subscription.

Table 94 – *USAGE PARAMETERs* – Meaning of prices

12.5.9 *Usage parameters: Eligibility*

Eligibility user parameters state who may use a given product.

12.5.10 *UsageParameter (Abstract)*

A parameter used to specify conditions on the use of a SALES OFFER PACKAGE or a FARE PRODUCT.

There are a number different specialisations for different purposes.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	USAGE PARAMETER inherits from PRICEABLE OBJECT**.**
«PK»	***id***	*UsageParameterIdType*	1:1	Identifier of USAGE PARAMETER.
	Url	*xsd:anyUri*	0:1	Url associated with parameter.
«FK»	***TypeOf-UsageParameter-Ref***	*TypeOfUsageParameterRef*	0:1	Type of USAGE PARAMETER.
«cntd»	***fareTables***	*FareTable*	0:*	FARE TABLEs for the USAGE PARAMETER.

«cntd»	***prices***	*UsageParameterPrice*	0:*	USAGE PARAMETER PRICEs for the USAGE PARAMETER.

Table 95 – *UsageParameter* – XML Element

12.5.11 *UserProfile*

The social profile of a passenger, based on age group, education, profession, social status, sex etc., often used for allowing discounts: 18-40 years old, graduates, drivers, unemployed, women etc.

EFP1>> User profiles should be classified with a ***UserType***. A set of predefined UK user types is included in the EFIP meta data, e.f.' efp:adult', 'efp:child'. If an operator has different age or qualifaction criteria, then they will need to specify additional user types.

Classifi-cation	Name	Type	Cardin-ality	Description
::>	::>	*UsageParameter*	::>	USER PROFILE inherits from USAGE PARAMETER.
«PK»	***id***	*UserProfileIdType*	1:1	Identifier of USER PROFILE.
«FK»	***BaseUserProfile-Ref***	*UserProfileIdType*	0:1	Base USER PROFILE which this profile refines.
«FK»	***TypeOf-ConcessionRef***	*TypeOfConcessionRef*	0:1	Classification by type of concession.
«enum»	***UserType***	*UserTypeEnum*	0:1	Classification of user type. • *anyone; adult; child; infant; senior; schoolPupil; student; youngPerson; disabled; disabledCompanion; employee; military; jobSeeker; guideDog; animal;*

	MinimumAge	*xsd:integer*	0:1	Minimum age for membership of USER PROFILE.
	MaximumAge	*xsd:integer*	0:1	Maximum age for membership of USER PROFILE.
	MonthDayOn-WhichAge-Applies	*xsd:gmonthDay*	0:1	Day / Month on which age applies. if any.
	MinimumHeight	*LengthType*	0:1	Minimum height for membership of USER PROFILE. For example, to restrict access for health and safety reasons.
	MaximumHeight	*LengthType*	0:1	Maximum weight for membership of USER PROFILE. This may be relevant for example for judging large dogs, or a limit on children.
	LocalResident	*xsd:boolean*	0:1	Whether user must be local resident. The default value is '*true*'.
«cntd»	***resides***	*ResidentialQualification*	0:*	RESIDENTIAL QUALIFICATIONs for USER PROFILE – if more than one, these will be logically ORed together.
«enum»	***Gender-Limitation***	*GenderLimitationList*	0:1	Gender required by USER PROFILE. Relevant for single sex accommodation products.
«enum»	***ProofRequired***	*ProofOfIdentityEnum*	0:*	Proof required for type of user. See allowed values below. • *noneRequired; passport; drivingLicence; birthCertificate; membershipCard; studentCard; identityDocument; creditCard; medicalDocument; letterWIthAddress; measurement; emailAccount; mobileDevice; other;*
«enum»	***DiscountBasis***	*DiscountBasisEnum*	0:1	Nature of discount for this type of user. See earlier

				for allowed values.
«cntd»	***companion-Profiles***	*CompanionProfile*	0:*	COMPANION PROFILEs describing users who may travel with user.

Table 96 – *UserProfile* – XML Element

12.5.12 *CompanionProfile*

The COMPANION PROFILE specifies the number and characteristics of persons entitled to travel in addition to the holder of an access right, for example children, wheelchair carer, etc.

It can be used to create a precise specification of a group,; for example that a family must include at least one adult or a maximum of two adults and between one and five children,

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*UsageParameter*	::>	COMPANION PROFILE inherits from USAGE PARAMETER.
«PK»	***id***	*GroupTicketUserIdType*	1:1	Identifier of COMPANION PROFILE.
	ParentRef	*UsageParameterRef+*	0:1	Parent USER PROFILE for whom this specifies an allowed companion type.
«FK»	***UserProfileRef***	*UserProfileRef*	0:1	Reference USER PROFILE defining a category of people eligible to be a companion.
«enum»	***Companion-Relationship***	*CompanionRelationship-Enum*	0:1	Required relationship of companion to eligible user. See allowed values below. +v1.1. • *anyone; parent; grandparent; child; grandchild; family; spouse; partner; dependent; colleague; pupil; teacher; carer*

	MinimumNumberOf-Persons	*xsd:integer*	0:1	Minimum number of persons overall allowed of this type.
	MaximumNumberOf-Persons	*xsd:integer*	0:1	Maximum number of persons overall allowed of this type.
«enum»	***DiscountBasis***	*DiscountBasisEnum*	0:1	Nature of discount for this type of user. See allowed values earlier.

Table 97 – CompanionProfile – XML Element

12.5.13 *GroupTicket*

The number and characteristics of persons entitled to travel in addition to the holder of an access right.

Classifi-cation	Name	Type	Cardin-ality	Description
::>	::>	*UsageParameter*	::>	GROUP TICKET inherits from USAGE PARAMETER**.**
«PK»	***id***	*GroupTicketIdType*	1:1	Identifier of GROUP TICKET.
«FK»	***TypeOf-ConcessionRef***	*TypeOfConcessionRef*		Type of concession to which this group applies.
	MinimumNumberOf-Persons	*NumberOfPersons*	0:1	Minimum number of persons overall allowed on GROUP TICKET.
	MaximumNumber-OfPersons	*NumberOfPersons*	0:1	Maximum number of persons overall allowed on GROUP TICKET.
	MinimumNumberOf-CardHolders	*NumberOfPersons*	0:1	Minimum number of card holders required to qualify for this GROUP TICKET.
«cntd»	***companionProfiles***	*CompanionProfile*	0:*	COMPANION OR GROUP allowed in each USER

				PROFILE category.
«enum»	***PricingBasis***	*PerBasisEnum*	0:1	Basis on which pricing is done - per whole group or per member. See allowed values below. • none; free; discountForAll; discountForFirstMemberOfGroup; discountForSecondAndSubsequentMembersOfGroup; stepDiscount
	MaximumPersons-Free	*NumberOfPassengers*	0:1	Number of persons allowed free on ticket.
	MaximumPersons-Discounted	*NumberOfPassengers*	0:1	Maximum number of persons for which a group discount is allowed.
	DiscountOnly-ForFirstPerson	*xsd:boolean*	0:1	Whether there is only a discount for the first person in the group.
	MinimumNumberOf CardHolders	*NumberOfPassengers*	0:1	Minimum number of persons in the group who must hold a qualifying railcard for the discount to be granted.
	OneForNPersons	*NumberOfPassengers*	0:1	Whether discount is on a one-for-n basis. Intermediate numbers are rounded down.
	GroupSizeChanges	*GroupSizeChanges-Enum*	0:1	Possibilities for changing the number of people in the group. See allowed values below. •
«enum»	***JointCheckIn***	*GroupCheckInEnum*	0:1	Whether joint check in is required. See allowed values below. • *none; required; allowed*
«enum»	***Ticketing***	*GroupTicketingEnum*	0:1	Nature of tickets issued for group. See allowed

				values +v1.1 • *allOnOneTicket; separateTickets* • *ticketWithCoupons; other*
«enum»	***GroupBooking-Facility***	*GroupBookingEnum*	0:1	Type of Group Booking allowed. See NeTEx Part1.

Table 98 – *GroupTicket* – XML Element

12.5.14 *ResidentialQualification [ADVANCED PROFILE ONLY]*

The RESIDENTIAL QUALIFICATION element describes a requirement to live in a certain area.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*VersionedChild*	::>	RESIDENTIAL QUALIFICATION inherits from VERSIONED CHILD. See NeTEx Part1.
«PK»	***id***	*ResidentialQualification-IdType*	1:1	Identifier of RESIDENTIAL QUALIFICATION.
	Name	*MultilingualString*	0:1	Name of RESIDENTIAL QUALIFICATION.
	Description	*MultilingualString*	0:1	Description of RESIDENTIAL QUALIFICATION.
«FK»	***ParentRef***	*UsageParameterRef+*	0:1	Parent USER PROFILE for whom this specifies a RESIDENTIAL QUALIFICATION.
	MustReside	*xsd:boolean*	0:1	Whether the user must or must not reside in specified TOPOGRAPHIC PLACE.
«FK»	***Topographical-PlaceRef***	*TopographicalPlaceRef*	0:1	TOPOGRAPHIC PLACE for which residency rule applies. See NeTEx Part1.

«enum»	***ResidenceType***	*ResidenceTypeEnum*	0:1	Classification of type of residence required, • work; *study*; *exchange*; *born*; *nonResident*.
	Minimum-Duration	*xsd:duration*	0:1	Minimum period of residency needed to qualify.

Table 99 – *ResidentialQualification* – XML Element

12.5.15 *CommercialProfile [ADVANCED PROFILE ONLY]*

A category of users depending on their commercial relations with the operator (frequency of use, amount of purchase etc.), often used for allowing discounts.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*UsageParameter*	::>	COMMERCIAL PROFILE inherits from USAGE PARAMETER.
«PK»	***id***	*CommercialProfileIdType*	1:1	Identifier of COMMERCIAL PROFILE.
«FK»	***TypeOf-ConcessionRef***	*TypeOfConcessionRef*	0:1	Reference to a TYPE OF CONCESSION.
	Consumption-Amount	*xsd:anyType*	0:1	Consumption amount associated with COMMERCIAL PROFILE.
	Consumption-Units	*xsd:anyType*	0:1	Units for Consumption amount associated with COMMERCIAL PROFILE.
	GeneralGroupOf EntitiesRef	*GeneralGroupOf-EntitiesRef*	0:1	GROUP OF ORGANISATIONs or other entities associated with the COMMERCIAL PROFILE.

Table 100 – *CommercialProfile* – XML Element

12.5.16 *Usage parameters: Travel*

Travel usage parameters describe limits on travel.

12.5.17 *RoundTrip*

Properties relating to single or return trip use of an access right.

It can be used to mark a product as single, return, hort trip etc, or to mark the separate VALIDABLE elements of a return journey as outwards, back etc.

If there is a standard discount for a return trip (e.g. 20%), then the discounted price can be attached to the ***ReturnTrip*** parameter as a ***UsageParameterPrice***

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*UsageParameter*	::>	ROUND TRIP inherits from USAGE PARAMETER.
«PK»	***id***	*RoundTripIdType*	1:1	Identifier of ROUND TRIP.
	TripType	*xsd:boolean*	0:1	Whether return trip is allowed. • *single;* • *return; returnOut; returnBack; returnOnly* • *multiple*
	Double-SingleFare	*xsd:boolean*	0:1	Whether fare for return trip is single fare doubled.
	ShortTrip	*xsd:boolean*	0:1	Whether trip is classified as a short trip for fares.
	IsRequired	*xsd:boolean*	0:1	Whether return trip is required.

Table 101 – *RoundTrip* – XML Element

12.5.18FrequencyOfUse

The limits of usage frequency for a FARE PRODUCT (or one of its components) or a SALES OFFER PACKAGE during a specific VALIDITY PERIOD. There may be different tariffs depending on how often the right is consumed during the period.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*UsageParameter*	::>	FREQUENCY OF USE inherits from USAGE PARAMETER.
«PK»	***id***	*FrequencyOfUseIdType*	1:1	Identifier of FREQUENCY OF USE.
«enum»	***Frequency-OfUseType***	*FrequencyOfUseEnum*	0:1	Type of Frequency of Use. See allowed values below. • *none; single; limited; unlimited; twiceADay*
	Minimal-Frequency	*xsd:integer*	0:1	Minimum number of times can be used.
	Maximal-Frequency	*xsd:integer*	0:1	Maximum number of times can be used.
	Frequency-Interval	*xsd:duration*	0:1	Interval within which frequency is measured. If absent forever.
«FK»	***TimeIntervalRef***	*TimeIntervalRef*	0:1	Interval within which frequency is measured. - as reference to arbitrary time interval.
«enum»	***DiscountBasis***	*DiscountBasisEnum*	0:1	Nature of discount for number of journeys. See allowed values below. • *none; free; discount*

Table 102 – *FrequencyOfUse* – XML Element

12.5.19 *Interchanging*

Limitations on making changes within a trip.

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*UsageParameter*	::>	INTERCHANGING inherits from USAGE PARAMETER.
«PK»	***id***	*InterchangingIdType*	1:1	Identifier of INTERCHANGING.
	CanInterchange	*xsd:boolean*	0:1	Whether an interchange can be made.
«enum»	***FromMode***	*VehicleModeEnum*	0:1	TRANSPORT MODE from which user is interchanging. See NeTEx Part1 for allowed values.
«enum»	***ToMode***	*VehicleModeEnum*	0:1	TRANSPORT MODE to which user is interchanging. See NeTEx Part1 for allowed values.
	Maximum-NumberOf-Changes	*xsd:integer*	0:1	Maximum number of transfers that can be made on a trip.
	MaximumTime-ToMakeA-Transfer	*xsd:duration*	0:1	Maximum time allowed to make a transfer.
	CanBreak-Journey	*xsd:boolean*	0:1	Whether the journey can be broken at an interchange point.
	CrossBorder	*xsd:boolean*	0:1	Whether the interchange is across a border.
«enum»	***RegisterBreakOf***	*RegisterBreakOfJourney-*	0:*	Whether the Journey can be interrupted, i.e. leave stop point and return. See allowed values below.

	Journey	*Enum*		+v1.1 • *none* • *markByStaff; markByValidator; markByMobileApp; other*

Table 103 – *Interchanging* – XML Element

12.5.20 *StepLimit*

Geographical parameter limiting the access rights by counts of stops, sections or zones.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*UsageParameter*	::>	STEP LIMIT inherits from USAGE PARAMETER.
«PK»	***id***	*StepLimitIdType*	1:1	Identifier of STEP LIMIT parameter.
	Restricted	*xsd:boolean*	0:1	Whether restricted to a number of stops.
«enum»	***StepUnits***	*StepUnitEnum*	0:`	Units in which steps are counted. • *stops; stopsIncludingPassThroughStops* • *sections; zones; networks; operators* • *countries*
	Minimum-NumberOfSteps	*xsd:integer*	0:1	Minimum number of steps allowed.
	Maximum-NumberOfSteps	*xsd:integer*	0:1	Maximum number of steps allowed.

	Maximum-NumberOfTrips	*xsd:integer*	0:1	Maximum number of trips allowed.

Table 104 – *StepLimit* – XML Element

12.5.21 *UsageValidityPeriod*

A time limitation for validity of a FARE PRODUCT or a SALES OFFER PACKAGE. It may be composed of a standard duration (e.g. 3 days, 1 month) and/or fixed start/end dates and times.

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*UsageParameter*	::>	USAGE VALIDITY PERIOD inherits from USAGE PARAMETER.
«PK»	***id***	*UsageValidityPeriodIdType*	1:1	Identifier of USAGE VALIDITY PERIOD.
«enum»	***ValidityType***	*ValidityTypeEnum*	0:*	Type of USAGE VALIDITY PERIOD. See allowed values below. • *singleRide; singleTrip; returnTrip; carnet* • *dayPass; weeklyPass; weekendPass; monthlyPass; annualPass; seasonTicket* • *profileMembership; subscription; openEnded* • *other*
«enum»	***UsageTrigger***	*UsageTriggerEnum*	0:1	Trigger event that starts validity period. See allowed values below. • *startOfPeriod; startOutboundRide; endOutboundRide; startReturnRide*

				• *enrolment; reservation; purchase; activation; fulfilment* • *specifiedStartDate; dayOffsetBeforeCalendarPeriod*
«enum»	***UsageEnd***	*UsageEndEnum*	0:1	Classification of when the end of the Usage validity period occurs. May be a specified period (Standard Duration) or an event, e.g. end of trip. See allowed values below. • *standardDuration;* • *endOfCalendarPeriod; endOfRide; endOfTrip; endOfFareDay; endOfFarePeriod; productExpiry; deregistration; profileExpiry; other;*
	Standard-Duration	*xsd:duration*	0:1	Duration of VALIDITY PERIOD after departure. or validation
«enum»	***ActivationMeans***	*ActivationMeansEnum*	0:1	Means of activatiing start of period. See allowed values below. +v1.1 • *noneRequired; checkIn; useOfValidator; useOfMobileDevice; automaticByTime; automaticByProximity; other;*
	StartDate	*xsd:date*	0:1	Start date for VALIDITY PERIOD.
	StartTime	*xsd:time*	0:1	Start time for VALIDITY PERIOD.
	EndDate	*xsd:date*	0:1	End date for VALIDITY PERIOD.
	EndTime	*xsd:time*	0:1	End time for VALIDITY PERIOD.
XGRP	***UsageValidity-PeriodStart-***	***xmlGroup***	0:1	Elements controlling the allowed start of period.

	ConstraintGroup			See below.
«enum»	***UsageStart-ConstraintType***	*UsageStartConstraintEnum*	0:1	Whether start type of trip or pass is variable or fixed. See allowed values below. +v1.1 • *variable; fixed; fixedWindow* • *noTravelWithinTimeband*
«cntd»	***startOnlyOn***	*DayType*	0:*	If ***UsageStartConstraintType*** is "fixedWindow", Dayes of week or month (specified as a DayType) on which usage of a period pass may start.
	Maximum-ServicesBefore	*xsd:nonNegativeInteger*	0:1	If ***UsageStartConstraintType*** is "fixedWindow", maximum number of services before the booked train that may also be used. +v1.1
	FlexiblePeriod-Before	*xsd:duration*	0:1	If UsageStartConstraintType is "fixedWindow", maximum period before the booked train during which other trains may also be caught. +v1.1
	Maximum-ServicesAfter	*xsd:nonNegativeInteger*	0:1	If UsageStartConstraintType is "fixedWindow", maximum number of services after the booked train that may also be used. +v1.1
	FlexiblePeriod-After	*xsd:duration*	0:1	If UsageStartConstraintType is "fixedWindow", maximum period after the booked train during which other trains may also be caught. +v1.1
«enum»	***BlackoutUse***	*BlackoutStartEnum*	0:1	When start of travel restriction applies. See allowed values below. • *noTravelWithinPeriod* • *mayCompleteIfStartedBefore* • *noTravelWithinTimeband*

Table 105 – *UsageValidityPeriod* – XML Element

12.5.22 *Routing [ADVANCED PROFILE ONLY]*

Limitations on routing of an access right.

Classifi-cation	Name	Type	Cardin-ality	Description
::>	::>	*UsageParameter*	::>	ROUTING inherits from USAGE PARAMETER.
«PK»	***id***	*RoutingIdType*	1:1	Identifier of ROUTING.
	Return-RouteIdentical	*xsd:boolean*	0:1	Whether return route must be same as outbound route.
	ForwardsOnly	*xsd:boolean*	0:1	Whether passenger may only take routes that proceed in a single direction. (They may not use product to achieve a return trip for the cost of a single trip).
	IsRestricted	*xsd:boolean*	0:1	Whether only allowed on certain routes or series.
	CrossBorder	*xsd:boolean*	0:1	Whether the routing is across a border.

Table 106 – *Routing* – XML Element

12.5.23 *Usage parameters: Entitlement*

Entitlement usage parameters describe prerequisite and dependent products.

12.5.24 *EntitlementRequired [ADVANCED PROFILE ONLY]*

Receiving of entitlement from another FARE PRODUCT.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*UsageParameter*	::>	ENTITLEMENT REQUIRED inherits from USAGE PARAMETER.
«PK»	***id***	*EntitlementRequired-IdType*	1:1	Identifier of ENTITLEMENT REQUIRED.
«FK»	***ServiceAccess-RightRef***	*ServiceAccessRightRef*	0:1	Entitlement comes from the referenced FARE PRODUCT.
	Minimum-Qualification-Period	*xsd:duration*	0:1	Minimum period that required product must be held in order to be eligible.
	Entitlement-Constraint	*EntitlementConstraint*	0:1	Constraints on related product or offer. +v1.1

Table 107 – *EntitlementRequired* – XML Element

12.5.25 *EntitlementGiven [ADVANCED PROFILE ONLY]*

Granting of entitlement to another FARE PRODUCT.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*UsageParameter*	::>	ENTITLEMENT GIVEN inherits from USAGE PARAMETER.
«PK»	***id***	*EntitlementGivenIdType*	1:1	Identifier of ENTITLEMENT GIVEN.
«FK»	***ServiceAccess-RightRef***	*ServiceAccessRightRef*	0:1	Entitlement comes from the referenced FARE PRODUCT.
	Minimum-Qualification-Period	*xsd:duration*	0:1	Minimum period that product must be held for entitlement to be granted.
	Entitlement-Constraint	*EntitlementConstraint*	0:1	Constraints on related product or offer. +v1.1
«enum»	***EntitlementType***	*EntitlementTypeEnum*	0:1	Type of entitlement. See allowed values below. • *use; purchase; none*

Table 108 – *EntitlementGiven* – XML Element

12.5.26 *LuggageAllowance [ADVANCED PROFILE ONLY]*

The number and characteristics (weight, volume) of luggage that a holder of an access right is entitled to carry.

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*UsageParameter*	::>	LUGGAGE ALLOWANCE inherits from USAGE PARAMETER.
«PK»	***id***	*LuggageAllowance-IdType*	1:1	Identifier of LUGGAGE ALLOWANCE.
«enum»	***BaggageUseType***	*BaggageUseEnum*	0:1	Use of baggage covered by the allowance. See allowed values below. • *carryOn; checkIn; oversizeCheckIn*
«enum»	***BaggageType***	*LuggageUseEnum*	0:1	Type of baggage covered by the allowance. See allowed values below. • *handbag; handLuggage; smallSuitcase; suitcase; trunk; oversizeItem; bicycle; sportingEquipment; skis; musicalInstrument; pushChair; motorizedWheelchair; largeMotorizedWheelchair; wheelchair; smallAnimal; animal; game; motorcycle; other;*
«enum»	***Luggage-AllowanceType***	*LuggageAllowance-Enum*	0:1	Classification of allowance type. See allowed values below. • *none; unlimited; single; limited*
	Maximum-NumberOfItems	*xsd:nonNegativeInteger*	0:1	Number of bags allowed.

	MaximumBagHeight	*LengthType*	0:1	Maximum bag height.
	MaximumBagWidth	*LengthType*	0:1	Maximum bag width.
	MaximumBagDepth	*LengthType*	0:1	Maximum bag depth.
	MaximumBagWeight	*WeightType*	0:1	Maximum bag weight.
	TotalWeight	*WeightType*	0:1	Total Weight limit of LUGGAGE ALLOWANCE.
«enum»	***Luggage-ChargingBasis***	*LuggageCharging-BasisEnum*	0:1	Basis on which luggage is charged. See allowed values below. • *free; chargedByItem; chargedByWeight; other*

Table 109 – *LuggageAllowance* – XML Element

12.6 Usage parameters: Booking

12.6.1 *Transferability*

The number and characteristics of persons entitled to use the public transport service instead of the original customer.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*UsageParameter*	::>	TRANSFERABILITY inherits from USAGE PARAMETER.
«PK»	***id***	*TransferabilityIdType*	1:1	Identifier of TRANSFERABILITY.
	CanTransfer	*xsd:boolean*	0:1	Whether ticket can be transferred to someone else.
	Maximum-NumberOf-Named-Transferees	*NumberOfPassengers*	0:1	Where a product can be used by a limited number of named users, maximum number of users allowed.
	HasTransferFee	*xsd:boolean*	0:1	Whether there is a charge for making a transfer.
«enum»	***SharedUsage***	*SharedUsageEnum*	0:1	Indicates the nature of the permitted sharing, if any, of products that can be shared, e.g. Trips from a multi-trip carnet. See allowed values +v1.1 • *singleUser* • *concurrent Users* • *concurrentDesignatedUsers*

Table 110 – *Transferability* – XML Element

12.6.2 *PurchaseWindow*

Period in which the product must be purchased.

Classifi-cation	Name	Type	Cardin-ality	Description
::>	::>	*UsageParameter*	::>	PURCHASE WINDOW inherits from USAGE PARAMETER.
«PK»	***id***	*PurchaseWindowIdType*	1:1	Identifier of PURCHASE WINDOW.
«enum»	***PurchaseAction***	*PurchaseActionEnum*	0:1	Action governed by Purchase Window. The default value is '*purchase*'. See allowed values below. +v1.1 • *purchase; orderWithoutPayment; reserve; payForPreviousOrder; subscribe; payInstallment; other*
«enum»	***PurchaseWhen***	*PurchaseWhenEnum*	0:1	When purchase may be made. See Part1 for allowed values. • *advanceOnly; untilPreviousDay; dayOfTravelOnly; advanceAndDayOfTravel; timeOfTravelOnly; subscriptionChargeMoment; other*
	LatestTime	*xsd:duration*	0:1	Latest time on specified last day when ticket can be purchased.
	MinimumPeriod-BeforeDeparture	*xsd:duration*	0:1	Minimum duration before departure that ticket may be purchased.
«FK»	***MinimumPeriod-***	*TimeIntervalRef*	0:1	Minimum period before departure that purchase

	IntervalRef			must be made - as arbitrary interval.
	MaximumPeriod-BeforeDeparture	*xsd:duration*	0:1	Maximum duration before departure that ticket may be purchased.
«FK»	***MaximumPeriod-IntervalRef***	*TimeIntervalRef*	0:1	Maximum period before departure that purchase must be made - as arbitrary interval.
«enum»	***PurchaseMoment***	*PurchaseMomentEnum*	0:1	Permitted moments of purchase. See Part1 for allowed values. +v1.1

Table 111 – *PurchaseWindow* – XML Element

12.6.3 *Reserving [ADVANCED PROFILE ONLY]*

Indicating whether the access right requires reservation and any limitations on making and changing reservations.

Classification	Name	Type	Cardinality	Description
::>	::>	*UsageParameter*	::>	RESERVING inherits from USAGE PARAMETER.
«PK»	***id***	*ReservingIdType*	1:1	Identifier of RESERVING.
«enum»	***Reserving-Requirements***	*ServiceReservation-FacilityEnum*	0:*	Nature of reservations required. See NeTEx Part1 for allowed values.
	Minimum-NumberToReserve	*NumberOfPassengers*	0:1	Minimum number of persons allowed on a reservation.
	MaximumNumber-ToReserve	*NumberOfPassengers*	0:1	Minimum number of persons allowed on a reservation.
	MustReserve-Whole-Compartment.	*xsd:boolean*	0:1	Whether a whole compartment must be reserved.
«enum»	***Reservation-ChargeType***	*Reservation-ChargeTypeEnum*	0:1	Nature of reservation fee. See allowed values below • *noFee; fee; singleFeeForReturnTrip; feeForEachDirection; feeForEachLeg*
«enum»	***FeeBasis***	*PerBasisEnum*	0:1	Basis on which refund is made. See allowed values below. • *perOffer; perPerson*

	HasFree-Connecting-Reservations	*xsd:boolean*	0:1	Whether connecting reservations are all free or not.
	NumberOfFree-Connecting-Reservations	*xsd:integer*	0:1	Number of free connecting reservations allowed.
	IsFeeRefundable	*xsd:boolean*	0:1	Whether reservation fees is refundable. +v1.1
«cntd»	***Booking-Arrangements***	*BookingArrangements*	0:1	Booking arrangements. See Part1 Service Restrictions Model.
«enum»	***SeatAllocation-Method***	*SeatAllocationMethod-Enum*	0:1	Method for allocating seat. See allowed values. • *autoAssignment*; *seatMap*; *openSeating*
	Reservation-ExpiryPeriod	*xsd:duration*	0:1	Period after which reservation without payment will expire if not paid for. +v1.1

Table 112 – *Reserving* – XML Element

12.6.4 *Cancelling [ADVANCED PROFILE ONLY]*

Requirements for cancelling a booking.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*UsageParameter*	::>	CANCELLING inherits from USAGE PARAMETER.
«PK»	***id***	*CancellingIdType*	1:1	Identifier of CANCELLING element.

	Booking-Arrangements	*BookingArrangements*	0:1	Arrangements for cancelling a booking. See Part1 Service restrictions Model

Table 113 – *Cancelling* – XML Element

12.6.5 *BookingArrangements [ADVANCED PROFILE ONLY]*

Information about booking to make a cancellation or other change. See also Part1 for details.

Classification	Name	Type	Cardinality	Description
	BookingContact	*Contact*	0:1	Contact for Booking.
«enum»	***BookingMethods***	*BookingMethodEnum*	0:*	Booking method for FLEXIBLE LINE.
«enum»	***BookingAccess***	*BookingAccessEnum*	0:1	Who can make a booking. See Part1.
«enum»	***BookWhen***	*PurchaseWhenEnum*	0:1	When Booking can be made. See Part1
«enum»	***BuyWhen***	*PurchaseMomentEnum*	0:*	When purchase can be made. See Part1.
	LatestBooking-Time	*xsd:time*	0:1	Latest time in day that booking can be made.
	MinimumBooking Period	*xsd:duration*	0:1	Minimum interval in advance of departure day or time that service may be ordered.
	BookingUrl	*xsd:anyURI*	0:1	URL for booking.
	BookingNote	*MultilingualString*	0:1	Note about booking the FLEXIBLE LINE.

Table 114 – *BookingArrangements* Group– XML Element

12.7 Usage parameters: After-sales

12.7.1 *Reselling (Abstract)]*

Common resale conditions (i.e. for exchange or refund) attaching to the product.

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*UsageParameter*	::>	RESELLING inherits from USAGE PARAMETER.
«PK»	***id***	*ResellingIdType*	1:1	Identifier of RESELLING.
«enum»	***Allowed***	*ResellTypeEnum*	0:1	Whether exchange or refund is allowed. See allowed values be • *none; partial; fulll.*
	CanChangeClass	*xsd:boolean*	0:1	Whether user can change class. ;
	UnusedTicketsOnly	*xsd:boolean*	0:1	Whether it is possible to exchange partially used tickets.
	OnlyAtCertain-DistributionPoints	*xsd:boolean*	0:1	Whether distribution is restricted to certain points.
«enum»	***ResellWhen***	*ResellWhenEnum*	0:1	Event marking when the is exchangeable status of the ticket changes. See allowed values below. • *never; withinPurchaseGracePeriod; beforeStartOfValidityPeriod; afterStartOfValidityParіod; afterEndOfValidityParіod; beforeFirstUse; afterFirstUse; beforeValidation; afterValidation*

			CHOICE		From when refund/exchange can be made
	a	***Exchangeable-FromAnyTime***	*EmptyType*	0:1	Can be exchanged or refunded from any point after purchase.
	b	***Exchangable-FromDuration***	*xsd:duration*	0:1	Duration to start of period before (negative) or after (positive) the trigger point, (i.e. either Start Of Validity or First Use) after which ticket may be exchanged or refunded.
	c	***Exchangable-FromPercentUse***	*xsd:decimal*	0:1	Can be exchanged once a certain percentage of duration or use has been achieved. +v1.1
«FK»	**Exchangeable-FromIntervalRef**		*TimeIntervalRef*	0:1	TimeInterval determining period from which exchange can be made relative to trigger point.
			CHOICE		Until when refund/exchange can be made
	a	***Exchangeable-UntilAnyTime***	*EmptyType*	0:1	Can be exchanged or refunded up until any point after purchase.
		Exchangable-UntilDuration	*xsd:duration*	0:1	Duration to end of period before (negative) or after (positive) the trigger point (i.e. either Start Of Validity or First Use) after which ticket may be exchanged or refunded.
		Exchangable-UntilPercentUse	*xsd:decimal*	0:1	Can be exchanged until a certain percentage of duration or use has been achieved. +v1.1
«FK»	**Exchangeable-UntilIntervalRef**		*TimeIntervalRef*	0:1	TimeInterval determining period up until which exchange can be made relative to trigger point.
«enum»	**EffectiveFrom**		*EffectiveFromEnum*		Constraint on when change can be made see allowed values. +v1.1

				• *never; nextInterval; nextInstallment; anyTime; other*
	NotificationPeriod	*xsd:duration*	0:1	Notice period needed before action is effective. +v1.1
	HasFee	*xsd:boolean*	0:1	Whether these is a fee for a refund or exchange.
«enum»	***RefundBasis***	*PerBasisEnum*	0:1	Basis on which refund is made. See allowed values below.
«enum»	***PaymentMethods***	*PaymentMethodEnum*	0:*	PAYMENT METHODs that may be used for transaction. See Part1 RC service Restriction model. +v1.1
«ctd»	***TypeOfPaymentMethod***	*TypeOfPaymentMethod*	0:*	PAYMENT METHODs that may be used for transaction. +v1.1

Table 115 – *Reselling* – XML Element

12.7.2 *Exchanging [ADVANCED PROFILE ONLY]*

Whether and how access rights may be exchanged for other access rights.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*Reselling*	::>	EXCHANGING inherits from RESELLING.
«PK»	***id***	*ExchangingIdType*	1:1	Identifier of EXCHANGING.
	NumberOf-Exchanges-Allowed	*xsd:integer*	0:1	Number of times a ticket may be exchanged.

«enum»	***ToFareClass***	*FareClassEnum*	0:1	Fare class to which can be exchanged. See NeTEx Part1. (From class would be expression as the Seat class on an ACCESS RIGHT PARAMETER ASSIGNMENT.)
«FK»	***ToClass-OfUseRef***	*ClassOfUseRef*	0:1	CLASS OF USE class to which can be exchanged.
«enum»	***ExchangableTo***	*ExchangableToEnum*	0:1	Type of exchange allowed. The default is '*anyProduct*', i.e. to any other fare. See allowed values below. • *anyProduct;* • *sameProductSameDay; sameProductLongerJourney; sameProductShorterJourney;sameProductAnyDay* • *upgradeToStandardFare; upgradeToSpecifiedFare; downgradeToSpecifiedFare; equivalentProduct* • *changeGroupSize; other*

Table 116 – *Exchanging* – XML Element

12.7.3 *Refunding [ADVANCED PROFILE ONLY]*

Whether and how the product may be refunded.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*Reselling*	::>	REFUNDING inherits from RESELLING.
«PK»	***id***	*RefundingIdType*	1:1	Identifier of REFUNDING.

«enum»	***RefundType***	*RefundTypeEnum*	0:1	Classification of REFUNDING. See allowed values below. • *unused; delay; cancellation* • *partialJourney; earlyTermination* • *changeOfGroupSize; other*
«enum»	***RefundPolicy***	*RefundPolicyEnum*	0:*	Reasons for giving refunds. See allowed values. +v1.1 • *any; illness; death; maternity; redundancy; changeOfEmployment; changeOfResidence ;* • *none; other*
«enum»	***PartialRefund-Basis***	*PartialRefundBasisEnum*	0:*	Basis on which partial refunds of period passes etc are calculated. See allowed values. +v1.1 • *unusedDays; unusedWeeks; unusedMonths; unusedSemesters; other*
«enum»	***PaymentMethod***	*PaymentMethodEnum*	0:*	DEPRECATED – Use ***PaymentMethods*** on RESELLING higher in hierarchy

Table 117 – *Refunding* – XML Element

12.7.4 *Replacing [ADVANCED PROFILE ONLY]*

Whether and how access rights may be replaced if lost or stolen.

Classifi-cation	Name	Type	Cardinality	Description

::>	::>	*Reselling*	::>	REPLACING inherits from RESELLING.
«PK»	***id***	*ReplacingIdType*	1:1	Identifier of REPLACING.

Table 118 – *Replacing* – XML Element

12.7.5 *Suspending [ADVANCED PROFILE ONLY]*

Conditions governing temporary suspension of a FARE PRODUCT, (i.e. period pass or subscription).

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*UsageParameter*	::>	SUSPENDING inherits from USAGE PARAMETER.
«PK»	***id***	*SuspendingIdType*	1:1	Identifier of USAGE VALIDITY PERIOD.
«enum»	***SuspensionPolicy***	*SuspensionPolicyEnum*	0:*	Allowed policies for suspending term of product. • *none* • *forCertifiedIllness* • *forParentalLeave* • *forHoliday* • *forAnyReason* • *weeklyPass* • *weekendPass*

				• *monthlyPass* • *seasonTicket* • *profileMembership* • *openEnded* • *other*
	QualificationPeriod	*duration*	0:1	Minimum duration that must have occurred before a suspension is allowed.
	QualificationPercent	*decimal*	0:1	Minimum proportion of term that must have occurred before a suspension is allowed.
	MinimumSuspension-Period	*duration*	0:1	Minimum duration allowed for a suspension.
	Maximum-SuspensionPeriod	*duration*	0:1	Maximum duration allowed for a suspension.
	MaximumNumberOf-SuspensionsPerTerm	*nonNegativeInteger*	0:1	Maximum duration allowed for a suspension. with the term of the fare product or subscription.

Table 119 – *Suspending* – XML Element

12.8 Usage parameters: Charging

12.8.1 *ChargingPolicy [ADVANCED PROFILE ONLY]*

Policy regarding different aspects of charging such as credit limits.

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*UsageParameter*	::>	CHARGING POLICY inherits from USAGE PARAMETER.
«PK»	***id***	*ChargingPolicyIdType*	1:1	Identifier of CHARGING POLICY.
«enum»	***CreditPolicy***	*CreditPolicyEnumeration*	0:1	Policy for traveling on credit – See allowed values below. • *allowTravel; blockPayAsYouGoTravel; blockAllTravel; other*
"	***ExpireAfter-Period***	*xsd:duration*	0:1	Any expiry period on collecting a rebate or adjustment.
	PaymentGrace-Period	*xsd:duration*	0:1	Period after purchase by which time payment must be settled. +v1.1
«enum»	***BillingPolicy***	*TravelBillingPolicy-Enumeration*	0:1	Policy for billing frequency – See Allowed values below. +v1.1 • *billAsYouGo; billOnThreshold; billAtFareDayEnd; billAtPeriodEnd*

Table 120 – *ChargingPolicy* – XML Element

12.8.2 *PenaltyPolicy [ADVANCED PROFILE ONLY]*

Policy regarding different aspects of penalty charges, for example repeated entry at the same station, no ticket etc.

Classifi-cation	Name	Type	Cardin-ality	Description

::>	::>	*UsageParameter*	::>	PENALTY POLICY inherits from USAGE PARAMETER.
«PK»	***id***	*PenaltyPolicyIdType*	1:1	Identifier of PENALTY POLICY.
«enum»	***PenaltyPolicyType***	*PenaltyPolicyEnum*	0:1	Classification of Penalty Policy. See below. • *noTicket; noCheckIn; noCheckOut; noValidation; other*
«enum»	***SameStation-EntryPolicy***	*SameStation-EntryPolicyEnum*	0:1	Policy for allowing re-entry at the same station within a certain time. See below. • *blocked; newFare; maximumFare; allowed*
	MinimumTime-BeforeRentry	*xsd:duration*	0:1	Minimum time before can re-enter at the same station before incurring penalty.
	Maximum-NumberOfFail-ToCheckOutEvents	*xsd:duration*	0:1	Limit on the number of fail-to-checkout events allowed before suspension. +v1.1

Table 121 – PenaltyPolicy – XML Element

12.8.3 *Subscribing [ADVANCED PROFILE ONLY]*

Parameters governing subscription to a product allowing payment at regular intervals.

Classifi-cation	Name	Type	Cardi nality	Description
::>	::>	*UsageParameter*	::>	SUBSCRIBING inherits from USAGE PARAMETER.
«PK»	***id***	*SubscribingIdType*	1:1	Identifier of SUBSCRIBING.
«enum»	***Subscription-TermType***	*SubscriptionTermTypeEnum*	0:1	Types of subscription term allowed. See allowed values below. • *fixed; variable; openEnded*
	Minimum-Subscription-Period	*duration*	0:1	Minimum duration allowed for a subscription.
	Maximum-Subscription-Period	*duration*	0:1	Maximum duration allowed for a subscription.
«enum»	***Subscription-RenewalPolicy***	*SubscriptionRenewalPolicy-Enum*	0:1	Policy on renewing subscription. See allowed values below. • *automatic; manual; automaticOnConfirmation; none; other*

«cntd»	***possible-Installment-Intervals***	*TimeIntervalRef*	0:*	Allowed billing Intervals for payment in installment.r
«enum»	***Installment-PaymentMethods***	*PaymentMethodsEnum*	0:1	Allowed means of payment of installations as standard value. See allowed values.
«cntd»	***installment-PaymentMethods***	*TypeOfPaymentMethod*	0:*	Allowed means of payment of installations as TYPE OF PAYMENT METHOD.

Table 122 - Subscribing

13 EFIP: Sales Description

The Sale Description model describes how FareProducts are made available for sale.

13.1 Ticket Media

The TYPE OF TRAVEL DOCUMENT MODEL indicates the available materialisations of products as tickets on media.

The TRAVEL DOCUMENT entity describes an individual physical support satisfying a TYPE OF TRAVEL DOCUMENT, which may be loaded with various contents: sold FARE PRODUCTs, results of VALIDATION ENTRies, CUSTOMER identification, etc.

TRAVEL DOCUMENTs are usually allocated to customers on the occasion of a SALE TRANSACTION.

TRAVEL DOCUMENTs are classified by a TYPE OF TRAVEL DOCUMENT, which expresses:

their general characteristics (type of medium, types of compatible fare products, etc.);

their local functional characteristics, specific to the operator or the authority (specific fare products stored on this type, type of retailer, etc.).

The classical general TYPEs OF TRAVEL DOCUMENTs include the following:

- single-use throw-away ticket, giving the right to consume only one VALIDABLE ELEMENT (e.g. one trip);
- throw-away ticket unit, for which the access right is granted by using a certain number of throw-away units (generally by punching them together in a validator);
- value card, debited by a certain amount for each consumption of VALIDABLE ELEMENTs;
- reloadable electronic purse, allowing access to the PT network; debited by each purchase.
- PT credit card, with post-payment on a central account;
- document attesting the right to benefit from a discount;

etc.

The following figure shows the physical model for TYPE OF TRAVEL DOCUMENTs.

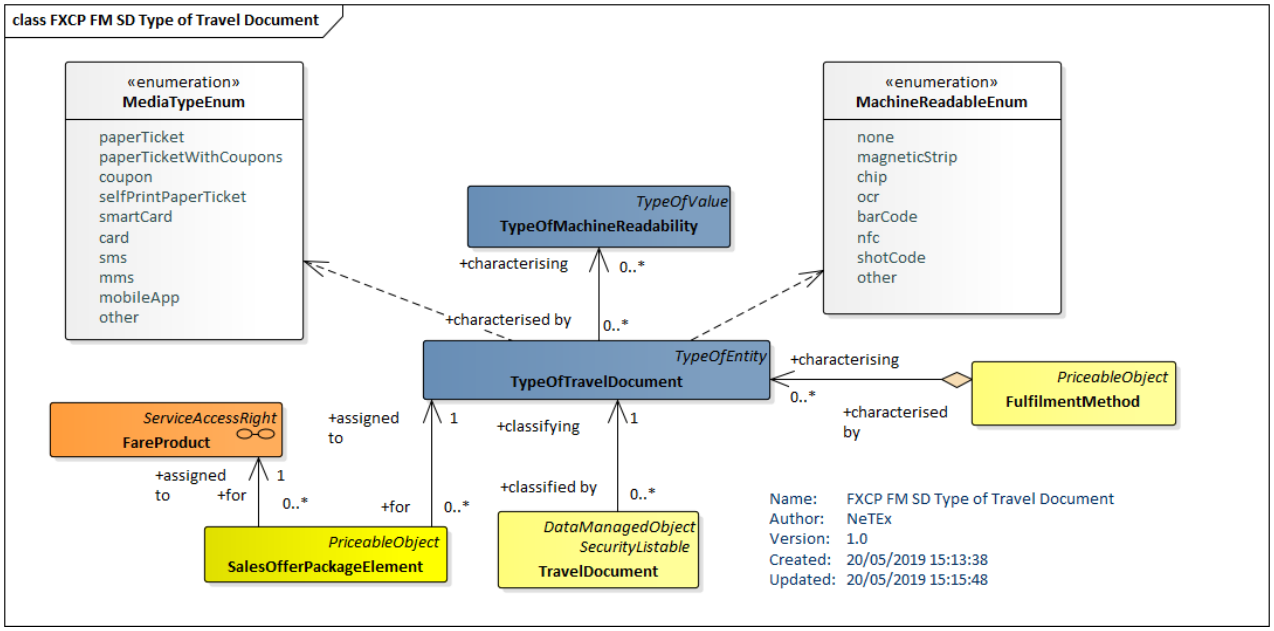


Figure 54 — Type of Travel Document (UML)

13.2 TypeOfTravelDocument

A classification of TRAVEL DOCUMENTs expressing their general function and local functional characteristics specific to the operator. Types of TRAVEL DOCUMENTs like e.g. throw-away ticket, throw-away ticket unit, value card, electronic purse allowing access, public transport credit card, etc. may be used to define these categories.

Table 123 – *TypeOfTravelDocument* – XML Element

Classifi-cation	Name	Type	Cardinality	Description

::>	::>	*TypeOfValue*	::>	TYPE OF TRAVEL DOCUMENT inherits from TYPE OF VALUE. See NeTEx Part1.
«PK»	***id***	*TypeOfTravel-DocumentIdType*	1:1	Identifier of TYPE OF TRAVEL DOCUMENT.
	IsCard	*xsd:boolean*	0:1	Whether the TRAVEL DOCUMENT is materialised as a card.
	IsSmartCard	*xsd:boolean*	0:1	Whether the TRAVEL DOCUMENT is materialised on a smart card or mobile device.
	HasPhoto	*xsd:boolean*	0:1	Whether the TRAVEL DOCUMENT has a photo.
«enum»	***MediaType***	*MediaTypeEnum*	0:1	Classification of the TRAVEL DOCUMENT by Media type. See allowed values below. • *paperTicket; paperTicketWithCoupons; coupon;* • *mobileApp ; selfPrintPaperTicket* • *smartCard; card*
«enum»	***Machine-Readable***	*MachineReadableEnum*	0:1	Classification of the TRAVEL DOCUMENT by Machine Readable mechanism standard values. See allowed values below. • *none* • *magneticStrip; chip; ocr; barCode; shotcode; nfc* • *other*
«cntd»	***typesOfMachine Readabilities***	*TypeOfMachine-ReadabilityRef*	0:*	Classification of the TRAVEL DOCUMENT by TYPE OF MACHINE READABILITY.

«cntd»	***alternative-Names***	*AlternativeName*	0:*	ALTERNATIVE NAMEs for element.

13.3 Sales Offers

A FARE SALES OFFER PACKAGE represents a products marketed to the user and available to purchase.

The FARE PRODUCTs are associated with TYPES OF TRAVEL DOCUMENTs in order to form packages suitable for selling to a customer. A SALES OFFER PACKAGE is defined as a package to be sold as a whole, consisting of one or more SALES OFFER PACKAGE ELEMENTs, each of which associates a specific FARE PRODUCT with a specific TYPE OF TRAVEL DOCUMENT

The user's actual purchase will be described by a TRAVEL SPECIFICATION associated with a CUSTOM CUSTOMER PURCHASE PACKAGE. The TRAVEL SPECIFICATION indicates which specific features of the SALES OFFER PACKAGE have been selected, for example the SALE PACKAGE might include a first class return, first class single, second class single, etc, of which only one of which will be selected in a TRAVEL SPECIFICATION.

The EFIP (is not concerned with sales data (Although it can be representedin NETEX) - nor is the EPIP - so SALES TRASACTIONS, TRAVEL SPECIFICATIONs and CUSTOMER PURCHASE PACKAGEs are all out of scope of the EFIP.

In most cases, a SALES OFFER PACKAGE will only consist of one FARE PRODUCT on one TRAVEL DOCUMENT, but more complex combinations are possible. For instance, a USAGE DISCOUNT RIGHT with its own TRAVEL DOCUMENT may be packaged with an AMOUNT OF PRICE UNIT on an electronic purse.

A SALES OFFER PACKAGE may be further characterised by VALIDITY PARAMETER ASSIGNMENTs.

A SALES OFFER PACKAGE may also indicate the DISTRIBUTION CHANNEL and FULFILMENT METHODs which a customer may use to buy a product. A DISTRIBUTION ASSIGNMENT assign the specific channels to a product.

Quite often there are a number of similar variants of SALES OFFER PACKAGES with many common properties. The GROUP OF SALES OFFER PACKAGEs an be used to specify such common properties: Any property specified on a GROUP OF SALES OFFER PACKAGES applies to all the SALES OFFER PACKAGES in the group.

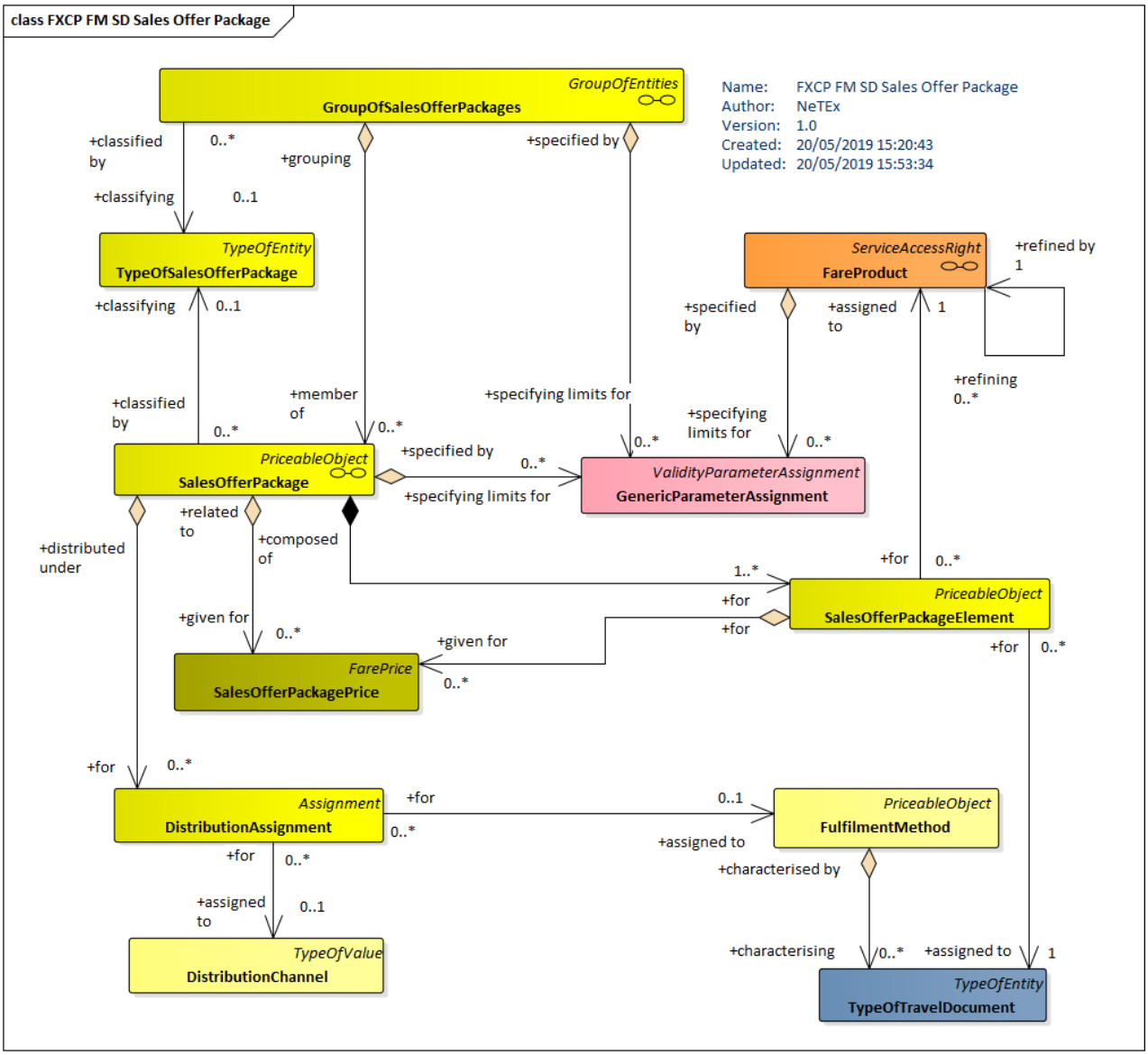


Figure 55 — EFIP - Sales Offer Package & Distribution (UML)

13.3.1 *SalesOfferPackage*

A package to be sold as a whole, consisting of one or several FARE PRODUCTs materialised thanks to one or several TRAVEL DOCUMENTs. The FARE PRODUCTs may be either directly attached to the TRAVEL DOCUMENTs, or may be reloadable on the TRAVEL DOCUMENTs.

Table 124 – *SalesOfferPackage* – XML Element

Classification	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	SALES OFFER PACKAGE inherits from PRICEABLE OBJECT.
«PK»	***id***	*SalesOfferPackageIdType*	1:1	Identifier of a SALES OFFER PACKAGE.
«AK»	***PrivateCode***	*PrivateCodeType*	0:1	Alternative identifier of an entity. can be used to associate with legacy systems.
«FK»	***TypeOfSales-OfferPackageRef***	*TypeOfSalesOfferPackageRef*	0:1	Type of SALES OFFER PACKAGE.
«cntd»	***Condition-Summary***	*ConditionSummary*	0:1	Summary description of conditions of a SALES OFFER PACKAGE that can be used to provide passenger information.
«cntd»	***validity-Parameter-Assignments***	*GenericAccessRightParameter-Assignment*	0:*	GENERIC PARAMETER ASSIGNMENTs (i.e. ACCESS RIGHT PARAMETER ASSIGNMENTs) associated with the SALES OFFER PACKAGE.
«cntd»	***distribution-Assignments***	*DistributionAssignment*	0:*	DISTRIBUTION ASSIGNMENTs for the SALES OFFER PACKAGE.
«FK»	***RoundingRef***	*RoundingRef*	0:1	Rounding to use on calculation

«cntd»	***prices***	*SalesOfferPackagePrice*	0:*	SALES OFFER PACKAGE PRICEs associated with the FARE
«cntd»	***salesOffer-Package-Elements***	*SalesOfferPackageElement*	0:*	SALES OFFER PACKAGE ELEMENTs associated with the SALES OFFER PACKAGE.
«cntd»	***groupsOfSales-OfferPackages***	*GroupOfSalesOffer-PackagesRef*	0:*	GROUPs OF SALES OFFER PACKAGEs with which this SALES OFFER PACKAGE shares common properties.
«cntd»	***salesOffer-Package-Substitutions***	*SalesOfferPackageSubstitution*	0:*	Allowed SALES OFFER PACKAGE SUBSTITUTIONs for the SALES OFFER PACKAGE.
«FK»	***TypeOfSales-OfferPackageRef***	*TypeOfSalesOfferPackageRef*	0:1	Type of SALES OFFER PACKAGE.
«cntd»	***Condition-Summary***	*ConditionSummary*	0:1	Summary description of conditions of a SALES OFFER PACKAGE that can be used to provide passenger information.
«cntd»	***validity-Parameter-Assignments***	*GenericAccessRightParameter-Assignment*	0:*	GENERIC PARAMETER ASSIGNMENTs (i.e. ACCESS RIGHT PARAMETER ASSIGNMENTs) associated with the SALES OFFER PACKAGE.

13.3.2 *SalesOfferPackageElement (Subcomponent of SalesOfferPackage)*

The assignment of a FARE PRODUCT to a TYPE OF TRAVEL DOCUMENT in order to define a SALES OFFER PACKAGE, realised as a fixed assignment (printing, magnetic storage etc.) or by the possibility for the FARE PRODUCT to be loaded onto the TYPE OF TRAVEL DOCUMENT.

Table 125 – *SalesOfferPackageElement* – XML Element

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*PriceableObject*	::>	SALES OFFER PACKAGE ELEMENT inherits from PRICEABLE OBJECT.
«PK»	***id***	*SalesOfferPackage-ElementIdType*	1:1	Identifier of SALES OFFER PACKAGE ELEMENT.
	Requires-Validation	*xsd:boolean*	0:1	Whether element requires validation before it can be used.
«cntd»	***Condition-Summary***	*ConditionSummary*	0:1	Summary description of SALES OFFER PACKAGE properties.
«FK»	***SalesOffer-PackageRef***	*SalesOfferPackageRef*	1:1	Reference to a SALES OFFER PACKAGE of which this is part. If not given by containing context must be specified.
«FK»	***TypeOfTravel-DocumenRef***	*TypeOfTravel-DocumentRef*	0:1	Reference to a TYPE OF TRAVEL DOCUMENT.
«FK»	***FareProductRef***	*FareProductRef+*	0:1	FARE PRODUCT associated with this SALES OFFER PACKAGE.
«cntd»	***validity-Parameter-Assignments***	*GenericParameter-Assignment*	0:*	GENERIC PARAMETER ASSIGNMENTs associated with the SALES OFFER PACKAGE ELEMENT.
«cntd»	***prices***	*SalesOfferPackagePrice*	0:*	SALES OFFER PACKAGE PRICEs associated with the SALES OFFER PACKAGE ELEMENT.

13.3.3 *GroupOfSalesOfferPackages [ADVANCED PROFILE ONLY]*

Quite often there are a number of variants of SALES OFFER PACKAGES. The GROUP OFf SALES OFFER PACKAGEs.can be used to specify common properties

Table 126 – *GroupOfSalesOfferPackages* – XML Element

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*GroupOfEntities*	::>	GROUP of SALES OFFER PACKAGEs inherits from GROUP OF ENTITIES. See NeTEx Part1.
«PK»	***id***	*GroupOfSalesOffer-PackagesIdType*	1:1	Identifier of GROUP of SALES OFFER PACKAGEs.
«cntd»	***alternative-Names***	*AlternativeName*	0:*	ALTERNATIVE NAMEs for GROUP of SALES OFFER PACKAGEs.
«cntd»	***notice-Assignments***	*NoticeAssignment*	0:*	NOTICE ASSIGNMENTs for GROUP of SALES OFFER PACKAGEs.
«FK»	***PricingService-Ref***	*PricingServiceRef*	0:1	PRICING SERVICE to use to fetch prices dynamically.
«FK»	***PricingRuleRef***	*PricingRuleRef+*	0:1	Default PRICING RULE to use to derive prices from this element.
«cntd»	***priceGroups***	*PriceGroup*	0:*	PRICE GROUPs associated with this element.
«cntd»	***fareTables***	*FareTable*	0:*	FARE TABLEs associated with this element.
«FK»	***TypeOfSales-OfferPackageRef***	*TypeOfSalesOffer-PackageRef*	0:1	Type of SALES OFFER PACKAGE.

«cntd»	***Condition-Summary***	*ConditionSummary*	0:1	Summary description of conditions of a SALES OFFER PACKAGE that can be used to provide passenger information.
«cntd»	***validity-Parameter-Assignments***	*GenericAccess-RightParameter-Assignment*	0:*	GENERIC PARAMETER ASSIGNMENTs (i.e. ACCESS RIGHT PARAMETER ASSIGNMENTs) associated with the SALES OFFER PACKAGE.
«cntd»	***members***	*SalesOfferPackageRef*	0:*	References to members of GROUP of SALES OFFER PACKAGEs. See above.

13.4 Fare Sales Distribution

Passenger information may need to include information on where particular products may be purchased for example over the counter, on-line, from self-service ticket machines, etc., as well as how they may be paid for, and how they may be collected.

SALES OFFER PACKAGEs can be restricted to specific DISTRIBUTION CHANNELs or GROUPs OF DISTRIBUTION CHANNELs using a DISTRIBUTION ASSIGNMENT.

Often only certain combinations of DISTRIBUTION CHANNEL and FULFILMENT METHOD are allowed, e.g. on-board cash paper ticket, or online to a mobile app. Certain FULFILMENT METHODS may have additional prices associated with them.

NeTEx provides number of standardised fixed value for channels and methods

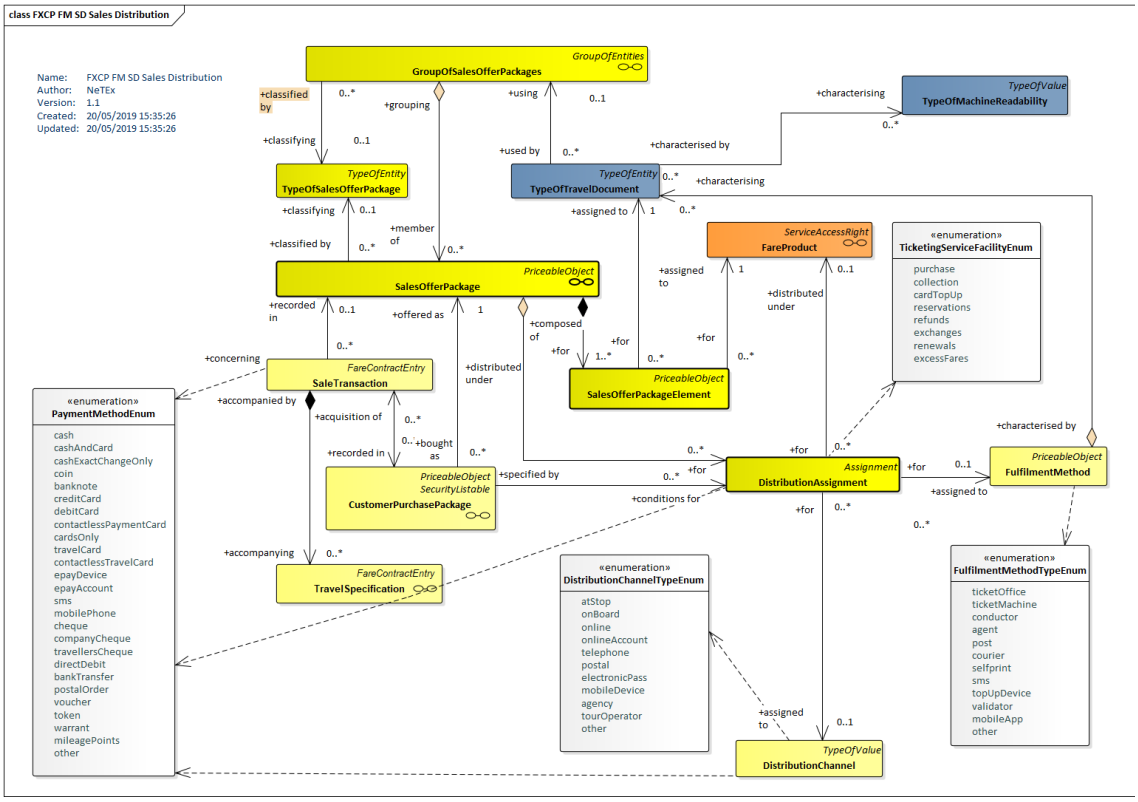

class FXCP FM SD Sales Distribution
Name: FXCP FM SD Sales Distribution
Author: NeTEx
Version: 1.1
Created: 20/05/2019 15:35:26
Updated: 20/05/2019 15:35:26
GroupOfEntities
GroupOfSalesOfferPackages
TypeOfValue
TypeOfMachineReadability
TypeOfEntity
TypeOfSalesOfferPackage
TypeOfEntity
TypeOfTravelDocument
ServiceAccessRight
FareProduct
«enumeration»
TicketingServiceFacilityEnum
purchase
collection
cardTopUp
reservations
refunds
exchanges
renewals
excessFares
PriceableObject
SalesOfferPackage
PriceableObject
SalesOfferPackageElement
FareContractEntry
SaleTransaction
«enumeration»
PaymentMethodEnum
cash
cashAndCard
cashExactChangeOnly
coin
banknote
creditCard
debitCard
contactlessPaymentCard
cardsOnly
travelCard
contactlessTravelCard
epayDevice
epayAccount
sms
mobilePhone
cheque
companyCheque
travellersCheque
directDebit
bankTransfer
postalOrder
voucher
token
warrant
mileagePoints
other
PriceableObject
SecurityListable
CustomerPurchasePackage
Assignment
DistributionAssignment
PriceableObject
FulfilmentMethod
FareContractEntry
TravelSpecification
«enumeration»
DistributionChannelTypeEnum
atStop
onBoard
online
onlineAccount
telephone
postal
electronicPass
mobileDevice
agency
tourOperator
other
«enumeration»
FulfilmentMethodTypeEnum
ticketOffice
ticketMachine
conductor
agent
post
courier
selfprint
sms
topUpDevice
validator
mobileApp
other
TypeOfValue
DistributionChannel
+classified by
+grouping
+using
+characterising
+classifying
+used by
+characterised by
+member of
+assigned to
+recorded in
+offered as
+composed of
+for
+distributed under
+concerning
+accompanied by
+acquisition of
+bought as
+recorded in
+specified by
+conditions for
+accompanying
+characterised by
+assigned to

Figure 56 — Sales Distribution (UML)

13.4.1 *DistributionAssignment Subcomponent of SalesOfferPackage)*

An assignment of the COUNTRY and/or DISTRIBUTION CHANNEL through which a product may or may not be distributed.

Table 127 – *DistributionAssignment* – XML Element

Classifi-cation	Name	Type	Cardinality	Description
::>	::>	*Assignment*	::>	DISTRIBUTION ASSIGNMENT inherits from ASSIGNMENT.
«PK»	***id***	*DistributionAssignment-IdType*	1:1	Identifier of a DISTRIBUTION ASSIGNMENT.
«FK»	***ServiceAccess-RightRef***	*ServiceAccessRightRef*	0:1	SERVICE ACCESS RIGHT (FARE PRODUCT) for which this specifies the DISTRIBUTION ASSIGNMENT.
«FK»	***SalesOffer-PackageRef***	*SalesOfferPackageRef*	0:1	SALES OFFER PACKAGE for which this specifies the DISTRIBUTION ASSIGNMENT.
«FK»	***GroupOfSales-OfferPackages-Ref***	*GroupOfSalesOffer-PackagesRef*	0:1	GROUP OF SALES OFFER PACKAGEs for which this specifies the DISTRIBUTION ASSIGNMENT.
	Distribution-Rights	***xmlGroup***	0:1	Distribution rights associated with this DISTRIBUTION ASSIGNMENT. See below.
		CHOICE		Country in which distribution may take place.
	a ***AllCountries***	*AllCountriesRef*	0:1	Distribution may be made in all countries.

		Ref			
«FK»	***b***	***CountryRef***	*CountryRef*	0:1	COUNTRY in which distribution can be made as described by this DISTRIBUTION ASSIGNMENT.
	AllowedIn-Country		*xsd:boolean*	0:1	Whether distribution is allowed or forbidden in the specified country.
«FK»	***Topographic-PlaceRef***		*TopographicPlaceRef*	0:1	TOPOGRAPHIC PLACE for which this specifies the DISTRIBUTION ASSIGNMENT.
			CHOICE		Channel by which distribution can be made.
	a	***All-Distribution-ChannelsRef***	*AllDistribution-ChannelsRef*	0:1	Distribution may be made through all channels.
«FK»	***b***	***Distribution-ChannelRef***	*DistributionChannelRef*	0:1	DISTRIBUTION CHANNEL for which this specifies the DISTRIBUTION ASSIGNMENT.
«FK»	***c***	***GroupOf-Distribution-ChannelsRef***	*GroupOfDistribution-ChannelsRef*	0:1	GROUP OF DISTRIBUTION CHANNELs for which this specifies the DISTRIBUTION ASSIGNMENT.
«enum»	***Distribution-ChannelType***		*DistributionChannel-TypeEnum*	0:1	Type of DISTRIBUTION CHANNEL. See earlier.
	AllowedIn-Channel		*xsd:boolean*	0:1	Whether distribution is allowed or forbidden by the specified DISTRIBUTION CHANNEL.
	RestrictedTo-Channel		*xsd:boolean*	0:1	Whether distribution is restricted to only the specified DISTRIBUTION CHANNELs.
	Mandatory-		*xsd:boolean*	0:1	Whether product is mandatory.

	Product				
	InitialCarrier		*xsd:boolean*	0:1	Distribution by carrier of first leg of trip.
	TransitCarrier		*xsd:boolean*	0:1	Distribution by carrier of middle of trip.
	FinalCarrier		*xsd:boolean*	0:1	Distribution by carrier of final leg of trip.
			Choice		Organisation who may distribute.
«FK»	***a***	***All-OrganisationsRef***	*AllOrganisationsRef*	0:1	All ORGANISATIONs may distribute.
«FK»	***b***	***OrganisationRef***	*(OrganisationRef)*	0:1	ORGANISATION for which this specifies the DISTRIBUTION ASSIGNMENT.
«FK»	***Responsibility-SetRef***		*ResponsibilitySetRef*	0:1	RESPONSIBILITY SET describing the DISTRIBUTION ASSIGNMENT.
«enum»	***Ticketing-ServiceFacility***		*TicketingService-FacilityEnum*	0:*	List of TICKETING SERVICE FACILITies, e.g. purchase, collection. top up. See Part1
«enum»	***PaymentMethods***		*PaymentMethodEnum*	0:*	Payment method supported on this distribution. See Part1
	Requires-Registration		*xsd:boolean*	0:1	Whether distribution requires the customer to register a personal identity either online or otherwise.
«FK»	***Fulfilment-MethodRef***		*FulfilmentMethodRef*	0:1	FULFILMENT METHOD to be used with this DISTRIBUTION CONDITION.
	notice-		*NoticeAssignment*	0:*	NOTICE ASSIGNMENTs associated with this

	Assignments			DISTRIBUTION ASSIGNMENT.

13.4.2 *DistributionChannel*

A type of outlet for selling a product.

Table 128 – *DistributionChannel* – XML Element

Classification	Name	Type	Cardinality	Description
::>	::>	*TypeOfValue*	::>	DISTRIBUTION CHANNEL inherits from TYPE OF VALUE. See NeTEx Part1.
«PK»	***id***	*DistributionChannelIdType*	1:1	Identifier of a DISTRIBUTION CHANNEL.
«cntd»	***alternative-Names***	*AlternativeName*	0:*	Alternative names for DISTRIBUTION CHANNEL.
«enum»	***Distribution-ChannelType***	*DistributionChannel-TypeEnum*	0:1	Type of DISTRIBUTION CHANNEL. See below for allowed values. • *onBoard*; *online*; *onlineAccount*; *telephone*; *electronicPass*; *mobileDevice* • *agency*; *tourOperator*; other
	IsObligatory	*xsd:boolean*	0:1	Whether the option to use the channel is obligatory, that is, must be allowed.
	RequiresEmail-Address	*xsd:boolean*	0:1	Whether to use the channel requires an email address.
«FK»	***OrganisationRef***	*(OrganisationRef)*	0:1	ORGANISATION associated with channel.
«enum»	***Payment-***	*PaymentMethodEnum*	0:*	Payment methods supported on this distribution.

	Methods			See NeTEx Part1 for allowed values.
«cntd»	***typesOfPayment Method***	*TypeOfPaymentMethodRef*	0:*	PAYMENT METHOD – open type. +v 1.1
«enum»	***Distribution-Rights***	*DistributionRightsEnum*	0:1	Default distribution rights for the DISTRIBUTION CHANNEL. • none; ;sell; refund; exchange' inform; private; other
«cntd»	***distribution-Points***	*PointRef+*	0:*	Points to which distribution is restricted, if any. For example, that a ticket can only be bought at a specific station.
«FK»	***Distribution-GroupRef***	*GroupOfEntitiesRef*	0:1	GROUP OF ENTITIes, e.g. places, organisations or other entities (E.g. on-board specific journeys or services places) to which distribution is restricted, if any. For example, that a ticket can only be bought at a specific station.

13.4.3 *FulfilmentMethod*

The means by which the ticket is delivered to the Customer. e.g. online, collection, etc.

Table 129 – *FulfilmentMethod* – XML Element

Classifi-cation	Name	Type	Cardin-ality	Description
::>	::>	*PriceableObject*	::>	FULFILMENT METHOD inherits from PRICEABLE OBJECT.
«PK»	***id***	*FulfillmentMethodIdType*	1:1	Identifier of FULFILMENT METHOD.

«enum»	***Fulfilment-MethodType***	*FulfilmentMethod-TypeEnum*	0:1	Type of FULFILMENT METHOD. See allowed values below. • ticketOffice; *ticketMachine; conductor;* • *agent; post; courier* • *selfprint; sms; topUpDevice; mobileApp* • *validator; other*
	RequiresCard	*xsd:boolean*	0:1	Whether collecting ticket requires credit card used to purchase.
	Requires-Booking-Reference	*xsd:boolean*	0:1	Whether collecting ticket requires booking reference.
«cntd»	***typesOf-Document***	*TypeOfTravel-DocumentRef*	0:*	Reference to TYPEs OF TRAVEL DOCUMENT allowed by method.
«cntd»	***prices***	*FulfilmentMethodPrice*	0:*	FULFILMENT METHOD PRICEs associated with the FULFILMENT METHOD.

14 EFIP: Frames for fares

NeTEx uses VERSION FRAMEs (see Part1) as a mechanism for grouping related instances of fare data into a single package for exchange.

The NeTEx schema uses an additional specific Frame – the ***FareFrame*** to hold the elements of a fare.

FARE FRAMEs have additional default value that are not used by other types of Frame.

The EFIPEFP makes three different selections of fare frame elements, nested in a particular way.

14.1 FareFrame contents: EU_FP_FARE_NETWORK

The 'EU_FP_FARE_NETWORK ***FareFrame*** has a subset of data elements for defining the fare related Network Elements.

14.2 FareFrame contents: EU_FP_FARE_PRODUCT

The 'EU_FP_FARE_PRODUCT ***FareFrame*** has a subset of data elements for defining the TARIFF, FARE PRODUCT and SALES OFFER PACKAGE elements.

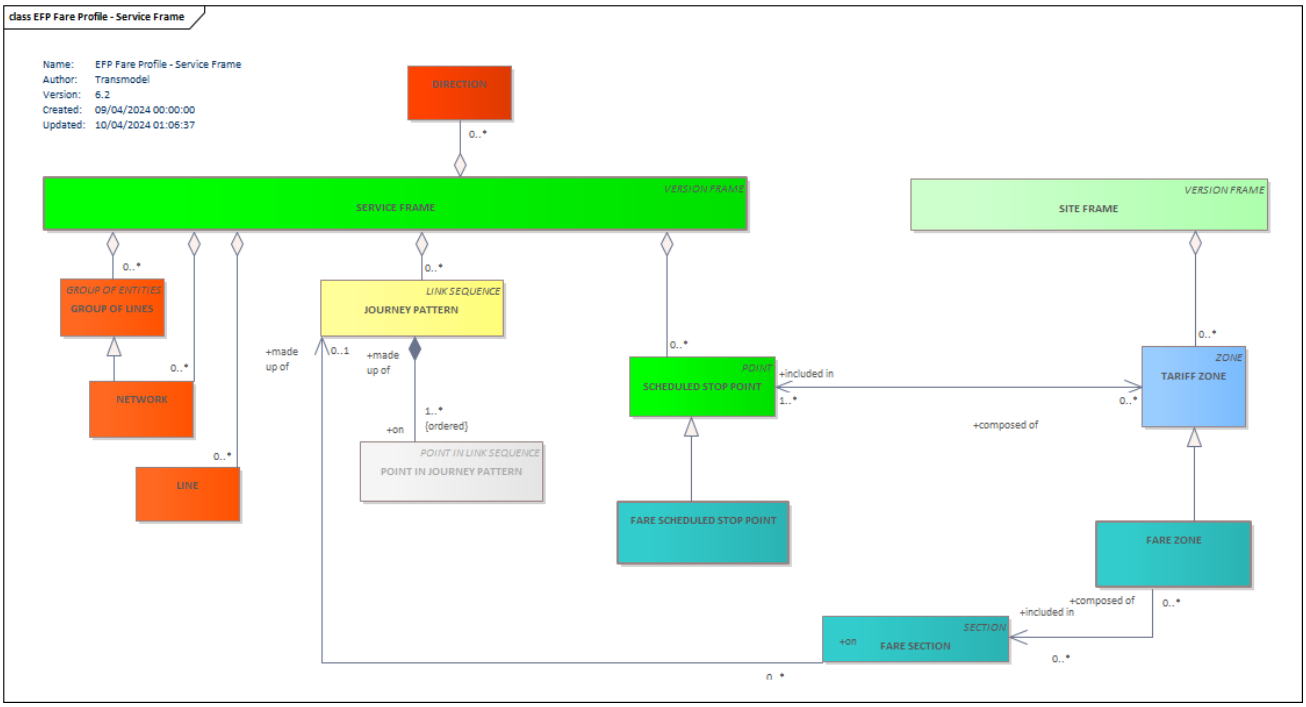


Figure 57 — Fares – Service and Site Frame Contents – EU_FP_FARE_PRODUCT (UML)

14.3 FareFrame contents: EU_FP_FARE_PRICE

The ‘EU_FP_FARE_PRICE ***FareFrame*** has a subset of data elements for defining the fare prices and price parameters

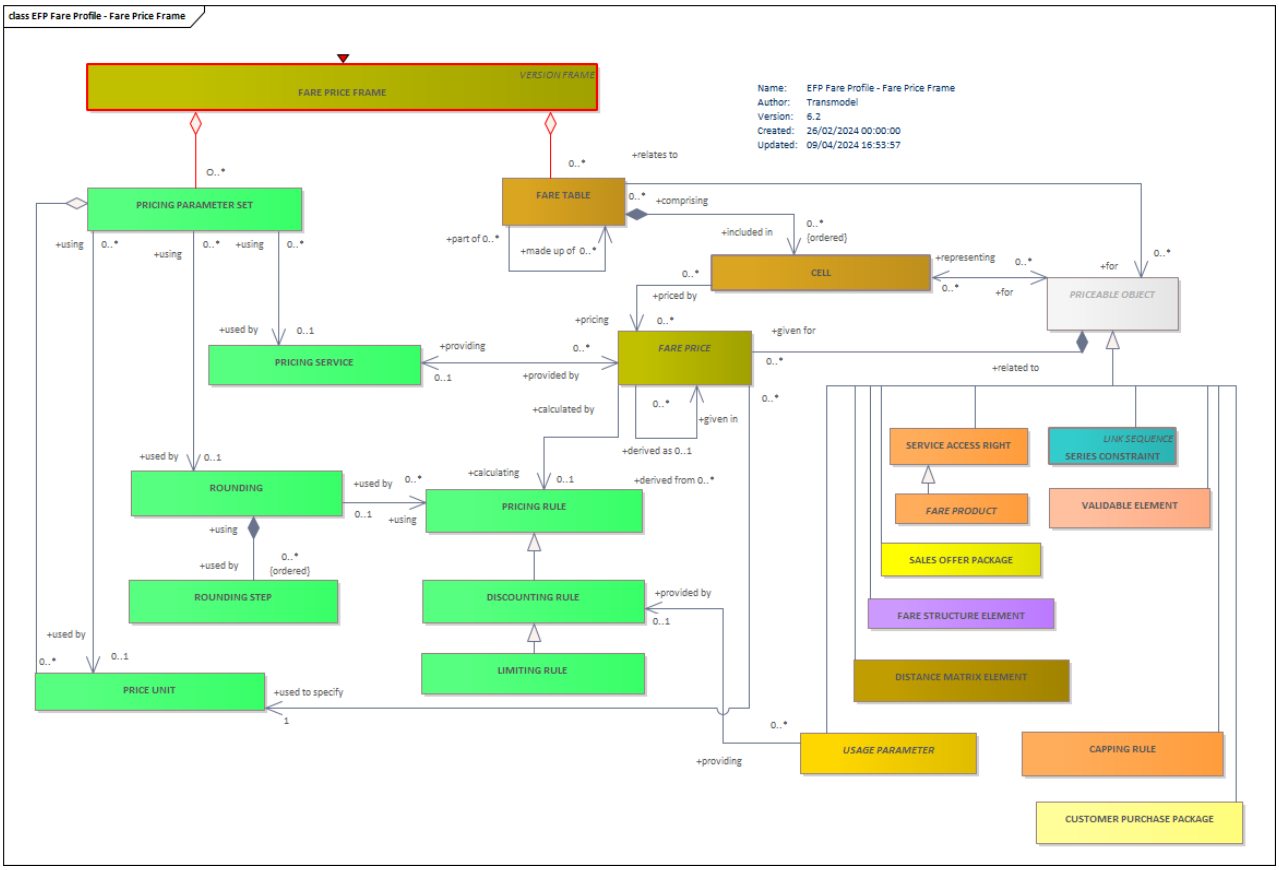


Figure 58 — Fare Frame Contents – EU_FP_FARE_PRICE (UML)

14.4 Nesting of frames

A COMPOSITE FRAME is used to group one or more other FRAME types.

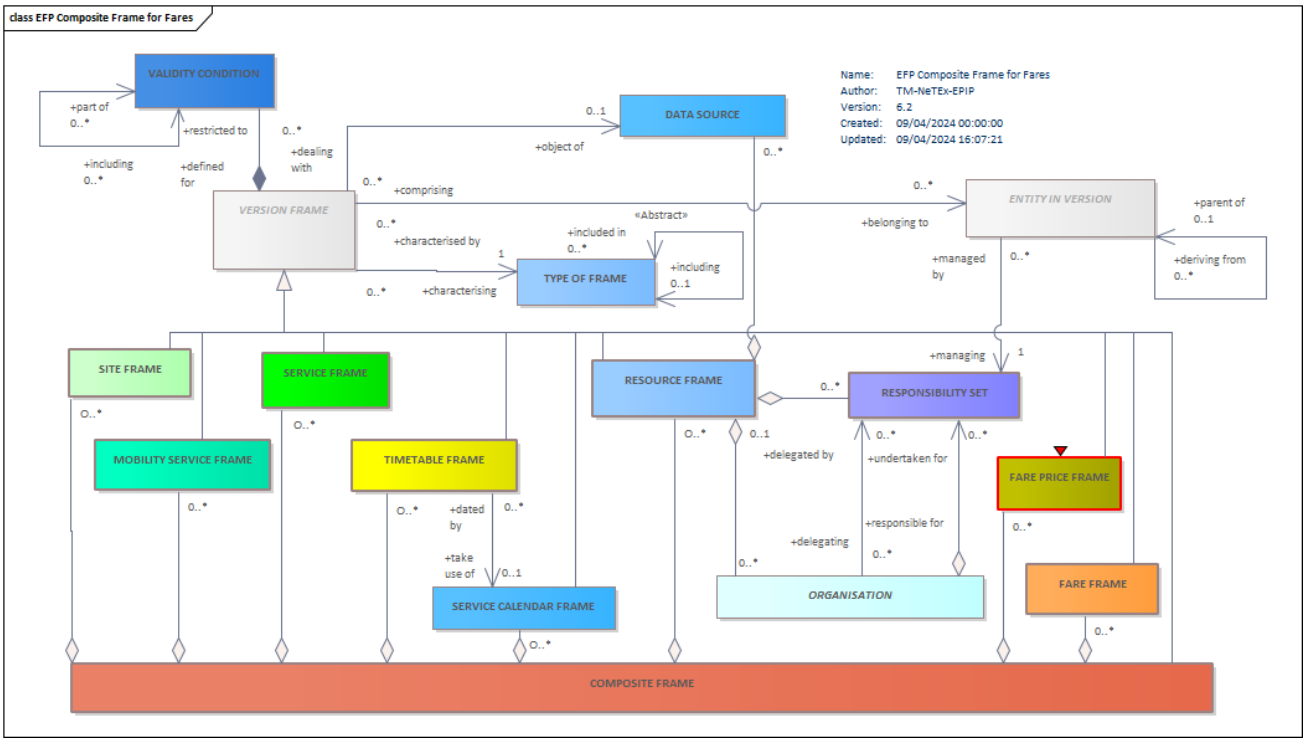


Figure 59 — Fare Frame Contents – EU_FP_FARE_PRICE (UML)

14.5 FareDefaultsGroup — (Subcomponent of FareFrame)

The set of default values for pricing etc., to be used for elements in a frame if not explicitly specified on individual elements.

Table 130 – *FareFrame / FareDefaults* – XML Element

Classifi-cation	Name	Type	Cardinality	Description
«enum»	***Mode***	*VehicleModeEnum*	0:1	Default vehicle MODE for FARE FRAME. See Part1.

«FK»	***Transport-OrganisationRef***	*(TransportOrganisationRef) OperatorRef | AuthorityRef*	0:1	Default TRANSPORT ORGANISATION for FARE FRAME. See Part1.
«FK»	***Pricing-ParameterSet***	*PricingParameterSet*	0:1	PRICING PARAMETER SET associated with FARE FRAME. See later.

15 Common rules for implementing the EFIP

15.1 Rules for Simplifying EU Fares Profile

This section sets out common rules for the fare profile (EFIP), to make the fare offer description as simple and as systematic as possible: NeTEx very often allows to have different ways of expressing the same information but, in order to ease the production and consumption of data, EFIP adds a set of rumes (some mandatory and some recommended) to avoid having too much diversity in the use of NeTEx across data propviders.

Also note that the fare profile follows all the same rules as to versions, identifiers, responsibilities, PublicationDelivery Headers, etc., as in the base EPIP profile – see the EFIP document for details.

The EFIP adds several further TYPEs OF FRAME for different types of ***FareFrame***, see below.

It also sets out rules for the organisation of data within each ***FareFrame***, see below

Table 131 – Summary of simplification rules of the fare profile

Category	Rule	Status
Conditions and Parameters	Specify Conditions and Parameters for FARE STRUCTURE ELEMENTS where possible (rather than for FARE PRODUCTs/ SALES OFFER PACKAGE)	Mandatory
	Specify conditions and Parameters: at FARE PRODUCT level (via VALIDABLE ELEMENT and FARE STRUCTURE ELEMENTS, see rule above), and SALES OFFER PACKAGE level via FARE TABLE only	Mandatory
	Embed VALIDABLE ELEMENTs elements in the FARE PRODUCT that uses them	Mandatory
	Avoid using of complex conditions (AND/OR for the assignment of	Recommended

	parameters)	
	Limit the organisation of data elements to a uniform approach	Recommended
	Require use of TARIFFs to organize access rights and limitations as FARE STRUCTURE ELEMENTs	Mandatory
	Use standard categories of TYPE OF FARE STRUCTURE ELEMENT (access, eligibility duration, conditions_of_travel, conditions_of_use) whenever it holds conditions/parameters (and tag the ID accordingly).	Recommended
Service description	Make sure information for a journey planner is there eg ZONEs, SCHEDULED STOP POINTs, etc.	Recommended
	Use FARE ZONE to specify topology etc	Mandatory
Prices	Separation of prices – Use of FARE TABLEs	Mandatory
	Nest tables to reduce repetition	Recommended
	Use precalculated prices where available (instead the PRICING RULE)	Recommended
	Use PRICING SERVICE for yield managed prices	Mandatory
Document to provide human readable	Provide textual Names and Description for products	Mandatory
	Use TYPE OF FARE STRUCTURE ELEMENT, TYPE OF ASSIGNMENT to set explicit meaning	Recommended

descriptions	Classify FARE PRODUCTs and TYPE OF FARE PRODUCT	Mandatory
	Include SummaryCondition on FARE PRODUCTs and SALES OFFER PACKAGEs	Mandatory
	Include links back to web pages text and graphics	Recommended
	Provide Notices to be printed by tools on the user interface	Recommended

15.2 Rules for simplifying use of parameters assignments and conditions

Parameter assignments (access right, scoping parameters, validity parameters, etc.) can be defined and assigned a pretty much any level of the fare offer model, according to Transmodel and NeTEx. While this is very flexible, it can quickly become an nightmare at implementation time, especially for the consumer who may face very different data structures depending on the providers, and this without any indication about the implementation choices.

Therefore, EFIP defines a set of mandatory rules to make the parameter assignments as systematic as possible. The purpose of these rules is not the optimization nor the strict compliance with the Transmodel rules, but only to facilitate the implementation for data producers and consumers.

> *Rule :* **Specify Conditions and Parameters within FARE STRUCTURE ELEMENTs where possible (rather than FARE PRODUCT/SALES OFFER PACKAGE)** *- Mandatory*
>
> *Rule :* **Specify conditions and Parameters : at FARE PRODUCT level (via VALIDABLE ELEMENT and FARE STRUCTURE ELEMENTS, see rule above), and SALES OFFER PACKAGE levEl via FARE TABLE only** *- Mandatory*

NeTEx allows the the parameter assignments to be hosted in a ***FareStructureElement***, even if this is not the most obvious location for these assignments, it is a very convenient and systematic way to describe them (see the example below). This rule is mandatory.

```
<!--  ==== Conditions of SALE  -================= -->
<FareStructureElement version="1.0" id="mytrain:my_single_trip@conditions_of_sale">
    <Name>Conditions of Sale</Name>
    <TypeOfFareStructureElementRef versionRef="efp:v1.0" ref="efp:conditions_of_sale"/>
    <GenericParameterAssignment version="1.0" id="
mytrain:my_single_trip@conditions_of_sale" order="1">
        <TypeOfAccessRightAssignmentRef versionRef="efp:v1.0"
ref="efp:conditions_of_sale"/>
        <limitations>
            <Refunding version="1.0" id=" mytrain:my_single_trip@conditions_of_sale">
                <Name>NoRefund</Name>
                <Description>Public transport regulations for the Ile-de-France region do
not provide for the reimbursement of unused or partially used transportation tickets
(purchased individually or in packages of 10)..</Description>
                <Url>https://www.ratp.fr/en/i-have-unused-tickets-can-i-have-them-
refunded</Url>
                <Allowed>none</Allowed>
            </Refunding>
        </limitations>
    </GenericParameterAssignment>
</FareStructureElement>
```

These parameter assignments will then be referred by FARE PRODUCT and their VALIDABLE ELEMENT (note that another rule is making mandatory to have the *ValidableElement* embeded inside the *FareProduct*, and not outside with a reference to it), as described below.

```
<ValidableElement version="1.0" id="mytrain:my_single_trip">
    <Name>Unlimited rides available for specified duration</Name>
```

```
    <fareStructureElements>
        <FareStructureElementRef version="1.0" ref="mytrain:my_single_trip@access"/>
        <FareStructureElementRef version="1.0" ref="tsln:my_noctilien@access"/>
        <FareStructureElementRef version="1.0" ref="mytrain:my_single_trip@eligibility"/>
        <FareStructureElementRef version="1.0"
ref="mytrain:my_single_trip@conditions_of_sale"/>
    </fareStructureElements>
</ValidableElement>
```

In some case, parameter assignments have to bas assigned to SALES OFFER PACKAGE. In EFIP this will be done via the FARE TABLE (note that FARE TABLEs are mandatory in EFIP), as described in the example below (this example assignes a sales condition and a price to the *SalesOfferPackage*).

```
<Cell version="any" id="lFR-Tarif-Example:Cell:002:LOC">
    <SalesOfferPackagePrice>
        <Amount>9.00</Amount>
    </SalesOfferPackagePrice>
    <FareStructureElementRef version="1.0"
ref="mytrain:my_single_trip@conditions_of_sale"/>
    <SalesOfferPackageRef ref="FR-Tarif-Example:SalesOfferPackage:001:LOC"
version="any"/>
</Cell>
```

As explained above EFIP is making mandatory to have the *ValidableElement* embeded inside de FareProduct, and not outside with a reference to it. This is a very simple rule, making data production and consumption more systematic, and corresponding to the fact that in most cases, VALIDABLE ELEMENT are used as a simple link between a FARE PRODUCT and the FARE STRUCTURE ELEMENTS (possibly in squence) that it needs to refer.

Rule : **Embed VALIDABLE ELEMENTs elements in the FARE PRODUCT that uses them**- *Mandatory*

Access right assignment can be very complex (in real life), therefore NeTEx allows complex combination (basically using AND/OR of multiple elements). If this kind of complex construction cannot be fully avoided due to the complexity of some real life fare offers (things like, with some residential eligiblity or a social eligibility, and a employment eligibility in some specific area, on non peak hours and specific working daytype … you are granteed a special fare, but it has some quota limitation and can't be used more than a certain amount of time ine a day, a week and a month … and no it's not only imaginitation !), it is strongly recommended to avoid them. Furthermore, since such complex fare will hardly be fully understood by data consumer, keeping a simplistic structured description associated with a fully detailed text NOTICE (via a NOTICE ASSIGNMENT), is most probably the best way to go.

Rule : **Avoid using complex conditions (AND/OR** for the assignment of parameters**)** - *recommended*

Grouping FARE STRUCTURE ELEMENTs in a TARIFFs as coherent sets which serve to describe the different aspects of the fare structure and relate them to the underlying transport network (for example the available tariff zones, stop-to-stop transitions, pass durations, etc.) is mandatory in EFIP (see 11.2.1 for more details abour TARRIFFs).

Rule : **Require use of TARIFFs to organize access rights and limitations as FARE STRUCTURE ELEMENTs** – *Mandatory*

As FARE STRUCTURE ELEMENTs are also used to host ACCESS RIGHT ASSIGNMENT, giving them an unambiguous type will be a great help for data consumers. EFIP predefines a set of types that can be used directly withouth the need of an explicit type description:

- *access*, for access rights (scoping parameters, etc.)
- *eligibility*, for eligibility conditions
- *access_duration*, for constrints on the duration of the access right

- *conditions_of_travel*,
- *conditions_of_use* , for usage parameters (companion, user profile,etc.)
- *conditions_of_sale*, for conditions lake refund, exchange, etc.

Rule : **Use standard categories of TYPE OF FARE STRUCTURE ELEMENTs (access, eligibility, access_duration, conditions_of_travel, conditions_of_use) whenever it holds conditions/parameters** - *recommended*

These predefined types must be used for *TypeOfFareStructureElementRef* as below.

```
<TypeOfFareStructureElementRef versionRef="efp:v1.0" ref="efp:access"/>
```

They can also, in addition, be used in IDs with a preceeding "@"as below:

```
<FareStructureElement version="1.0" id="mytrain:my_single_trip@access">
```

Or

```
<GenericParameterAssignment version="1.0" id="mytrain:my_single_trip@access"
order="1">
```

15.3 Rules regarding the Network Structure

EFIP describes the fare offer applicable to a a set of services. Therefore having the fares, without a comprehensive description if the service (timetables, lines, stops, etc.) is useless. In most cases, the service offer will be provided in a separated NeTEx data set (the EPIP Profile should be considered here).

NeTEx is offering a nice prerequite mechanism that helps referring other data sets on which the current dataset is relying. In EFIP, it is strongly recommended to make sure that the service offer is fully decribed and properly referenced (including Id and version) with prerequisites.

Rule : **Make sure information for a journey planner is there eg ZONEs, SCHEDULED STOP POINTs** - *Recommended*

These prerequisites can be described as below.

```
<FareFrame version="1.0" id="mytrain:SVC_01@fares">
    <FrameDefaults>
```

```
      …
   </FrameDefaults>
   <prerequisites>
      <InfrastructureFrameRef version="1.0"
ref="mytrain:SVC_01@infrastructure"/>
   </prerequisites>
```

The service offer can contain TARIFF ZONEs (see EPIP). However, at the fare level, EFIP makes mandatory to use the FARE ZONE, that must be build from the TARIFF ZONE, enhancing it with additional infomration like the zone topology, it's scoping method (is it built from its boundaries, or from the contained stops), or it's neigbours and contained areas.

15.4 *Rule :* Use FARE ZONE to specify topology etc - *Mandator*Rules regarding the Prices

EFIP make mandatory to systematically present the prices through a FARE TABLE. The objective here is to systematize the production and interpretation of fare data, but also to avoid too great a disparity in the choices and modeling options that would inevitably arise (especially in the fare domain) if they were not somewhat constrained.

The presentation of prices in the form of a fare table remains a great classic in the field of transport, and this choice will also allow the simplification of the presentation of the offer to the traveler (which is consistent with the profile's objective of providing traveler information).

In addition, there are cases where an additional price information may be provided outside of FARE TABLEs (it is a complement, not a replacement): for example, if a product offers a fixed reduction, for example of €5, this may be explicitly indicated in the description of the fare product so as to be able to give a complete and relevant description to the traveler.

EFIP chooses not to provide cumulative discounts (is this really happens, they have to be grouped in a single "coumpound" discount).

A FARE TABLE allows the representation of price groups for combinations of fare elements. It defines a multidimensional matrix of CELLS, each of which can indicate a price for a combination of one or more fare

elements. For example, we can have USER PROFILE + DISTANCE MATRIX ELEMENT + CLASS OF USE references on each cell in order to define adult and child fares for first and second class.

The tables can be nested (allowing to defined reusable tables that can be included by several others); All objects inheriting from "PRICEABLE OBJECT" (which is the case for the majority of objects in this profile) can be used within a FARE TABLE.

The FARE TABLE allows to define a whole set of relatively independent and effectively reusable components that will be assembled (by reference) within the table to provide a corresponding price (for example: Ticket from A to B + single ticket + young traveler (18-25 years old) => Price), therefore keeping a very modular construction.

The FARE TABLE makes references to all the elements that constitute it with the exception of the prices themselves which must be completely defined within the FARE TABLE.

Rule : **Systematically use of FARE TABLEs, including the price information** *- Mandatory*

Rule : **Nest tables to reduce repetition** *- Recommendation*

Rule : **Use precalculated prices where available (instead the PRICING RULE)** *– Recommendation*

Rule : **Use PRICING SERVICE for yield managed prices** *- Mandatory*

15.5 Document to provide human readable descriptions

It is fundamental that the fare offer description contains a comprehensive description in order to provide a useful information when displayed to end users. But this is also important for data consumer to be confident on the fact that some data elements will be provided as expected. Also they are not about the data structure itself, the following rules are highly important.

Rule : **Provide textual names for FARE PRODUCTs** *- Mandatory*

Names are mandatory for all elements that can be faced to the public, including *FARE PRODUCT (and associated subtype)*, *SALES OFFER PACKAGE*, TYPE OF TRAVEL DOCUMENT, DISTRIBUTION CHANNEL.

Rule : **Classify FARE PRODUCTs with TYPE OF FARE PRODUCT** *- Mandatory*

The ProductType element shall be filled

Rule : **Include ConditionSummary** *- Mandatory*

ConditionsSummary are mandatory (in FARE PRODUCTs and SALES OFFER PACKAGE). The following table is a recommendation to dispatch the summary information between FARE PRODUCTs and SALES OFFER PACKAGE.

Name	
FareStructureType	FARE PRODUCT
TariffBasis	FARE PRODUCT
HasNotices	All
ProvidesCard	SALES OFFER PACKAGE
GoesOnCard	SALES OFFER PACKAGE
IsPersonal	SALES OFFER PACKAGE
RequiresPhoto	SALES OFFER PACKAGE
MustCarry	SALES OFFER PACKAGE

RequiresAccount	SALES OFFER PACKAGE
IsSupplement	FARE PRODUCT
RequiresEntitlement	FARE PRODUCT
GivesEntitlement	FARE PRODUCT
HasOperatorRestrictions	FARE PRODUCT
HasTravelTimeRestrictions	FARE PRODUCT
HasRouteRestrictions	FARE PRODUCT
HasTrainRestrictions	FARE PRODUCT
HasZoneRestrictions	FARE PRODUCT
CanBreakJourney	FARE PRODUCT
ReturnTripsOnly	FARE PRODUCT
CanChangeClass	FARE PRODUCT
IsRefundable	SALES OFFER PACKAGE
IsExchangable	SALES OFFER PACKAGE
HasExchangeFee	SALES OFFER PACKAGE
HasDiscountedFares	FARE PRODUCT

AllowAdditionalDiscounts	FARE PRODUCT
AllowCompanionDiscount	FARE PRODUCT
HasMinimumPrice	FARE PRODUCT
RequiresPositiveBalance	SALES OFFER PACKAGE
PenaltyWIthoutTicket	SALES OFFER PACKAGE
AvailableOnSubscription	SALES OFFER PACKAGE
HasPurchaseConditions	SALES OFFER PACKAGE
HasDynamicPricing	FARE PRODUCT
RequiresReservation	SALES OFFER PACKAGE
HasReservationFee	SALES OFFER PACKAGE
HasQuota	SALES OFFER PACKAGE

Table 132 - *ConditionSummary* dispatch recommendtaion– Element

Rule : **Include links back to web pages text and graphics** *- Recommendation*

It is trongly recommended to include to web pages and graphics providing to the end users a simplified way of understand the fare offer. This can easlily be done using the InfoLink element as described in the example below.

```
<infoLinks>
```

```
   <InfoLink>https://www.iledefrance-mobilites.fr/en/the-network/mobility-
services/bus/noctilien-night-buses</InfoLink>
</infoLinks>
```

Rule : **Provide Notices to be printed by tools on the user interface** *- Recommendation*

Notices is very important to add an explanation for the end users. It is highly recommended that the providers include such notices in their datasets, and it is **mandatory** for consumer to display them the the end users.

Another important element in the context of EFIP is that the Notices shall be exclusively attached (assigned) to a FARE TABLE (***FareTable***), which will in fact allow it to be associated with a SALES OFFER PACKAGE (***SalesOfferPackage***), or possibly with a FARE PRODUCT (***PreassignedFareProduct***). Theoretically, a note can be attached to any object, but attaching notes without a rule to many places would make the exploitation of the data very delicate.

It will also often be useful to offer translations of the notices in different languages.

15.6 Allow simplified fare offer description

As explained above Notices are an important element in communicating about pricing (have to look at a price sheet to see that it is riddled with asterisks and references to a whole series of notices).

It should also be noted that, the implementation of EFIP may be more delicate than the other profiles (EPIP, EPIAP) because there are very few fare offers already fully described in a structured and digital way (still mostly web pages and PDF documents to present the fare offer): the initial structured representation work will therefore be relatively long and substantial and it will be useful to proceed in stages.

One solution to proceed step by step could be to start by structuring only the key elements of products: describe the fundamentals (***FareStructureElement*** as the origin-destinations or a timing structures for example), the fare product (***PreassignedFareProduct***) and the sales offer (***SalesOfferPackage***) but with very

simple parameters (***ValidityParameter***, ***UsageParameter***, etc.), and assemble the whole in a ***FareTable*** in which are added notices specifying the rights. The idea here is typically to provide enough elements so that an information system (trip calculator, service in a MaaS context, etc.) can offer the fare offer description, when it has a good chance of being relevant and that the associated notices allow the user to decide whether the ticket suits them or not.

The figure below shows a TER (Bourgogne-Franche-Compté) ticket that can be described in detail with NeTEx but which requires a number of combined user profile elements (employee of companies, administrative agent, other professionals, etc.), which will also require a registration of the professional or his company, etc. These access rights and user profile elements can, as a first approach, be inserted in a note: by reusing the “standard” title (***PreassignedFareProduct***), a dedicated sales offer (***SalesOfferPackage***) will be created that will be associated, in the ***FareTable***, with a note (***Notice***) and information on the corresponding reduction (via a ***FarePrice***), here 30%.


LE TARIF ENTREPRISE
30 % DE RÉDUCTION
POUR LES PROS
POUR QUI ? Les salariés des entreprises, les agents des administrations et tous les autres professionnels.
QUELS TRAJETS ? Pour les voyages réguliers ou occasionnels sur un parcours défini, en TER en Bourgogne-Franche-Comté.
COMMENT ? En achetant un carnet de 10 billets. Non nominatifs, ces billets sont utilisables indifféremment par toute personne de l'entreprise ou de l'administration concernée pendant un an.
30% DE RÉDUCTION valable dans les deux sens du parcours choisi au moment de l'achat du carnet de 10 billets. Par exemple Dijon-Besançon 10,50€ avec le tarif entreprise au lieu de 15€ au tarif normal.
ACHETEZ VOS CARNETS
DE 10 BILLETS TARIF ENTREPRISE
AUTOMATES RÉGIONAUX OU AUX GUICHETS
dans les gares de Bourgogne-Franche-Comté
Prix au 29 août 2017.

Figure 60 — Example of a special rate that can justify the use of a notice to simplify it

15.7 Guidelines for presenting EFIP data in tables and spreadsheets

It is often useful to have fare price data available as a spreadsheet or other simple table format. It is straightforward to establish mappings between the EFIP core fare elements and such tables and generic principles can be used to identify data elements and to combine normalised model elements in a smaller number of tables as show below.

The qualified name of the NeTEx schema attribute shall be used as a column heading for a given attribute in a table. Columns may include both values, references to elements in other tables (since NeTEx uses a unified system of identifiers, codespaces and references) and “view” elements, derived from other tables through the reference, that annotate or explain the references o that data in a given table can be understood. Each table shall have a default codespace: data identifiers and references in the table that are not in the default codespace shall be qualified with the codespace prefix.

As well as a table of prices, tables to describe the tariff components making up the prices in the table can also be defined, for example, TARIFF ZONES, GEOGRAPHIC INTERVALS, TIME INTERVALS, FARE PRODUCTS, SALES PACKAGES, etc. Commercial conditions not relevant for access or pricing are omitted in the example (e.g. USAGE PARAMETERS for EXCHANGING. REFUNDING, etc) but could be added.

The essential data for many basic fare types can thus be defined by a small number of tables.

Prices

- FARE TABLE – Prices.

Networks

- FARE ZONEs. Tariff zones –
- FARE ZONE members – Stops in a zone –

- FARE POINTs IN PATTERN – Stages on a JOURNEY PATTERN.

Tariffs

Spatial aspects

- Access: TARIFF / DISTANCE MATRIX – Point-to-point, zone-to-zone., or zonal
- Access: TARIFF / GEOGRAPHIC INTERVALs.
- Access: NETWORK, OPERATOR, LINE, GROUP OF LINEs, GROUP OF OPERATORs,

Temporal aspects

- Durations: TARIFF / TIME INTERVALs.

Other quality aspects

- Access: When TARIFF / FARE DEMAND FACTORs – Peak, off peak etc.
- Carnet Units: TARIFF / FARE QUALITY FACTORs – Carnet Quantities numbers.

Parameters

- Eligibility: TARIFF / USER PROFILEs.
- Groups: TARIFF / GROUP TICKETs.

Fare Products

- FARE PRODUCT.
- FARE PRODUCT / ACCESS RIGHTS IN PRODUCT.
- VALIDABLE ELEMENT/ TARIFF / FARE STRUCTURE ELEMENT.

Sales Packages

- TYPEs OF TRAVEL DOCUMENT.
- SALES OFFER PACKAGEs.
- SALES OFFER PACKAGE ELEMENTs.

Framework

- CODESPACEs
- OPERATORS

The rest of this section provides examples of all of the above tables. Only a minimum et of attributes I included.

15.8 Codespaces and identifiers for fare data

The use of codespaces and identifiers on fare data is the same as that for timetable data, that is, the fare data itself will generally be assigned to the provider's (i.e. operator's) codespace, and use their identifier system, but will by preference use reference data from national data sets such as operator codes ( ), stops ( ) places ( ) and tariff zones ( ).

The identifiers of fare data elements must be stable and unique within their codespace.

The identifiers for tariff zones may in particular need to be shared by many operators; a single organisation should be assigned responsibility for issuing the identifiers for a Network.

Note: a recommended convention for naming DISTANCE MATRIX ELEMENTS is to use a '+' to connect the identifiers of the origin and destination zone or stops. For example

```
<DistanceMatrixElement version="1.0" id="myb:4400CY0037+4400CY0038">
                <Name>Alpha+Beta</Name>
                <StartStopPointRef version="naptStop:any" ref="naptStop:4400CY0037"/>
                <EndStopPointRef version="naptStop:any" ref="naptStop:4400CY0038"/>
</DistanceMatrixElement>


<DistanceMatrixElement version="1.0" id="myb:Zone1+Zone3>
```

```
            <Name>Alpha+Beta</Name>
            <StartTariffZoneRef version="any" ref="myb:Zone1"/>
            <EndTariffZoneRef version="any" ref="myb:Zone3"/>
    </DistanceMatrixElement>
```

15.9 Identifiers for version frames containing fare data

Exactly the same principles may be used for the identifiers of version frames containing fare data as those described for other types of data in the profile – see the EPIP for details..

To recap -The structure of a EFIP VERSION FRAME identifier is:

[country-code] : [local-code] : [NeTEx-frame-type] _ [EFIP-type] : [frame-topic]

15.10 Names for documents containing fare data

Exactly the same principles may be used for the names of documents containing fare data as those described for other types of data in the EFIP- in the profile – see the EPIP sspecification for details.

Thus: The structure of the file name for a EFIP conformant document is:

[prefix] - [EFIP-version] _ [country-code] _ [provider-code] _ [profile-type] _ [doc-topic] _[creation-date]

Example file names:

```
FX-PI-01_FR_METR_LINE-FARE _L01_20160302.xml
FX-PI-01_FR_FEBR_NETWORK-FARE_WOE-060+086+110_20230909.xml
FX-PI-01_FR_DFT_STOP-FARE_NPTG-Plusbus-UK_20150705.xml
```

15.11 Version management of elements

The versioning and validity condition management in the context of the European passenger information profile is simple and mainly based on the FRAME concept.

The main rules for object versioning are the following:

- All objects having an ***Id*** shall also have a ***version*** (if the version is not managed or unknown, fill the version information with "*any*");
- Also note that the ***Id*** shall be stable: an update of an object shall not change the ***Id***;
- As required by FRAMEs, all the versions of objects within a FRAME shall be consistent all together (but their version value can be decided on a per object base);
- The versions of the FRAMEs are mandatory and must be :
  - A number (sequence of digits)
  - Any FRAME version provided after another one (update of a FRAME with the same Id) shall have a version number higher than the previous one.

The main rules for VALIDITY CONDITIONs are the following:

- The main VALIDITY CONDITION is the one of the FRAME: any VALIDITY CONDITION of an object being out of the VALIDITY CONDITION of the FRAME will not be taken into account;
- If a version of a FRAME has part of all its VALIDITY CONDITION overlapping the VALIDITY CONDITION of a previous version of the same FRAME (same Id), the new version completely overrides the previous one in the overlapping period (therefore cancelling all the previous information and replacing them with the new ones)
- A FRAME shall have VALIDITY CONDITIONs
- VALIDITY CONDITION of COMPOSITE FRAME is acting on the contained FRAME as the FRAME's VALIDITY CONDITION is acting on the contained objects.

15.12 Versions of the profile

The EFIP itself is given a version number to allow implementors to determine the version of the profile being used (as opposed to the version of the NeTEx schema)

The profile's version codification is the following:

- for *PublicationDelivery*: x.y:EU-FP-a.0 (e.g. "`2.0:EU_FP-1.0`").
- for FRAME (via TypeOfFrame): **x.y:EU_FP_nnnn-a.b** (e.g. "`2.0: EU_FP_COMMON-1.0`").

The codification components are the following:

1) **x.y** is the NeTEx version (mandatory, it is the XSD version)
2) "**:**" is a mandatory delimiter
3) **EU_FP** (European Passenger Fare Profile**y**) and **EU_FP_*nnnn*** where ***nnnn*** is "EU_FP_COMMON", "EU_FP_STOP", "EU_FP_NETWORK", "EU_FP_TIMETABLE", " EU_FP_CALENDAR" or "EU_FP_LINE_OFFER" for TYPE OF FRAME
1) "**-**" is a mandatory delimiter
2) **a.b** is the profile version (mandatory). a and b are integer
   - **a**=main version
   - **b**=secondary version number (always 0 in ***PublicationDelivery***)

EXEMPLE

```
<PublicationDelivery version="2.01:EU_FP-1.0" ...

<TypeOfFrameRef
ref="FR1:TypeOfFrame:NETEX_COMMUN:">version="2.02:EU_FP_LINE_OFFER-
1.3"</TypeOfFrameRef>
```

15.13 Structure of IDs

When exchanging information, it is of high importance to make sure that identifiers are stable and unique. In order to ease interoperability and ensure uniqueness, the European passenger information profile provides some rules to built identifiers. These rules are defined so they can easily be applied on internal identifiers of a system at export time, and processed at import time. They also take advantage of the codespace mechanism proposed by NeTEx.

Important Note: identifiers are structured for uniqueness purpose ONLY, there shall never be any attribute extraction from the identifiers.

The ID structure of objects is the following:

[Country code]:[Local Code]:[object type]:[technical-identifier]:[ID provider for shared IDs]

- ***[Country code]*** is an ISO 3166-1 code of the country (2 characters)
- ***[Local Code]*** is a code uniquely identifying the locality or the provider within the country (may be a region code like the European NUTS code, an authority code, etc.).The Europen NUTS code is recommended here. However this code is not mandatory if the other elements make the code unique (but surrounding ":" must be present).
- ***[object type]*** is the NeTEx object type (ServiceJourney, **PassengerStopAssignment**, **Line**, etc. using exactly the tag format, UpperCamelCase and no space) corresponding to the XML tag, and is provided to avoid any collision of single identifiers being used for different types of objects.
  A small exception is defined for **StopPlace** in order to differentiate between monomodal and General STOP PLACEs, the ***[object type]*** will be **MonomodalStopPlace** or **GeneralStopPlace** instead of **StopPlace** (see Error: Reference source not found-Error: Reference source not found)

- ***[technical-identifier]*** is a technical identifier for the object, it can be whatever code the system define (built of upper-case or lower case non-accented characters, numbers "-" and "_") but shall be unique for the object and durable (a single object can't change its identifier)
- ***[ID provider for share IDs]*** is used to provide a reference to the ID provider (reference data system) when there is one. When this non-mandatory field is present, it ensures by itself the uniqueness and stability of the ***[technical-identifier]***.
- All the "**:**" separating characters are mandatory (even if a field is empty).

Example *(purely fictional, just illustrating the coding rule)*

FR:SMIRT:MonomodalStopPlace:454e74:

GE*:*79189:Line:78-a-1:

NO::Route:985a:ENTUR

FR::MonomodalStopPlace:43289:FR1

UK:: GeneralStopPlace:199G98765431:NAPTAN

15.13.1 *Frame ID*

The identifier of FRAMEs follows the identifier pattern defined above. However it specialises it in order facilitate the implementation of the rules defined in ***Error: Reference source not found-***

Error: Reference source not found and ***15.11-Version management***

The FRAME identifier basic structure is:

[Country code]:[Local Code]:GeneralFrame/CompositeFrame:EU_FP_nnnn_[VersionNumber]:

Where ***EU_FP_nnnn_[VersionNumber]*** is:

- EU_FP_STOP-[VersionNumber] for Stop FRAME
- EU_FP_NETWORK-[VersionNumber] for Network FRAME
- EU_FP_TIMETABLE-[DaVersionNumberte] for Timetable FRAME
- EU_FP_LINE_OFFER-[VersionNumber] for Line FRAME
- EU_FP_CALENDAR-[VersionNumber] for Calendar FRAME
- **[*VersionNumber*]** is a number associated with the version of the Frame

EXAMPLE : **`id="FR:RATP:GeneralFrame:EU_FP_LINE_OFFER_20170816103000Z:"`**

15.13.2 *Other specific coded or ID*

The ***ParticipantRef*** for National Access Points will simply be the country ISO 3166-1 code.

15.14 Available types of frame

EFP1>> All frames in an EFIP conformant XML document containing fare data must be assigned to one of the standard EFIP **TypeOfFrame** types. The available frame types are given below.

15.14.1 *EPIP Basic Profile frames*

The following TYPEs OF FRAME from the base profile EPIP are available for use in the EFIP to describe stop and network data that is referenced by the fare definitions – see the Part2 EPIP for details.

Ancillary data:

- EU_FP_COMMON
- EU_FP_CALENDAR

Payload data:

- EU_FP_STOP
- EU_FP_NETWORK
- EU_FP_TIMETABLE

Composites:

- EU_FP_LINE_OFFER
- EU_FP_NETWORK_OFFER
- EU_FP_STOP_OFFER

A further TYPE OF FRAME can be used to declare EFIP specific metadata for validating the other frame types.

- EU_FP_METADATA

15.14.2 *Fare Profile frames*

The following additional frames are defined in the EFIP specifically to hold fare data of different types:

Payload:

- EU_FP_FARE_NETWORK
- EU_FP_FARE_PRODUCT
- EU_FP_FARE_PRICE

Composite:

- EU_FP_FARE_OFFER
- EU_FP_NETWORK_FARE_OFFER
- EU_FP_STOP_FARE_OFFER

The following tables summarise the content of each TYPE OF FRAME.

15.14.3 *Classes for SPECIFIC FARE FRAMEs*

The following notation is used:

	Unless nesting is indicated by an arrow, elements are contained directly in their frame type.
è	Subelement is a component of preceding composite element: it must be embedded in-line within wrapper tags as per the NeTEx schema. (additional arrows indicated additional subcomponent levels).
à	Subelement is an independent element referenced by another element: In the EFIP and the EPIP it should be nested within the referencing element that it annotates.
*	Parent is generic – the subelement may be nested within more than one type of element (If no parent is indicated, element is nested directly within a FRAME.
()	Classes in brackets are abstract
	The most important “payload” entities for EFIP in each frame are highlighted in pink.
‘epip’	Epip_metadata namespace =http:/nete-cen.eu/epip

15.14.3.1 *EU_FP_FARE_NETWORK*

Table 133 - TypeOfFrame: *EU_FP_FARE_NETWORK*

Name	XML Type	Content XML Classes & Nesting	TM Entities & Composition
id	TypeOfFrameIdType	'efp:*EU_FP_FARE_NETWORK*'	TYPE OF FRAME
FrameClassRef	*NameOfClass*	***FareFrame***	FARE FRAME
classes	*ClassInContextRef*	***FareScheduledStopPoint***	FARE SCHEDULED STOP POINT
		TariffZone ***FareZone*** ***àFareSection*** ***àèPointOnSection***	TARIFF ZONE FARE ZONE FARE SECTION POINT ON SECTION
		SeriesConstraint ***èFarePointInPattern***	SERIES CONSTRAINT **è**FARE POINT IN PATTERN
Data Scope: PAYLOAD	A ***ServiceFrame*** instance conforming to '*EU_FP_NETWORK*' shall contain schedule data for only a single LINE.		

15.14.3.2 *EU_FP_FARE_PRODUCT*

Table 134 - TypeOfFrame: *EU_FP_FARE_PRODUCT*

Name	XML Type	Content XML Classes & Nesting	TM Entities & Composition
id	TypeOfFrameIdType	'efp:*EU_FP_FARE_PRODUCT*'	TYPE OF FRAME
FrameClassRef	*NameOfClass*	***FareFrame***	FARE FRAME
classes	*ClassInContextRef*	***Tariff*** ***àGeographicalInterval*** ***àTimeInterval*** ***àDistanceMatriElement*** ***àGroupOfDistanceMatrixElements*** ***àQualityFareStructureFactor*** ***àFareDemandFactor*** ***àFareStructureElement*** ***àà GenericParameterAssignment*** ***àà | UserProfile | CompanionProfile*** ***àà | GroupTicket*** ***àà | RoundTrip | Interchanging*** ***àà | Frequency | StepLimit*** ***àà | UsageValidityPeriod*** ***àà | Transferability*** ***àà | PurchaseWIndow***	***TARIFF*** ***GEOGRAPHICAL INTERVAL*** ***TIME INTERVAL*** ***DISTANCE MATRIX ELEMENT*** GROUP OF DISTANCE MATRIX ELEMENTS QUALITY FARE STRUCTURE FACTOR FARE DEMAND FACTOR FARE STRUCTURE ELEMENT àGENERIC PARAMETER ASSIGNMENT USER PROFILE | COMPANION PROFILE | GROUP TICKET | ROUND TRIP | INTERCHANGING | FREQUENCY | STEP LIMIT USAGE VALIDITY PERIOD | TRANSFERABILITY | PURCHASE WINDOW
		PreassIgnedFareProduct ***AmountOfPriceUnit*** ***SalesDiscountRight*** ***SupplementProduct***	PREASSIGNED FARE PRODUCT AMOUNT OF PRICE UNIT SALES DISCOUNT RIGHT SUPPLEMENTPRODUCT

		èAccessRightInProduct ***àValidableElement*** ***àèGenericParameterAssignment***	èACCESS RIGHT IN PRODUCT VALIDABLE ELEMENT **è**GENERIC PARAMETER ASSIGNMENT
		TypeOfTravelDocument	TYPE OF TRAVEL DOCUMENT
		SalesOfferPackage ***èSalesOfferPackageElement*** ***èDistributionAssignment***	SALES OFFER PACKAGE èSALES OFFER PACKAGE ELEMENT èDISTRIBUTION ASSIGNMENT
		GroupOfSalesOfferPackages ***èSalesOfferPackageElement*** ***èDistributionAssignment***	GROUP OF SALES OFFER PACKAGEs èSALES OFFER PACKAGE ELEMENT DISTRIBUTION ASSIGNMENT
		TypeOfTravelDocument	TYPE OF TRAVEL DOCUMENT
		DistributionChannele	DISTRIBUTION CHANNEL
		FulfilmentMethod	FULFILMENT METHOD
Data Scope: PAYLOAD	A ***FareeFrame*** instance conforming to '*EU_FP_FARE_PRODUCT'* shall contain product data for a single LINE.or NETWORK		

15.14.3.3 *EU_FP_FARE_PRICE*

Table 135 - TypeOfFrame: *EU_FP_FARE_PRICE*

Name	XML Type	Content XML Classes & Nesting	TM Entities & Composition
id	TypeOfFrameIdType	'efp:*EU_FP_FARE_PRICE*	TYPE OF FRAME
FrameClassRef	*NameOfClass*	***FareFrame***	FARE FRAME
classes	*ClassInContextRefP*	***PricingParameterSet*** ***èPricingRule*** ***èDiscountingRule*** ***èLimitingRule*** ***èPriceUnit*** ***èRounding***	PRICING PARAMETER SET èPRICING RULE èDISCOUNTING RULE èLIMITING RULE èPRICE UNIT èROUNDING
		FareTable ***èFareColumn*** ***èFareRow*** ***èCell*** ***èFarePrice*** ***è{FareTable}***	FARE TABLE èFARE COLUMN èFARE ROW èCELL àFARE PRICE è{FARE TABLE}
		PriceGroup ***èFarePrice***	PRICE GROUP àFARE PRICE

Data Scope: PAYLOAD	A ***FareFrame*** instance conforming to '*EU_FP_FARE PRICE*' shall contain price data for a corresponding FARE_PRODUCT ***FareFrame***.

15.15

15.16 Available types of frame

As explained in Error: Reference source not found the available list of TYPEs OF FRAME for the European fare profile is:

- EU_FP_COMMON
- EU_FP_CALENDAR
- EU_FP_STOP
- EU_FP_NETWORK
- EU_FP_TIMETABLE
- EU_FP_FARE_OFFER

The following tables provide the content of each TYPE OF FRAME. The **TypeOfFrame** may not be exchanged together with the dataset and considered as a common agreement (just use their identifiers to reference them).

15.16.1 *Types for SPECIFIC FRAMEs*

Table 136 - TypeOfFrame EU_FP_COMMON for ResourceFrame

Name	Type	Content
id	TypeOfFrameIdType	

classes	*ClassInContextRef*	VALIDITY CONDITION ALTERNATIVE NAME ALTERNATIVE TEXT RESPONSIBILITY SET ORGANISATION TYPE OF FRAME TYPE OF VALUE

Table 137 - TypeOfFrame EU_FP_CALENDAR for CalendarFrame

Name	Type	Content
id	TypeOfFrameIdType	
classes	*ClassInContextRef*	DAY TYPE OPERATING DAY SERVICE CALENDAR

Table 138 - TypeOfFrame EU_FP_STOP for SiteFrame

Name	Type	Content
id	TypeOfFrameIdType	
classes	*ClassInContextRef*	STOP PLACE STOP PLACE ENTRANCE TOPOGRAPHIC PLACE POINT OF INTEREST

Table 139 - TypeOfFrame EU_FP_NETWORK for ServiceFrame

Name	Type	Content

id	TypeOfFrameIdType	
classes	*ClassInContextRef*	LINE DIRECTION GROUP OF LINE NETWORK ROUTE ROUTE POINT ROUTE LINK FLEXIBLE LINE TARIFF ZONE DESTINATION DISPLAY SERVICE JOURNEY PATTERN POINT IN JOURNEY PATTERN SCHEDULED STOP POINT CONNECTION PASSENGER STOP ASSIGNMENT NOTICE

Table 140 - TypeOfFrame EU_FP_TIMETABLE for TimetableFrame

Name	Type	Content

id	TypeOfFrameIdType	
classes	*ClassInContextRef*	SERVICE JOURNEY JOURNEY PART JOURNEY PART COUPLE TRAIN NUMBER NOTICE

15.16.2 *Types for COMPOSITE FRAMEs*

Table 141 - TypeOfFrame for EU_FP_FARE_OFFER

Name	Type	Content
id	TypeOfFrameIdType	
includes	*TypeOfFrameRef*	EU_FP_COMMON EU_FP_STOP EU_FP_NETWORK EU_FP_TIMETABLE EU_FP_CALENDAR EU_FP_FARE EU_FP_FAREPRIC Note: shall not contain information related to more than one single LINE

15.16.3 *Classes for COMPOSITE FARE FRAMEs*

15.16.3.1 *EU_FP_FARE_OFFER*

Table 142 - TypeOfFrame for EU_FP_FARE_OFFER - ("EFIP-FL")

Name	Type		Content XML Classes & Nesting	TM Entities & Composition
id	TypeOfFrameIdType			TYPE OF FRAME
FrameClassRef	*NameOfClass*		***CompositeFrame***	COMPOSITE FRAME
includes	*TypeOfFrameRef*	*1:1*	• *'efp:EU_FP_COMMON'*	TYPE OF FRAME (RESOURCE FRAME)
		0:1	• *'efp:EU_FP_STOP'*	TYPE OF FRAME (SITE FRAME)
		1:1	• *'efp:EU_FP_NETWORK'*	TYPE OF FRAME (SERVICE FRAME)
		1:1	• *efp:XX_PF_FARE_NETWORK'*	TYPE OF FRAME (FARE FRAME)
		*1:**	• *'efp:XX_PF_FARE_PRODUCT'*	TYPE OF FRAME (FARE FRAME)
		*1:**	• *'efp:XX_PF_FARE PRICE'*	TYPE OF FRAME (FARE FRAME)
		0:1	• *'efp:EU_FP_CALENDAR'*	TYPE OF FRAME (SERVICE CALENDAR FRAME
Data Scope: CONTAINER	A ***CompositeFrame*** instance conforming to '*EU_FP_FARE_OFFER*' shall contain fare data for only a single LINE.			

15.16.3.2 *EU_FP_NETWORK_FARE_OFFER*

Table 143 - TypeOfFrame for EU_FP_NETWORK_FARE_OFFER - ("EFIP-FN")

Name	Type		Content XML Classes & Nesting	TM Entities & Composition
id	TypeOfFrameIdType			TYPE OF FRAME
FrameClassRef	*NameOfClass*		***CompositeFrame***	COMPOSITE FRAME
includes	*TypeOfFrameRef*	*1:1*	• *'efp:EU_FP_COMMON'*	TYPE OF FRAME (RESOURCE FRAME)
		0:1	• *efp:EU_FP_STOP'*	TYPE OF FRAME (SITE FRAME)
		1:1	• *'efp:EU_FP_NETWORK'*	TYPE OF FRAME (SERVICE FRAME)
		*1:**	• *'efp:XX_PF_FARE_NETWORK'*	TYPE OF FRAME (FARE FRAME)
		*1:**	• *'efp:XX_PF_FARE_PRODUCT'*	TYPE OF FRAME (FARE FRAME)
		*1:**	• *efp:XX_PF_FARE_PRICE'*	TYPE OF FRAME (FARE FRAME)
		0:1	• *'EFIP:EU_FP_CALENDAR'*	TYPE OF FRAME (SERVICE CALENDAR FRAME
Data Scope: CONTAINER	A ***CompositeFrame*** instance conforming to '*EU_FP_FARE_NETWORK OFFER'* shall contain fare data for a NETWORK or part of a NETWORK (E.g. a GROUP OF LINEs).			

15.16.3.3 *EU_FP_STOP_FARE_OFFER*

Table 144 - TypeOfFrame for EU_FP_FARE STOP_OFFER – ("EFIP-FS")

Name	Type		Content XML Classes & Nesting	TM Entities & Composition
id	TypeOfFrameIdType			TYPE OF FRAME
FrameClassRef	*NameOfClass*		***CompositeFrame***	COMPOSITE FRAME
includes	*TypeOfFrameRef*	*1:1*	• *'efp:EU_FP_COMMON'*	TYPE OF FRAME (RESOURCE FRAME)
		0:1		TYPE OF FRAME (SITE FRAME)
		1:1	• *efp:EU_FP_NETWORK'*	TYPE OF FRAME (SERVICE FRAME)
		*1:**	• *'efp:XX_PF_FARE NETWORK'*	TYPE OF FRAME (FARE FRAME)
Data Scope: CONTAINER	A ***CompositeFrame*** instance conforming to '*EU_FP_STOP_FARE_OFFER'* shall contain dat about the network for defining fares data for an operator or an area.			

15.17 Organisation of fare data

The EFIP sets out a specific structure of organising data within EFIP document in order to facilitate implementation. See also example below.

15.17.1 *Use of frames*

Data should be assigned to one of the specified types of EFIP frame, each classified with a ***TypeOfFrame***, as described above, and indicating the prerequisite frames.

1. Basic profile frames as described in Part2 (EPIP) should be used to define any prerequisite data elements such as OPERATORs, BRANDING and RESPONSIBILITY SETs (in a ***ResourceFrame*** of type EU_FP_COMMON), SCHEDULED STOP POINT, TARIFF ZONEs, etc (in a ***ServiceFrame*** of type EU_FP_NETWORK) etc.

2. Fare network data elements, such as a FARE ZONES, FARE SECTIONs, etc (which may be shared between multiple products) should be placed in a separate ***FareFrame*** of type EU_FP_FARE_NETWORK. The frame should declare any prerequisite EU_FP_NETWORK frames upon which it depends.

3. Elements to define a single product in particular TARIFFs, FARE STRUCTURE ELEMENTS, FARE PRODUCTS, and SALES OFFER PACKAGEs should be placed in a separate ***FareFrame*** of type EU_FP_FARE_PRODUCT. The frame should declare any prerequisite EU_FP_FARE_NETWORK frames upon which it depends.

4. Prices for the SALES OFFER PACKAGE should be placed in a separate ***FareFrame*** of type EU_FP_FARE_PRICE; there should be a PRICE FRAME corresponding to each PRODUCT frame giving the prices for the PRODUCTs. The frame should declare the prerequisite EU_FP_FARE_PRODUCT frame for which it states the prices.

15.17.2 *Organisation of fare product definitions*

A EU_FP_FARE_PRODUCT frame should group together the elements needed to define a single fare product or set of closely related products (e.g. a trip, a pass, a carnet that are based on the same tariff structure), along with the ***SalesOfferPackage*** instances that define how they are marketed. A product frame should therefore contain.

- A Tariff, embedding FareStructureElement, DistanceMatrixElement, TimeInterval, GeographicInterval, and FareQualityFactor instances.
- One or more ***FareProduct*** instances:
- One or more ***SalesOfferPackage*** instances, referencing the fare products defined in the frame.

There is usually just one ***Tariff*** per product, and one or more ***SalesOfferPackage*** per fare product.

Furthermore:

- ***UsageParameter*** instances should be grouped within ***FareStructureElement*** instances in a ***Tariff*** (and not be contained directly in the ***FareFrame***). If necessary, a ***FareStructureElement*** may reference a ***UsageParameter*** of another ***Tariff.***

- ***ValidableElement*** instances should be nested within a ***FareProduct*** (and not be contained directly in the ***FareFrame***). If necessary, a ***FareProduct*** may reference a ***ValidableElement*** of another ***FareProduct.***

- Each ***FareProduct*** should be categorised with a ***ProductType*** and a ***TariffStructureType.***

15.17.3 *Fare table nesting rules*

When creating FARE TABLEs for the EFIP, the following principles shall be followed

g. FARE PRICEs shall be nested in FARE TABLEs for clarity and to reduce data volumes.

h. Each NESTING LAYER shall correspond to one further pricing factor

i. The type of fare price used shall be for that of lowest level element, For example if the lowest level is the USER PROFILE (a type of USAGE PARAMETER), such as 'Adult' or 'Child', then it shall be a USAGE PARAMETER PRICE, if it is a TYPE OF TRAVEL DOCUMENT (as associated with a SALES OFFER PACKAGE) it shall be a SALES OFFER PACKAGE PRICE, etc

j. Where the price is for a specific fare element such as a DISTANCE MATRIX ELEMENT, TIME INTERVAL, GEOGRAPHICAL INTERVAL etc, the relevant price shall be embedded in line and must include a reference to the priced element.

k. Where the price is for an additional option out of several specified for a single FARE STRUCTURE ELEMENT (as say may be the case for a FARE ZONE or a CLASS OF USE, a reference to the specific option shall be included in the CELL, for example: a ***FareZoneRef***, or ***ClassOfUseRef***, so that a consuming system can determine its meaning.

l. The preferred order for nesting tables is

   SALES OFFER PACKAGE / {spatial Factor] / (TIME INTERVAL) / USER PROFILE / CLASS OF USE /

   Where Spatial factor is either

   (1) DISTANCE MATRIX ELEMENT or

   (2) FARE STRUCTURE ELEMENT + ZONE

(3) GEOGRAPHICAL INTERVAL

Table 145 – *Common fare pricing factors*

(1)	(2) Pricing factor	(3) Tariff element	(4) FARE PRICE	(5) Cell Assignment
(6) Point-to-point fare	*(7) DISTANCE MATRIX ELEMENT*	*(8) DISTANCE MATRIX ELEMENT*	*(9) DISTANCE MATRIX ELEMENT PRICE*	***(10) [DistanceMatrixElementRef)***
(11) Zonal fare	*(12) FARE ZONE*	*(13) FARE STRUCTURE ELEMENT*	*(14) FARE STRUCTURE ELEMENT PRICE*	***(15) [FareStructureElementRef)*** ***(16) FareZoneRef***
(17) Different media	*(18) TYPE OF TRAVEL DOCUMENT*	*(19) SALES OFFER PACKAGE ELEMENT*	*(20) SALES OFFER PACKAGE PRICE*	***(21) (SalesOfferPackageRef)*** ***(22) TypeOfTravelDocumentRef*** ***(23) FareProductRef***
(24) Pass	*(25) TIME INTERVAL*	*(26) TIME INTERVAL*	*(27) TIME INTERVAL PRICE*	***(28) (TimeIntervalRef)***
(29) Stage fare	*(30) GEOGRAPHICAL INTERVAL*	*(31) GEOGRAPHICAL INTERVAL*	*(32) GEOGRAPHICAL INTERVAL PRICE*	***(33) (GeographicIntervalRef)***
(34) Carnet	*(35) FARE QUALITY FACTOR*	*(36) FARE QUALITY FACTOR*	*(37) FARE STRUCTURE FACTOR PRICE*	***(38) (FareQualityFactorRef)***
(39) Peak/OffPeak	*(40) FARE DEMAND FACTOR*	*(41) FARE DEMAND FACTOR*	*(42) FARE STRUCTURE FACTOR PRICE*	***(43) (FareDemandFactorRef)***

(44) *Class of Use*	*(45)* *CLASS OF USE*	*(46)* *FARE STRUCTURE ELEMENT*	*(47)* *FARE STRUCTURE ELEMENT PRICE*	***(48)*** ***(SalesOfferPackageRef)*** ***(49)*** ***ClassOfUsetRef***
(50) *Delivery options*	*(51)* *FULFILMENT METHOD*	*(52)* *FULFILMENT METHOD*	*(53)* *FULFILMENT METHOD PRICE*	***(54)***
(55)	*(56)* *SUPPLEMENT FARE PRODUCT*	*(57)* *SUPPLEMENT FARE PRODUCT*	*(58)* *FARE PRODUCT PRICE*	***(59)*** ***(SupplementFareProductRef)***

15.17.4 *Nested objects and references*

It is always preferred to have the description of an object whithin the XML tag hierarchy rather to a reference to an XML object located at the root level of the XML hierarchy. Therefore nested objects shall be used each time it is possible (as opposed to referenced objects). References should only be used when a single object is used several time (to avoid repeating the same information multiple times).

This is specifically true for *NoticeAssignment* (always prefer the *Notice* to the *NoticeRef*).

15.18 Classification of fare elements

Data elements should be classified with the NeTEx built in enumerated values and in some cases also the appropriate EFIP ***TypeOfValue***.

15.18.1 *Classifications of Tariffs*

Each EFIP Tariff should be categorised with a ***TariffBasis*** and a ***TypeOfTariff*** fxc :value

Each EFIP ***FareStructureElement*** should be classified with a ***TypeOfFareStructureElement*** efp:value

Each EFIP ***AccessRightAssignments*** should be classified with a ***TypeOfAccessRightAssignment***

Classifications of FareProduct

Each EFIP FARE PRODUCT should be categorised with a ***ProductType*** values

- Each ***FareStructureElement*** should be categorised with one of the 'fxc TypeOfFareStructureElement' values.
- A ***TypeOfFareStructureElement*** can be used to further indicate to a consumer system the meaning of particular fare structure elements.
- Each ***ValidityParameterAssignments*** should be categorised with one of the 'efp:***TypeOfAccessRightAssignment'*** values.

Table 146 – Classifications of FareStructureElements

(60) ***Tariff aspect***	***(61)*** ***FareStructureElement*** contents or references	(62) ***TypeOfFareStructure-ElementRef***	(63) ***TariffBasis***
(64) Point-to-point fare	***(65)*** ***DistanceMatrixElement***	*(66)* *'efp:access'*	(67) *pointToPoint*
(68) Zone-to-zone fare	***(69)*** ***DistanceMatrixElement***	(70) *'efp:access'*	(71) *pointToPoint*
(72) Zonal fare	***(73)*** ***TariffZone***	(74) *'efp:access'*	(75) *zoneToZone*
(76) Stage/Section fare	***(77)*** ***GeographicalInterval, GeographicalStructureFactor***	(78) *'efp:access'*	(79) *unitSection*

(80) Flat fare	***(81)*** ***Network, Groupof Lines, Line***	*(82)* *'efp:access'*	*(83)* *flat*
(84) Period pass	***(85)*** ***TimeInterval,*** ***TimeStructureFactor***	*(86)* *'efp: durations*	*(87)* *period*
(88) Peak/offpeak, etc	***(89)*** ***FareDemandFactor***	*(90)* *'efp:access_when'*	*(91)*
(92) Carnet	***(93)*** ***QualityStructureFactor***	*(94)* *'efp: carnet_units*	*(95)* *pointToPoint, zoneTone, zone*
(96)	***(97)*** ***Operator, Authority,*** ***GroupOfOperators***	*(98)* *'efp: accepted_by*	*(99)*
(100) Quota	***(101)*** ***FareQuotaFactor***	*(102)* *'efp: quotas'*	*(103)* *pointToPoint*
(104) User Type	***(105)*** ***UserProfile***	*(106)* *'efp: carnet_units'*	*(107)* *other*
(108) Group Ticket	***(109)*** ***GroupTicket***	*(110)* *'efp: groups'*	*(111)*
(112) Discount Rights	***(113)*** ***EntitlementGiven***	*(114)* *'efp: entitles'*	*(115)* *other*
(116) Discount Rights	***(117)*** ***EntitlementRequiredn***	*(118)* *'efp: prerequisites'*	*(119)* *other*

(120) Validity	***(121)*** ***FareStructureElementInSequence***	*(122)* *'efp: steps'*	*(123)* *other*
(124) Validity	***(125)*** ***RoundTrip, StepLimit, Frequency, Transferring, Interchanging, ValidityPeriod, etc***	*(126)* *'efp: travel_conditions'*	*(127)* *other*
(128) Commercial	***(129)*** ***PurchaseWindow, Transferability, Refunding, Exchanging, Replacing, etc***	*(130)* *'efp: sales_conditions'*	*(131)* *other*
(132) Avalability	***(133)*** ***FareQuotaFactor***	*(134)* *'efp: quotas*	*(135)* *other*

Only certain combinations of ***FareStructureElement*** and ***TypeOfAccessRightAssignment*** **are meaningful.**

Table 147 – Classifications of FareStructureElements and TypeOfAccessRightAssignment

(136) ***TypeOfFare-StructureElement***	**(137)** **Description**	***(138)*** ***TypeOfAccessRigh-tAssignment***	(139) De`scription
(140) *efp:eligibility*	(141) Eligibility to purchase	*(142)* *efp:eligible*	(143) Eligible for a product or discount
(144) *efp:groups*	(145) Available Group	*(146)* *efp:can_access*	(147) Grants access rights to use or travel on
(148) *efp:access*	(149) Access rights	*(150)* *efp:can_access*	(151) Grants access rights to use or travel on

(152) *efp:access*	(153) Access rights	*(154)* *efp:cannot_access*	(155) Revokes access rights to use or travel on
(156) *efp:access*	(157) Access rights	*(158)* *efp:must_access*	(159) Requires access rights to use or travel on
(160) *efp:accepted_by*	(161) Accepted by operator	*(162)* *efp:can_access*	(163) Grants access rights to use or travel on
(164) *efp:access_when*	(165) Fare demand :peak /off peak etc access rights to use or travel during some period	*(166)* *efp:can_access_when*	(167) Grants access rights to use or travel during some period
(168) *efp:durations*	(169) Available Time intervals	*(170)* *efp:can_purchase_as*	(171) Available duration of travel
(172) *efp:intervals*	(173) Available Geographical intervals	*(174)* *efp:can_purchase_as*	(175) Available interval
(176) *efp:carnet_units*	(177) Available units for purchase	*(178)* *efp:can_purchase_as*	(179) Gives right to purchase
(180) *efp:quotas*	(181) Available Quotas	*(182)* *efp:quota_applies*	(183) Quota applies
(184) *efp:media*	(185) Condition oin Media	*(186)* *efp:can_use_with_type-_of_travel_document*	(187) Right defines tye travel document condition
(188) *efp:steps*	(189) Steps in sequence	*(190)* *efp:condition_of_use*	(191)
(192) *efp:validity*	(193) Validity	*(194)* *efp:validity_chaining*	(195)

(196) *efp:travel_conditions*	(197) Condition on use	*(198)* *efp:use_before*	(199) Takes properties from group package
(200) *efp:travel_conditions*	(201) Condition on use	*(202)* *efp:condition_of_use*	(203) Defines a condition or restriction on use
(204) *efp:travel_conditions*	(205) Condition on use	*(206)* *efp:penalties*	(207) Penalties for misuuse
(208) *efp:sale_conditions*	(209) Defines a condition or restriction on purchase	*(210)* *efp:condition_of_sale*	(211) Defines a condition or restriction on purchase
(212) *Efp:booking*	(213) Condition for reservation	*(214)* *efp:reservation_condition*	(215) Indicates need for reservation
(216) *efp:sale_conditions*	(217) Defines a condition or restriction on purchase	*(218)* *efp:can_purchase_as*	(219) Purchasable item
(220) *efp:sale_conditions*	(221) Defines a condition or restriction on purchase	*(222)* *efp:can_purchase_when*	(223) Specifies when a purchase can be made
(224) *efp:sale_conditions*	(225) Defines a condition or restriction on purchase	*(226)* *efp:distribution_channel_restriction*	(227) Constrains availability for distribution by a channel
(228) *efp:sale_conditions*	(229) Defines a condition or restriction on purchase	*(230)* *efp:can_use_to_pay_for*	(231) Gives ability to purchase automatically using
(232) *efp:sale_conditions*	(233) Defines a condition or restriction on purchase	*(234)* *efp:promotion_applies*	(235) Promotion applies
(236) *efp:outlets*	(237) Specifies where a purchase can be made	*(238)* *efp:can_purchase_where*	(239) Specifies where a purchase can be made
(240)	(241)	*(242)*	(243)

efp:payment_schedules	Payment Schedules	*efp:payment_schedules*	Specifies nature ofs ubcription
(244) *efp:prerequisites*	(245) Defines required entitlements	*(246)* *efp:prerequisites*	(247) Requires right to other products
(248) *efp:entitles*	(249) Defines rights to other products	*(250)* *efp:gives_entitlement*	(251) Gives right to other products

15.19 Delivery structure (headers)

The NeTEx publication schema describes an XML document suitable for exchange either as a stand-alone document, or embedded in a SIRI Delivery (that is the response to a SIRI request for NETEX data). Only the stand-alone document is accepted by the passenger information profile. A publication wraps one or more NeTEx VERSION FRAMEs inside a ***PublicationDelivery*** element. Each ***PublicationDelivery*** acts as a container and is annotated with a time stamp.

The primary element in the publication schema is a ***PublicationDelivery*****,** comprising a simple header, followed by a payload of VERSION FRAMES containing NeTEx data objects (limited to the ones selected in the profile: any other NeTEx object remains valid. However it can be safely ignored without breaking the semantic of the dataset). The delivery may also include some optional metadata indicating the classes present in the payload.

Table 148 – *PublicationDelivery* –Element

Classification	Name	Type	Cardinality	Description
	version	*xsd:NMTOKEN*		Version identifier of schema. Fixed.
PublicationHeaderGroup	***PublicationTimestamp***	*xsd:dateTime*	1:1	Time of output of data. Preferred format UTC. For example, `2017-07-26T13:52:00Z`
	ParticipantRef	*ParticipantCodeType*	1:1	Identifier of system providing data. Often this will be the same as the DATA SOURCE but it is possible there may be multiple participant systems belonging to a single data source.
	Description	*xsd:normalizedString*	0:1	Description of contents.
PayloadGroup	***dataObjects***	*dataObjects_RelStructure*	0:1	NeTEx VERSION FRAMES making up publication.

Delivery example

```
<PublicationDelivery
version="2.04:EU_PI-1.0"
xsi:schemaLocation="http://www.netex.org.uk/netex
NeTEx_publication.xsd"
xmlns="http://www.netex.org.uk/netex"
xmlns:xlink="http://www.w3.org/1999/xlink"
xmlns:ifopt="http://www.ifopt.org.uk/ifopt"
xmlns:gml="http://www.opengis.net/gml/3.2"
xmlns:core="http://www.govtalk.gov.uk/core"
xmlns:siri="http://www.siri.org.uk/siri"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
```

15.20 Structure of References

It is very common in NeTEx to have a reference to another object. Referenced objects can be in the same data set or in another dataset (sent separately). The difference between internal (same dataset) and external object (different datasets) can be made using the **version** attribute (internal) or **versionRef** attribute (external). Also note that the use of the **version** attribute will trigger the NeTEx's XML automatic consistency check (the referenced object must exist in the dataset, if not a validation error is raised). This automatic consistency check will not occur when using **versionRef**.

Table 149 – VersionOfObjectRefStructure –structure

Classifi-cation	Nom	Type	Cardinalité	Description
	NameOfRefClass	NameOfClass	0:1	
«FK»	version	VersionRef	0:1	Reference to version of ENTITY that is in a VERSION FRAME of the current dataset. Mutually exclusive with **versionRef**. Use "any" if the version is unknown (note that this will trigger the NeTEx's XML automatic consistency check).
«FK»	**ref**	ObjectRefObjectIdType	1:1	
«FK»	**versionRef**	VersionIdType	0:1	Identifier of version of the referenced entity. For use when reference is external (i.e in another dataset) and a specific version is required. Mutually exclusive with **version.** Use "any" if the version is unknown. This will also make clear that the referenced object is external.

15.21 Empty values

A lot of NeTEx elements and attributes are not mandatory. This leads to the questions of what should be done with these fields when there is no information to fill them. In XML, three different options are available:

1) No XML tag
2) Tag with no content `<type object></type object>`
3) Empty tag `<type object/>`

For the European passenger information profile, the first option (no XML tag) is the preferred one, and the second one is not accepted.

16 Example XML: Code organisation for a simple fare product.

The following example encodes a simple point-to-point tariff for a line with just three stops, Alpha, Beta and Gamma, along with a fare product, scales offer package and prices. The elements are organised into frames and nested as required by the EFIP.

Table 150 – *Example Price matrix for Mybus line 1*

(1) **Alpha**	(2)	(3)
(4) £1.60	**(5)** **Beta**	(6)
(7) £2.40	(8) £1.80	**(9)** **Gamma**

16.1.1 *Example XML: EU_FP_FARE_OFFER CompositeFrame*

The following XML fragment shows the top levels of a COMPOSITE FRAME of type EU_FP_FARE_OFFER that contains all the other frames. The frames may be in any order.

A frame also allows us to set defaults for the frame contents such as a default currency for financial amounts.

```
<CompositeFrame version="1.0" id="epd:UK:myb:CompositeFrame_EU_FP_FARE_OFFER:Trip@Line_3"
dataSourceRef="myb:Mybus" responsibilitySetRef="myb:tariffs">

    <ValidBetween>

        <FromDate>2019-01-01T00:00:00</FromDate>

        <ToDate>2020-12-31T12:00:00</ToDate>
```

```
</ValidBetween>
<Name>Mybus 1 0 - Fares Example</Name>
<Description>This is a basic example of point to point fares for e</Description>
<TypeOfFrameRef ref="efp:UK:DFT:TypeOfFrame_EU_FP_FARE_OFFER:EFIP" versionRef="efp:v1.0"/>
<!--==== CODESPACEs ==== -->
<codespaces>
    <CodespaceRef ref="fxc_metadata"/>
    <CodespaceRef ref="noc_data"/>
    <CodespaceRef ref="naptStop_data"/>
    <CodespaceRef ref="myb_data"/>
</codespaces>
<FrameDefaults>
    <DefaultCodespaceRef ref="myb_data"/>
    <DefaultDataSourceRef ref="myb:Mybus" version="1.0"/>
    <DefaultCurrency>GBP</DefaultCurrency>
</FrameDefaults>
<frames>
    <!--==== FARE PAYLOAD ==== -->
    <ServiceFrame version="1.0" id="epd:UK:myb:ServiceFrame_EU_FP_NETWORK:Line_3:myb"
            dataSourceRef="myb:Mybus" responsibilitySetRef="myb:tariffs">
    :::: FRAME CONTENTS::::::
    </ServiceFrame>
    <FareFrame version="1.0" id="epd:UK:myb:FareFrame_EU_FP_FARE_PRODUCT:Trip@Line_3:myb"
            dataSourceRef="myb:Mybus" responsibilitySetRef="myb:tariffs">
    :::: FRAME CONTENTS::::::
```

```
        </FareFrame>
        <FareFrame version="1.0" id="epd:UK:myb:FareFrame_EU_FP_FARE_PRICE:Trip@Line_3:myb"
                dataSourceRef="myb:Mybus" responsibilitySetRef="myb:tariffs">
        :::: FRAME CONTENTS::::::
        </FareFrame>
                <!--==== OPERATOR COMMON RESOURCES==== -->
        <ResourceFrame version="1.0" id="epd:UK:MYBUS:ResourceFrame_EU_FP_COMMON:MYBUS:myb"
                    dataSourceRef="myb:Mybus" responsibilitySetRef="myb:network_data">
            <Name>Mybus Operator specific  common resources</Name>
            <TypeOfFrameRef ref="efp:UK:DFT:TypeOfFrame_EU_FP_COMMON:EFIP" versionRef="efp:v1.0"/>
    :::: FRAME CONTENTS::::::
        </ResourceFrame>
    </frames>
</CompositeFrame>
```

16.1.2 *Example XML: EU_FP_COMMON ResourceFrame*

The following XML fragment shows a definition of a RESOURCE FRAME of type EU_FP_COMMON that provides common details about the OPERATOR, RESPONSIBILITY SET DATA SOURCE etc for all the other frames. The same frame contents will probably suffice for all data sets supplied by the OPERATOR.

A NOC code is used to udentfiy the operator.

Note that in this and other examples the ‘myb:’ prefix could be omitted as it is the defaut codespace for the fram.

```
    <ResourceFrame version="1.0" id="epd:UK:MYBUS:ResourceFrame_EU_FP_COMMON:MYBUS:myb"
            dataSourceRef="myb:Mybus" responsibilitySetRef="myb:network_data">
        <Name>Mybus Operator specific  common resources</Name>
```

```
<TypeOfFrameRef ref="efp:UK:DFT:TypeOfFrame_EU_FP_COMMON:EFIP" versionRef="efp:v1.0"/>
<codespaces>
    <Codespace id="myb_data">
        <Xmlns>myb</Xmlns>
        <XmlnsUrl>http://www.mybus.co.uk/</XmlnsUrl>
        <Description>Mybus data</Description>
    </Codespace>
</codespaces>
<dataSources>
    <DataSource id="myb:Mybus" version="1.0">
        <Email>feedback@mybus.co.uk</Email>
    </DataSource>
</dataSources>
<responsibilitySets>
    <ResponsibilitySet version="1.0" id="myb:tariffs">
        <Name>Operator Tariffs</Name>
        <roles>
            <ResponsibilityRoleAssignment version="1.0" id="myb:tariff_data@creates">
                <DataRoleType>creates</DataRoleType>
                <StakeholderRoleType>FareManagement</StakeholderRoleType>
                <ResponsibleOrganisationRef ref="noc:MYBUS"
                        version="1.0">Mybus</ResponsibleOrganisationRef>
            </ResponsibilityRoleAssignment>
        </roles>
    </ResponsibilitySet>
```

```
</responsibilitySets>
            <!--==== CODE VALUES  ==== -->
<typesOfValue>
    <Branding version="1.0" id="myb:Mybus@brand">
        <Name>Mybus</Name>
        <Url>https://www.Mybus.co.uk/static/images/colorways/Mybus/logo.png</Url>
    </Branding>
</typesOfValue>
<organisations>
    <!-- ==== ORGANISATIONS ==== -->
    <Operator version="1.0" id="noc:MYBUS">
        <PublicCode>MYBU</PublicCode>
        <PrivateCode type="nocn">nocn:987122</PrivateCode>
        <Name>Mybus</Name>
        <ShortName>Mybus</ShortName>
        <TradingName>Mybus Ltd</TradingName>
        <ContactDetails>
            <Phone>01283 449191</Phone>
        </ContactDetails>
        <OrganisationType>operator</OrganisationType>
        <Address>
            <Street>The Close</Street>
            <Town>Barchester</Town>
            <PostCode>BT10 9XA</PostCode>
            <PostalRegion>Barsetshire</PostalRegion>
```

```
            </Address>
            <PrimaryMode>bus</PrimaryMode>
        </Operator>
    /organisations>
</ResourceFrame>
```

16.1.3 *Example XML: EU_FP_ NETWORK ServiceFrame*

The following XML fragment shows a definition of a SERVICE FRAME of type EU_FP_COMMON that defines the LINE and three SCHEDULED STOP POINTs for the *Mybus Line 3* tariff. The stop definitions have NaPTAN identifiers and and content, including external references to TOPOGRAPHIC PLACEs corresponding to ***NptgLocality*** instances.

```
<ServiceFrame version="1.0" id="epd:UK:myb:ServiceFrame_EU_FP_NETWORK:Line_3:myb"
        dataSourceRef="myb:Mybus" responsibilitySetRef="myb:tariffs">
    <Name>Netwrok elements for Mybus Line 3</Name>
    <TypeOfFrameRef ref="efp:UK:DFT:TypeOfFrame_EU_FP_NETWORK:EFIP" versionRef="efp:v1.0"/>
    <codespaces>
        <CodespaceRef ref="naptStop_data"/>
    </codespaces>
    <prerequisites>
        <ResourceFrameRef version="1.0"
                ref="epd:UK:MYBUS:ResourceFrame_EU_FP_COMMON:MYBUS:myb"/>
    </prerequisites>
    <!--==== LINEs ==== -->
    <lines>
        <Line version="1.0" id="myb:Line_3">
```

```
            <Name>Mybus 3</Name>
            <Description>Alpha - -Gamma</Description>
            <PublicCode>3</PublicCode>
            <PrivateCode type="noc_data">METR_1</PrivateCode>
            <OperatorRef version="1.0" ref="noc:MYBUS">137122</OperatorRef>
            <LineType>local</LineType>
            <PaymentMethods>cash</PaymentMethods>
        </Line>
    </lines>
    <!--==== STOP POINTs ==== -->
    <scheduledStopPoints>
        <ScheduledStopPoint version="naptStop:any" id="naptStop:4400CY0037">
            <Name>Alpha</Name>
            <TopographicPlaceView>
                <TopographicPlaceRef versionRef="nptg:any" ref="nptgLocality:E0056633"/>
                <Name>Barchester</Name>
                <QualifierName>Barsetshire</QualifierName>
            </TopographicPlaceView>
        </ScheduledStopPoint>
        <ScheduledStopPoint version="naptStop:any" id="naptStop:4400CY0038">
            <Name>Beta</Name>
            <TopographicPlaceRef versionRef="nptg:any" ref="nptgLocality:E0056633"/>
        </ScheduledStopPoint>
        <ScheduledStopPoint version="naptStop:any" id="naptStop:4400CY0039">
            <Name>Gama</Name>
```

```
            <TopographicPlaceRef versionRef="nptg:any" ref="nptgLocality:E0056633"/>
        </ScheduledStopPoint>
    </scheduledStopPoints>
</ServiceFrame>
```

16.1.4 *Example XML: EU_FP_FARE_PRODUCT FareFrame*

The following XML fragment shows a definition of a SERVICE FRAME of type EU_FP_FARE_PRODUCT that provides the TARIFF, FARE PRODUCT and SALE OFFER PACKAGEs and SCHEDULED STOP POINTs for the *MyBus Line 3* tariff.

The service frame from above is declared as a prerequisite.

```
<FareFrame version="1.0" id="epd:UK:myb:FareFrame_EU_FP_FARE_PRODUCT:Trip@Line_3:myb"
        <dataSourceRef="myb:Mybus" responsibilitySetRef="myb:tariffs">
    <Name>Mybus 1</Name>
    <TypeOfFrameRef ref="efp:UK:DFT:TypeOfFrame_EU_FP_FARE_PRODUCT:EFIP"
        <versionRef="efp:v1.0"/>
    <prerequisites>
        <ServiceFrameRef version="1.0"
                ref="epd:UK:myb:ServiceFrame_EU_FP_NETWORK:Line_3:myb"/>
    </prerequisites>
    <!--==== Fare Structure ==== -->
    <tariffs>
        <Tariff version="1.0" id="myb:Tariff@single@Line_3">
        ::::: TARIFF DEFINITION
        </Tariff>
    </tariffs>
```

```
    <!--==== Fare Product ==== -->
    <fareProducts>
        <PreassignedFareProduct version="1.0" id="myb:Trip@single">
        ::::: FARE PRODUCT DEFINITIONs
        </PreassignedFareProduct>
        <!--==== Other Products ==== -->
    </fareProducts>
    <!--==== Sales Packages==== -->
    <salesOfferPackages>
        <SalesOfferPackage version="1.0" id="myb:Trip@single-SOP@p-ticket">
        ::::: SALES OFFFPACKAGE DEFINITIONs
    </SalesOfferPackage>
    </salesOfferPackages>
</FareFrame>
```

16.1.4.1 *Example XML: Tariff in a EU_FP_PRODUCT frame*

The following XML fragment shows a definition of a TARIFF for the *Mybus Line3* example.

The TARIFF defines three separate FARE STRUCTURE ELEMENTs. Each is used to specify a independent aspects of the fare structure

a) **Access rights** (classified as *'efp:access'* with a ***TypeOfFareStructureElementRef*** element), defined with a list of DISTANCE MATRIX ELEMENTs, one for each possible stop combination. This is specifying which network access rights may be consumed. Since this is a point to point fare, a list of DISTANCE MATRX ELEMENTs is given, each specify an origibn/ destination pair. When the user buys the productt, one of the specified DISTANCE MATRX ELEMENTs must be selected and indicates the start and end stop for a trip.

b) **Eligibility** (classified as *'efp:eligibility'* with a ***TypeOfFareStructureElementRef*** element), defined with a list of USER PROFILE usage parameter; in this case just 'adult'. The specified user type can be validated.

c) **Travel** (classified as *'efp:travel_conditions'* with a ***TypeOfFareStructureElementRef*** element), restricted to a single trip with no interchanging with a ROUND TRIP and INTERCHANGIING usage parameters. The specified travel parametesr indicate how the product can be used are can be validated as such

defines a VALDIABLE ELEMENT to specif its ACCESS RIGHTS IN PRODUCT. The VALIDABLE ELEMENT references the FARE STRUCTURE ELEMENTs defined by the TARIFF on which the product is based.

The DISTANCE MATRIX ELEMENTs reference the three NaPTAN stops defined above .

```
<Tariff version="1.0" id="myb:Tariff@single@Line_3">
    <validityConditions>
        <ValidBetween>
            <FromDate>2017-01-01T00:00:00Z</FromDate>
            <ToDate>2018-01-01T00:00:00Z</ToDate>
        </ValidBetween>
    </validityConditions>
    <Name>Mybus Line 3 - Single  Fares</Name>
    <OperatorRef version="1.0" ref="noc:MYBUS">137122</OperatorRef>
    <LineRef version="1.0" ref="myb:Line_3"/>
    <fareStructureElements>
        <FareStructureElement version="1.0" id="myb:Tariff@single@access">
            <Name>O/d pairs for Line 3</Name>
            <TypeOfFareStructureElementRef versionRef="efp:v1.0" ref="efp:access"/>
            <!--Here we encode the actual fare triangle elements
```

```
            Alpha
            160 Beta
            240 180 Gamma -->
        <distanceMatrixElements>
            <DistanceMatrixElement version="1.0" id="myb:4400CY0037+4400CY0038">
                <Name>Alpha+Beta</Name>
                <StartStopPointRef version="naptStop:any" ref="naptStop:4400CY0037"/>
                <EndStopPointRef version="naptStop:any" ref="naptStop:4400CY0038"/>
            </DistanceMatrixElement>
            <DistanceMatrixElement version="1.0" id="myb:4400CY0037+4400CY0039">
                <Name>Alpha+Gamma</Name>
                <StartStopPointRef version="naptStop:any" ref="naptStop:4400CY0037"/>
                <EndStopPointRef version="naptStop:any" ref="naptStop:4400CY0039"/>
            </DistanceMatrixElement>
            <DistanceMatrixElement version="1.0" id="myb:4400CY0038+4400CY0039">
                <Name>Beta+Gamma</Name>
                <StartStopPointRef version="naptStop:any" ref="naptStop:4400CY0038"/>
                <EndStopPointRef version="naptStop:any" ref="naptStop:4400CY0039"/>
            </DistanceMatrixElement>
        </distanceMatrixElements>
        <GenericParameterAssignment version="1.0" order="01" id="myb:Tariff@single@access">
            <TypeOfAccessRightAssignmentRef versionRef="efp:v1.0" ref="efp:can_access"/>
            <ValidityParameterAssignmentType>EQ</ValidityParameterAssignmentType>
            <validityParameters>
                <LineRef version="1.0" ref="myb:Line_3"/>
```

```
            </validityParameters>
        </GenericParameterAssignment>
    </FareStructureElement>
    <FareStructureElement id="myb:Tariff@single@eligibility" version="1.0">
        <Name>eligible user types</Name>
        <TypeOfFareStructureElementRef versionRef="efp:v1.0" ref="efp:eligibility"/>
        <GenericParameterAssignment order="1"
                id="myb:Tariff@single@eligibility" version="1.0">
            <TypeOfAccessRightAssignmentRef versionRef="efp:v1.0" ref="efp:eligible"/>
            <LimitationGroupingType>XOR</LimitationGroupingType>
            <limitations>
                <UserProfile version="1.0" id="myb:adult">
                    <Name>Adult</Name>
                </UserProfile>
            </limitations>
        </GenericParameterAssignment>
    </FareStructureElement>
    <FareStructureElement id="myb:Tariff@single@conditions_of_travel" version="1.0">
        <Name>COnditions on travel</Name>
        <TypeOfFareStructureElementRef versionRef="efp:v1.0" ref="efp:travel_conditions"/>
        <GenericParameterAssignment version="1.0" order="1"
                id="myb:Tariff@single@travel_conditions">
            <Name>Conditions of travel</Name>
            <TypeOfAccessRightAssignmentRef versionRef="efp:v1.0" ref="efp:condition_of_use"/>
            <LimitationGroupingType>AND</LimitationGroupingType>
```

```
            <limitations>
                <RoundTrip version="1.0" id="myb:Trip@travel@condition@direction">
                    <TripType>single</TripType>
                </RoundTrip>
                <Interchanging version="1.0" id="myb:Trip@single@NoTransfers">
                    <MaximumNumberOfInterchanges>0</MaximumNumberOfInterchanges>
                </Interchanging>
            </limitations>
        </GenericParameterAssignment>
    </FareStructureElement>
</fareStructureElements>
</Tariff>
```

16.1.4.2 *Example XML: Fare product in a EU_FP_PRODUCT frame*

The following XML fragment shows a definition of a FARE PRODUCT for the *Mybus Line3* example.

The PREASSIGNED FARE PRODUCT defines a VALDIABLE ELEMENT to specif its ACCESS RIGHTS IN PRODUCT. The VALIDABLE ELEMENT references the FARE STRUCTURE ELEMENTs defined by the TARIFF on which the product is based.

```
<PreassignedFareProduct version="1.0" id="myb:Trip@single">
    <Name> Single Ticket</Name>
    <ChargingMomentRef versionRef="efp:v1.0" ref="efp:prepayment"/>
    <ChargingMomentType>beforeTravel</ChargingMomentType>
    <TypeOfFareProductRef versionRef="efp:v1.0" ref="efp:standard_product@trip@single"/>
    <OperatorRef version="1.0" ref="noc:MYBUS">137122</OperatorRef>
    <ConditionSummary>
        <FareStructureType>pointToPointFare</FareStructureType>
```

```
        <TariffBasis>pointToPoint</TariffBasis>
    </ConditionSummary>
    <!--==== VALIDABLE ELEMENTs ==== -->
    <validableElements>
        <ValidableElement version="1.0" id="myb:Trip@single@travel">
            <Name>Single  ride</Name>
            <fareStructureElements>
                <FareStructureElementRef version="1.0" ref="myb:Tariff@single@access"/>
                <FareStructureElementRef version="1.0" ref="myb:Tariff@single@eligibility"/>
                <FareStructureElementRef version="1.0"
                        ref="myb:Tariff@single@conditions_of_travel"/>
            </fareStructureElements>
        </ValidableElement>
    </validableElements>
    <!--==== ACCESS RIGHTs ===== -->
    <accessRightsInProduct>
        <AccessRightInProduct version="1.0" id="myb:Trip@single" order="1">
            <ValidableElementRef version="1.0" ref="myb:Trip@single@travel"/>
        </AccessRightInProduct>
    </accessRightsInProduct>
    <ProductType>singleTrip</ProductType>
</PreassignedFareProduct>
```

16.1.4.3 *Example XML: Sales offer package in a EU_FP_PRODUCT frame*

The following XML fragment shows a definition of a SALES OFFER PACKAGE for the *Mybus Line3* example. It references the PREASSIGNED FARE PRODUCT defined above as well a a TYPE O FTRAVEL DOCUMENT specifying the type of media.

```
<SalesOfferPackage version="1.0" id="myb:Trip@single-SOP@p-ticket">
    <BrandingRef version="1.0" ref="myb:Mybus@brand"/>
    <Name>Mybus Cash</Name>
    <TypeOfSalesOfferPackageRef versionRef="efp:v1.0" ref="efp:standard_product@operator"/>
    <distributionAssignments>
        <DistributionAssignment version="1.0" id="myb:Trip@single-SOP@p-ticket@onBoard" order="2">
            <Name>Onboard</Name>
            <Description>Pay for  ticket  onboard</Description>
            <DistributionChannelRef versionRef="efp:v1.0" ref="efp:on_board"/>
            <DistributionChannelType>onBoard</DistributionChannelType>
            <TicketingServiceFacilityList>purchase</TicketingServiceFacilityList>
            <PaymentMethods>cashAndCard</PaymentMethods>
            <FulfilmentMethodRef ref="efp:collect_on_board" versionRef="efp:v1.0"/>
        </DistributionAssignment>
    </distributionAssignments>
    <salesOfferPackageElements>
        <SalesOfferPackageElement version="1.0" id="myb:Trip@single-SOP@p-ticket" order="1">
            <TypeOfTravelDocumentRef versionRef="efp:v1.0" ref="efp:printed_ticket"/>
            <PreassignedFareProductRef version="1.0" ref="myb:Trip@single"/>
        </SalesOfferPackageElement>
```

```
    </salesOfferPackageElements>
</SalesOfferPackage>
```

16.1.5 *Example XML: EU_FP_FARE_PRICE FareFrame*

The following XML fragment shows a definition of a FARE FRAME of type EU_FP_FARE_PRICE that gives the three FARE PRiCEs for the DISTANCE MATRIX ELEMENTs defined in the previous example. The prices are listed simply in a FARE TABLE without any consideration for presentation.

Note that FARE TABLEs can also be used to provide more elaborate organisation with guidance for presentation by placing prices in CELLs and assigning tehm to ROWs and COLUMNs.

```
    <FareFrame version="1.0" id="epd:UK:myb:FareFrame_EU_FP_FARE_PRICE:Trip@Line_3:myb"
            dataSourceRef="myb:Mybus" responsibilitySetRef="myb:tariffs">
        <TypeOfFrameRef ref="efp:UK:DFT:TypeOfFrame_EU_FP_FARE_PRICE:EFIP" versionRef="efp:v1.0"/>
        <FrameDefaults>
            <DefaultCurrency>GBP</DefaultCurrency>
        </FrameDefaults>
        <prerequisites>
            <FareFrameRef version="1.0"
                    ref="epd:UK:myb:FareFrame_EU_FP_FARE_PRODUCT:Trip@Line_3:myb"/>
        </prerequisites>
        <fareTables>
            <FareTable version="1.0" id="myb:Trip@single-SOP@p-ticket@Line_3@adult">
                <Description>Adult Single Fares - Organised as a fare triangle</Description>
                <!--
```

```
160 Beta
240 180 Gamma    -->
                <pricesFor>
                    <PreassignedFareProductRef version="1.0" ref="myb:Trip@single"/>
                    <SalesOfferPackageRef version="1.0" ref="myb:Trip@single-SOP@p-ticket"/>
                    <UserProfileRef version="1.0" ref="myb:adult"/>
                </pricesFor>
                <usedIn>
                    <TariffRef version="1.0" ref="myb:Tariff@single@Line_3"/>
                </usedIn>
                <specifics>
                    <LineRef version="1.0" ref="myb:Line_3"/>
                    <TypeOfTravelDocumentRef versionRef="efp:v1.0" ref="efp:printed_ticket"/>
                </specifics>
                <prices>
                    <DistanceMatrixElementPrice version="1.0"
                            id="myb:Trip@single-SOP@p-ticket@Line_3@adult@4400CY0037+4400CY0038">
                        <Amount>1.60</Amount>
                        <DistanceMatrixElementRef version="1.0" ref="myb:4400CY0037+4400CY0038"/>
                    </DistanceMatrixElementPrice>
                    <DistanceMatrixElementPrice version="1.0"
                            id="myb:Trip@single-SOP@p-ticket@Line_3@adult@4400CY0037+4400CY0039">
                        <Amount>2.40</Amount>
                        <DistanceMatrixElementRef version="1.0" ref="myb:4400CY0037+4400CY0039"/>
                    </DistanceMatrixElementPrice>
```

```
                <DistanceMatrixElementPrice version="1.0"
                        id="myb:Trip@single-SOP@p-ticket@Line_3@adult@4400CY0038+4400CY0039">
                    <Amount>1.80</Amount>
                    <DistanceMatrixElementRef version="1.0" ref="myb:4400CY0038+4400CY0039"/>
                </DistanceMatrixElementPrice>
            </prices>
        </FareTable>
    </fareTables>
</FareFrame>
```

17 Profile compliance and data quality checking

The checking of profile conformance and of data quality is extremely important for achieving successful interoperability. Data producers need a means to verify that their output is correct and usable. Data consumers need to know that any EFIP compliant document meets a reasonable set of quality criteria before they even attempt to use it. There also needs to be a process for resolving issues in exported data with the producer.

See the EPIP specification for a discussion of validation and data quality.

17.1 Summary of data quality rules for EFIP

The same quality rules apply to the fare profile as to the base profile – See the EPIP specification. In addition, a number of additional semantic rules can be applied that are specific to fare. Some examples are given below.

Table 151 – *Primary quality checking rules*

Type	Checking rule category	Description
D	**Common NeTEx semantic rules conformance** (General Profile validator)	Rules relating to the use of NeTEx elements arising from the real-world semantics of the objects that they represent; that cannot be or are not expressed in the XML but which apply to any NeTEx schema. – The fare structures and product must be a realistic combination, e.g. DayPass will normally be Zonal, Season tickets, SingleTrip products will be point to point, zonal or stage fares, etc. – A ***TariffZone*** must have a at least one member ***ScheduledStopPoint***; – If the ***FareZone*** topology is given as disjoint, then each ***ScheduledStopPoints*** should be in only one ***TariffZone*** – There should be a separate SalesOfferPackage for each

		TypeOfTravelDocument; • Etc.
E	**Specific profile conformance** (EFIPEFP validators)	The populating of data values in a profile conformant document shall fully uphold the specific semantics of the profile: • The rules for encoding EFIPEFP identifiers shall be followed. • A ***Tariff*** must be categorised with a ***TariffType*** • A ***FareProduct*** must be categorised with a ***ProductType*** • Etc.
F	**Data Completeness** (EFIPEFP validators)	All data required to make the data consistent and usable as a coherent set shall be available. The notion of 'completeness' depends on the intended use case; for example, whether all tariff zones or just the tariff zones that are used by a line should be present; or whether only current data is needed or future data as well. Examples for the EFIP include: • All the ***ScheduledStopPoints*** in a ***FareZone*** should be specified as members**.** • A ***DistanceMatrixElement*** must be given for each permitted Origin/Destination pair of a point-to-point tariff. • All the ***TariffZone*** instances must be given for a zonal tariff. • A ***UserProfile*** must be defined for each type of user. • A ***FarePrice*** should be given for every combination of price that is available**.** • Etc.
G	**Data Plausibility**	Values of parameters should be plausible against common sense measures. (Thus, individual bus trips do not cost £100**,** tariff distances for buses are not

	(General Profile validator)	measured in hundreds of miles, child fares are cheaper than adult ones**)**. • Prices increase with distance, usually progressively (i.e. more is cheaper) • Period pass prices increase with ***TimeInterval*** length. (A month cost more than a day). • The spatial coordinates of ***ScheduledStopPoints*** should lie within those of their ***TariffZone***. • ***TariffZone*** instances should lie within the bounds of their ***TopographicPlace***. • Dates should lie within a few years of the present, durations should be plausible for their usage (e.g. season passes are only available in simple interval multiples of a day, week or month, etc). • The public identifiers used to label tariff zones should be unique within their expected scope of use. (For example, many towns and operators will have a '*Town Centre*, but there should only be one '*Town Centre* (at least per ***Operator***) in a given town). • Instances of ***PriceRule, Roundings***, etc referenced other elements and needed to calculate prices should be defined.
H	**Data Relevance** (Profile validator)	Documents should not contain irrelevant or unused instances of data. (this requires a notion of which elements in a frame are the primary payload for the profile). For example, for the FXC fares: • Every ***ScheduledStopPoint*** present in a document, or one of its components should be used by at least one ***TariffZone, FarePointInPattern*** or ***DistanceMatrixElement***. • Every TariffZone should be used by at least one DistanceMatrixElement or

		FareStructureElement/ GenericParameterAssignment. • Etc.
I	**Data correctness** (Manual inspection against real-world evidence)	Data values should accurately describe the real-world entities they are meant to represent. • For an EU_FP_NETWORK_FARE_OFFER, all the lines of a ***Network*** shall be described. • Names of Sales Offer Packages, etc, should be correct and spelled right. • For a stage fare, all the Fare Stages for a ***Line*** shall be specified. • For a zonal fare, all the instances of a ***TariffZone*** for a given ***Network*** shall be specified. • Prices should be correct. • Etc.

18 Data Exchange

In the context of the European passenger information profile, data exchange will be implemented through XML file exchange. Each file will contain a single **PublicationDelivery**, itself containing a single **CompositeFrame** or specific frames **RessourceFrame, SiteFrame, ServiceFrame, TimetableFrame, CalendarFrame** (of one of the previously defined types).

Three different types of files will be accepted:

- Line XML files containing a single EU_FP_LINE_OFFER CompositeFrame (single whole line description with stops, topology and timetable).

- Network XML files containing a single EU_FP_NETWORK_OFFER CompositeFrame (set of multiple whole line description).
- Stop XML files containing a single EU_FP_STOP SiteFrame (comprehensive description of all the stops of a country, an authority or an operator).

Note that this file structure result in possible redundancy across files for information like STOP PLACE, CONNECTION, etc. This is, of course, valid and will be managed with a simple rule: the latest version (most recent **changed** attributes or higher **version** number when it is a number, if the version is not managed, the last received ones will be used) always wins !

18.1 File naming

The following file name structure is recommended for any file dissemination:

- Prefix of the file names which have been established and agreed upon by the exchange partners, here **NETEX_FP**
- Major version of the profile; currently 01
- country ISO code
- Provider name (use NAP if coming from a national access point)
- File type (LINE, NETWORK or STOP)
- For LINE only: line number or name if no number is available (short name, fewer than 15 characters, made only of letters (uppercase and lowercase), numbers and "-")
- Date of the creation of the data package (YYYMMDD)
- All fields are separated by "_"

Example file name:

```
NETEX_FP_01_GE_BVB_LINE_3_20161201.xml
NETEX_FP_02_FR_FR1_NETWORK_20171105.xml
NETEX_FP_01_UK_NAPTAN_STOP_20171212.xml
```

Annex A
(Normative)**115226**

Data Dictionary

This data dictionary repeats the definitions from Transmodel (v6.0) and NeTEx (v1.1), but selects only the concepts incuded in the EPIP. Concepts written in uppercase are from the conceptual model (i.e. Transmodel); those in camel-case are specific to the NeTEx physical model. See Transmodel and NeTEx for further information.

- Basic – In simple EFIP
- Further – In enhanced EFIP
- Additional; - Not in profile (may be used in an extended data set)

ACCESS RIGHT PARAMETER ASSIGNMENT	Basic	The assignment of a fare collection parameter (referring to geography, time, quality or usage) to an element of a fare system (access right, validated access, control mean, etc.).
AMOUNT OF PRICE UNIT	Basic	A FARE PRODUCT consisting in a stored value of PRICE UNITs: an amount of money on an electronic purse, amount of units on a value card etc.
BLACKLIST	ADDITIONAL	A list of identified TRAVEL DOCUMENTs or CONTRACTs the validity of which has been cancelled temporarily or permanently, for a specific reason like loss of the document, technical malfunction, no credit on bank account, offences committed by the customer, etc.
BORDER POINT	Further	A POINT on the Network marking a boundary for the fare

		calculation. May or may not be a SCHEDULED STOP POINT.
CANCELLING	Further	Parameter giving conditions for cancelling of a purchased access right.
CAPPED DISCOUNT RIGHT	Further	A specialisation of SALE DISCOUNT RIGHT where the discount is expressed as a rule specifying a ceiling for a given time interval. For example, the London Oyster card fare, which charges for each journey until travel equivalent to a day pass has been consumed after which further travel is free at that day.
CAPPING RULE	Further	A capping limit for a given time interval, where the capping is expressed by another product. For example, the London Oyster card fare, which charges for each journey until travel equivalent to a day pass for the mode of travel has been consumed.
CAPPING RULE PRICE	Further	A set of all possible price features of a CAPPING RULE: default total price, discount in value or percentage etc.
CELL	Basic	An unique individual combination of features within a FARE TABLE, used to associate a FARE PRICE with a fare element.
CHARGING MOMENT	Basic	A classification of FARE PRODUCTs according to the payment method and the account location: pre-payment with cancellation (throw-away), pre-payment with debit on a value card, pre-payment without consumption registration (pass), post-payment etc.
CHARGING POLICY	Basic	Parameter governing minimum amount and credit allowed when consuming a FARE PRODUCT.
COMMERCIAL	Further	A category of users depending on their commercial relations with

PROFILE		the operator (frequency of use, amount of purchase etc.), often used for allowing discounts.
COMMERCIAL PROFILE ELIGIBILITY	ADDITIONAL	Whether a specific TRANSPORT CUSTOMER is eligible for a FARE PRODUCT with a specific COMMERCIAL PROFILE as a validity parameter.
COMPANION PROFILE	Basic	The number and characteristics of the persons entitled to travel in a group or as companions to another USER PROFILE.
CONTROLLABLE ELEMENT	ADDITIONAL	A CONTROLLABLE ELEMENT as a part of a FARE STRUCTURE ELEMENT, including its possible order in the sequence of CONTROLLABLE ELEMENTs grouped together to form that FARE STRUCTURE ELEMENT, and its possible quantitative limitation.
CONTROLLABLE ELEMENT PRICE	ADDITIONAL	A set of all possible price features of a CONTROLLABLE ELEMENT: default total price, discount in value or percentage etc.
CUSTOMER	ADDITIONAL	An identified person or organisation involved in a fare process. There may be a CONTRACT between the CUSTOMER and the OPERATOR or the AUTHORITY ruling the consumption of services.
CUSTOMER ACCOUNT	ADDITIONAL	A registration of the CUSTOMER with an ACCOUNT PROVIDER to obtain travel services.
CUSTOMER ACCOUNT SECURITY LISTING	ADDITIONAL	The presence of a CUSTOMER ACCOUNT on a SECURITY LIST.
CUSTOMER ACCOUNT	ADDITIONAL	A classification of CUSTOMER ACCOUNT.

STATUS		
CUSTOMER ELIGIBILITY	ADDITIONAL	Whether a specific TRANSPORT CUSTOMER is eligible for a FARE PRODUCT with a specific validity Parameter. This may be subject to a particular VALIDITY CONDITION
CUSTOMER PURCHASE PACKAGE	ADDITIONAL	A purchase of a SALES OFFER PACKAGE by a CUSTOMER, giving access rights to one or several FARE PRODUCTs materialised as one or several TRAVEL DOCUMENTs.
CUSTOMER PURCHASE PACKAGE ELEMENT	ADDITIONAL	The assignment of a SALES OFFER PACKAGE ELEMENT, for use in a CUSTOMER SALES PACKAGE.
CUSTOMER PURCHASE PACKAGE ELEMENT ACCESS	ADDITIONAL	Access to a VALIDABLE ELEMENT by a specific CUSTOMER PURCHASE PACKAGE through use of CUSTOMER PURCHASE PACKAGE.
CUSTOMER PURCHASE PACKAGE PRICE	ADDITIONAL	A specialisation of FARE PRICE that defines the price of a CUSTOMER PURCHASE PACKAGE
CUSTOMER PURCHASE PARAMETER ASSIGNMENT	ADDITIONAL	A VALIDITY PARAMETER ASSIGNMENT specifying practical parameters chosen for a CUSTOMER PURCHASE PACKAGE within a given fare structure (e.g. the origin or destination zone in a zone-counting system).
CUSTOMER SECURITY LISTING	ADDITIONAL	The presence of a CUSTOMER on a SECURITY LIST.

DISCOUNTING RULE	Basic	A price calculation rule determined by a set of discounts, depending upon a USAGE PARAMETER, to be applied to a FARE PRICE.
DISTANCE MATRIX ELEMENT	Basic	A cell of an origin-destination matrix for TARIFF ZONEs or STOP POINTs, expressing a fare distance for the corresponding trip: value in km, number of fare units etc.
DISTANCE MATRIX ELEMENT PRICE	Basic	A set of all possible price features of a DISTANCE MATRIX ELEMENT: default total price etc.
DISTRIBUTION ASSIGNMENT	Basic	An assignment of the COUNTRY and/or DISTRIBUTION CHANNEL through which a product may or may not be distributed.
DISTRIBUTION CHANNEL	Basic	A type of outlet for selling of a product.
ELIGIBILITY CHANGE POLICY	Further	Parameter indicating the action to be taken when a user's eligibility status changes.
ENTITLEMENT CONSTRAINT	Further	Constraints on choices for an dependent entitled product relative to the required choices for the prerequisite entitling product.
ENTITLEMENT GIVEN	Further	Parameter indicating whether a particular FARE PRODUCT provides an entitlement to buy or use an access right.
ENTITLEMENT PRODUCT	ADDITIONAL	A precondition to access a service or to purchase a FARE PRODUCT issued by an organisation that may not be a PT operator (e.g. military card).
ENTITLEMENT	Further	Parameter indicating whether a particular FARE PRODUCT requires

REQUIRED		an entitlement to by or use an access right.
EXCHANGING	Further	Whether and how the access right may be exchanged for another access right.
FARE CONTRACT	ADDITIONAL	A contract with a particular (but possibly anonymous) customer, ruling the consumption of transport services (and joint services). A FARE CONTRACT may be designed for a fixed SALES OFFER PACKAGE (e.g. ticket) or to allow successive purchases of SALES OFFER PACKAGEs.
FARE CONTRACT ENTRY	ADDITIONAL	A log entry describing an event referring to the life of a FARE CONTRACT: initial contracting, sales, validation entries, etc. A subset of a FARE CONTRACT ENTRY is often materialised on a TRAVEL DOCUMENT.
FARE CONTRACT SECURITY LISTING	ADDITIONAL	The presence of a FARE CONTRACT on a SECURITY LIST.
FARE DAY TYPE	Basic	A type of day used in the fare collection domain, characterised by one or more properties which affect the definition of access rights and prices in the fare system.
FARE DEMAND FACTOR	Basic	A named set of parameters defining a period of travel with a given price, for example off peak, peak, super off peak, etc.
FARE ELEMENT IN SEQUENCE	ADDITIONAL	A FARE ELEMENT as a part of an ELEMENT, including its possible order in the sequence of FARE ELEMENTs.
FARE FRAME	Basic	The set of all fare data defined for a specific VEHICLE MODE to which the same VALIDITY CONDITIONs have been assigned.

FARE FRAME DEFAULTS	Basic	Set of pricing parameters and values to apply to an individual element in the frame if no explicit value is specified on the element.
FARE INTERVAL	Basic	An interval based aspect of the fare structure.
FARE POINT IN PATTERN	Basic	A POINT IN PATTERN which represents the start or end of a FARE SECTION, or a point used to define a SERIES CONSTRAINT.
FARE PRICE	Basic	Price features DEFINED BY DEFAULT characterizing different PRICE GROUPs.
FARE PRODUCT	Basic	An immaterial marketable element (access rights, discount rights, etc.), specific to a CHARGING MOMENT.
FARE PRODUCT PRICE	Basic	A set of all possible price features of a FARE PRODUCT: default total price, discount in value or percentage etc.
FARE QUOTA FACTOR	ADDITIONAL	A named set of parameters defining a number of quota fares available of a given denomination.
FARE SCHEDULED STOP POINT	Basic	A specialisation of SCHEDULED STOP POINT describing a stop with fare accounting and routing characteristics.
FARE SECTION	Basic	A subdivision of a JOURNEY PATTERN consisting of consecutive POINTs IN JOURNEY PATTERN, used to define an element of the fare structure.
Fare structure	Basic	Set of parameters that determine the basic tariffs.
FARE STRUCTURE ELEMENT	Basic	A sequence or set of CONTROLLABLE ELEMENTs to which rules for limitation of access rights and calculation of prices (fare structure)

		are applied.
FARE STRUCTURE ELEMENT IN SEQUENCE	Basic	A FARE STRUCTURE ELEMENT as a part of a VALIDABLE ELEMENT, including its possible order in the sequence of FARE STRUCTURE ELEMENTs forming that VALIDABLE ELEMENT, and its possible quantitative limitation.
FARE STRUCTURE ELEMENT PRICE	Basic	A set of all possible price features of a FARE STRUCTURE ELEMENT: default total price, discount in value or percentage etc.
FARE STRUCTURE FACTOR	Basic	A factor influencing access rights definition or calculation of prices.
FARE TABLE	Basic	A grouping of prices (specialization of PRICE GROUP) that may be associated with all or any of DISTANCE MATRIX ELEMENT, FARE STRUCTURE ELEMENT GEOGRAPHICAL INTERVAL, GROUP OF ACCESS RIGHT PARAMETER, CLASS OF USE, OPERATOR, VEHICLE MODE, FARE PRODUCT.
FARE UNIT	Basic	A unit associated with a FARE STRUCTURE FACTOR.
FARE ZONE	Basic	A specialization of TARIFF ZONE to include FARE SECTIONs.
FREQUENCY OF USE	Basic	The limits of usage frequency for a FARE PRODUCT (or one of its components) or a SALES OFFER PACKAGE during a specific VALIDITY PERIOD. There may be different tariffs depending on how often the right is consumed during the period.
FULFILMENT METHOD	Basic	The means by which the ticket is delivered to the CUSTOMER, e.g. online, collection, etc.
FULFILMENT	Basic	A set of all possible price features of a FULFILMENT METHOD

METHOD PRICE		default total price etc.
GENERIC PARAMETER ASSIGNMENT	Basic	A VALIDITY PARAMETER ASSIGNMENT specifying generic access rights for a class of products (e.g. a time band limit - 7 to 10 a.m. - for trips made with a student pass).
GEOGRAPHICAL INTERVAL	Basic	A geographical interval specifying access rights for the FARE STRUCTURE ELEMENTs within the range of this interval: 0-5 km, 4-6 zones etc.
GEOGRAPHICAL INTERVAL PRICE	Basic	A set of all possible price features of a GEOGRAPHICAL INTERVAL: default total price etc.
GEOGRAPHICAL STRUCTURE FACTOR	Basic	The value of a GEOGRAPHICAL INTERVAL or a DISTANCE MATRIX ELEMENT expressed by a GEOGRAPHICAL UNIT.
GEOGRAPHICAL UNIT	Basic	A unit for calculating geographical graduated fares.
GEOGRAPHICAL UNIT PRICE	Basic	A set of all possible price features of a GEOGRAPHICAL UNIT: default total price etc.
GROUP OF DISTANCE MATRIX ELEMENTs	Basic	A grouping of DISTANCE MATRIX ELEMENTS. May be used to provide reusable Origin / Destination pairs (and associate them with a PRICE).
GROUP OF DISTRIBUTION CHANNELs	Further	A grouping of DISTRIBUTION CHANNELs.

GROUP OF SALES OFFER PACKAGEs	Further	A grouping of SALES OFFER PACKAGEs.
GROUP TICKET	Basic	The number and characteristics of persons entitled to travel in addition to the holder of an access right.
INTERCHANGING	Basic	Limitations on making changes within a trip.
LIMITING RULE	Basic	Rule for limiting the results of a price calculation.
LUGGAGE ALLOWANCE	Further	The number and characteristics (weight, volume) of luggage that a holder of an access right is entitled to carry.
MINIMUM STAY	ADDITIONAL	Details of any minimum stay at the destination required to use the product.
MONTH VALIDITY OFFSET	ADDITIONAL	Days before (negative) or after (positive) the start of the month that a product with a calendar period driven activation becomes valid.
NETWORK VALIDITY PARAMETER	Basic	A type of VALIDITY PARAMETER related to the network structure.
OFFERED TRAVEL SPECIFICATION	ADDITIONAL	A set of parameters giving details of the consumption of access rights associated with an offer or a purchase. (e.g. origin and destination of a travel, class of travel, etc.).
ORGANISATIONAL VALIDITY PARAMETER	Basic	A type of VALIDITY PARAMETER related to organisational issues.

PARKING CHARGE BAND	ADDITIONAL	Parking charges that describe the cost of using a PARKING or PARKING AREA for a given period.
PARKING PRICE	ADDITIONAL	A specialisation of FARE PRICE that defines the price of a PARKING CHARGE BAND.
PARKING TARIFF	ADDITIONAL	A set of parking CHARGE BANDs that describe the cost of using a PARKING or PARKING AREA.
FARE CONTRACT	ADDITIONAL	A contract with a particular (but possibly anonymous) customer, ruling the consumption of transport services (and joint services). A FARE CONTRACT may be designed for a fixed SALES OFFER PACKAGE (e.g. ticket) or to allow successive purchases of SALES OFFER PACKAGEs.
FARE CONTRACT ENTRY	ADDITIONAL	A log entry describing an event referring to the life of a FARE CONTRACT: initial contracting, sales, validation entries, etc. A subset of a FARE CONTRACT ENTRY is often materialised on a TRAVEL DOCUMENT.
PENALTY POLICY	ADDITIONAL	Policy regarding different aspects of penalty charges, for example repeated entry at the same station, not having a ticket etc.
Post-paid ticketing	Basic	The user is charged sometime after using the transport service
PRE-ASSIGNED FARE PRODUCT	Basic	A FARE PRODUCT consisting of one or several VALIDABLE ELEMENTs, specific to a CHARGING MOMENT.
Prepaid ticketing	Basic	The user is charged for either a fare product (ticket) or a deposit prior to riding (detailed description of process see below).

Price	Basic	Value of fare or tariff.
PRICE GROUP	Basic	A grouping of prices, allowing the grouping of numerous possible consumption elements into a limited number of price references, or to apply grouped increase, in value or percentage.
PRICE UNIT	Basic	A unit to express prices: amount of currency, abstract fare unit, ticket unit or token etc.
PRICEABLE OBJECT	Basic	An element which may have a FARE PRICE.
PRICING PARAMETER SET	Basic	A set of parameters controlling pricing calculations.
PRICING RULE	Basic	A rule used for the calculation of FARE PRICE, determined either by a set of parameters to be applied to a reference price or by a more complex algorithm.
PRICING SERVICE	Basic	A web service used to provide prices dynamically at time of booking or purchase.
PRODUCT VALIDITY PARAMETER	Basic	A type of VALIDITY PARAMETER linked to fare products and/or their distribution.
PURCHASE WINDOW	Additional	Period in which the product must be purchased.

QUALITY STRUCTURE FACTOR	Basic	A factor influencing access rights definition or calculation of prices, based on the quality: traffic congestion threshold, early/late reservation etc.
QUALITY STRUCTURE FACTOR PRICE	Basic	A set of all possible price features of a QUALITY STRUCTURE FACTOR, e.g. default total price etc.
REFUNDING	Further	Whether and how the product may be refunded.
REPLACING	Further	whether and how the access right may be replaced.
REQUESTED TRAVEL SPECIFICATION	ADDITIONAL	A set of parameters giving details of an intended consumption of access rights requested by a TRANSPORT CUSTOMER (e.g. origin and destination of a travel, class of travel, etc.
RESELLING	Further	Common resale conditions (i.e. for exchange or refund) attached to the product.
RESERVING	Further	indicating whether the access right requires reservation.
RESIDENTIAL ELIGIBILITY	ADDITIONAL	Whether a specific TRANSPORT CUSTOMER is eligible for a FARE PRODUCT with a specific RESIDENTIAL QUALIFICATION as a validity parameter.
RESIDENTIAL QUALIFICATION	Further	A parameter providing an authorisation to consider a user as being characterised by a USER PROFILE.
RETAIL CONSORTIUM	ADDITIONAL	A group of ORGANISATIONs formally incorporated as a retailer of fare products.
RETAIL DEVICE	ADDITIONAL	A retail device used to sell fare products. Its identity can be used to

		record fulfilment and support security processes.
RETAIL DEVICE SECURITY LISTING	ADDITIONAL	The presence of a RETAIL DEVICE on a SECURITY LIST..
ROUND TRIP	Basic	Properties relating to single or return trip use of an access right.
ROUNDING	Basic	Parameters directing the rounding of values that are the result of calculations.
ROUNDING STEP	Basic	A rounding step to use to round a range of values. If step stable rounding is used, any value larger than the step key and smaller that the next step key should be rounded to this value.
ROUTING	Basic	Limitations on routing of an access right.
ROUTING VALIDITY PARAMETER	Basic	A type of VALIDITY PARAMETER linked to specific routing.
RULE STEP RESULT	Basic	Interim amounts for any pricing rules applied when deriving one price from another, for example VAT amount charged.
SALE DISCOUNT RIGHT	Further	A FARE PRODUCT allowing a customer to benefit from discounts when purchasing SALES OFFER PACKAGEs.
SALE TRANSACTION	ADDITIONAL	A SALE of a FIXED PACKAGE or a SALE of a RELOADABLE PACKAGE.
SALES NOTICE ASSIGNMENT	Basic	The assignment of a NOTICE to a SALES OFFER PACKAGE or a GROUP OF SALES OFFER PACKAGEs.

SALES OFFER ENTITLEMENT GIVEN	Further	Parameter indicating whether a particular SALES OFFER PACKAGE provides an entitlement to buy or use an access right.
SALES OFFER ENTITLEMENT REQUIRED	Further	Parameter indicating whether a particular SALES OFFER PACKAGE requires an entitlement to by or use an access right.
SALES OFFER PACKAGE	Basic	A package to be sold as a whole, consisting of one or several FARE PRODUCTs materialised thanks to one or several TRAVEL DOCUMENTs. The FARE PRODUCTs may be either directly attached to the TRAVEL DOCUMENTs, or may be reloadable on the TRAVEL DOCUMENTs.
SALES OFFER PACKAGE ELEMENT	Basic	The assignment of a FARE PRODUCT to a TYPE OF TRAVEL DOCUMENT in order to define a SALES OFFER PACKAGE, realised as a fixed assignment (printing, magnetic storage etc.) or by the possibility for the FARE PRODUCT to be reloaded on the TYPE OF TRAVEL DOCUMENT.
SALES OFFER PACKAGE PRICE	Basic	A set of all possible price features of a SALES OFFER PACKAGE: default total price etc.
SALES OFFER PACKAGE SUBSTITUTION	ADDITIONAL	Information on the preferred substitution of packages with other packages if a quota restricted product is no longer available.
SALES TRANSACTION FRAME	ADDITIONAL	A set of SALES TRANSACTION data elements (SALES TRANSACTIONs, etc.) to which the same VALIDITY CONDITIONs have been assigned.
SCOPING	Basic	Grouping of assignments to elements.

VALIDITY PARAMETER		
SERIES CONSTRAINT	ADDITIONAL	An extension of a DISTANCE MATRIX ELEMENT (a cell of an origin-destination matrix for TARIFF ZONEs or SCHEDULED STOP POINTs) expressing a fare distance for the corresponding trip (value in km, number of fare units etc.), constrained to specific routes.SERIES CONSTRAINTs are mainly used for rail fares.
SERIES CONSTRAINT PRICE	ADDITIONAL	A set of all possible price features of a SERIES CONSTRAINT: default total price etc.
SERVICE ACCESS RIGHT	Basic	An immaterial marketable element dedicated to accessing some services.
SERVICE VALIDITY PARAMETER	Basic	A type of VALIDITY PARAMETER related to service characteristics (e.g. class).
SPECIFIC PARAMETER ASSIGNMENT	Basic	A VALIDITY PARAMETER ASSIGNMENT specifying practical parameters during a TRAVEL SPECIFICATION, within a given fare structure (e.g. the origin or destination zone in a zone-counting system).
START TIME AT STOP POINT	ADDITIONAL	A time at which a Fare time band (time band peak, off peak) is deemed to begin for trips starting at a particular station.
STEP LIMIT	Basic	Geographical parameter limiting the access rights by counts of stops, sections or zones.
Subscription	Further	Purchase of a product by staged payments made on a regular basis.

SUBSCRIBING	Further	Parameter specifying conditions relating to paying for a product by subscription.
SUPPLEMENT PRODUCT	Further	A PRE-ASSIGNED FARE PRODUCT that will provide additional right when used with (as a complement of) another (reserved seat, second to first class upgrade, etc.). SUPPLEMENT PRODUCT also usually means supplement price.
SUSPENDING	ADDITIONAL	Parameter specifying conditions relating to suspending use of a season pass.
TARIFF	Basic	A particular tariff, described by a combination of parameters. From a planner perspective: the set of discrete elements to be used according to the fare calculation rules to calculate the fare.
TEMPORAL VALIDITY PARAMETER	Basic	Grouping of temporal validity parameters.
THIRD PARTY PRODUCT	Further	A FARE PRODUCT that is marketed together with a Public Transport FARE PRODUCT.
TIME INTERVAL	Basic	A time-based interval specifying access rights for the FARE STRUCTURE ELEMENTs within the range of this interval: 0-1 hour, 1-3 days etc.
TIME INTERVAL PRICE	Basic	A set of all possible price features of a TIME INTERVAL, e.g. default total price etc.
TIME STRUCTURE FACTOR	Basic	The value of a TIME INTERVAL expressed by a TIME UNIT.

TIME UNIT	Basic	A unit for calculating time-based graduated fares.
TIME UNIT PRICE	Basic	A set of all possible price features of a TIME UNIT: default total price etc.
TRANSFERABILITY	Further	The number and characteristics of persons entitled to use the public transport service instead of the original customer.
TRAVEL DOCUMENT	ADDITIONAL	A particular physical support (ticket, card, etc.) to be held by a customer, allowing the right to travel or to consume joint-services, to proof a payment (including possible discount rights), to store a subset of the CONTRACT liabilities or a combination of those.
TRAVEL DOCUMENT SECURITY LISTING	ADDITIONAL	The presence of a TRAVEL DOCUMENT on a SECURITY LIST.
TRAVEL SPECIFICATION	ADDITIONAL	The recording of a specification by a customer of parameters giving details of an intended consumption (e.g. origin and destination of a travel).
TYPE OF CONCESSION	Basic	A classification of USER PROFILE by type of person eligible to use it.
TYPE OF CUSTOMER ACCOUNT CONTRACT	ADDITIONAL	A classification of CUSTOMER ACCOUNT CONTRACT
TYPE OF FARE PRODUCT	Basic	A classification of FARE PRODUCTs.

TYPE OF FARE CONTRACT	Basic	A classification of FARE CONTRACT.
TYPE OF FARE CONTRACT ENTRY	Basic	A classification of FARE CONTRACT ENTRies.
TYPE OF RETAIL DEVICE	ADDITIONAL	A classification of RETAIL DEVICEs.
TYPE OF SALES OFFER PACKAGE	Basic	A classification of SALES OFFER PACKAGEs.
TYPE OF TARIFF	Basic	A classification of TARIFFs to express the different classes of fares.
TYPE OF TRAVEL DOCUMENT	Basic	A classification of TRAVEL DOCUMENTs expressing their general functionalities and local functional characteristics specific to the operator. Types of TRAVEL DOCUMENTs like e.g. throw-away ticket, throw-away ticket unit, value card, electronic purse allowing access, public transport credit card etc. may be used to define these categories.
TYPE OF USAGE PARAMETER	Basic	A classification of USAGE PARAMETERs to express the nature of parameters.
USAGE DISCOUNT RIGHT	Basic	A FARE PRODUCT allowing a customer to benefit from discounts when consuming VALIDABLE ELEMENTs.
USAGE PARAMETER	Basic	A parameter used to specify the use of a SALES OFFER PACKAGE or a FARE PRODUCT.
USAGE PARAMETER	Basic	A set of all possible price features of a USAGE PARAMETER:

PRICE		discount in value or percentage etc.
USAGE VALIDITY PERIOD	Basic	A time limitation for validity of a FARE PRODUCT or a SALES OFFER PACKAGE. It may be composed of a standard duration (e.g. 3 days, 1 month) and/or fixed start/end dates and times.
USER PROFILE	Basic	The social profile of a passenger, based on age group, education, profession, social status, sex etc., often used for allowing discounts: 18-40 years old, graduates, drivers, unemployed, women etc.
USER PROFILE ELIGIBILITY	ADDITIONAL	Whether a specific TRANSPORT CUSTOMER is eligible for a FARE PRODUCT with a specific USER PROFILE as a validity parameter.
VALIDABLE ELEMENT	Basic	A sequence or set of FARE STRUCTURE ELEMENTs, grouped together to be validated in one go.
VALIDABLE ELEMENT PRICE	Basic	A set of all possible price features of a VALIDABLE ELEMENT: default total price, discount in value or percentage etc.
VALIDITY PARAMETER ASSIGNMENT	Basic	An ACCESS RIGHT PARAMETER ASSIGNMENT relating a fare collection parameter to a theoretical FARE PRODUCT (or one of its components) or a SALES OFFER PACKAGE.

Annex B TODO

(Normative, future developments)**215327**

Facility code list

This section lists enumerations relevant for fares – For future development

- **AccessibilityInfoFacilityEnumeration**
- **CateringFacilityEnumeration**
- **ScopeOfTicketEnumeration**

Annex C
(informative)**315428**

Related EU Directive

ITS Directive Delegated Regulation: EU-wide Multimodal Travel Information Services

The ITS Directive is a framework for the coordinated and effective deployment and use of intelligent transport systems. Its main objectives are:

- Establish a framework for coordinated and effective deployment and use of ITS within MS and across borders
- Develop specifications necessary to ensure the compatibility, interoperability and continuity for the deployment and operational use of ITS for priority actions

The ITS directive defines the following Priority Action.

a) the provision of EU-wide multimodal travel information services

b) the provision of EU-wide real-time traffic information services;

c) data and procedures for the provision, where possible, of road safety related minimum universal traffic information free of charge to users;

d) the harmonised provision for an interoperable EU-wide eCall

e) the provision of information services for safe and secure parking places for trucks and commercial vehicles;

f) the provision of reservation services for safe and secure parking places for trucks and commercial vehicles.

This document is designed to answer to the expectations of Priority Action a) which is expecting the definition of the necessary requirements to make EU-wide multimodal travel information services accurate and available across borders based on one or more of the following types of provisions:

- functional provisions describing the roles of various stakeholders and the information flow between them
- technical provisions providing for the technical means to fulfil functional provisions
- organisational provisions describing procedural obligations of the various stakeholders
- service provisions describing the levels of services and their content

The follow are two excerpt article of this commission delegated regulation (eu) of 31.5.2017

> *(17) To ensure the optimal use and full interoperability of the aforementioned standards between Member States, **a common minimum profile that identifies the different key elements of the standard should be established** and used within national access points. Member States' national profiles must be based on a common minimum European profile when it exists.*
>
> *(25) Through the Connecting Europe Facility the Commission will support different technical requirements established within this Regulation through a programme support action19, notably the establishment of the national access point, the conversion to prescribed data exchange standards and **the use of common minimum profiles** within national access points and the linkage of travel information services where relevant.*

References:

- Europe on the Move: Commission takes action for clean, competitive and connected mobility

https://ec.europa.eu/transport/modes/road/news/2017-05-31-europe-on-the-move_en

- Commission Delegated Regulation on the provision of EU-wide multimodal travel information service

http://ec.europa.eu/info/law/better-regulation/initiatives/c-2017-3574_en